Discrete Data Analysis and Lab

Lecture notes, Chapters 1–11

Based on A. Agresti, *Categorical Data Analysis*, 2nd ed. (Wiley, 2002)

Course map

The course develops methods for categorical responses, mainly from the biomedical and social sciences, and shows how to carry them out with statistical software in the weekly lab. A single generalized linear model framework links logistic regression and loglinear models for discrete data with normal regression for continuous data. The syllabus topics correspond to the chapters of these notes as follows.

	Syllabus topic	Chapters
1.	Introduction	1
2.	Contingency table analysis: chi-squared tests, measures of association	2, 3
3.	Generalized linear models	4
4.	Logistic model	5, 6, 7
5.	Loglinear model: three-way and four-way contingency tables	8, 9
6.	Matched pairs data	10
7.	Repeated measurement data analysis: generalized estimating equations (GEE)	11

Contents

Course map 1

Chapter 1. Distributions and Inference for Categorical Data 10

1.1 Categorical response data 10
1.1.1 Response–explanatory variable distinction . . . 10
1.1.2 Nominal–ordinal scale distinction . . . 10
1.1.3 Continuous–discrete variable distinction . . . 10
1.1.4 Quantitative–qualitative variable distinction . . . 10
1.1.5 Organization of this book . . . 10

1.2 Distributions for categorical data 11
1.2.1 Binomial distribution . . . 11
1.2.2 Multinomial distribution . . . 11
1.2.3 Poisson distribution . . . 12
1.2.4 Overdispersion . . . 12
1.2.5 Connection between Poisson and multinomial distributions . . . 13

1.3 Statistical inference for categorical data 13
1.3.1 Likelihood functions and maximum likelihood estimation . . . 13
1.3.2 Likelihood function and ML estimate for binomial parameter . . . 14
1.3.3 Wald–likelihood ratio–score test triad . . . 14
1.3.4 Constructing confidence intervals . . . 16

1.4 Statistical inference for binomial parameters 16
1.4.1 Tests about a binomial parameter . . . 16
1.4.2 Confidence intervals for a binomial parameter . . . 16
1.4.3 Proportion of vegetarians example . . . 17

1.5 Statistical inference for multinomial parameters 17
1.5.1 Estimation of multinomial parameters . . . 18
1.5.2 Pearson statistic for testing a specified multinomial . . . 18
1.5.3 Example: testing Mendel's theories . . . 18
1.5.4 Chi-squared theoretical justification . . . 18
1.5.5 Likelihood-ratio chi-squared . . . 19
1.5.6 Testing with estimated expected frequencies . . . 19

Chapter 2. Describing Contingency Tables 21

2.1 Probability structure for contingency tables 21
2.1.1 Contingency tables and their distributions . . . 21
2.1.2 Sensitivity and specificity . . . 21
2.1.3 Independence of categorical variables . . . 22
2.1.4 Poisson, binomial, and multinomial sampling . . . 22
2.1.5 Seat belt example . . . 23
2.1.6 Types of studies . . . 23

2.2 Comparing two proportions 24
2.2.1 Difference of proportions . . . 24
2.2.2 Relative risk . . . 24

2.2.3 Odds ratio . . . 24
2.2.4 Properties of the odds ratio . . . 24
2.2.5 Aspirin and heart attacks revisited . . . 25
2.2.6 Case–control studies and the odds ratio . . . 25
2.2.7 Relationship between odds ratio and relative risk . . . 26

2.3 Partial association in stratified 2×2 tables 26
2.3.1 Partial tables . . . 26
2.3.2 Death penalty example . . . 26
2.3.3 Conditional and marginal odds ratios . . . 28
2.3.4 Marginal versus conditional independence . . . 28
2.3.5 Homogeneous association . . . 29

2.4 Extensions for $I \times J$ tables 30
2.4.1 Odds ratios in $I \times J$ tables . . . 30
2.4.2 Summary measures of association . . . 31
2.4.3 Ordinal trends: concordant and discordant pairs . . . 31
2.4.4 Ordinal measure of association: gamma . . . 32
2.4.5 Gamma for job satisfaction example . . . 33

Chapter 3. Inference for Contingency Tables 34

3.1 Confidence intervals for association parameters 34
3.1.1 Interval estimation of odds ratios . . . 34
3.1.2 Aspirin and myocardial infarction example . . . 34
3.1.3 Interval estimation of difference of proportions . . . 34
3.1.4 Interval estimation of relative risk . . . 35
3.1.5 Deriving standard errors with the delta method* . . . 35
3.1.6 Delta method applied to sample logit* . . . 35
3.1.7 Delta method for log odds ratio* . . . 36

3.2 Testing independence in two-way contingency tables 37
3.2.1 Pearson and likelihood-ratio chi-squared tests . . . 37
3.2.2 Education and religious fundamentalism example . . . 38

3.3 Following-up chi-squared tests 38
3.3.1 Pearson and standardized residuals . . . 38
3.3.2 Education and religious fundamentalism revisited . . . 39
3.3.3 Partitioning chi-squared . . . 39
3.3.4 Origin of schizophrenia example . . . 40
3.3.5 Rules for partitioning . . . 40
3.3.6 Limitations of chi-squared tests . . . 40
3.3.7 Why consider independence? . . . 40

3.4 Two-way tables with ordered classifications 41
3.4.1 Linear trend alternative to independence . . . 41
3.4.2 Job satisfaction example revisited . . . 41
3.4.3 Monotone trend alternatives to independence . . . 41
3.4.4 Extra power with ordinal tests . . . 42
3.4.5 Choice of scores . . . 42
3.4.6 Trend tests for $I \times 2$ and $2 \times J$ tables . . . 42
3.4.7 Nominal–ordinal tables . . . 42

3.5 Small-sample tests of independence **42**
3.5.1 Fisher's exact test for 2×2 tables . . . 42
3.5.2 Fisher's tea drinker . . . 43
3.5.3 Two-sided P-values for Fisher's exact test . . . 43
3.5.4 Discreteness and conservatism issues . . . 44

3.6 Extensions for multiway tables and nontabulated responses **44**
3.6.1 Categorical data need not be contingency tables . . . 44

Chapter 4. Introduction to Generalized Linear Models **45**

4.1 Generalized linear model **45**
4.1.1 Components of generalized linear models . . . 45
4.1.2 Binomial logit models for binary data . . . 45
4.1.3 Poisson loglinear models for count data . . . 45
4.1.4 Generalized linear models for continuous responses . . . 45
4.1.5 Deviance . . . 46
4.1.6 Advantages of the GLM formulation . . . 46

4.2 Generalized linear models for binary data **46**
4.2.1 Linear probability model . . . 47
4.2.2 Snoring and heart disease example . . . 47
4.2.3 Logistic regression model . . . 47
4.2.4 Binomial GLM for 2×2 contingency tables . . . 48
4.2.5 Probit and inverse cdf link functions . . . 48

4.3 Generalized linear models for counts **48**
4.3.1 Poisson loglinear models . . . 48
4.3.2 Horseshoe crab mating example . . . 48
4.3.3 Overdispersion for Poisson GLMs . . . 49
4.3.4 Negative binomial GLMs . . . 49
4.3.5 Poisson regression for rates . . . 50
4.3.6 Poisson GLM of independence in $I \times J$ contingency tables . . . 50

4.4 Moments and likelihood for generalized linear models **50**
4.4.1 Mean and variance functions for the random component . . . 51
4.4.2 Mean and variance functions for Poisson and binomial . . . 51
4.4.3 Systematic component and link function . . . 51
4.4.4 Likelihood equations for a GLM . . . 52
4.4.5 Likelihood equations for binomial GLMs . . . 52
4.4.6 Asymptotic covariance matrix of model parameter estimators . . . 53
4.4.7 Likelihood equations and covariance matrix for Poisson loglinear model . . . 54

4.5 Inference for generalized linear models **54**
4.5.1 Deviance and goodness of fit . . . 54
4.5.2 Deviance for Poisson models . . . 55
4.5.3 Deviance for binomial models: grouped and ungrouped data . . . 55
4.5.4 Likelihood-ratio model comparison using the deviance . . . 56
4.5.5 Residuals for GLMs . . . 56

4.6 Fitting generalized linear models **57**
4.6.1 Newton–Raphson method . . . 57
4.6.2 Fisher scoring method . . . 58

4.6.3 ML as iterative reweighted least squares . . . 58
4.6.4 Simplifications for canonical links . . . 59

Chapter 5. Logistic Regression 60

5.1 Interpreting parameters in logistic regression 60
5.1.1 Interpreting β: odds, probabilities, and linear approximations . . . 60
5.1.2 Looking at the data . . . 61
5.1.3 Horseshoe crabs revisited . . . 61
5.1.4 Logistic regression with retrospective studies . . . 62

5.2 Inference for logistic regression 63
5.2.1 Types of inference . . . 63
5.2.2 Inference for horseshoe crab data . . . 64
5.2.3 Checking goodness of fit: ungrouped and grouped data . . . 64
5.2.4 Goodness of fit of model for horseshoe crabs . . . 64
5.2.5 Checking goodness of fit with ungrouped data by grouping . . . 65

5.3 Logit models with categorical predictors 66
5.3.1 ANOVA-type representation of factors . . . 66
5.3.2 Dummy variables in logit models . . . 66
5.3.3 Alcohol and infant malformation example revisited . . . 67
5.3.4 Linear logit model for $I \times 2$ tables . . . 67
5.3.5 Cochran–Armitage trend test . . . 67

5.4 Multiple logistic regression 69
5.4.1 Logit models for multiway contingency tables . . . 69
5.4.2 AIDS and AZT example . . . 70
5.4.3 Goodness of fit as a likelihood-ratio test . . . 70
5.4.4 Horseshoe crab example revisited . . . 71
5.4.5 Model comparison . . . 71
5.4.6 Quantitative treatment of ordinal predictor . . . 71
5.4.7 Standardized and probability-based interpretations . . . 72

5.5 Fitting logistic regression models 72
5.5.1 Likelihood equations . . . 72
5.5.2 Asymptotic covariance matrix of parameter estimators . . . 73
5.5.3 Distribution of probability estimators . . . 73
5.5.4 Newton–Raphson method applied to logistic regression . . . 73
5.5.5 Convergence and existence of finite estimates . . . 74

Chapter 6. Building and Applying Logistic Regression Models 76

6.1 Strategies in model selection 76
6.1.1 Horseshoe crab example revisited . . . 76
6.1.2 Stepwise procedures . . . 76
6.1.3 Backward elimination for the horseshoe crab example . . . 76
6.1.4 AIC, model selection, and the "correct" model . . . 77
6.1.5 Using causal hypotheses to guide model building . . . 78
6.1.6 New model-building strategies for data mining . . . 78

6.2 Logistic regression diagnostics 78

6.2.1 Pearson, deviance, and standardized residuals . . . 78
6.2.2 Heart disease example . . . 79
6.2.3 Graduate admissions example . . . 80
6.2.4 Influence diagnostics for logistic regression . . . 80
6.2.5 Summarizing predictive power: R and R-squared measures . . . 81
6.2.6 Summarizing predictive power: classification tables and ROC curves . . . 81

6.3 Inference about conditional associations in $2 \times 2 \times K$ tables 82
6.3.1 Using logit models to test conditional independence . . . 83
6.3.2 Cochran–Mantel–Haenszel test of conditional independence . . . 83
6.3.3 Multicenter clinical trial example . . . 84
6.3.4 Estimation of common odds ratio . . . 84
6.3.5 Testing homogeneity of odds ratios . . . 85
6.3.6 Summarizing heterogeneity in odds ratios . . . 86

Chapter 7. Logit Models for Multinomial Responses 87

7.1 Nominal responses: baseline-category logit models 87
7.1.1 Baseline-category logits . . . 87
7.1.2 Alligator food choice example . . . 87
7.1.3 Estimating response probabilities . . . 88

7.2 Ordinal responses: cumulative logit models 89
7.2.1 Cumulative logits . . . 89
7.2.2 Proportional odds model . . . 89
7.2.3 Mental impairment example . . . 90
7.2.4 More complex models . . . 91

7.3 Ordinal responses: cumulative link models 91
7.3.1 Types of cumulative links . . . 92
7.3.2 Estimation for cumulative link models . . . 92
7.3.3 Life table example . . . 92

Chapter 8. Loglinear Models for Contingency Tables 94

8.1 Loglinear models for two-way tables 94
8.1.1 Independence model . . . 94
8.1.2 Interpretation of parameters . . . 94
8.1.3 Saturated model . . . 95
8.1.4 Alternative parameter constraints . . . 96
8.1.5 Multinomial models for cell probabilities . . . 96

8.2 Loglinear models for independence and interaction in three-way tables 96
8.2.1 Types of independence . . . 97
8.2.2 Homogeneous association and three-factor interaction . . . 98
8.2.3 Interpreting model parameters . . . 98
8.2.4 Alcohol, cigarette, and marijuana use example . . . 99

8.3 Inference for loglinear models 100
8.3.1 Chi-squared goodness-of-fit tests . . . 100
8.3.2 Inference about conditional associations . . . 101

8.4 Loglinear models for higher dimensions 101

8.4.1 Four-way contingency tables . . . 101
8.4.2 Automobile accident example . . . 102
8.4.3 Large samples and statistical versus practical significance . . . 103
8.4.4 Dissimilarity index . . . 103

8.5 Loglinear–logit model connection 104
8.5.1 Using logit models to interpret loglinear models . . . 104
8.5.2 Auto accident example revisited . . . 104
8.5.3 Correspondence between loglinear and logit models . . . 104
8.5.4 Generalized loglinear model . . . 105

8.6 Loglinear model fitting: likelihood equations 106
8.6.1 Minimal sufficient statistics . . . 106
8.6.2 Likelihood equations for loglinear models . . . 106
8.6.3 Birch's results for loglinear models . . . 107
8.6.4 Direct versus iterative calculation of fitted values . . . 107
8.6.5 Connection between multinomial and Poisson loglinear models . . . 108

Chapter 9. Building and Extending Loglinear–Logit Models 110

9.1 Association graphs and collapsibility 110
9.1.1 Association graphs . . . 110
9.1.2 Collapsibility in three-way contingency tables . . . 111
9.1.3 Collapsibility and logit models . . . 111
9.1.4 Collapsibility and association graphs for multiway tables . . . 112

9.2 Model selection and comparison 112
9.2.1 Considerations in model selection . . . 112
9.2.2 Model building for the Dayton student survey . . . 113
9.2.3 Loglinear model comparison statistics . . . 113
9.2.4 Partitioning chi-squared with model comparisons . . . 114
9.2.5 Identical marginal and conditional tests of independence . . . 114

9.3 Diagnostics for checking models 115
9.3.1 Residuals for loglinear models . . . 115
9.3.2 Student survey example revisited . . . 115
9.3.3 Correspondence between loglinear and logit residuals . . . 115

9.4 Modeling ordinal associations 116
9.4.1 Linear-by-linear association in two-way tables . . . 116
9.4.2 Corresponding logit model for adjacent responses . . . 117
9.4.3 Likelihood equations and model fitting . . . 118
9.4.4 Sex opinions example . . . 118
9.4.5 Directed ordinal test of independence . . . 118

9.5 Poisson regression for rates 119
9.5.1 Analyzing rates using loglinear models with offsets . . . 119
9.5.2 Modeling death rates for heart valve operations . . . 119

Chapter 10. Models for Matched Pairs 121

10.1 Comparing dependent proportions 121
10.1.1 Inference for dependent proportions . . . 121

10.1.2 Prime Minister approval rating example . . . 122
10.1.3 Increased precision with dependent samples . . . 122
10.1.4 Small-sample test comparing matched proportions . . . 122
10.1.5 Connection between McNemar and Cochran–Mantel–Haenszel tests . . . 123

10.2 Conditional logistic regression for binary matched pairs 123
10.2.1 Marginal versus conditional models for matched pairs . . . 123
10.2.2 A logit model with subject-specific probabilities . . . 124
10.2.3 Conditional ML inference for binary matched pairs . . . 124
10.2.4 Random effects in binary matched-pairs model . . . 125
10.2.5 Logistic regression for matched case–control studies . . . 126
10.2.6 Conditional ML for matched pairs with multiple predictors . . . 126
10.2.7 Marginal models and conditional models: extensions . . . 126

10.3 Marginal models for square contingency tables 127
10.3.1 Marginal models for ordinal classifications . . . 127
10.3.2 Premarital and extramarital sex example . . . 127
10.3.3 Marginal models for nominal classifications . . . 127
10.3.4 Migration example . . . 128

10.4 Symmetry, quasi-symmetry, and quasi-independence 128
10.4.1 Symmetry as logit and loglinear models . . . 128
10.4.2 Quasi-symmetry . . . 129
10.4.3 Quasi-independence . . . 131
10.4.4 Migration revisited . . . 131
10.4.5 Marginal homogeneity and quasi-symmetry . . . 131
10.4.6 Ordinal quasi-symmetry model . . . 132
10.4.7 Premarital and extramarital sex revisited . . . 132
10.4.8 Other ordinal models for square tables . . . 133

10.5 Measuring agreement between observers 133
10.5.1 Agreement: departures from independence . . . 133
10.5.2 Using quasi-independence to analyze agreement . . . 133
10.5.3 Quasi-symmetry and agreement modeling . . . 134
10.5.4 Kappa measure of agreement . . . 134
10.5.5 Weighted kappa: quantifying disagreement . . . 135
10.5.6 Extensions to multiple observers . . . 135

Chapter 11. Analyzing Repeated Categorical Response Data 136

11.1 Comparing marginal distributions: multiple responses 136
11.1.1 Binary marginal models and marginal homogeneity . . . 136
11.1.2 Crossover drug comparison example . . . 136
11.1.3 Modeling margins of a multicategory response . . . 136
11.1.4 Wald and generalized CMH score tests of marginal homogeneity . . . 137

11.2 Marginal modeling: maximum likelihood approach 137
11.2.1 Longitudinal mental depression example . . . 138
11.2.2 Modeling a repeated multinomial response . . . 138
11.2.3 Insomnia example . . . 138
11.2.4 Comparisons that control for initial response . . . 138

11.3 Marginal modeling: generalized estimating equations (GEE) approach 138

11.3.1 Generalized estimating equation methodology: basic ideas . . . 139
11.3.2 Longitudinal mental depression example . . . 139
11.3.3 GEE approach for multinomial responses: insomnia example . . . 140

11.4 Quasi-likelihood and its GEE multivariate extension: details 140
11.4.1 Properties of quasi-likelihood estimators . . . 140
11.4.2 GEE methodology: technical details . . . 142

11.5 Markov chains: transitional modeling 144
11.5.1 Transitional models . . . 144
11.5.2 First-order Markov chains . . . 145
11.5.3 Respiratory illness example . . . 146
11.5.4 Transitional models with explanatory variables . . . 146
11.5.5 Child's respiratory illness and maternal smoking . . . 147

Chapter 1. Distributions and Inference for Categorical Data

1.1 Categorical response data

Motivation. The measurement scale of a categorical response decides which models and summaries are meaningful.

A *categorical variable* has a measurement scale consisting of a set of categories (political philosophy: liberal / moderate / conservative; a mammogram diagnosis: normal, benign, suspicious, malignant).

1.1.1 Response–explanatory variable distinction

We distinguish a *response* variable from *explanatory* variables; these notes concern *categorical responses*, with explanatory variables of any type.

1.1.2 Nominal–ordinal scale distinction

Definition (Nominal, ordinal, interval)

(a) A categorical variable is *nominal* if its categories have no natural ordering (religious affiliation, mode of transport).
(b) It is *ordinal* if the categories are ordered but the distances between them are unknown (patient condition good/fair/serious/critical).
(c) An *interval* variable has numerical distances between any two values (blood pressure, income); it is a *ratio* variable if ratios are also meaningful.

The classification depends on how a variable is measured (education: public/private school, highest degree, years of schooling). In the hierarchy interval > ordinal > nominal, methods for a lower level can be used at a higher level (ignoring information) but not conversely.

1.1.3 Continuous–discrete variable distinction

All measurement is in practice discrete; the useful distinction is between variables taking many values and variables taking few values, which are the subject here.

1.1.4 Quantitative–qualitative variable distinction

Nominal variables are *qualitative*, interval variables *quantitative*. Ordinal variables sit in between, often with an underlying continuous variable; analysts exploit this by assigning *scores* or assuming a latent continuous distribution.

1.1.5 Organization of this book

The models replace normality by binomial, multinomial or Poisson distributions; the central families are *logistic regression* and *loglinear models*. Chapters 1–3 cover one- and two-way data; Chapters 4–9 generalized linear models; Chapters 10–13 clustered responses; Chapters 14–16 theory and history.

1.2 Distributions for categorical data

Motivation. Binomial, multinomial and Poisson play the role for counts that the normal plays for continuous responses.

1.2.1 Binomial distribution

Let $Y_1, \dots, Y_n$ be independent Bernoulli trials with $P(Y_i = 1) = \pi$. Then $Y = \sum_i Y_i \sim \text{bin}(n, \pi)$:

$$p(y) = \binom{n}{y} \pi^y (1-\pi)^{n-y}, \qquad y = 0, 1, \dots, n. \tag{1.1}$$

Proposition 1.1 (Binomial moments)

If $Y \sim \text{bin}(n, \pi)$ then $\mu = E(Y) = n\pi$, $\sigma^2 = \text{Var}(Y) = n\pi(1-\pi)$, and

$$\frac{E(Y-\mu)^3}{\sigma^3} = \frac{1-2\pi}{\sqrt{n\pi(1-\pi)}}.$$

In particular the distribution becomes symmetric (and, by the CLT, normal) as $n \to \infty$ for fixed π.

Proof.

Step 1. For one trial $E(Y_i) = E(Y_i^2) = \pi$, so $\text{Var}(Y_i) = \pi(1-\pi)$; by linearity and independence $E(Y) = n\pi$, $\text{Var}(Y) = n\pi(1-\pi)$.

Step 2. Third central moments of independent variables add (cumulants are additive), and $E(Y_i - \pi)^3 = (1-\pi)^3\pi - \pi^3(1-\pi) = \pi(1-\pi)(1-2\pi)$. Hence $E(Y-\mu)^3 = n\pi(1-\pi)(1-2\pi)$; dividing by $\sigma^3 = [n\pi(1-\pi)]^{3/2}$ gives the formula, which tends to 0. ■

Without independent, identical trials the binomial fails (e.g. sampling without replacement gives the hypergeometric, Section 3.5).

1.2.2 Multinomial distribution

Let each of n independent, identical trials fall in one of c categories, $y_{ij} = 1$ if trial i falls in category j, $n_j = \sum_i y_{ij}$ and $\pi_j = P(y_{ij} = 1)$. Then $(n_1, \dots, n_c)$ is *multinomial*:

$$p(n_1, \dots, n_{c-1}) = \frac{n!}{n_1!\, n_2! \cdots n_c!} \pi_1^{n_1} \pi_2^{n_2} \cdots \pi_c^{n_c}, \tag{1.2}$$

a $(c-1)$-dimensional distribution since $\sum_j n_j = n$; $c = 2$ gives the binomial.

Proposition 1.2 (Multinomial moments)

For the multinomial $(n, \boldsymbol{\pi})$ distribution,

$$E(n_j) = n\pi_j, \qquad \text{Var}(n_j) = n\pi_j(1-\pi_j), \qquad \text{Cov}(n_j, n_k) = -n\pi_j\pi_k \ (j \neq k),$$

and each n_j is marginally $\text{bin}(n, \pi_j)$.

Proof. [Supplement]

Step 1. n_j is a sum of n independent Bernoulli(π_j) indicators y_{ij}, so $n_j \sim \text{bin}(n, \pi_j)$ (Proposition 1.1).

Step 2. For $j \neq k$, $y_{ij}y_{ik} = 0$, so $\mathrm{Cov}(y_{ij}, y_{ik}) = -\pi_j\pi_k$; summing over independent trials gives $\mathrm{Cov}(n_j, n_k) = -n\pi_j\pi_k$. ■

In matrix form, $\mathrm{Cov}(\mathbf{n}) = n[\mathrm{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top]$.

1.2.3 Poisson distribution

Counts without a fixed upper limit are often modelled as *Poisson* with mean μ:

$$p(y) = \frac{e^{-\mu}\mu^y}{y!}, \qquad y = 0, 1, 2, \ldots \tag{1.3}$$

Its skewness is $1/\sqrt{\mu}$, so it approaches normality as $\mu \to \infty$.

Proposition 1.3 (Poisson moments and binomial limit)

(a) If $Y \sim \mathrm{Poisson}(\mu)$ then $E(Y) = \mathrm{Var}(Y) = \mu$.
(b) If $n \to \infty$ and $\pi \to 0$ with $n\pi \to \mu$, then $\mathrm{bin}(n, \pi)$ probabilities converge to $\mathrm{Poisson}(\mu)$ probabilities.

Proof. [Supplement]

Step (a). $E(Y) = \sum_{y\geq 1} e^{-\mu}\mu^y/(y-1)! = \mu$ and $E[Y(Y-1)] = \sum_{y\geq 2} e^{-\mu}\mu^y/(y-2)! = \mu^2$, so $\mathrm{Var}(Y) = \mu^2 + \mu - \mu^2 = \mu$.

Step (b). $\binom{n}{y}\pi^y(1-\pi)^{n-y} = \frac{(n\pi)^y}{y!} \cdot \frac{n(n-1)\cdots(n-y+1)}{n^y} \cdot (1-\pi)^{n-y}$. For fixed y the middle factor $\to 1$, and $(1-\pi)^{n-y} = \exp\{-n\pi + O(n\pi^2)\}(1-\pi)^{-y} \to e^{-\mu}$. ■

1.2.4 Overdispersion

Real counts often vary more than the binomial or Poisson allows (*overdispersion*), because the success probability or rate itself varies with unmeasured factors.

Proposition 1.4 (Mixing creates overdispersion)

Let Y given λ have mean $E(Y \mid \lambda)$ and variance $\mathrm{Var}(Y \mid \lambda)$, where λ is random. Then

$$E(Y) = E[E(Y \mid \lambda)], \qquad \mathrm{Var}(Y) = E[\mathrm{Var}(Y \mid \lambda)] + \mathrm{Var}[E(Y \mid \lambda)].$$

If $Y \mid \lambda \sim \mathrm{Poisson}(\lambda)$ with $E(\lambda) = \mu$, then $E(Y) = \mu$ and $\mathrm{Var}(Y) = \mu + \mathrm{Var}(\lambda) > \mu$ whenever λ is nondegenerate.

Proof. [Supplement]

Step 1. The first identity is the tower property. For the second, $\mathrm{Var}(Y) = E[\mathrm{Var}(Y \mid \lambda) + E(Y \mid \lambda)^2] - (E[E(Y \mid \lambda)])^2 = E[\mathrm{Var}(Y \mid \lambda)] + \mathrm{Var}[E(Y \mid \lambda)]$.

Step 2. In the Poisson case $E(Y \mid \lambda) = \mathrm{Var}(Y \mid \lambda) = \lambda$. ■

Mixtures of binomials likewise give extra-binomial variation (negative binomial, Section 4.3.4; random effects, Chapters 12–13).

1.2.5 Connection between Poisson and multinomial distributions

Theorem 1.5 (Poisson conditioned on the total is multinomial)

Let $Y_1, \dots, Y_c$ be independent Poisson with means $\mu_1, \dots, \mu_c$. Then $n = \sum_i Y_i \sim \text{Poisson}(\sum_i \mu_i)$, and conditional on n,

$$(Y_1, \dots, Y_c) \mid \textstyle\sum_i Y_i = n \;\sim\; \text{multinomial}(n, \{\pi_i\}), \qquad \pi_i = \mu_i / \textstyle\sum_j \mu_j.$$

Proof.

Step 1. The product of the pgfs $e^{\mu_i(s-1)}$ is $e^{(\sum \mu_i)(s-1)}$, the pgf of Poisson$(\sum \mu_i)$.

Step 2. For $\sum n_i = n$, the conditional probability is

$$\frac{\prod_i e^{-\mu_i}\mu_i^{n_i}/n_i!}{e^{-\sum_j \mu_j}(\sum_j \mu_j)^n/n!} = \frac{n!}{\prod_i n_i!}\prod_i \Big(\frac{\mu_i}{\sum_j \mu_j}\Big)^{n_i},$$

using $\sum_i n_i = n$ to distribute $(\sum_j \mu_j)^n$; this is (1.2). ■

Since the Poisson and multinomial likelihoods share the kernel in $\{\pi_i\}$, the two models usually give the same estimates.

1.3 Statistical inference for categorical data

Motivation. Maximum likelihood and the Wald, likelihood-ratio and score tests, with the intervals obtained by inverting them, form the toolkit for the whole course.

1.3.1 Likelihood functions and maximum likelihood estimation

Definition (Likelihood, ML estimate, information)

The *likelihood* $\ell(\beta)$ is the probability of the observed data as a function of β, and $L(\beta) = \log \ell(\beta)$. The *ML estimate* $\hat\beta$ maximizes ℓ, typically solving $\partial L/\partial\beta = 0$. The *information matrix* has (j,k) element

$$-E\Big(\frac{\partial^2 L(\boldsymbol\beta)}{\partial\beta_j\,\partial\beta_k}\Big). \tag{1.4}$$

Under regularity conditions $\text{Cov}(\hat{\boldsymbol\beta})$ is asymptotically the inverse information; greater curvature of L means smaller SEs.

Lemma 1.6 (Score identities)

Let $f(y;\beta)$ be a density or pmf such that differentiation under $\int dy$ (or $\sum_y$) is allowed twice. For one observation let $u(\beta) = \partial \log f(Y;\beta)/\partial\beta$. Then

$$E_\beta[u(\beta)] = 0, \qquad \text{Var}_\beta[u(\beta)] = -E_\beta\Big[\frac{\partial^2 \log f(Y;\beta)}{\partial\beta^2}\Big] =: i(\beta).$$

Proof. [Supplement]

Step 1. Differentiate $\int f\,dy = 1$: $0 = \int (\partial_\beta \log f)\, f\, dy = E[u]$.

Step 2. Differentiate again: $0 = \int [(\partial_\beta^2 \log f) f + (\partial_\beta \log f)^2 f]\, dy$, so $\text{Var}(u) = E[u^2] = -E[\partial_\beta^2 \log f]$. ■

Theorem 1.7 (Asymptotic normality of the ML estimator)

Let $Y_1, \dots, Y_n$ be i.i.d. $f(y;\beta_0)$ with β_0 interior to the parameter space, satisfying Lemma 1.6, $0 < i(\beta_0) < \infty$, and $|\partial_\beta^3 \log f(y;\beta)| \le M(y)$ near β_0 with $E\,M(Y) < \infty$. If $\hat\beta$ is a consistent root of the likelihood equation, then

$$\sqrt{n}(\hat\beta - \beta_0) \xrightarrow{d} N\big(0,\, 1/i(\beta_0)\big).$$

For a vector $\boldsymbol\beta$ the same holds with $i(\boldsymbol\beta_0)$ the per-observation information matrix: $\sqrt{n}(\hat{\boldsymbol\beta} - \boldsymbol\beta_0) \xrightarrow{d} N(\mathbf{0}, i(\boldsymbol\beta_0)^{-1})$, and in fact $\sqrt{n}(\hat{\boldsymbol\beta} - \boldsymbol\beta_0) = i(\boldsymbol\beta_0)^{-1}\, n^{-1/2} \nabla L(\boldsymbol\beta_0) + o_p(1)$.

Proof. [Supplement]

Step 1. Expanding $0 = L'(\hat\beta) = L'(\beta_0) + (\hat\beta - \beta_0)L''(\beta_0) + \tfrac{1}{2}(\hat\beta - \beta_0)^2 L'''(\beta^*)$, with β^* between $\hat\beta$ and β_0,

$$\sqrt{n}(\hat\beta - \beta_0) = \frac{n^{-1/2} L'(\beta_0)}{-n^{-1}L''(\beta_0) - \tfrac{1}{2}(\hat\beta - \beta_0)\, n^{-1} L'''(\beta^*)}.$$

Step 2. $L'(\beta_0)$ is a sum of i.i.d. terms with mean 0 and variance $i(\beta_0)$ (Lemma 1.6), so $n^{-1/2}L'(\beta_0) \xrightarrow{d} N(0, i(\beta_0))$.

Step 3. $-n^{-1}L''(\beta_0) \xrightarrow{p} i(\beta_0)$ (WLLN), and with probability $\to 1$, $|n^{-1}L'''(\beta^*)| \le n^{-1}\sum_i M(Y_i) = O_p(1)$, which times $\hat\beta - \beta_0 = o_p(1)$ vanishes. So the denominator $\xrightarrow{p} i(\beta_0)$.

Step 4. Slutsky gives $N(0, i(\beta_0))/i(\beta_0) = N(0, 1/i(\beta_0))$ and the stated representation. The vector case replaces L'' by the Hessian and division by matrix inversion. ∎

1.3.2 Likelihood function and ML estimate for binomial parameter

Constants such as $\binom{n}{y}$ do not affect the maximizer; only the *kernel* matters.

Proposition 1.8 (Binomial ML estimate and information)

For y successes in n trials, the log likelihood, score and information are

$$L(\pi) = y \log \pi + (n - y)\log(1 - \pi), \tag{1.5}$$

$$\frac{\partial L}{\partial \pi} = \frac{y}{\pi} - \frac{n - y}{1 - \pi} = \frac{y - n\pi}{\pi(1 - \pi)}, \tag{1.6}$$

$$-E\Big(\frac{\partial^2 L}{\partial \pi^2}\Big) = \frac{n}{\pi(1 - \pi)}. \tag{1.7}$$

The ML estimate is $\hat\pi = y/n$, with $E(\hat\pi) = \pi$ and $\sigma(\hat\pi) = \sqrt{\pi(1 - \pi)/n}$, the inverse square root of the information.

Proof.

Step 1. (1.6) vanishes iff $\pi = y/n$, and $\partial^2 L/\partial\pi^2 = -y/\pi^2 - (n - y)/(1 - \pi)^2 < 0$, so y/n is the global maximizer (for $y = 0$ or n, L is monotone with maximum at the boundary, again y/n).

Step 2. With $E(Y) = n\pi$: $E[Y/\pi^2 + (n - Y)/(1 - \pi)^2] = n/[\pi(1 - \pi)]$; and $E(\hat\pi) = \pi$, $\mathrm{Var}(\hat\pi) = \pi(1 - \pi)/n$. ∎

1.3.3 Wald–likelihood ratio–score test triad

For $H_0\colon \beta = \beta_0$ write $L_0 = L(\beta_0)$ and $L_1 = L(\hat\beta)$.

Definition (The three likelihood-based statistics)

(a) *Wald*: $z_W = (\hat\beta - \beta_0)/\mathrm{SE}$ with SE evaluated at $\hat\beta$; for a vector, $W = (\hat{\boldsymbol\beta} - \boldsymbol\beta_0)^\top[\widehat{\mathrm{Cov}}(\hat{\boldsymbol\beta})]^{-1}(\hat{\boldsymbol\beta} - \boldsymbol\beta_0)$.
(b) *Likelihood ratio (LR)*: $\Lambda = \ell_0/\ell_1 \le 1$ and $-2\log\Lambda = -2(L_0 - L_1)$.
(c) *Score*: with $u(\beta_0) = \partial L/\partial\beta|_{\beta_0}$ and $\iota(\beta_0) = -E[\partial^2 L/\partial\beta^2]|_{\beta_0}$, $z_S = u(\beta_0)/\iota(\beta_0)^{1/2}$; in the vector case $\mathbf{u}^\top\iota^{-1}\mathbf{u}$ at $\boldsymbol\beta_0$.

Wald uses L at $\hat\beta$, score its slope and curvature at β_0, and LR both values (Figure 1).

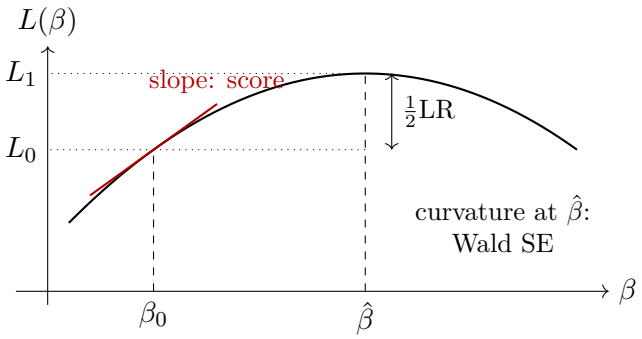


Figure 1: Information used by the three tests of $H_0\colon \beta = \beta_0$.

Lemma 1.9 (Normal quadratic forms)

(a) If $\mathbf{z} \sim N_q(\mathbf{0}, \Sigma)$ with Σ nonsingular, then $\mathbf{z}^\top\Sigma^{-1}\mathbf{z} \sim \chi^2_q$.
(b) If $\mathbf{z} \sim N_c(\mathbf{0}, P)$ with P symmetric idempotent of rank r, then $\mathbf{z}^\top\mathbf{z} \sim \chi^2_r$.

Proof. [Supplement]

Step (a). $\mathbf{w} = \Sigma^{-1/2}\mathbf{z} \sim N_q(\mathbf{0}, I)$ and $\mathbf{z}^\top\Sigma^{-1}\mathbf{z} = \sum_{k=1}^q w_k^2$.

Step (b). $P = Q\Delta Q^\top$ with Q orthogonal and Δ diagonal with r ones and zeros elsewhere (eigenvalues of an idempotent are 0 or 1). Then $\mathbf{w} = Q^\top\mathbf{z} \sim N(\mathbf{0}, \Delta)$, so $\mathbf{z}^\top\mathbf{z} = \mathbf{w}^\top\mathbf{w} = \sum_{k\le r} w_k^2 \sim \chi^2_r$. ■

Theorem 1.10 (Asymptotic null distributions of the triad)

Under the conditions of Theorem 1.7 and $H_0\colon \boldsymbol\beta = \boldsymbol\beta_0$ ($\boldsymbol\beta \in \mathbb{R}^q$), the Wald, LR and score statistics each converge in distribution to χ^2_q, and the difference between any two of them converges in probability to 0. For $q = 1$, z_W and z_S are asymptotically $N(0,1)$.

Proof. [Supplement] Write $\iota_1 = i(\boldsymbol\beta_0)$ and $\mathbf{s}_n = n^{-1/2}\nabla L(\boldsymbol\beta_0) \xrightarrow{d} N(\mathbf{0}, \iota_1)$ (CLT, Lemma 1.6).

Step 1 (score). The score statistic is $\nabla L^\top(n\iota_1)^{-1}\nabla L = \mathbf{s}_n^\top\iota_1^{-1}\mathbf{s}_n \xrightarrow{d} \chi^2_q$ by Lemma 1.9(a).

Step 2 (Wald). By Theorem 1.7, $\mathbf{z}_n = \sqrt{n}(\hat{\boldsymbol\beta} - \boldsymbol\beta_0) = \iota_1^{-1}\mathbf{s}_n + o_p(1)$, and the Wald statistic is $\mathbf{z}_n^\top\hat\iota_1\mathbf{z}_n$ with $\hat\iota_1 \xrightarrow{p} \iota_1$; so it equals $\mathbf{s}_n^\top\iota_1^{-1}\mathbf{s}_n + o_p(1)$.

Step 3 (LR). Expanding about $\hat{\boldsymbol\beta}$, where $\nabla L = \mathbf{0}$, $L(\boldsymbol\beta_0) = L(\hat{\boldsymbol\beta}) + \frac{1}{2}(\boldsymbol\beta_0 - \hat{\boldsymbol\beta})^\top\nabla^2 L(\tilde{\boldsymbol\beta})(\boldsymbol\beta_0 - \hat{\boldsymbol\beta})$ with $\tilde{\boldsymbol\beta}$ on the segment. Hence $-2(L_0 - L_1) = \mathbf{z}_n^\top[-n^{-1}\nabla^2 L(\tilde{\boldsymbol\beta})]\mathbf{z}_n = \mathbf{z}_n^\top\iota_1\mathbf{z}_n + o_p(1)$ (WLLN and the third-derivative bound), the Wald statistic plus $o_p(1)$.

Step 4. All three are $\mathbf{s}_n^\top\iota_1^{-1}\mathbf{s}_n + o_p(1) \xrightarrow{d} \chi^2_q$. For $q = 1$, $z_S = \mathbf{s}_n/\sqrt{\iota_1} \xrightarrow{d} N(0,1)$ and $z_W = z_S + o_p(1)$. ■

For composite H_0, Wilks' theorem gives $-2\log\Lambda \xrightarrow{d} \chi^2_{\mathrm{df}}$, df being the difference in parameter-space dimensions (Theorem 1.18 proves this for the multinomial). For small to moderate n the LR test is usually more reliable than Wald.

1.3.4 Constructing confidence intervals

Inverting a test gives an interval: the 95% interval is the set of β_0 whose test has $P > 0.05$.

(a) *Wald interval*: $\hat\beta \pm z_{\alpha/2}\mathrm{SE}$.

(b) *LR interval*: $\{\beta_0 : -2[L(\beta_0) - L(\hat\beta)] < \chi^2_1(\alpha)\}$, where $\chi^2_1(\alpha) = z^2_{\alpha/2}$.

(c) *Score interval*: $\{\beta_0 : |z_S(\beta_0)| < z_{\alpha/2}\}$.

If L is exactly parabolic (normal $\hat\beta$) the three coincide; with small categorical samples they can disagree markedly. The LR interval is preferable for small to moderate n; Wald is the most common.

1.4 Statistical inference for binomial parameters

Motivation. Near the boundary the three large-sample methods already behave very differently for the binomial.

1.4.1 Tests about a binomial parameter

Proposition 1.11 (Binomial test statistics)

For $H_0\colon \pi = \pi_0$ with $\hat\pi = y/n$:

$$z_W = \frac{\hat\pi - \pi_0}{\sqrt{\hat\pi(1-\hat\pi)/n}}, \tag{1.8}$$

$$z_S = \frac{\hat\pi - \pi_0}{\sqrt{\pi_0(1-\pi_0)/n}}, \tag{1.9}$$

$$-2(L_0 - L_1) = 2\sum \text{observed}\,\log\frac{\text{observed}}{\text{fitted}} = 2\Big[y\log\frac{y}{n\pi_0} + (n-y)\log\frac{n-y}{n-n\pi_0}\Big]. \tag{1.10}$$

Each is asymptotically $N(0,1)$ or χ^2_1 under H_0.

Proof.

Step 1. Wald uses $\mathrm{SE} = \sqrt{\hat\pi(1-\hat\pi)/n}$ (Proposition 1.8 at $\pi = \hat\pi$).

Step 2. Score: by (1.6)–(1.7), $u(\pi_0)/\iota(\pi_0)^{1/2} = \frac{y-n\pi_0}{\pi_0(1-\pi_0)}\sqrt{\frac{\pi_0(1-\pi_0)}{n}} = \frac{y-n\pi_0}{\sqrt{n\pi_0(1-\pi_0)}}$, which is (1.9).

Step 3. LR: $-2(L_0 - L_1) = 2[y\log(\hat\pi/\pi_0) + (n-y)\log((1-\hat\pi)/(1-\pi_0))]$, which is (1.10). The asymptotics are Theorem 1.10 with $q = 1$. ∎

The score statistic uses the null SE, and its null distribution is closer to $N(0,1)$.

1.4.2 Confidence intervals for a binomial parameter

Inverting the Wald test gives

$$\hat\pi \pm z_{\alpha/2}\sqrt{\hat\pi(1-\hat\pi)/n}, \tag{1.11}$$

whose coverage is usually below nominal, much below for π near 0 or 1 (adding about 2 successes and 2 failures helps greatly).

Proposition 1.12 (Wilson score interval)

The score interval $\{\pi_0 : |\hat\pi - \pi_0| < z\sqrt{\pi_0(1-\pi_0)/n}\}$, $z = z_{\alpha/2}$, has endpoints

$$\hat\pi\Big(\frac{n}{n+z^2}\Big) + \frac{1}{2}\Big(\frac{z^2}{n+z^2}\Big) \;\pm\; z\sqrt{\frac{1}{n+z^2}\Big[\hat\pi(1-\hat\pi)\Big(\frac{n}{n+z^2}\Big) + \frac{1}{2}\cdot\frac{1}{2}\Big(\frac{z^2}{n+z^2}\Big)\Big]}.$$

Its midpoint is $(y + z^2/2)/(n + z^2)$, the sample proportion after adding $z^2/2$ successes and $z^2/2$ failures.

Proof. [Supplement]

Step 1. The set $|z_S| < z$ is where $(\hat\pi - \pi_0)^2 - z^2\pi_0(1-\pi_0)/n < 0$, i.e. between the roots of $(1 + z^2/n)\pi_0^2 - (2\hat\pi + z^2/n)\pi_0 + \hat\pi^2 = 0$.

Step 2. Multiplying by n, the quadratic formula gives $\pi_0 = \big[n\hat\pi + z^2/2 \pm z\sqrt{n\hat\pi(1-\hat\pi) + z^2/4}\big]/(n + z^2)$; dividing numerator terms by $n + z^2$ gives the stated centre and half-width. ∎

The squared coefficient of z averages the variance of a proportion at $\hat\pi$ and at $1/2$, with sample size $n + z^2$. Wilson performs much better than Wald.

1.4.3 Proportion of vegetarians example

Example 1.13 (Proportion of vegetarians)

Of $n = 25$ students, $y = 0$ were vegetarian, so $\hat\pi = 0$.
(a) Wald: $(0, 0)$, useless at the boundary.
(b) Score: $(0.0,\ 0.133)$; e.g. $\pi_0 = 0.10$ gives $z_S = -1.67$ (inside).
(c) LR: $L(\pi) = 25\log(1-\pi)$, $L(\hat\pi) = 0$; the upper bound solves $-50\log(1-\pi_0) = 3.84$, giving $(0.0,\ 0.074)$.
The methods differ greatly because $\hat\pi$ is highly skewed near 0.

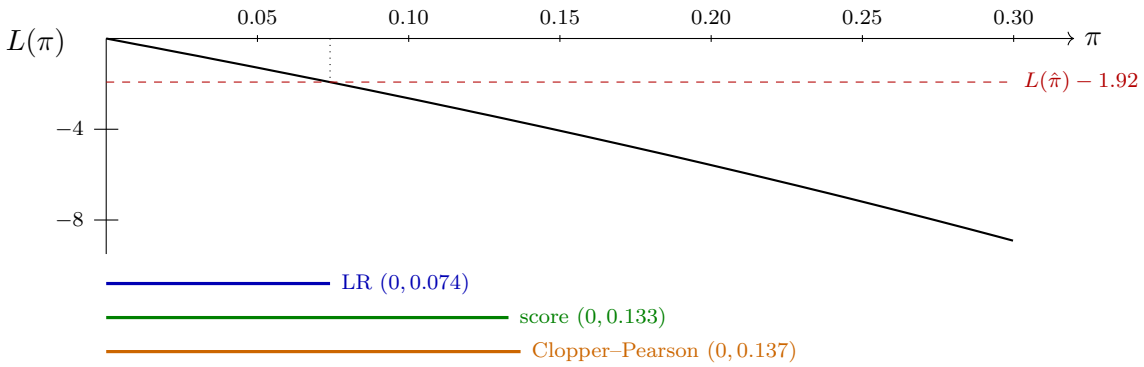


Figure 2: Binomial log likelihood for $y = 0$, $n = 25$; the LR interval is where L is within $3.84/2$ of its maximum.

1.5 Statistical inference for multinomial parameters

Motivation. Goodness of fit of a multinomial leads to Pearson's X^2 and the likelihood-ratio G^2, the workhorse statistics of the subject.

1.5.1 Estimation of multinomial parameters

Proposition 1.14 (Multinomial ML estimates)

The multinomial kernel

$$\prod_j \pi_j^{n_j}, \qquad \pi_j \geq 0, \quad \sum_j \pi_j = 1, \tag{1.12}$$

is maximized uniquely by the sample proportions $\hat{\pi}_j = p_j = n_j/n$.

Proof.

Step 1 (likelihood equations). With $\pi_c = 1 - \sum_{j<c} \pi_j$, $L = \sum_j n_j \log \pi_j$ has $\partial L/\partial \pi_j = n_j/\pi_j - n_c/\pi_c = 0$, $j < c$. So $\hat{\pi}_j/\hat{\pi}_c = n_j/n_c$; summing, $\hat{\pi}_c = n_c/n$ and $\hat{\pi}_j = n_j/n$.

Step 2 (global maximum). [Supplement] For any admissible $\boldsymbol{\pi}$ (sum over j with $n_j > 0$), Jensen's inequality gives

$$L(\boldsymbol{\pi}) - L(\mathbf{p}) = n \sum_j p_j \log \frac{\pi_j}{p_j} \leq n \log \sum_j p_j \frac{\pi_j}{p_j} \leq n \log 1 = 0,$$

with equality iff $\pi_j = p_j$ for all j. ■

1.5.2 Pearson statistic for testing a specified multinomial

For $H_0 \colon \pi_j = \pi_{j0}$ (Pearson, 1900), with *expected frequencies* $\mu_j = n\pi_{j0}$,

$$X^2 = \sum_j \frac{(n_j - \mu_j)^2}{\mu_j}, \tag{1.13}$$

and for large n, $X^2 \approx \chi^2_{c-1}$ (Theorem 1.16).

1.5.3 Example: testing Mendel's theories

Example 1.15 (Mendel's peas)

Mendel predicted 75% yellow seeds; of $n = 8023$, 6022 were yellow. Then $\mu_1 = 6017.25$, $X^2 = 0.015$ (df $= 1$), $P = 0.90$. Fisher (1936), summing independent statistics over Mendel's experiments, got $X^2 = 42$ with df $= 84$ (mean 84, SD 13): left-tail P is tiny, so the fit was *too good*, suggesting adjusted data.

1.5.4 Chi-squared theoretical justification

Theorem 1.16 (Asymptotic null distribution of Pearson's X^2)

Under $H_0 \colon \boldsymbol{\pi} = \boldsymbol{\pi}_0$ with all $\pi_{j0} > 0$, $X^2 \xrightarrow{d} \chi^2_{c-1}$ as $n \to \infty$. Moreover X^2 is the score statistic for H_0.

Proof. [Supplement] Let $\hat{\boldsymbol{\pi}} = (p_1, \ldots, p_{c-1})^\top$, $\boldsymbol{\pi}_0 = (\pi_{10}, \ldots, \pi_{c-1,0})^\top$ and $\Sigma_0 = \operatorname{diag}(\boldsymbol{\pi}_0) - \boldsymbol{\pi}_0 \boldsymbol{\pi}_0^\top$.

Step 1 (CLT). $\hat{\boldsymbol{\pi}}$ is a mean of i.i.d. indicator vectors with covariance Σ_0 (Proposition 1.2), so $\sqrt{n}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}_0) \xrightarrow{d} N_{c-1}(\mathbf{0}, \Sigma_0)$.

Step 2 (inverse). $\Sigma_0^{-1} = \operatorname{diag}(1/\pi_{j0}) + \mathbf{1}\mathbf{1}^\top/\pi_{c0}$: using $\boldsymbol{\pi}_0^\top \mathbf{1} = 1 - \pi_{c0}$, the product is $I + \boldsymbol{\pi}_0 \mathbf{1}^\top [1 - \pi_{c0} - (1 - \pi_{c0})]/\pi_{c0} = I$.

Step 3 (algebra). With $d_j = p_j - \pi_{j0}$, $j = 1, \dots, c$, and $d_c = -\sum_{j<c} d_j$,

$$n\,\mathbf{d}^\top \Sigma_0^{-1}\mathbf{d} = n\Big[\sum_{j<c}\frac{d_j^2}{\pi_{j0}} + \frac{(\sum_{j<c} d_j)^2}{\pi_{c0}}\Big] = n\sum_{j=1}^{c}\frac{d_j^2}{\pi_{j0}} = X^2. \tag{1.14}$$

Step 4 (limit). By continuous mapping and Lemma 1.9(a), $X^2 \xrightarrow{d} \chi^2_{c-1}$.

Step 5 (score test). By Proposition 1.14, Step 1, the score at $\boldsymbol{\pi}_0$ is $u_j = n(d_j/\pi_{j0} - d_c/\pi_{c0}) = n(\Sigma_0^{-1}\mathbf{d})_j$, and the information is $n\Sigma_0^{-1}$. So the score statistic is $\mathbf{u}^\top(n\Sigma_0^{-1})^{-1}\mathbf{u} = n\mathbf{d}^\top\Sigma_0^{-1}\mathbf{d} = X^2$. ■

Heuristically (Fisher), independent Poisson n_j give $\sum_j (n_j - \mu_j)^2/\mu_j \approx \chi^2_c$, and conditioning on $\sum n_j = n$ (Theorem 1.5) loses one df. For $c = 2$, $X^2 = z_S^2$ with z_S from (1.9).

1.5.5 Likelihood-ratio chi-squared

The kernel (1.12) is maximized at $\boldsymbol{\pi}_0$ under H_0 and at p_j in general, so

$$G^2 = -2\log\Lambda = 2\sum_j n_j \log\frac{n_j}{n\pi_{j0}}, \tag{1.15}$$

of the form (1.10), with $G^2 \xrightarrow{d} \chi^2_{c-1}$ under H_0 (Theorem 1.10, $q = c - 1$).

Theorem 1.17 (Asymptotic equivalence of X^2 and G^2)

Let μ_j be (possibly estimated) expected frequencies with $\sum_j \mu_j = n$ and $(n_j - \mu_j)/\mu_j = O_p(n^{-1/2})$ for each j, as holds under H_0. Then $G^2 - X^2 \xrightarrow{p} 0$; in particular $G^2 \xrightarrow{d} \chi^2_{c-1}$ under $H_0\colon \boldsymbol{\pi} = \boldsymbol{\pi}_0$.

Proof. [Supplement]

Step 1. Write $n_j = \mu_j(1 + x_j)$, $x_j = O_p(n^{-1/2})$. Since $(1+x)\log(1+x) = x + x^2/2 + R(x)$ with $|R(x)| \le C|x|^3$ for $|x| \le 1/2$,

$$G^2 = 2\sum_j \mu_j x_j + \sum_j \mu_j x_j^2 + 2\sum_j \mu_j R(x_j).$$

Step 2. $\sum_j \mu_j x_j = \sum_j (n_j - \mu_j) = 0$, $\sum_j \mu_j x_j^2 = X^2$, and $\mu_j R(x_j) = O(n)O_p(n^{-3/2}) \xrightarrow{p} 0$. Slutsky with Theorem 1.16 gives the limit. ■

Under H_a both grow like n but need not be close. X^2 usually approaches chi-squared faster; G^2 is often poor when $n/c < 5$.

1.5.6 Testing with estimated expected frequencies

Often $\pi_{j0} = \pi_j(\boldsymbol{\theta})$ with $\dim\boldsymbol{\theta} = p$ unknown; then $\hat\mu_j = n\pi_j(\hat{\boldsymbol{\theta}})$, and estimation reduces the df.

Theorem 1.18 (Degrees of freedom with estimated parameters)

Assume $\boldsymbol{\pi}(\boldsymbol{\theta})$ has continuous second derivatives, $\pi_j(\boldsymbol{\theta}_0) > 0$, the $c \times p$ Jacobian $\partial\boldsymbol{\pi}/\partial\boldsymbol{\theta}^\top$ has rank p at $\boldsymbol{\theta}_0$, and $\hat{\boldsymbol{\theta}}$ is a consistent ML estimate. If $H_0\colon \boldsymbol{\pi} = \boldsymbol{\pi}(\boldsymbol{\theta})$ holds, then

$$X^2 = \sum_j \frac{(n_j - \hat{\mu}_j)^2}{\hat{\mu}_j} \xrightarrow{d} \chi^2_{c-1-p}, \qquad \text{and } G^2 - X^2 \xrightarrow{p} 0.$$

Proof. [Supplement] Let $\boldsymbol{\pi}_0 = \boldsymbol{\pi}(\boldsymbol{\theta}_0)$, $D = \operatorname{diag}(\boldsymbol{\pi}_0)$, $\mathbf{s} = (\sqrt{\pi_{10}}, \ldots, \sqrt{\pi_{c0}})^\top$ (a unit vector), and $A = D^{-1/2}\,\partial\boldsymbol{\pi}/\partial\boldsymbol{\theta}^\top$ at $\boldsymbol{\theta}_0$ ($c \times p$, rank p).

Step 1. Since $\sum_j \pi_j(\boldsymbol{\theta}) = 1$, $\mathbf{s}^\top A = \sum_j \partial\pi_j/\partial\boldsymbol{\theta}^\top = \mathbf{0}$.

Step 2. With $\mathbf{p} = (p_1, \ldots, p_c)^\top$ and $\operatorname{Cov}(\sqrt{n}\,\mathbf{p}) = D - \boldsymbol{\pi}_0\boldsymbol{\pi}_0^\top$, the CLT gives $\mathbf{z}_n := \sqrt{n}\,D^{-1/2}(\mathbf{p} - \boldsymbol{\pi}_0) \xrightarrow{d} \mathbf{z} \sim N_c(\mathbf{0},\, I - \mathbf{s}\mathbf{s}^\top)$.

Step 3. $L(\boldsymbol{\theta}) = n\sum_j p_j \log \pi_j(\boldsymbol{\theta})$ has score at $\boldsymbol{\theta}_0$

$$\nabla L = n\,A^\top D^{-1/2}\mathbf{p} = n\,A^\top D^{-1/2}(\mathbf{p} - \boldsymbol{\pi}_0) = \sqrt{n}\,A^\top \mathbf{z}_n,$$

using $A^\top D^{-1/2}\boldsymbol{\pi}_0 = A^\top \mathbf{s} = \mathbf{0}$. The per-observation information is $\sum_j (\partial\pi_j/\partial\boldsymbol{\theta})(\partial\pi_j/\partial\boldsymbol{\theta})^\top/\pi_j = A^\top A$, so by Theorem 1.7 $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = (A^\top A)^{-1}A^\top \mathbf{z}_n + o_p(1)$.

Step 4. By the delta method, $\sqrt{n}\,D^{-1/2}(\boldsymbol{\pi}(\hat{\boldsymbol{\theta}}) - \boldsymbol{\pi}_0) = A\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + o_p(1)$. So with $H = A(A^\top A)^{-1}A^\top$,

$$\mathbf{e}_n := \sqrt{n}\,D^{-1/2}(\mathbf{p} - \boldsymbol{\pi}(\hat{\boldsymbol{\theta}})) = (I - H)\mathbf{z}_n + o_p(1) \xrightarrow{d} (I - H)\mathbf{z}.$$

Step 5. X^2 differs from $\mathbf{e}_n^\top \mathbf{e}_n$ only by $\pi_j(\hat{\boldsymbol{\theta}})$ in place of π_{j0} in the denominators, a factor $\xrightarrow{p} 1$; so $X^2 = \mathbf{e}_n^\top \mathbf{e}_n + o_p(1)$.

Step 6. $(I - H)\mathbf{z}$ has covariance $(I - H)(I - \mathbf{s}\mathbf{s}^\top)(I - H) = I - H - \mathbf{s}\mathbf{s}^\top$, as $H\mathbf{s} = \mathbf{0}$. H and $\mathbf{s}\mathbf{s}^\top$ project onto orthogonal subspaces of dimensions p and 1, so $I - H - \mathbf{s}\mathbf{s}^\top$ is idempotent of rank $c - p - 1$; Lemma 1.9(b) gives $X^2 \xrightarrow{d} \chi^2_{c-1-p}$.

Step 7. Step 4 gives $(n_j - \hat{\mu}_j)/\hat{\mu}_j = O_p(n^{-1/2})$ and $\sum_j \hat{\mu}_j = n$, so Theorem 1.17 gives $G^2 - X^2 \xrightarrow{p} 0$. ■

Example 1.19 (Pneumonia infections in calves)

156 calves were classified by primary infection and, if infected, secondary infection; (no primary, yes secondary) is a *structural zero*. Counts (fitted): (yes,yes) 30 (38.1), (yes,no) 63 (39.0), (no,no) 63 (78.9). If $P(\text{primary}) = \pi = P(\text{secondary} \mid \text{primary})$, the cell probabilities are $(\pi^2, \pi(1-\pi), 1-\pi)$, the kernel is $\pi^{2n_{11}+n_{12}}(1-\pi)^{n_{12}+n_{22}}$, and

$$\hat{\pi} = \frac{2n_{11} + n_{12}}{2n_{11} + 2n_{12} + n_{22}} = \frac{123}{249} = 0.494.$$

$X^2 = 19.7$, df $= 3 - 1 - 1 = 1$ ($P < 0.00001$): primary infection seems to have an immunizing effect.

Chapter 2. Describing Contingency Tables

2.1 Probability structure for contingency tables

Motivation. The relationship between two categorical variables is entirely contained in their joint distribution, from which the marginal and conditional distributions follow. How the data were sampled determines which of these distributions the observed table can estimate.

2.1.1 Contingency tables and their distributions

Let X have I categories and Y have J categories. A rectangular table with I rows and J columns whose IJ cells hold frequency counts is a *contingency table* (Karl Pearson, 1904), or $I \times J$ cross-classification table.

Definition (Joint, marginal and conditional distributions)

(a) The *joint distribution* is $\{\pi_{ij}\}$, $\pi_{ij} = P(X = i, Y = j)$.
(b) The *marginal distributions* are $\pi_{i+} = \sum_j \pi_{ij}$ and $\pi_{+j} = \sum_i \pi_{ij}$, with $\sum_i \pi_{i+} = \sum_j \pi_{+j} = \sum_i \sum_j \pi_{ij} = 1$ (a "+" denotes summation over that index).
(c) When Y is a response, the *conditional distribution* of Y in row i is $\pi_{j|i} = P(Y = j \mid X = i) = \pi_{ij}/\pi_{i+}$, with $\sum_j \pi_{j|i} = 1$.

If X is fixed by design (not random), only the conditional distributions are meaningful. A principal aim of many studies is to compare the conditional distributions of Y across levels of X.

Example 2.1 (Aspirin and heart attacks)

The Physicians' Health Study was a five-year randomized, blind trial of whether regular aspirin intake reduces cardiovascular mortality. Every other day physicians took aspirin or a placebo.

	Myocardial infarction			
Group	Fatal	Nonfatal	No attack	Total
Placebo	18	171	10,845	11,034
Aspirin	5	99	10,933	11,037

Of 11,034 placebo takers 18 had fatal attacks, against 5 of 11,037 aspirin takers.

2.1.2 Sensitivity and specificity

For a diagnostic test let X = true disease status and Y = diagnosis (positive/negative).

Definition (Sensitivity, specificity)

Sensitivity $= P(Y = \text{positive} \mid X = \text{diseased}) = \pi_{1|1}$; *specificity* $= P(Y = \text{negative} \mid X = \text{not diseased}) = \pi_{2|2}$.

For mammography combined with clinical breast examination, a literature survey estimated sensitivity 0.82 and specificity 0.99: 82% of women with breast cancer, and 99% of those without, are diagnosed correctly.

2.1.3 Independence of categorical variables

Definition (Independence)

X and Y are *independent* if $\pi_{ij} = \pi_{i+}\pi_{+j}$ for all i, j.

Proposition 2.2 (Independence as homogeneity)

If all $\pi_{i+} > 0$, then X and Y are independent if and only if $\pi_{j|1} = \pi_{j|2} = \cdots = \pi_{j|I}$ for every j; the common value is then π_{+j}.

Proof.

Step 1. If $\pi_{ij} = \pi_{i+}\pi_{+j}$ then $\pi_{j|i} = \pi_{ij}/\pi_{i+} = \pi_{+j}$ for every i.

Step 2. Conversely, if $\pi_{j|i} = c_j$ for all i, then $\pi_{+j} = \sum_i \pi_{i+}\pi_{j|i} = c_j \sum_i \pi_{i+} = c_j$, so $\pi_{ij} = \pi_{i+}\pi_{j|i} = \pi_{i+}\pi_{+j}$. ■

When Y is a response this *homogeneity of conditional distributions* is the more natural definition. Sample analogues: $p_{ij} = n_{ij}/n$, $p_{j|i} = n_{ij}/n_{i+}$, $n = \sum_i \sum_j n_{ij}$; estimates may also be written $\hat{\pi}_{ij}$.

2.1.4 Poisson, binomial, and multinomial sampling

(a) *Poisson sampling*: the IJ counts Y_{ij} are independent Poisson(μ_{ij}), with joint pmf $\prod_i \prod_j e^{-\mu_{ij}} \mu_{ij}^{n_{ij}}/n_{ij}!$; nothing is fixed.

(b) *Multinomial sampling*: n is fixed and $\{n_{ij}\}$ is multinomial over the IJ cells, $\frac{n!}{\prod\prod n_{ij}!} \prod\prod \pi_{ij}^{n_{ij}}$.

(c) *Independent (product) multinomial sampling*: the row totals $n_i = n_{i+}$ are fixed and each row is an independent multinomial sample with probabilities $\{\pi_{j|i}\}$:

$$\prod_i \Big[\frac{n_i!}{\prod_j n_{ij}!} \prod_j \pi_{j|i}^{n_{ij}} \Big]. \tag{2.1}$$

(d) When both margins are fixed the distribution is hypergeometric (Section 3.5).

Theorem 2.3 (Conditioning on row totals gives product multinomial)

If $\{n_{ij}\}$ arise from independent Poisson sampling with means $\{\mu_{ij}\}$, or from multinomial sampling with probabilities $\pi_{ij} = \mu_{ij}/n$, then conditional on the row totals $\{n_{i+}\}$ the rows are independent multinomial samples with probabilities $\pi_{j|i} = \mu_{ij}/\mu_{i+}$.

Proof. [Supplement]

Step 1 (Poisson). The rows are independent sets of independent Poisson counts. By Theorem 1.5 applied to row i, given n_{i+} the counts $(n_{i1}, \ldots, n_{iJ})$ are multinomial$(n_{i+}, \{\mu_{ij}/\mu_{i+}\})$. Since row i's counts and total depend only on row i's Poisson variables, conditioning on all row totals keeps the rows independent.

Step 2 (multinomial). The row totals $(n_{1+}, \ldots, n_{I+})$ are multinomial$(n, \{\pi_{i+}\})$ (collapsing categories of a multinomial). Hence for tables with these row totals,

$$P(\{n_{ij}\} \mid \{n_{i+}\}) = \frac{\frac{n!}{\prod_i \prod_j n_{ij}!} \prod_i \prod_j \pi_{ij}^{n_{ij}}}{\frac{n!}{\prod_i n_{i+}!} \prod_i \pi_{i+}^{n_{i+}}} = \prod_i \Big[\frac{n_{i+}!}{\prod_j n_{ij}!} \prod_j \Big(\frac{\pi_{ij}}{\pi_{i+}} \Big)^{n_{ij}} \Big],$$

using $\pi_{i+}^{n_{i+}} = \prod_j \pi_{i+}^{n_{ij}}$. This is (2.1) with $\pi_{j|i} = \pi_{ij}/\pi_{i+} = \mu_{ij}/\mu_{i+}$. ■

So when X is explanatory it is sensible to analyse conditional on the row totals, as if they formed separate independent samples, even when the design did not fix them.

2.1.5 Seat belt example

Example 2.4 (Seat belts and crash outcome)

A study of seat-belt use (yes/no) by crash outcome (fatal/nonfatal) on the Massachusetts Turnpike illustrates the sampling schemes:
(a) recording all accidents in the coming year: n random, the four counts independent Poisson;
(b) sampling 200 police records from last year: $n = 200$ fixed, multinomial over four cells;
(c) sampling 100 fatal and 100 nonfatal records: column totals fixed, each column an independent binomial;
(d) the (unethical) experiment randomizing 100 of 200 subjects to wear belts and forcing crashes: row totals fixed, independent binomial rows.

2.1.6 Types of studies

Example 2.5 (Smoking and lung cancer: a case–control study)

Doll and Hill queried 709 lung cancer patients in 20 London hospitals about smoking, and for each a matched noncancer control (same hospital, gender, 5-year age group).

	Lung cancer	
Smoker	Cases	Controls
Yes	688	650
No	21	59
Total	709	709

The design fixes the lung cancer margin; we can estimate $P(\text{smoker} \mid \text{case}) = 688/709 = 0.970$ and $P(\text{smoker} \mid \text{control}) = 650/709 = 0.917$, but not $P(\text{cancer} \mid \text{smoking})$ without knowing the population prevalence (Bayes' theorem).

Definition (Study designs)

(a) *Retrospective* (case–control): samples are taken at fixed levels of the outcome Y and the explanatory X is looked up in the past; each column is a multinomial sample on X.
(b) *Prospective*: groups defined by X are followed into the future; *clinical trials* randomize subjects to X, *cohort studies* let subjects choose. Each row is a multinomial sample on Y.
(c) *Cross-sectional*: subjects are sampled and classified on both variables at once; the IJ counts are one multinomial sample.

Case–control, cohort and cross-sectional studies are *observational*; a clinical trial is *experimental*, and randomization balances the groups on other variables. Observational studies are common but more prone to bias.

2.2 Comparing two proportions

Motivation. With two groups and a binary response the association is a comparison of two success probabilities π_1, π_2. Three summaries are used—difference, ratio, and odds ratio—and they behave very differently under change of sampling design.

In a 2×2 table write $\pi_i = \pi_{1|i}$ for the success probability in row i, so $\pi_{2|i} = 1 - \pi_i$.

2.2.1 Difference of proportions

The *difference of proportions* $\pi_1 - \pi_2 \in [-1, 1]$ is the same whether we compare successes or failures, since $(1-\pi_1) - (1-\pi_2) = -(\pi_1 - \pi_2)$. It is 0 iff the rows have identical conditional distributions, i.e. iff Y is independent of the row classification. The difference comparing columns is in general different.

2.2.2 Relative risk

A difference of fixed size may matter more near 0 or 1: $0.010 - 0.001$ and $0.410 - 0.401$ are both 0.009, but the ratios are 10.0 and 1.02.

Definition (Relative risk)

The *relative risk* is $\pi_1/\pi_2 \in [0, \infty)$; it equals 1 under independence. Comparing the second response category gives a different value $(1-\pi_1)/(1-\pi_2)$.

2.2.3 Odds ratio

Definition (Odds, odds ratio)

For success probability π the *odds* are $\Omega = \pi/(1-\pi)$, inverted by $\pi = \Omega/(\Omega+1)$. With $\Omega_i = \pi_i/(1-\pi_i) = \pi_{i1}/\pi_{i2}$, the *odds ratio* is

$$\theta = \frac{\Omega_1}{\Omega_2} = \frac{\pi_1/(1-\pi_1)}{\pi_2/(1-\pi_2)} = \frac{\pi_{11}\pi_{22}}{\pi_{12}\pi_{21}}, \tag{2.2}$$

also called the *cross-product ratio* (Yule). The sample odds ratio is $\hat{\theta} = n_{11}n_{22}/(n_{12}n_{21})$.

E.g. $\pi = 0.75$ gives $\Omega = 3$: a success is three times as likely as a failure; $\Omega = 1/3$ gives $\pi = 0.25$.

2.2.4 Properties of the odds ratio

Proposition 2.6 (Properties of the odds ratio)

Assume all $\pi_{ij} > 0$.
(a) $\theta = 1$ iff $\Omega_1 = \Omega_2$ iff $\pi_1 = \pi_2$ (independence); $\theta > 1$ iff $\pi_1 > \pi_2$.
(b) Reversing the order of the rows (or of the columns) replaces θ by $1/\theta$, i.e. $\log\theta$ by $-\log\theta$; transposing the table leaves θ unchanged.
(c) θ is the same whichever conditional direction is used:

$$\theta = \frac{P(Y=1 \mid X=1)/P(Y=2 \mid X=1)}{P(Y=1 \mid X=2)/P(Y=2 \mid X=2)} = \frac{P(X=1 \mid Y=1)/P(X=2 \mid Y=1)}{P(X=1 \mid Y=2)/P(X=2 \mid Y=2)}. \tag{2.3}$$

(d) $\hat{\theta}$ is unchanged when both counts in a row, or both counts in a column, are multiplied by a positive constant.

Proof.

Step (a). $x \mapsto x/(1-x)$ is strictly increasing on $(0,1)$, so $\Omega_1 \gtreqless \Omega_2$ iff $\pi_1 \gtreqless \pi_2$.

Step (b). Swapping rows maps $\pi_{11}\pi_{22}/(\pi_{12}\pi_{21})$ to $\pi_{21}\pi_{12}/(\pi_{22}\pi_{11}) = 1/\theta$; similarly for columns. Transposing swaps π_{12} and π_{21}, which leaves the cross-product ratio unchanged.

Step (c). The first ratio is $(\pi_{11}/\pi_{12})/(\pi_{21}/\pi_{22})$ after cancelling π_{1+} and π_{2+}; the second is $(\pi_{11}/\pi_{21})/(\pi_{12}/\pi_{22})$ after cancelling π_{+1}, π_{+2}. Both equal $\pi_{11}\pi_{22}/(\pi_{12}\pi_{21})$.

Step (d). Multiplying row 1 by c multiplies numerator $n_{11}n_{22}$ and denominator $n_{12}n_{21}$ each by c; columns likewise. ■

Thus $\theta = 4$ means the odds (not the probability) in row 1 are four times those in row 2, and $\log 4 = 1.39$ and $\log 0.25 = -1.39$ represent equally strong associations. By (c) and (d), θ is equally valid for prospective, retrospective and cross-sectional designs: $\hat{\theta}$ estimates the same parameter whether a flu–vaccination case–control study samples $100 + 100$ or $40 + 160$ subjects. The sample difference and relative risk are invariant to scaling within rows, but not within columns or under transposition.

2.2.5 Aspirin and heart attacks revisited

Example 2.7 (Three measures for the aspirin study)

Combining fatal and nonfatal attacks in Example 2.1: placebo 189/11,034 = 0.0171, aspirin 104/11,037 = 0.0094.
(a) Difference: $0.0171 - 0.0094 = 0.0077$.
(b) Relative risk: $0.0171/0.0094 = 1.82$.
(c) Odds ratio: $\hat{\theta} = \dfrac{189 \times 10{,}933}{10{,}845 \times 104} = 1.83$; the odds of heart attack were 83% higher with placebo.

2.2.6 Case–control studies and the odds ratio

In a case–control study we can estimate $P(X = i \mid Y = j)$ but usually not $P(Y = j \mid X = i)$, hence neither the difference nor the relative risk of the outcome. By (2.3), however, the odds ratio is determined by conditional probabilities in either direction.

Example 2.8 (Odds ratio from Doll and Hill's data)

From Example 2.5,

$$\hat{\theta} = \frac{(688/709)/(21/709)}{(650/709)/(59/709)} = \frac{688 \times 59}{650 \times 21} = 3.0.$$

Using the direction of interest: the estimated odds of lung cancer for smokers were 3.0 times those for nonsmokers.

2.2.7 Relationship between odds ratio and relative risk

Proposition 2.9 (Odds ratio versus relative risk)

$$\theta = \frac{\pi_1}{\pi_2} \times \frac{1-\pi_2}{1-\pi_1}.$$

Hence $\theta \approx$ relative risk when π_1 and π_2 are both close to 0, and θ is farther from 1 than the relative risk (on the same side).

Proof.

Step 1. Directly from (2.2): $\theta = \frac{\pi_1}{1-\pi_1} \cdot \frac{1-\pi_2}{\pi_2}$.

Step 2. If $\pi_1 > \pi_2$ then $(1-\pi_2)/(1-\pi_1) > 1$ so $\theta > \pi_1/\pi_2 > 1$; if $\pi_1 < \pi_2$ the factor is < 1 and $\theta < \pi_1/\pi_2 < 1$. As $\pi_1, \pi_2 \to 0$ the factor tends to 1. ■

In the aspirin study ($\pi_i < 0.02$) the relative risk 1.82 and odds ratio 1.83 almost agree. Hence in case–control studies of rare diseases $\hat{\theta}$ roughly estimates the unestimable relative risk (Cornfield, 1951): smokers had roughly 3 times the lung cancer rate of nonsmokers.

2.3 Partial association in stratified 2×2 tables

Motivation. In observational studies an apparent effect of X on Y may reflect a third variable Z that affects both (confounding). Controlling for Z by holding it fixed can change the association dramatically, even reversing its direction.

To study a passive-smoking effect on lung cancer, for example, one must control age and socioeconomic status: spouses of nonsmokers may be younger, and younger people get lung cancer less often.

2.3.1 Partial tables

Definition (Partial and marginal tables)

Fixing the levels of Z, the two-way cross-sections of an $X \times Y \times Z$ table are the *partial tables*; their associations are *conditional associations*. Summing the partial tables gives the XY *marginal table*, which ignores Z rather than controlling it.

2.3.2 Death penalty example

Example 2.10 (Death penalty verdict by defendant's and victims' race)

674 defendants in Florida multiple-murder indictments, 1976–1987. Y = death penalty, X = defendant's race, Z = victims' race.

Victims' race	Defendant's race	Yes	No	Percent yes
White	White	53	414	11.3
	Black	11	37	22.9
Black	White	0	16	0.0
	Black	4	139	2.8
Total	White	53	430	11.0
	Black	15	176	7.9

(a) Controlling for victims' race, black defendants received the death penalty more often: by $22.9 - 11.3 = 11.6$ points with white victims and 2.8 points with black victims.
(b) Ignoring victims' race, white defendants received it more often (11.0% vs. 7.9%): the association reverses.
(c) Explanation: victims' race is strongly associated with defendant's race (marginal odds ratio $467 \times 143/(48 \times 16) = 87.0$), and the death penalty is much more likely when the victims are white. Whites mostly kill whites, and killing whites more often brings the death penalty.

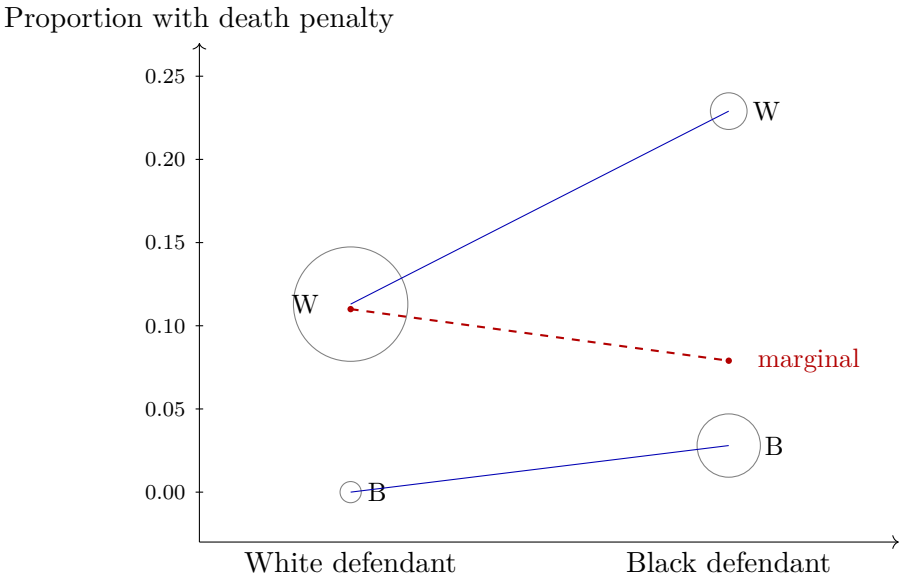


Figure 3: Simpson's paradox in the death penalty data. Letters give victims' race; circle area is proportional to the number of cases. Both conditional lines rise, the marginal line falls, since the summary proportions lie near the large circles.

Definition (Simpson's paradox)

A marginal association whose direction differs from that of each conditional association is an instance of *Simpson's paradox* (Yule 1903, Simpson 1951).

It warns against causal interpretation of an X–Y association. Fisher argued that some factor (e.g. genetic) might explain away the smoking–lung cancer association; Cornfield showed that for a very strong association, the confounder would itself need a very strong association with both X and Y.

2.3.3 Conditional and marginal odds ratios

For a $2 \times 2 \times K$ table with expected frequencies μ_{ijk}:

$$\theta_{XY(k)} = \frac{\mu_{11k}\mu_{22k}}{\mu_{12k}\mu_{21k}} \quad \text{(conditional)}, \qquad \theta_{XY} = \frac{\mu_{11+}\mu_{22+}}{\mu_{12+}\mu_{21+}} \quad \text{(marginal)},$$

with $\mu_{ij+} = \sum_k \mu_{ijk}$; sample versions use counts.

Example 2.11 (Odds ratios for the death penalty data)

(a) White victims: $\hat{\theta}_{XY(1)} = \dfrac{53 \times 37}{414 \times 11} = 0.43$; the odds of the death penalty for white defendants were 43% of those for black defendants.
(b) Black victims: $\hat{\theta}_{XY(2)} = \dfrac{0 \times 139}{16 \times 4} = 0.0$ (no white defendant with black victims got the death penalty).
(c) Marginal: $\hat{\theta}_{XY} = \dfrac{53 \times 176}{430 \times 15} = 1.45$; the odds were 45% higher for white defendants.

2.3.4 Marginal versus conditional independence

Definition (Conditional independence)

X and Y are *conditionally independent at level* k of Z if they are independent in partial table k, i.e. $P(Y = j \mid X = i, Z = k) = P(Y = j \mid Z = k)$ for all i, j. They are *conditionally independent given* Z if this holds for every k.

Proposition 2.12 (Factorization under conditional independence)

For a single multinomial over the three-way table with $\pi_{ijk} = P(X = i, Y = j, Z = k)$ and all $\pi_{++k} > 0$, X and Y are conditionally independent given Z iff

$$\pi_{ijk} = \frac{\pi_{i+k}\,\pi_{+jk}}{\pi_{++k}} \qquad \text{for all } i, j, k.$$

Conditional independence does not imply marginal independence: in general $\pi_{ij+} = \sum_k \pi_{i+k}\pi_{+jk}/\pi_{++k} \neq \pi_{i++}\pi_{+j+}$.

Proof.

Step 1. Always $\pi_{ijk} = P(X = i, Z = k)\,P(Y = j \mid X = i, Z = k) = \pi_{i+k}\,P(Y = j \mid X = i, Z = k)$.

Step 2. Under conditional independence the last factor equals $P(Y = j \mid Z = k) = \pi_{+jk}/\pi_{++k}$, giving the factorization. Conversely, the factorization gives $P(Y = j \mid X = i, Z = k) = \pi_{ijk}/\pi_{i+k} = \pi_{+jk}/\pi_{++k}$ whenever $\pi_{i+k} > 0$.

Step 3. Summing over k gives $\pi_{ij+} = \sum_k \pi_{i+k}\pi_{+jk}/\pi_{++k}$, which involves k in every factor and does not reduce to $\pi_{i++}\pi_{+j+}$; Example 2.13 is an explicit counterexample. ■

For $2 \times 2 \times K$ tables, conditional independence means $\theta_{XY(k)} = 1$ for all k.

Example 2.13 (Conditional but not marginal independence)

Expected frequencies for Y = response, X = treatment, Z = clinic:

Clinic	Treatment	Success	Failure
1	A	18	12
	B	12	8
2	A	2	8
	B	8	32
Total	A	20	20
	B	20	40

(a) $\theta_{XY(1)} = \dfrac{18 \times 8}{12 \times 12} = 1.0$ and $\theta_{XY(2)} = \dfrac{2 \times 32}{8 \times 8} = 1.0$: conditional independence.

(b) Marginal: $\theta_{XY} = \dfrac{20 \times 40}{20 \times 20} = 2.0$.

(c) The XZ and YZ conditional odds ratios all equal 6.0, e.g. $\dfrac{18 \times 8}{12 \times 2} = 6.0$: clinic 1 uses treatment A more and also has more successes (perhaps its patients are healthier). Concluding from the marginal table that A is better would be misleading.

2.3.5 Homogeneous association

Definition (Homogeneous association)

A $2 \times 2 \times K$ table has *homogeneous XY association* if $\theta_{XY(1)} = \theta_{XY(2)} = \cdots = \theta_{XY(K)}$. Conditional independence is the special case where the common value is 1. If the conditional odds ratios vary with k there is *interaction*, and Z is an *effect modifier*.

Proposition 2.14 (Homogeneous association is symmetric)

For a $2 \times 2 \times 2$ table with positive probabilities,

$$\frac{\theta_{XY(1)}}{\theta_{XY(2)}} = \frac{\theta_{XZ(1)}}{\theta_{XZ(2)}} = \frac{\theta_{YZ(1)}}{\theta_{YZ(2)}},$$

where $\theta_{XZ(j)}$ is the XZ odds ratio at level j of Y and $\theta_{YZ(i)}$ the YZ odds ratio at level i of X. Hence, in a $2 \times 2 \times K$ table, homogeneous XY association holds iff for every pair of levels k, k' of Z the XZ (and the YZ) odds ratio formed with k, k' is the same at both levels of Y (of X).

Proof. [Supplement]

Step 1. Directly,

$$\frac{\theta_{XY(1)}}{\theta_{XY(2)}} = \frac{\pi_{111}\pi_{221}}{\pi_{121}\pi_{211}} \cdot \frac{\pi_{122}\pi_{212}}{\pi_{112}\pi_{222}}, \qquad \frac{\theta_{XZ(1)}}{\theta_{XZ(2)}} = \frac{\pi_{111}\pi_{212}}{\pi_{112}\pi_{211}} \cdot \frac{\pi_{122}\pi_{221}}{\pi_{121}\pi_{222}}.$$

Both equal $\dfrac{\pi_{111}\pi_{122}\pi_{212}\pi_{221}}{\pi_{112}\pi_{121}\pi_{211}\pi_{222}}$ (numerator: the cells whose index sum is odd; denominator: those whose index sum is even). The same expression arises for $\theta_{YZ(1)}/\theta_{YZ(2)}$ by the symmetry of this formula in the three indices.

Step 2. For $2 \times 2 \times K$, apply Step 1 to the $2 \times 2 \times 2$ subtable formed by levels k, k' of Z: $\theta_{XY(k)} = \theta_{XY(k')}$ iff the XZ odds ratio for (k, k') is the same at $Y = 1$ and $Y = 2$, and likewise for YZ. ■

So homogeneous association is a property of the three variables jointly ("no three-factor interaction"). For instance, if for smoking and lung cancer the odds ratios were 1.2, 3.9, 8.8 in age groups

< 45, 45–65, > 65, age would be an effect modifier. For the death penalty data $\hat{\theta}_{XY(1)} = 0.43$ and $\hat{\theta}_{XY(2)} = 0.0$; the latter is unstable (zero count), and adding 1/2 to each cell gives 0.94, so the data need not contradict homogeneity (tests in Section 6.3).

2.4 Extensions for $I \times J$ tables

Motivation. Beyond 2×2 tables a single number rarely captures the association. A minimal set of odds ratios describes it completely, while summary indices (entropy-based for nominal, concordance-based for ordinal variables) trade completeness for interpretability.

2.4.1 Odds ratios in $I \times J$ tables

Any pair of rows a, b and columns c, d gives an odds ratio $\pi_{ac}\pi_{bd}/(\pi_{bc}\pi_{ad})$; there are $\binom{I}{2}\binom{J}{2}$ of them, highly redundant.

Definition (Local and baseline odds ratios)

The *local odds ratios* are

$$\theta_{ij} = \frac{\pi_{ij}\,\pi_{i+1,j+1}}{\pi_{i,j+1}\,\pi_{i+1,j}}, \qquad i = 1, \ldots, I-1,\ j = 1, \ldots, J-1. \tag{2.4}$$

An alternative basic set uses the last row and column as baseline:

$$\alpha_{ij} = \frac{\pi_{ij}\,\pi_{IJ}}{\pi_{Ij}\,\pi_{iJ}}, \qquad i = 1, \ldots, I-1,\ j = 1, \ldots, J-1. \tag{2.5}$$

Proposition 2.15 ($(I-1)(J-1)$ odds ratios describe the association)

Assume all $\pi_{ij} > 0$.
(a) Every odds ratio $\pi_{ac}\pi_{bd}/(\pi_{bc}\pi_{ad})$ with $a < b$, $c < d$ is the product $\prod_{i=a}^{b-1}\prod_{j=c}^{d-1}\theta_{ij}$.
(b) Given the marginal distributions $\{\pi_{i+}\}, \{\pi_{+j}\}$, the local odds ratios determine the cell probabilities uniquely.
(c) Independence holds iff all $(I-1)(J-1)$ local odds ratios equal 1.

Proof. [Supplement]

Step (a). Let $\lambda_{ij} = \log \pi_{ij}$ and $\log \theta_{ij} = \lambda_{ij} - \lambda_{i,j+1} - \lambda_{i+1,j} + \lambda_{i+1,j+1}$. Summing over $j = c, \ldots, d-1$ telescopes to $\lambda_{ic} - \lambda_{id} - \lambda_{i+1,c} + \lambda_{i+1,d}$; summing that over $i = a, \ldots, b-1$ telescopes to $\lambda_{ac} - \lambda_{ad} - \lambda_{bc} + \lambda_{bd}$, the log of the stated odds ratio.

Step (b). Suppose $\{\pi_{ij}\}$ and $\{\pi'_{ij}\}$ have the same margins and the same local odds ratios. Let $\delta_{ij} = \log(\pi'_{ij}/\pi_{ij})$. Its local contrasts $\delta_{ij} - \delta_{i,j+1} - \delta_{i+1,j} + \delta_{i+1,j+1}$ vanish, so by the telescoping in (a) so does $\delta_{ij} - \delta_{iJ} - \delta_{Ij} + \delta_{IJ}$. With $r_i = \delta_{iJ} - \delta_{IJ}$ and $c_j = \delta_{Ij}$ this gives $\delta_{ij} = r_i + c_j$. Hence

$$\sum_i \sum_j (\pi'_{ij} - \pi_{ij}) \log \frac{\pi'_{ij}}{\pi_{ij}} = \sum_i r_i(\pi'_{i+} - \pi_{i+}) + \sum_j c_j(\pi'_{+j} - \pi_{+j}) = 0.$$

Each term $(\pi' - \pi)\log(\pi'/\pi)$ is ≥ 0, with equality iff $\pi' = \pi$ (log is increasing). Hence $\pi'_{ij} = \pi_{ij}$ for all i, j.

Step (c). If $\pi_{ij} = \pi_{i+}\pi_{+j}$ then every $\theta_{ij} = 1$. Conversely, the independence table $\{\pi_{i+}\pi_{+j}\}$ has the same margins and all local odds ratios 1; by (b) it is the only such table. ∎

In this sense $(I-1)(J-1)$ parameters describe any association in an $I \times J$ table. For Example 2.1 the sample local odds ratios are $\frac{18\times 99}{171\times 5} = 2.08$ (fatal vs. nonfatal) and $\frac{171\times 10{,}933}{99\times 10{,}845} = 1.74$ (nonfatal vs. none); their product 3.63 is the odds ratio for the first and third columns. In each case the more serious outcome was more prevalent under placebo. For $I \times J \times K$ tables, homogeneous XY association means every conditional XY odds ratio is the same at each level of Z.

2.4.2 Summary measures of association

A *proportional reduction in variation* measure, with $V(Y)$ a variation measure for the marginal distribution $\{\pi_{+j}\}$ and $V(Y \mid i)$ that for $\{\pi_{j|i}\}$, is

$$\frac{V(Y) - E[V(Y \mid X)]}{V(Y)}, \qquad E[V(Y \mid X)] = \sum_i \pi_{i+} V(Y \mid i), \tag{2.6}$$

generalizing R^2. For a nominal response Theil used the *entropy* $V(Y) = -\sum_j \pi_{+j} \log \pi_{+j}$, giving the *uncertainty coefficient*

$$U = -\frac{\sum_i \sum_j \pi_{ij} \log(\pi_{ij}/(\pi_{i+}\pi_{+j}))}{\sum_j \pi_{+j} \log \pi_{+j}}. \tag{2.7}$$

Proposition 2.16 (Range of the uncertainty coefficient)

If more than one $\pi_{+j} > 0$, then $0 \le U \le 1$; $U = 0$ iff X and Y are independent, and $U = 1$ iff for each i some $\pi_{j|i} = 1$.

Proof. [Supplement] Write

$$H(Y) = -\sum_j \pi_{+j} \log \pi_{+j} > 0, \qquad H(Y \mid X) = -\sum_i \pi_{i+} \sum_j \pi_{j|i} \log \pi_{j|i} \ge 0.$$

Step 1. (2.7) is (2.6) with entropy: since $\log \pi_{j|i} = \log \pi_{ij} - \log \pi_{i+}$, $H(Y) - H(Y \mid X) = \sum_i \sum_j \pi_{ij} \log \frac{\pi_{ij}}{\pi_{i+}\pi_{+j}}$, so $U = [H(Y) - H(Y \mid X)]/H(Y)$.

Step 2. By Jensen, $-\sum\sum \pi_{ij} \log \frac{\pi_{i+}\pi_{+j}}{\pi_{ij}} \ge -\log \sum\sum \pi_{i+}\pi_{+j} \ge 0$, with equality iff $\pi_{ij} = \pi_{i+}\pi_{+j}$ on all cells (then the sum is 1). So $U \ge 0$, with equality iff independence.

Step 3. $U \le 1$ iff $H(Y \mid X) \ge 0$, true since each $-\pi \log \pi \ge 0$; equality iff every conditional distribution is degenerate. ■

It is hard to develop intuition for such measures (what is a "30% reduction in entropy"?); summaries are easier to interpret when both variables are ordinal.

2.4.3 Ordinal trends: concordant and discordant pairs

For ordinal X and Y a monotone trend is common. A pair of subjects is *concordant* if the subject ranking higher on X also ranks higher on Y, *discordant* if it ranks lower on Y, and *tied* if they agree on X and/or Y.

Example 2.17 (Job satisfaction and income)

Black males in the 1996 General Social Survey ($n = 96$); VD = very dissatisfied, LD = a little dissatisfied, MS = moderately satisfied, VS = very satisfied.

Income ($)	VD	LD	MS	VS
< 15,000	1	3	10	6
15,000–25,000	2	3	10	7
25,000–40,000	1	6	14	12
> 40,000	0	1	9	11

Each cell's count times the total count below and to the right gives concordant pairs:

$$\begin{aligned} C = {} & 1(3+10+7+6+14+12+1+9+11) + 3(10+7+14+12+9+11) \\ & + 10(7+12+11) + 2(6+14+12+1+9+11) + 3(14+12+9+11) + 10(12+11) \\ & + 1(1+9+11) + 6(9+11) + 14(11) = 1331, \end{aligned}$$

and similarly (below and to the left) $D = 3(2+1+0)+10(2+3+1+6+0+1)+\cdots+12(0+1+9) = 849$. Since $C > D$, low income tends to occur with low satisfaction.

For two independent observations from $\{\pi_{ij}\}$ the probabilities of concordance and discordance are

$$\Pi_c = 2\sum_i\sum_j \pi_{ij}\Big(\sum_{h>i}\sum_{k>j}\pi_{hk}\Big), \qquad \Pi_d = 2\sum_i\sum_j \pi_{ij}\Big(\sum_{h>i}\sum_{k<j}\pi_{hk}\Big), \tag{2.8}$$

the factor 2 because either observation may be the one in cell (i, j).

2.4.4 Ordinal measure of association: gamma

Definition (Goodman–Kruskal gamma)

Given that a pair is untied on both variables, $\Pi_c/(\Pi_c+\Pi_d)$ is the probability of concordance. *Gamma* is

$$\gamma = \frac{\Pi_c - \Pi_d}{\Pi_c + \Pi_d}, \qquad \text{sample version } \hat{\gamma} = \frac{C-D}{C+D}. \tag{2.9}$$

Proposition 2.18 (Properties of gamma)

(a) $-1 \le \gamma \le 1$; $\gamma = 1$ iff $\Pi_d = 0$ and $\gamma = -1$ iff $\Pi_c = 0$ (perfect monotonicity, not linearity).
(b) Reversing the category order of one variable changes the sign of γ; γ is symmetric in X and Y.
(c) Independence implies $\gamma = 0$; the converse is false.

Proof. [Supplement]

Step (a). $\Pi_c, \Pi_d \ge 0$, so $|\Pi_c - \Pi_d| \le \Pi_c + \Pi_d$, with the extreme cases as stated.

Step (b). Reversing one ordering turns concordant pairs into discordant ones and vice versa, swapping Π_c and Π_d. Interchanging the roles of X and Y does not change whether a pair is concordant.

Step (c). Let (X_1, Y_1), (X_2, Y_2) be independent draws. Then $\Pi_c = 2P(X_1 > X_2, Y_1 > Y_2)$ and $\Pi_d = 2P(X_1 > X_2, Y_1 < Y_2)$. Under independence of X and Y, (X_1, X_2) is independent of (Y_1, Y_2), so $\Pi_c = 2P(X_1 > X_2)P(Y_1 > Y_2)$ and $\Pi_d = 2P(X_1 > X_2)P(Y_1 < Y_2)$; since Y_1, Y_2 are i.i.d., $P(Y_1 > Y_2) = P(Y_1 < Y_2)$, so $\Pi_c = \Pi_d$ and $\gamma = 0$. For the converse, a U-shaped joint distribution can have $\Pi_c = \Pi_d$ without independence, e.g. the 3×3 table with $\pi_{11} = \pi_{31} = \pi_{22} = 1/3$ has $\Pi_c = 2\pi_{11}\pi_{22} = 2/9 = 2\pi_{22}\pi_{31} = \Pi_d$. ∎

2.4.5 Gamma for job satisfaction example

Example 2.19 (Gamma for income and job satisfaction)

From Example 2.17, $\hat{\gamma} = (1331-849)/(1331+849) = 0.221$: of the untied pairs, the proportion concordant exceeds the proportion discordant by 0.221, a weak tendency for satisfaction to increase with income.

Chapter 3. Inference for Contingency Tables

3.1 Confidence intervals for association parameters

Motivation. Standard errors for association measures come from the delta method; better intervals come from inverting score or LR tests.

3.1.1 Interval estimation of odds ratios

The sample odds ratio $\hat{\theta} = n_{11}n_{22}/(n_{12}n_{21})$ is 0 or ∞ when some $n_{ij} = 0$, so $E(\hat{\theta})$ and $\mathrm{Var}(\log\hat{\theta})$ do not exist; the amended $\tilde{\theta} = \frac{(n_{11}+0.5)(n_{22}+0.5)}{(n_{12}+0.5)(n_{21}+0.5)}$ has the same asymptotic distribution and behaves better. $\hat{\theta}$ is very skewed, while $\log\hat{\theta}$ approaches normality faster, with estimated standard error (Corollary 3.7)

$$\hat{\sigma}(\log\hat{\theta}) = \Big(\frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}\Big)^{1/2}. \tag{3.1}$$

Definition (Woolf interval)

$\log\hat{\theta} \pm z_{\alpha/2}\,\hat{\sigma}(\log\hat{\theta})$ is a Wald interval for $\log\theta$; exponentiating its endpoints gives the Woolf interval for θ.

It is usually slightly conservative. If $\hat{\theta} = 0$ (or ∞), use 0 (or ∞) as one limit and compute the other with $n_{ij} + 0.5$.

3.1.2 Aspirin and myocardial infarction example

Example 3.1 (Swedish aspirin study: odds ratio)

Of 684 placebo patients 28 died of MI; of 676 aspirin patients, 18. Then $\hat{\theta} = \frac{28\times658}{656\times18} = 1.56$, $\log\hat{\theta} = 0.445$, $\hat{\sigma}(\log\hat{\theta}) = 0.307$, and the 95% interval for θ is $(e^{-0.157}, e^{1.047}) = (0.85, 2.85)$: equal odds are plausible.

3.1.3 Interval estimation of difference of proportions

Treat the rows as independent binomials $y_i \sim \mathrm{bin}(n_i, \pi_i)$, $\hat{\pi}_i = y_i/n_i$.

Proposition 3.2 (Difference of proportions)

$E(\hat{\pi}_1 - \hat{\pi}_2) = \pi_1 - \pi_2$ and

$$\sigma(\hat{\pi}_1 - \hat{\pi}_2) = \Big[\frac{\pi_1(1-\pi_1)}{n_1} + \frac{\pi_2(1-\pi_2)}{n_2}\Big]^{1/2}. \tag{3.2}$$

As $n_1, n_2 \to \infty$, $(\hat{\pi}_1 - \hat{\pi}_2 - (\pi_1 - \pi_2))/\hat{\sigma} \xrightarrow{d} N(0,1)$, where $\hat{\sigma}$ replaces π_i by $\hat{\pi}_i$; so $(\hat{\pi}_1 - \hat{\pi}_2) \pm z_{\alpha/2}\hat{\sigma}$ is a Wald interval.

Proof.

Step 1. Linearity and independence give the mean and variance, using $\mathrm{Var}(\hat{\pi}_i) = \pi_i(1-\pi_i)/n_i$ (Proposition 1.8).

Step 2. [Supplement] The two standardized proportions are independent and asymptotically $N(0,1)$ (CLT), so is the standardized difference; $\hat{\sigma}/\sigma \xrightarrow{p} 1$ and Slutsky completes the proof. ■

As with (1.11), coverage is usually below nominal near 0 or 1.

3.1.4 Interval estimation of relative risk

The sample relative risk $r = \hat{\pi}_1/\hat{\pi}_2$ is also better handled on the log scale, with

$$\sigma(\log r) = \Big[\frac{1-\pi_1}{\pi_1 n_1} + \frac{1-\pi_2}{\pi_2 n_2}\Big]^{1/2} \tag{3.3}$$

(Corollary 3.5(b)); exponentiate $\log r \pm z_{\alpha/2}\hat{\sigma}(\log r)$.

Example 3.3 (Swedish aspirin study: relative risk and difference)

Proportions 0.0409 (placebo), 0.0266 (aspirin).
(a) $r = 1.54$, interval (0.86, 2.75).
(b) Difference 0.014, interval $(-0.005, 0.033)$.
No effect is plausible; the larger study of Example 2.7 shows a benefit.

3.1.5 Deriving standard errors with the delta method*

Theorem 3.4 (Delta method)

Suppose $\sqrt{n}(T_n - \theta) \xrightarrow{d} N(0,\sigma^2)$ and g is differentiable at θ. Then

$$\sqrt{n}\,[g(T_n) - g(\theta)] \xrightarrow{d} N(0,\,[g'(\theta)]^2\sigma^2).$$

If moreover g' is continuous at θ, $g'(\theta) \neq 0$, and $\hat{\sigma} \xrightarrow{p} \sigma > 0$, then $\sqrt{n}[g(T_n) - g(\theta)]/(|g'(T_n)|\hat{\sigma}) \xrightarrow{d} N(0,1)$.

Proof. [Supplement]

Step 1. Let $h(t) = \frac{g(t)-g(\theta)}{t-\theta} - g'(\theta)$ for $t \neq \theta$, $h(\theta) = 0$; h is continuous at θ. Since $T_n \xrightarrow{p} \theta$, $h(T_n) \xrightarrow{p} 0$.

Step 2. $\sqrt{n}[g(T_n) - g(\theta)] = [g'(\theta) + h(T_n)]\sqrt{n}(T_n - \theta) \xrightarrow{d} g'(\theta)N(0,\sigma^2)$ by Slutsky.

Step 3. $|g'(T_n)|\hat{\sigma} \xrightarrow{p} |g'(\theta)|\sigma > 0$, and Slutsky again gives the standardized version. ■

Locally g is linear with slope $g'(\theta)$, so the spread of $g(T_n)$ is $|g'(\theta)|$ times that of T_n (Figure 4).

3.1.6 Delta method applied to sample logit*

Corollary 3.5 (Sample logit and log proportion)

Let $\hat{\pi} = y/n$ with $y \sim \text{bin}(n,\pi)$, $0 < \pi < 1$.
(a) The sample logit satisfies

$$\sqrt{n}\Big[\log\frac{\hat{\pi}}{1-\hat{\pi}} - \log\frac{\pi}{1-\pi}\Big] \xrightarrow{d} N\Big(0,\,\frac{1}{\pi(1-\pi)}\Big).$$

(b) $\sqrt{n}(\log\hat{\pi} - \log\pi) \xrightarrow{d} N(0,(1-\pi)/\pi)$; hence for independent samples the log relative risk

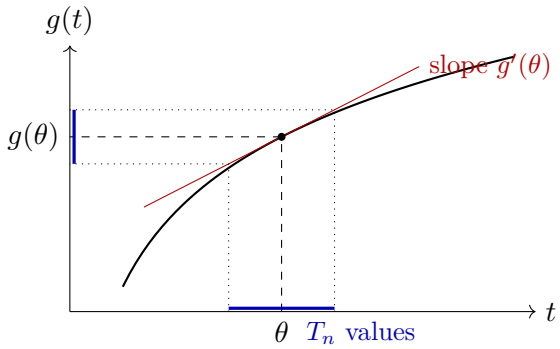


Figure 4: The delta method: near θ, g is almost linear, so a band of T_n values maps to a band of $g(T_n)$ values about $|g'(\theta)|$ times as wide.

has asymptotic variance (3.3).

Proof.

Step (a). $\sqrt{n}(\hat{\pi} - \pi) \xrightarrow{d} N(0, \pi(1-\pi))$, and $g(\pi) = \log[\pi/(1-\pi)]$ has $g'(\pi) = 1/[\pi(1-\pi)]$; Theorem 3.4 gives variance $\pi(1-\pi)[\pi(1-\pi)]^{-2}$.

Step (b). With $g = \log$, $g'(\pi) = 1/\pi$ and the variance is $(1-\pi)/\pi$. Dividing by n_i and adding over the two independent samples gives (3.3). ■

The asymptotic variance is that of the limiting normal; the true variance does not exist, since $\hat{\pi} \in \{0, 1\}$ with positive probability.

3.1.7 Delta method for log odds ratio*

Theorem 3.6 (Multivariate delta method for multinomial proportions)

Let $(n_1, \ldots, n_c)$ be multinomial$(n, \boldsymbol{\pi})$, $\hat{\boldsymbol{\pi}} = \mathbf{n}/n$, and let g be differentiable at $\boldsymbol{\pi}$ with $\phi_i = \partial g/\partial \pi_i$. Define

$$\sigma^2 = \sum_i \pi_i \phi_i^2 - \Big(\sum_i \pi_i \phi_i\Big)^2. \tag{3.4}$$

If $\sigma > 0$ then $\sqrt{n}[g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi})]/\sigma \xrightarrow{d} N(0,1)$, and the same holds with σ replaced by its ML estimate $\hat{\sigma}$ (substituting $\hat{\pi}_i$ for π_i in (3.4)).

Proof. [Supplement]

Step 1. By the multivariate CLT, $\sqrt{n}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}) \xrightarrow{d} \mathbf{z} \sim N_c(\mathbf{0}, \Sigma)$, $\Sigma = \operatorname{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top$ (Proposition 1.2).

Step 2. Differentiability gives $\sqrt{n}[g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi})] = \boldsymbol{\phi}^\top \sqrt{n}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}) + o_p(1) \xrightarrow{d} \boldsymbol{\phi}^\top \mathbf{z}$, since $\|\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}\| = O_p(n^{-1/2})$.

Step 3. $\boldsymbol{\phi}^\top \mathbf{z} \sim N(0, \boldsymbol{\phi}^\top \Sigma \boldsymbol{\phi})$, and $\boldsymbol{\phi}^\top \Sigma \boldsymbol{\phi} = \sum_i \pi_i \phi_i^2 - (\sum_i \pi_i \phi_i)^2 = \sigma^2$.

Step 4. If the ϕ_i are continuous, $\hat{\sigma} \xrightarrow{p} \sigma$ (WLLN), and Slutsky applied to the product of the standardized statistic and $\sigma/\hat{\sigma} \xrightarrow{p} 1$ gives the limit $N(0,1)$. ■

Corollary 3.7 (Standard error of the log odds ratio)

For a multinomial 2×2 table, $\log \hat{\theta}$ has asymptotic standard error $\left(\sum_i \sum_j 1/(n\pi_{ij})\right)^{1/2}$, estimated by (3.1).

Proof.

Step 1. $g(\boldsymbol{\pi}) = \log \pi_{11} + \log \pi_{22} - \log \pi_{12} - \log \pi_{21}$ has $\phi_{ij} = \pm 1/\pi_{ij}$ (+ on the diagonal).

Step 2. $\sum\sum \pi_{ij}\phi_{ij} = 1 - 1 - 1 + 1 = 0$ and $\sum\sum \pi_{ij}\phi_{ij}^2 = \sum\sum 1/\pi_{ij}$; Theorem 3.6 gives $\sigma/\sqrt{n} = (\sum\sum 1/(n\pi_{ij}))^{1/2}$, estimated by $(\sum\sum 1/n_{ij})^{1/2}$. ■

The same holds for independent binomial rows (a difference of two sample logits, Corollary 3.5(a)). A Wald interval for θ itself is not recommended.

3.2 Testing independence in two-way contingency tables

Motivation. Independence, $\pi_{ij} = \pi_{i+}\pi_{+j}$, is the starting point of contingency-table analysis, tested by X^2 and G^2.

Under independent multinomial rows the hypothesis is homogeneity; the same tests apply.

3.2.1 Pearson and likelihood-ratio chi-squared tests

Proposition 3.8 (ML estimates under independence)

Under H_0, the ML estimates are $\hat{\pi}_{i+} = n_{i+}/n$, $\hat{\pi}_{+j} = n_{+j}/n$, so the estimated expected frequencies are

$$\hat{\mu}_{ij} = n\hat{\pi}_{i+}\hat{\pi}_{+j} = \frac{n_{i+}n_{+j}}{n}.$$

Proof. The kernel $\prod_i \prod_j (\pi_{i+}\pi_{+j})^{n_{ij}} = \prod_i \pi_{i+}^{n_{i+}} \cdot \prod_j \pi_{+j}^{n_{+j}}$ is a product of two multinomial kernels in separate parameters, each maximized by sample proportions (Proposition 1.14). ■

$$X^2 = \sum_i \sum_j \frac{(n_{ij} - \hat{\mu}_{ij})^2}{\hat{\mu}_{ij}}, \qquad G^2 = 2\sum_i \sum_j n_{ij} \log \frac{n_{ij}}{\hat{\mu}_{ij}}. \tag{3.5}$$

Theorem 3.9 (Null distribution of X^2 and G^2 for independence)

Under H_0 with all $\pi_{i+}, \pi_{+j} > 0$, X^2 and G^2 both converge in distribution to $\chi^2_{(I-1)(J-1)}$, and $X^2 - G^2 \xrightarrow{p} 0$.

Proof. [Supplement] Apply Theorem 1.18 to the $c = IJ$ cells with the $p = (I-1) + (J-1)$ parameters $\pi_{1+}, \ldots, \pi_{I-1,+}, \pi_{+1}, \ldots, \pi_{+,J-1}$.

Step 1 (rank). With $a_i = \pi_{i+}$, $b_j = \pi_{+j}$, a tangent direction is $d\pi_{ij} = b_j\, da_i + a_i\, db_j$ with $\sum_i da_i = \sum_j db_j = 0$. If all $d\pi_{ij} = 0$, summing over j gives $da_i = 0$, and then $db_j = 0$; so the Jacobian has rank p.

Step 2 (df). Hence $X^2 \xrightarrow{d} \chi^2_{IJ-1-p}$ with $IJ - 1 - (I-1) - (J-1) = (I-1)(J-1)$, and $G^2 - X^2 \xrightarrow{p} 0$. ■

Fisher (1922) corrected Pearson's df $= IJ - 1$. The limits also hold for Poisson and product multinomial sampling; X^2 converges faster than G^2 (small-sample methods: Section 3.5).

3.2.2 Education and religious fundamentalism example

Example 3.10 (Education and religious beliefs)

1996 GSS: count, $\hat{\mu}_{ij}$, and standardized residual (3.6).

Highest degree	Fundamentalist		Moderate		Liberal		Total
< High school	178 (137.8)	4.5	138 (161.5)	−2.6	108 (124.7)	−1.9	424
HS / junior college	570 (539.5)	2.6	648 (632.1)	1.3	442 (488.4)	−4.0	1660
Bachelor / graduate	138 (208.7)	−6.8	252 (244.5)	0.7	252 (188.9)	6.3	642
Total	886		1038		802		2726

$X^2 = 69.2$, $G^2 = 69.8$, df $= 4$, $P < 0.0001$.

3.3 Following-up chi-squared tests

Motivation. Residuals, partitions of G^2 and odds ratios reveal the nature of an association that a test only detects.

3.3.1 Pearson and standardized residuals

Definition (Pearson and standardized residuals)

The *Pearson residual* and *standardized Pearson residual* for independence are

$$e_{ij} = \frac{n_{ij} - \hat{\mu}_{ij}}{\hat{\mu}_{ij}^{1/2}}, \qquad r_{ij} = \frac{n_{ij} - \hat{\mu}_{ij}}{[\hat{\mu}_{ij}(1 - p_{i+})(1 - p_{+j})]^{1/2}}, \tag{3.6}$$

with $p_{i+} = n_{i+}/n$, $p_{+j} = n_{+j}/n$. Note $\sum_i \sum_j e_{ij}^2 = X^2$.

Theorem 3.11 (Asymptotic variance of Pearson residuals)

Under independence, $e_{ij} \xrightarrow{d} N(0, (1 - \pi_{i+})(1 - \pi_{+j}))$; hence $r_{ij} \xrightarrow{d} N(0, 1)$. The average asymptotic variance of the e_{ij} is $(I - 1)(J - 1)/(IJ) < 1$.

Proof. [Supplement] Use the notation of the proof of Theorem 1.18 with cells (i, j), $a_i = \pi_{i+}$, $b_j = \pi_{+j}$, and unit vectors $\mathbf{r} = (\sqrt{a_i})_i$, $\mathbf{s}_c = (\sqrt{b_j})_j$, so $\mathbf{s} = \mathbf{r} \otimes \mathbf{s}_c$.

Step 1. By that proof the Pearson residuals are $(I - H)\mathbf{z}_n + o_p(1)$ with limiting covariance $I - H - \mathbf{s}\mathbf{s}^\top$, H the projection onto the column space of A.

Step 2. A tangent direction $d\pi_{ij} = b_j\, da_i + a_i\, db_j$ gives $D^{-1/2} d\boldsymbol{\pi}$ with entries $\sqrt{b_j}\, x_i + \sqrt{a_i}\, y_j$, $x_i = da_i/\sqrt{a_i}$, $y_j = db_j/\sqrt{b_j}$, where $\mathbf{x} \perp \mathbf{r}$, $\mathbf{y} \perp \mathbf{s}_c$. So the column space is $(\mathbf{r}^\perp \otimes \mathbf{s}_c) \oplus (\mathbf{r} \otimes \mathbf{s}_c^\perp)$, two orthogonal subspaces, and

$$H = (I_I - \mathbf{r}\mathbf{r}^\top) \otimes \mathbf{s}_c\mathbf{s}_c^\top + \mathbf{r}\mathbf{r}^\top \otimes (I_J - \mathbf{s}_c\mathbf{s}_c^\top).$$

Step 3. The (ij, ij) entries of H and $\mathbf{s}\mathbf{s}^\top$ are $(1 - a_i)b_j + a_i(1 - b_j)$ and $a_i b_j$, so e_{ij} has asymptotic variance $1 - a_i - b_j + a_i b_j = (1 - a_i)(1 - b_j)$.

Step 4. Slutsky gives $r_{ij} \xrightarrow{d} N(0, 1)$, and $\frac{1}{IJ} \sum_i \sum_j (1 - a_i)(1 - b_j) = \frac{(I-1)(J-1)}{IJ}$. ■

Comparing e_{ij} with $N(0, 1)$ is conservative; $|r_{ij}|$ above about 2 or 3 indicates lack of fit in that cell.

3.3.2 Education and religious fundamentalism revisited

Example 3.12 (Residuals for education and religion)

$r_{11} = (178 - 137.8)/[137.8(1 - 0.156)(1 - 0.325)]^{1/2} = 4.5$. Large positive residuals at (less than HS, fundamentalist) and (bachelor+, liberal), large negative at the other two corners; the corner 2×2 odds ratio is $\frac{178 \times 252}{108 \times 138} = 3.0$.

3.3.3 Partitioning chi-squared

Since independent $\chi^2_{\nu_1}$ and $\chi^2_{\nu_2}$ sum to $\chi^2_{\nu_1+\nu_2}$, a statistic can be split into components showing which categories drive an association.

Theorem 3.13 (Exact partition of G^2 by successive columns)

For an $I \times J$ table, for $j = 1, \ldots, J-1$ let G^2_j be the G^2 statistic for independence in the $I \times 2$ table whose first column merges columns $1, \ldots, j$ and whose second column is column $j+1$. Then

$$G^2 = \sum_{j=1}^{J-1} G^2_j,$$

and each G^2_j has $\mathrm{df} = I - 1$, so the df also add up to $(I-1)(J-1)$.

Proof. [Supplement]

Step 1. With $\Phi(x) = x \log x$, for any table with counts m_{ij}, margins m_{i+}, m_{+j} and total m, (3.5) gives

$$G^2/2 = \sum_i \sum_j \Phi(m_{ij}) - \sum_i \Phi(m_{i+}) - \sum_j \Phi(m_{+j}) + \Phi(m),$$

since $\log(m_{ij}/\hat{\mu}_{ij}) = \log m_{ij} + \log m - \log m_{i+} - \log m_{+j}$.

Step 2. Let $S_{ij} = \sum_{l \le j} n_{il}$ and $S_{+j} = \sum_i S_{ij}$. The jth subtable has cells S_{ij}, $n_{i,j+1}$, row totals $S_{i,j+1}$, column totals S_{+j}, $n_{+,j+1}$ and total $S_{+,j+1}$, so $G^2_j/2$ equals

$$\sum_i [\Phi(S_{ij}) + \Phi(n_{i,j+1}) - \Phi(S_{i,j+1})] - \Phi(S_{+j}) - \Phi(n_{+,j+1}) + \Phi(S_{+,j+1}).$$

Step 3. Summing over j, $\Phi(S_{ij}) - \Phi(S_{i,j+1})$ telescopes to $\Phi(n_{i1}) - \Phi(n_{i+})$, which with $\sum_{j \ge 2} \Phi(n_{ij})$ gives $\sum_j \Phi(n_{ij}) - \Phi(n_{i+})$; likewise the column terms give $\Phi(n) - \sum_j \Phi(n_{+j})$. The total is $G^2/2$ by Step 1. ■

Repeating this on rows gives Lancaster's partition into $(I-1)(J-1)$ components with $\mathrm{df} = 1$, from the 2×2 tables

$$\begin{matrix} \sum_{a<i} \sum_{b<j} n_{ab} & \sum_{a<i} n_{aj} \\ \sum_{b<j} n_{ib} & n_{ij} \end{matrix} \qquad i = 2, \ldots, I, \; j = 2, \ldots, J. \tag{3.7}$$

Comparing each column with the last does *not* give an additive partition (Section 9.2.4 views partitions as nested models).

3.3.4 Origin of schizophrenia example

Example 3.14 (School of psychiatric thought and origin of schizophrenia)

Schools (eclectic, medical, psychoanalytic) by ascribed origin (biogenic, environmental, combination): $G^2 = 23.04$, df $= 4$. By (3.7), Ecl vs. Med components are 0.29 and 1.36; (Ecl+Med) vs. Psy components are 12.95 (Bio vs. Env) and 8.43 ((Bio+Env) vs. Com), summing to 23.04. Psychoanalysts more often ascribe a combined or environmental origin.

3.3.5 Rules for partitioning

Necessary conditions for independent components (Goodman; Lancaster):

(a) the subtable df sum to the df of the full table;

(b) each cell count of the full table is a cell count in exactly one subtable;

(c) each marginal total of the full table is a marginal total of exactly one subtable.

Exact partitions hold for G^2 but not X^2.

3.3.6 Limitations of chi-squared tests

They only measure evidence, need large samples, and are invariant to reordering categories, so they ignore ordinal information (Section 3.4).

3.3.7 Why consider independence?

If independence approximately holds, $p_{i+}p_{+j}$ can beat p_{ij} in MSE: biased, but with fewer parameters.

Proposition 3.15 (MSE of the two estimators)

For multinomial sampling,

$$\sum_i \sum_j E(p_{ij} - \pi_{ij})^2 = \frac{1 - \sum_i \sum_j \pi_{ij}^2}{n}, \qquad \sum_i \sum_j E(p_{i+}p_{+j} - \pi_{ij})^2 \to \sum_i \sum_j (\pi_{i+}\pi_{+j} - \pi_{ij})^2.$$

Under independence, $n \sum \sum E(p_{i+}p_{+j} - \pi_{ij})^2 \to \sum_i \sum_j [\pi_{+j}^2 \pi_{i+}(1 - \pi_{i+}) + \pi_{i+}^2 \pi_{+j}(1 - \pi_{+j})]$.

Proof. [Supplement]

Step 1. p_{ij} is unbiased with $\mathrm{Var}(p_{ij}) = \pi_{ij}(1 - \pi_{ij})/n$; sum over cells.

Step 2. With $a = p_{i+} - \pi_{i+}$, $b = p_{+j} - \pi_{+j}$, $p_{i+}p_{+j} - \pi_{ij} = (\pi_{i+}\pi_{+j} - \pi_{ij}) + \pi_{+j}a + \pi_{i+}b + ab$. Since $E(a^2), E(b^2) = O(n^{-1})$ and $E(a^2b^2) \le [E(a^4)E(b^4)]^{1/2} = O(n^{-2})$, all but the first term vanish in mean square.

Step 3. Under independence the first term is 0, and $nE(\pi_{+j}a + \pi_{i+}b)^2 = \pi_{+j}^2 \pi_{i+}(1-\pi_{i+}) + \pi_{i+}^2 \pi_{+j}(1-\pi_{+j}) + 2\pi_{i+}\pi_{+j}\, n\,\mathrm{Cov}(a, b)$, where $n\,\mathrm{Cov}(p_{i+}, p_{+j}) = \pi_{ij} - \pi_{i+}\pi_{+j} = 0$; the terms involving ab are $o(n^{-1})$ by Cauchy–Schwarz. ∎

Example 3.16 (A 3×3 illustration)

Let $\pi_{i+} = \pi_{+j} = 1/3$ and $\pi_{ij} = \frac{1}{9}[1 + \delta(i-2)(j-2)]$.
(a) At $\delta = 0$ the total MSE is $8/(9n)$ for p_{ij} and $\approx 4/(9n)$ for $p_{i+}p_{+j}$.

(b) For $\delta \neq 0$ the latter tends to $4\delta^2/81$, yet still wins for moderate n (e.g. $n = 500$, $\delta = 0.1$).

3.4 Two-way tables with ordered classifications

Motivation. For ordinal variables, one-parameter trend statistics give df $= 1$ tests with much more power against monotone alternatives.

3.4.1 Linear trend alternative to independence

With row scores $u_1 \leq \cdots \leq u_I$ and column scores $v_1 \leq \cdots \leq v_J$, let r be the sample correlation of the scores over the n subjects:

$$r = \frac{\sum\sum u_i v_j n_{ij} - (\sum_i u_i n_{i+})(\sum_j v_j n_{+j})/n}{\left[\sum_i u_i^2 n_{i+} - \frac{(\sum_i u_i n_{i+})^2}{n}\right]^{1/2}\left[\sum_j v_j^2 n_{+j} - \frac{(\sum_j v_j n_{+j})^2}{n}\right]^{1/2}}.$$

Definition (Linear trend statistic)

$M^2 = (n-1)r^2$ (Mantel, 1963), referred to χ^2_1; for one-sided alternatives use $M = \sqrt{n-1}\, r$ with the normal tail.

Theorem 3.17 (Null distribution of M^2)

Under independence of X and Y (multinomial sampling, each score variable nondegenerate), $\sqrt{n}\, r \xrightarrow{d} N(0,1)$; hence $M \xrightarrow{d} N(0,1)$ and $M^2 \xrightarrow{d} \chi^2_1$.

Proof. [Supplement] Let (U_k, V_k), $k \leq n$, be the subjects' score pairs: i.i.d., U independent of V, with means μ, ν and variances $\sigma^2, \tau^2 > 0$.

Step 1. $S_{UV} = \frac{1}{n}\sum_k W_k - (\bar{U} - \mu)(\bar{V} - \nu)$ with $W_k = (U_k - \mu)(V_k - \nu)$ i.i.d., $E(W_k) = 0$ and $\mathrm{Var}(W_k) = \sigma^2\tau^2$ by independence; so $n^{-1/2}\sum_k W_k \xrightarrow{d} N(0, \sigma^2\tau^2)$.

Step 2. $\sqrt{n}(\bar{U} - \mu)(\bar{V} - \nu) = O_p(1)o_p(1) \xrightarrow{p} 0$, so $\sqrt{n}S_{UV} \xrightarrow{d} N(0, \sigma^2\tau^2)$.

Step 3. $S_{UU} \xrightarrow{p} \sigma^2$, $S_{VV} \xrightarrow{p} \tau^2$, so by Slutsky $\sqrt{n}\, r \xrightarrow{d} N(0,1)$; as $\sqrt{(n-1)/n} \to 1$, $M \xrightarrow{d} N(0,1)$ and $M^2 \xrightarrow{d} \chi^2_1$. ■

3.4.2 Job satisfaction example revisited

Example 3.18 (Linear trend test for income and job satisfaction)

For Example 2.17 ($n = 96$), $X^2 = 6.0$, $G^2 = 6.8$, df $= 9$ ($P \approx 0.7$). With scores $1, 2, 3, 4$ and income midpoints $7.5, 20, 32.5, 60$, $r = 0.200$, $M^2 = 3.81$ ($P = 0.051$) and $M = 1.95$ (one-sided $P = 0.026$): much stronger evidence.

3.4.3 Monotone trend alternatives to independence

Ordinal measures such as gamma (Section 2.4.4) give $z = \hat{\gamma}/\mathrm{SE}$; for Example 2.17, $\hat{\gamma} = 0.221$, SE $= 0.117$, $z = 1.89$, 95% interval $(-0.01, 0.45)$.

3.4.4 Extra power with ordinal tests

X^2 and G^2 spread power over $(I-1)(J-1)$ parameters; M^2 and $(\hat{\gamma}/\mathrm{SE})^2$ concentrate it on one, so a comparable value lies much farther in the χ^2_1 tail (Section 6.4).

3.4.5 Choice of scores

Any scores chosen before seeing the data give a valid test, but poor scores give an insensitive one; linear transformations leave M^2 unchanged.

Example 3.19 (Maternal drinking and congenital malformations)

Malformation (present/absent) by drinks per day (0, < 1, 1–2, 3–5, ≥ 6), with present counts $48, 38, 5, 1, 1$ among 17,114, 14,502, 793, 127, 38 subjects.
(a) Scores $0, 0.5, 1.5, 4.0, 7.0$: $M^2 = 6.57$, $P = 0.010$.
(b) Scores $1, 2, 3, 4, 5$: $M^2 = 1.83$, $P = 0.18$.
(c) Midranks: $M^2 = 0.35$, $P = 0.55$; they treat the sparse last three categories as nearly equal, which is inappropriate.

Use scores reflecting distances and check sensitivity. With midranks for both variables, r is *Spearman's rho*.

3.4.6 Trend tests for $I \times 2$ and $2 \times J$ tables

For $2 \times J$ with $u = (0, 1)$, M^2 compares the two row means of the Y scores; with midranks it is the *Wilcoxon–Mann–Whitney* test. For $I \times 2$ it is the *Cochran–Armitage trend test* (Section 5.3.5).

3.4.7 Nominal–ordinal tables

With a nominal variable of more than two categories, compare mean scores of the ordinal variable across categories (Section 7.5.3).

3.5 Small-sample tests of independence

Motivation. Conditioning on the margins eliminates nuisance parameters and gives the exact hypergeometric null distribution.

3.5.1 Fisher's exact test for 2×2 tables

Theorem 3.20 (Hypergeometric null distribution)

Under independence, for Poisson, multinomial, or independent binomial sampling, conditional on both sets of marginal totals,

$$p(t) = P(n_{11} = t) = \frac{\binom{n_{1+}}{t}\binom{n_{2+}}{n_{+1}-t}}{\binom{n}{n_{+1}}}, \qquad m_- \le t \le m_+, \tag{3.8}$$

where $m_- = \max(0,\, n_{1+} + n_{+1} - n)$ and $m_+ = \min(n_{1+},\, n_{+1})$.

Proof.

Step 1. By Theorem 2.3, given row totals, $n_{11} \sim \text{bin}(n_{1+}, \pi)$ and $n_{21} \sim \text{bin}(n_{2+}, \pi)$ independently under H_0, and $n_{+1} \sim \text{bin}(n, \pi)$.

Step 2. So $P(n_{11} = t \mid n_{+1})$ is $\binom{n_{1+}}{t}\binom{n_{2+}}{n_{+1}-t}\pi^{n_{+1}}(1-\pi)^{n-n_{+1}}$ divided by $\binom{n}{n_{+1}}\pi^{n_{+1}}(1-\pi)^{n-n_{+1}}$; the powers cancel, leaving (3.8).

Step 3. $0 \le t \le n_{1+}$ and $0 \le n_{+1} - t \le n_{2+}$ give $m_- \le t \le m_+$. ■

For $H_a\colon \theta > 1$, $P = P(n_{11} \ge t_o)$.

Proposition 3.21 (Orderings of 2×2 tables with fixed margins)

With all margins fixed, the sample odds ratio and the difference of proportions $n_{11}/n_{1+} - n_{21}/n_{2+}$ are increasing in n_{11}, and

$$X^2 = \frac{n^3\big(n_{11} - E(n_{11})\big)^2}{n_{1+}n_{2+}n_{+1}n_{+2}}, \qquad E(n_{11}) = \frac{n_{1+}n_{+1}}{n}.$$

Hence one-sided P-values agree for all three orderings, and $P(X^2 \ge X_o^2) = P(|n_{11} - E(n_{11})| \ge |t_o - E(n_{11})|)$.

Proof. [Supplement]

Step 1. With $n_{12} = n_{1+} - n_{11}$, $n_{21} = n_{+1} - n_{11}$, $n_{22} = n - n_{1+} - n_{+1} + n_{11}$, the numerator of $\hat\theta$ increases and its denominator decreases in n_{11}, and $n_{11}/n_{1+} - (n_{+1} - n_{11})/n_{2+}$ increases.

Step 2. $n_{11}n_{22} - n_{12}n_{21} = n\,n_{11} - n_{1+}n_{+1} = n[n_{11} - E(n_{11})]$ by expanding.

Step 3. In (3.5) all four $n_{ij} - \hat\mu_{ij}$ equal $\pm(n_{11}n_{22} - n_{12}n_{21})/n$ and $\sum_{i,j} 1/\hat\mu_{ij} = n^3/(n_{1+}n_{2+}n_{+1}n_{+2})$, so $X^2 = n(n_{11}n_{22} - n_{12}n_{21})^2/(n_{1+}n_{2+}n_{+1}n_{+2})$; Step 2 gives the display. ■

3.5.2 Fisher's tea drinker

Example 3.22 (Fisher's tea-tasting experiment)

A lady tasted 8 cups, 4 milk-first, and named 4 as milk-first, getting 3 right ($n_{11} = 3$, all margins 4, fixed by design). For $H_a\colon \theta > 1$,

$$P(n_{11} = 3) = \frac{\binom{4}{3}\binom{4}{1}}{\binom{8}{4}} = \frac{16}{70} = 0.229, \qquad P(n_{11} = 4) = \frac{1}{70} = 0.014,$$

so $P = 0.243$: too small an experiment to establish her ability.

3.5.3 Two-sided P-values for Fisher's exact test

Two-sided criteria include:

(a) sum $p(t)$ over t with $p(t) \le p(t_o)$ (the most popular);

(b) $P(|n_{11} - E(n_{11})| \ge |t_o - E(n_{11})|)$, which equals $P(X^2 \ge X_o^2)$ by Proposition 3.21;

(c) $2\min[P(n_{11} \ge t_o), P(n_{11} \le t_o)]$, which can exceed 1;

(d) the smaller tail plus the largest attainable probability in the other tail not exceeding it (Blaker).

3.5.4 Discreteness and conservatism issues

In the tea experiment one-sided P-values can only be 0.014, 0.243, 0.757, 0.986, 1, so rejecting at $P \leq 0.05$ has actual size 0.014: the test is *conservative*.

Example 3.23 (Randomized test of exact size 0.05)

Reject if $n_{11} = 4$, and with probability 0.157 if $n_{11} = 3$: $P(\text{reject}) = 0.014 + 0.157(0.229) = 0.05$.

Randomization is unacceptable in practice; instead report the *mid-P-value* $\frac{1}{2}P(n_{11} = t_o) + P(n_{11} > t_o)$ (0.129 for the tea data), rarely much above nominal size.

3.6 Extensions for multiway tables and nontabulated responses

Motivation. Models extend these tests to conditional independence (Chapter 6) and to continuous explanatory variables.

3.6.1 Categorical data need not be contingency tables

Data files usually have one line per subject; the generalized linear models of the next chapter contain this chapter's methods as special cases.

Chapter 4. Introduction to Generalized Linear Models

Generalized linear models (GLMs) contain logistic regression, Poisson loglinear models and ordinary regression. Sections 4.1–4.3 present the models; Sections 4.4–4.6 give the likelihood theory.

4.1 Generalized linear model

Motivation. For counts and proportions neither normality nor a linear mean is natural; a GLM keeps the linear predictor but chooses the response distribution and the linearizing scale separately.

4.1.1 Components of generalized linear models

Definition (Natural exponential family)

A family of the form

$$f(y_i;\theta_i) = a(\theta_i)\, b(y_i) \exp\{y_i\, Q(\theta_i)\} \tag{4.1}$$

is a *natural exponential family*; $Q(\theta)$ is the *natural parameter*.

Definition (Generalized linear model)

(i) *Random component*: independent $y_1, \ldots, y_N$, each from (4.1) (more generally, the exponential dispersion family of Section 4.4).
(ii) *Systematic component*: a *linear predictor* $\eta_i = \sum_{j=1}^{p} \beta_j x_{ij}$ (usually $x_{i1} \equiv 1$, an intercept α).
(iii) *Link function*: a monotone differentiable g with

$$g(\mu_i) = \eta_i = \sum_j \beta_j x_{ij}, \qquad \mu_i = E(Y_i). \tag{4.2}$$

The link with $g(\mu_i) = Q(\theta_i)$ is the *canonical link*; $g(\mu) = \mu$ is the *identity link*.

4.1.2 Binomial logit models for binary data

For $Y \in \{0,1\}$ with $P(Y=1) = \pi$, $f(y;\pi) = (1-\pi)\exp\{y \log[\pi/(1-\pi)]\}$, which is (4.1) with $a(\pi) = 1-\pi$, $b(y) = 1$, $Q(\pi) = \operatorname{logit} \pi$. The logit is the canonical link; GLMs with it are *logit models*.

4.1.3 Poisson loglinear models for count data

For a Poisson count, $f(y;\mu) = e^{-\mu}(1/y!)\exp(y \log \mu)$, so $Q(\mu) = \log \mu$ and the canonical link is the log link:

$$\log \mu_i = \sum_j \beta_j x_{ij}, \qquad i = 1, \ldots, N, \tag{4.3}$$

the *Poisson loglinear model*.

4.1.4 Generalized linear models for continuous responses

The normal distribution (with a dispersion parameter) has the mean as natural parameter, so regression, ANOVA and ANCOVA are GLMs with identity link.

Random component	Link	Systematic	Model
Normal	identity	cont./categ./mixed	regression / ANOVA / ANCOVA
Binomial	logit	mixed	logistic regression (Ch. 5–6)
Poisson	log	mixed	loglinear (Ch. 8–9)
Multinomial	generalized logit	mixed	multinomial response (Ch. 7)

Unlike transforming Y and using least squares, the link need not stabilize variance or produce normality: fitting maximizes the likelihood of the chosen distribution.

4.1.5 Deviance

Let $L(\boldsymbol{\mu}; \mathbf{y})$ be the log likelihood in terms of the means. The *saturated model* ($\hat{\mu}_i = y_i$, one parameter per observation) attains the largest value $L(\mathbf{y}; \mathbf{y})$.

Definition (Deviance)

For a Poisson or binomial GLM, $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = -2[L(\hat{\boldsymbol{\mu}}; \mathbf{y}) - L(\mathbf{y}; \mathbf{y})]$, the LR statistic for the model against the saturated alternative.

With N fixed and growing counts it is asymptotically χ^2_{N-p} (Theorem 4.19).

Proposition 4.1 (Deviance of the homogeneity model)

Let $y_i \sim \text{bin}(n_i, \pi_i)$ independently, $i = 1, \ldots, N$, and consider the model $\pi_i = \pi$ for all i ($p = 1$). Its deviance equals the statistic G^2 for testing independence in the $N \times 2$ table of successes and failures, and has df $= N - 1$.

Proof.

Step 1. Let $n = \sum n_i$, $s = \sum y_i$. With $L = \sum_i [y_i \log \pi_i + (n_i - y_i) \log(1 - \pi_i)]$, the MLEs are $\hat{\pi} = s/n$ (model) and $\tilde{\pi}_i = y_i/n_i$ (saturated).

Step 2. Hence

$$D = 2 \sum_i \left[y_i \log \frac{y_i}{n_i s/n} + (n_i - y_i) \log \frac{n_i - y_i}{n_i(n-s)/n} \right].$$

Step 3. The independence fitted values of the $N \times 2$ table are $n_i s/n$ and $n_i(n-s)/n$, so $D = 2 \sum \text{obs} \log(\text{obs}/\text{fit}) = G^2$, with df $= N - 1$ (N saturated parameters minus one). ■

4.1.6 Advantages of the GLM formulation

One theory and one algorithm (IRLS, Section 4.6) cover every exponential-family response and every link; deviance components serve as residuals and deviance comparisons generalize ANOVA (Section 4.5).

4.2 Generalized linear models for binary data

Motivation. For binary Y, $\text{Var}(Y) = \pi(x)[1 - \pi(x)]$, and the choice of which function of $\pi(x)$ is linear in x (identity, logit, probit) determines how effects are interpreted.

4.2.1 Linear probability model

The *linear probability model* $\pi(x) = \alpha + \beta x$ is the binomial GLM with identity link; β is the change in probability per unit x. Its defect is that $\alpha + \beta x$ eventually leaves $[0, 1]$. It should be fitted by binomial ML, not least squares, since the variance depends on x; the two agree only when fitted values stay in about 0.3–0.7.

4.2.2 Snoring and heart disease example

Example 4.2 (Snoring and heart disease)

2484 subjects, snoring scores $x = (0, 2, 4, 5)$, independent binomial rows.

Snoring	Yes	No	Proportion	Linear fit	Logit fit
Never ($x = 0$)	24	1355	0.017	0.017	0.021
Occasionally ($x = 2$)	35	603	0.055	0.057	0.044
Nearly every night ($x = 4$)	21	192	0.099	0.096	0.093
Every night ($x = 5$)	30	224	0.118	0.116	0.132

The ML linear fit is $\hat{\pi}(x) = 0.0172 + 0.0198x$ ($\widehat{\text{se}} = 0.0028$): risk 0.02 for nonsnorers, rising about 0.04 per two score units. The logistic fit is $\text{logit}\,\hat{\pi}(x) = -3.87 + 0.40x$; over this narrow range the fits are similar (Figure 5).

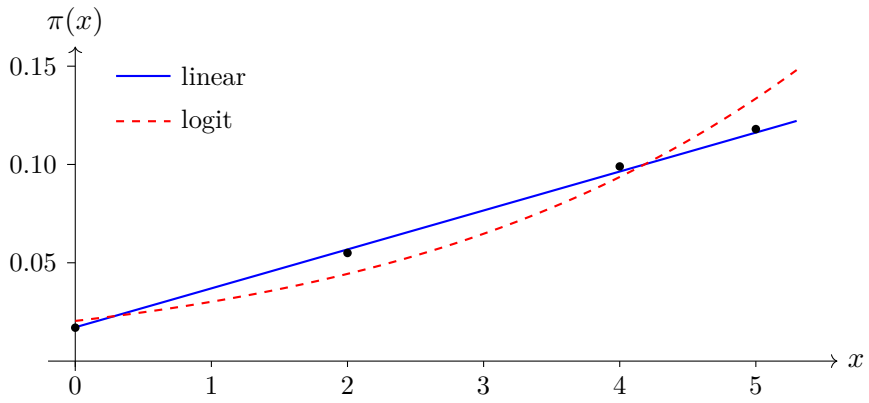


Figure 5: Snoring data: sample proportions with the linear probability and logistic fits.

4.2.3 Logistic regression model

Usually a fixed change in x matters less when $\pi(x)$ is near 0 or 1, giving S-shaped curves such as

$$\pi(x) = \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)}. \tag{4.4}$$

Proposition 4.3 (Logistic regression is the binomial GLM with logit link)

Model (4.4) holds if and only if

$$\log \frac{\pi(x)}{1 - \pi(x)} = \alpha + \beta x.$$

Hence logistic regression is the binomial GLM with the canonical (logit) link.

Proof. From (4.4), $1-\pi(x) = 1/[1+\exp(\alpha+\beta x)]$, so the odds are $\exp(\alpha+\beta x)$; take logs. Conversely, solving $\log[\pi/(1-\pi)] = \eta$ gives $\pi = e^\eta/(1+e^\eta)$. The logit is the binomial natural parameter (Section 4.1.2). ■

The logit ranges over $\mathbb{R}$, so the structural problem of the linear model disappears; for the snoring data $\hat{\beta} = 0.40 > 0$.

4.2.4 Binomial GLM for 2×2 contingency tables

Proposition 4.4 (Effect parameters for a binary predictor)

Let $x \in \{0,1\}$ and $\text{link}[\pi(x)] = \alpha + \beta x$. Then $\beta = \text{link}[\pi(1)] - \text{link}[\pi(0)]$; in particular
(a) identity link: $\beta = \pi(1) - \pi(0)$, the difference of proportions;
(b) log link: $\beta = \log[\pi(1)/\pi(0)]$, the log relative risk;
(c) logit link: $\beta = \log \dfrac{\pi(1)/[1-\pi(1)]}{\pi(0)/[1-\pi(0)]}$, the log odds ratio.

Proof. Subtract $\text{link}[\pi(0)] = \alpha$ from $\text{link}[\pi(1)] = \alpha + \beta$ and specialize the link; for (c), a difference of log odds is a log odds ratio. ■

4.2.5 Probit and inverse cdf link functions

A monotone curve has the shape of a cdf: for a standard cdf Φ of a location–scale class,

$$\pi(x) = \Phi(\alpha+\beta x) \quad \Longleftrightarrow \quad \Phi^{-1}[\pi(x)] = \alpha + \beta x, \tag{4.5}$$

with link Φ^{-1}. The standard normal Φ gives the *probit model* (Section 6.6). The standard logistic cdf $e^x/(1+e^x)$ (sd $\pi/\sqrt{3}$, slightly heavier tails than the normal) has inverse logit π, so the logit is the inverse cdf link of the logistic distribution.

4.3 Generalized linear models for counts

Motivation. Counts are nonnegative with variance growing in the mean; the Poisson log-linear GLM is basic, and overdispersion leads to the negative binomial.

4.3.1 Poisson loglinear models

With one predictor,

$$\log \mu = \alpha + \beta x, \qquad \mu = \exp(\alpha+\beta x) = e^\alpha (e^\beta)^x, \tag{4.6}$$

so a one-unit increase in x multiplies the mean by e^β.

4.3.2 Horseshoe crab mating example

Example 4.5 (Horseshoe crab satellites)

For 173 female crabs, Y = number of satellite males; predictors color, spine condition, weight and width x (mean 26.3 cm). Grouped by width:

Width (cm)	< 23.25	-24.25	-25.25	-26.25	-27.25	-28.25	-29.25	> 29.25
Cases	14	14	28	39	22	24	18	14
Mean	1.00	1.43	2.39	2.69	2.86	3.87	3.94	5.14
Variance	2.77	8.88	6.54	11.38	6.88	8.81	16.88	8.29

(a) Log link: $\log \hat{\mu} = -3.305 + 0.164x$ ($\widehat{\text{se}} = 0.020$); $\hat{\mu} = 2.74$ at $x = 26.3$, and $e^{0.164} = 1.18$: each cm raises the mean by 18%.
(b) Identity link: $\hat{\mu} = -11.53 + 0.55x$ ($\widehat{\text{se}} = 0.06$). The fits agree where most data lie.

4.3.3 Overdispersion for Poisson GLMs

Above, the variances far exceed the means: *overdispersion*, typically from unobserved heterogeneity.

Proposition 4.6 (Mixtures of Poissons are overdispersed)

If $Y \mid \lambda \sim \text{Poisson}(\lambda)$ and λ is random with $E(\lambda) = \mu$, then $E(Y) = \mu$ and $\text{Var}(Y) = \mu + \text{Var}(\lambda) \geq \mu$, with equality only when λ is degenerate.

Proof. [Supplement] $E(Y) = E[E(Y \mid \lambda)] = \mu$ and $\text{Var}(Y) = E[\text{Var}(Y \mid \lambda)] + \text{Var}[E(Y \mid \lambda)] = \mu + \text{Var}(\lambda)$. ■

If the mean model is right but the distribution is not Poisson, the ML estimates stay consistent but the standard errors are wrong.

4.3.4 Negative binomial GLMs

The negative binomial mass function is

$$f(y; k, \mu) = \frac{\Gamma(y+k)}{\Gamma(k)\Gamma(y+1)} \Big(\frac{k}{\mu+k}\Big)^k \Big(1 - \frac{k}{\mu+k}\Big)^y, \qquad y = 0, 1, 2, \ldots \tag{4.7}$$

Proposition 4.7 (Negative binomial as a gamma mixture)

If $Y \mid \lambda \sim \text{Poisson}(\lambda)$ and λ has the gamma density with shape k and mean μ, then Y has the mass function (4.7), with $E(Y) = \mu$ and $\text{Var}(Y) = \mu + \mu^2/k$. For fixed k, (4.7) is a natural exponential family with natural parameter $\log[\mu/(\mu+k)]$, so negative binomial regression is a GLM. As $k^{-1} \to 0$ it converges to the Poisson.

Proof. [Supplement]

Step 1. With $h(\lambda) = (k/\mu)^k \lambda^{k-1} e^{-k\lambda/\mu}/\Gamma(k)$,

$$P(Y = y) = \frac{(k/\mu)^k}{\Gamma(k)\, y!} \int_0^\infty \lambda^{y+k-1} e^{-\lambda(1+k/\mu)} d\lambda = \frac{\Gamma(y+k)}{\Gamma(k)\, y!} \Big(\frac{k}{\mu+k}\Big)^k \Big(\frac{\mu}{\mu+k}\Big)^y,$$

which is (4.7).

Step 2. The gamma law has mean μ, variance μ^2/k; Proposition 4.6 gives the moments.

Step 3. With $Q = \log[\mu/(\mu+k)]$, (4.7) is $(1 - e^Q)^k\, b(y) \exp(yQ)$, $b(y) = \Gamma(y+k)/[\Gamma(k)y!]$: the form (4.1).

Step 4. As $k \to \infty$, $\Gamma(y+k)/[\Gamma(k)(\mu+k)^y] \to 1$ and $(1 + \mu/k)^{-k} \to e^{-\mu}$, so $f(y) \to e^{-\mu}\mu^y/y!$. ■

k^{-1} is a *dispersion parameter* (Section 13.4).

Example 4.8 (Crab satellites: negative binomial fit)

Identity link: Poisson $\hat{\mu} = -11.53 + 0.55x$ ($\widehat{\text{se}} = 0.06$); negative binomial $\hat{\mu} = -11.15 + 0.53x$ ($\widehat{\text{se}} = 0.11$), $\hat{k}^{-1} = 0.98$. Similar fits, but the honest SE is nearly doubled.

4.3.5 Poisson regression for rates

For a rate μ_i/t_i with exposure t_i, $\log(\mu_i/t_i) = \sum_j \beta_j x_{ij}$ is the Poisson GLM $\log \mu_i = \log t_i + \sum_j \beta_j x_{ij}$ with *offset* $\log t_i$ (Section 9.7).

4.3.6 Poisson GLM of independence in $I \times J$ contingency tables

Proposition 4.9 (Poisson form of independence)

Let Y_{ij} be independent Poisson with means $\mu_{ij} = \mu\alpha_i\beta_j$, where $\alpha_i, \beta_j > 0$ and $\sum_i \alpha_i = \sum_j \beta_j = 1$; equivalently

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y, \qquad \lambda = \log \mu,\ \lambda_i^X = \log \alpha_i,\ \lambda_j^Y = \log \beta_j. \tag{4.8}$$

Conditional on $n = \sum_{i,j} Y_{ij}$, the cell counts are multinomial with probabilities $\pi_{ij} = \alpha_i\beta_j$, which satisfy $\pi_{i+} = \alpha_i$, $\pi_{+j} = \beta_j$ and $\pi_{ij} = \pi_{i+}\pi_{+j}$.

Proof.

Step 1. $n \sim \text{Poisson}(\sum \mu_{ij}) = \text{Poisson}(\mu)$.

Step 2. For counts $\{n_{ij}\}$ with total n,

$$P(\{Y_{ij} = n_{ij}\} \mid n) = \frac{\prod_{ij} e^{-\mu_{ij}} \mu_{ij}^{n_{ij}}/n_{ij}!}{e^{-\mu}\mu^n/n!} = \frac{n!}{\prod_{ij} n_{ij}!} \prod_{ij} \Big(\frac{\mu_{ij}}{\mu}\Big)^{n_{ij}},$$

multinomial with $\pi_{ij} = \alpha_i\beta_j$.

Step 3. $\pi_{i+} = \alpha_i \sum_j \beta_j = \alpha_i$ and $\pi_{+j} = \beta_j$, so $\pi_{ij} = \pi_{i+}\pi_{+j}$. ∎

So Chapter 3 independence inferences are inferences for (4.8); Chapters 8–9 extend them.

4.4 Moments and likelihood for generalized linear models

Motivation. Mean and variance come from derivatives of one function $b(\theta)$, and the likelihood equations and information depend on the distribution only through them.

Definition (Exponential dispersion family)

Independent $y_1, \ldots, y_N$ with

$$f(y_i; \theta_i, \phi) = \exp\Big\{\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)\Big\}; \tag{4.9}$$

θ_i is the *natural* and ϕ the *dispersion parameter*; usually $a(\phi) = \phi/\omega_i$ with known weight ω_i. For known ϕ this is (4.1) with $Q(\theta) = \theta/a(\phi)$.

4.4.1 Mean and variance functions for the random component

With $L_i = \log f(y_i; \theta_i, \phi)$ and $L = \sum_i L_i$,

$$L_i = \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi). \tag{4.10}$$

Theorem 4.10 (Moments of the exponential dispersion family)

For (4.9), with θ_i interior to the natural parameter space,

$$\mu_i = E(Y_i) = b'(\theta_i), \qquad \mathrm{Var}(Y_i) = b''(\theta_i)\, a(\phi). \tag{4.11}$$

Proof. [Supplement]

Step 1. (Bartlett identities.) Let $\ell = \log f$. For θ interior, differentiation under $\int f\, dy = 1$ is permitted (support free of θ, locally dominated derivatives), giving $E(\partial\ell/\partial\theta) = 0$; differentiating again with $\partial f/\partial\theta = f\,\partial\ell/\partial\theta$ gives $-E(\partial^2\ell/\partial\theta^2) = E(\partial\ell/\partial\theta)^2$ (sums for discrete Y).

Step 2. From (4.10), $\partial L_i/\partial\theta_i = [y_i - b'(\theta_i)]/a(\phi)$ and $\partial^2 L_i/\partial\theta_i^2 = -b''(\theta_i)/a(\phi)$.

Step 3. The first identity gives $\mu_i = b'(\theta_i)$; the second gives $b''(\theta_i)/a(\phi) = \mathrm{Var}(Y_i)/a(\phi)^2$, i.e. $\mathrm{Var}(Y_i) = b''(\theta_i)a(\phi)$. ■

Since $b'' > 0$, b' is strictly increasing, so θ_i and μ_i determine each other.

4.4.2 Mean and variance functions for Poisson and binomial

Proposition 4.11 (Poisson and binomial in dispersion form)

(a) Poisson: $\theta_i = \log\mu_i$, $b(\theta_i) = e^{\theta_i}$, $a(\phi) = 1$, $c = -\log y_i!$; then $E(Y_i) = \mathrm{Var}(Y_i) = \mu_i$.
(b) Binomial proportion: if $n_iY_i \sim \mathrm{bin}(n_i, \pi_i)$, then $\theta_i = \log[\pi_i/(1-\pi_i)]$, $b(\theta_i) = \log(1+e^{\theta_i})$, $a(\phi) = 1/n_i$, $c = \log\binom{n_i}{n_iy_i}$; then $E(Y_i) = \pi_i$ and $\mathrm{Var}(Y_i) = \pi_i(1-\pi_i)/n_i$.

Proof.

(a) $e^{-\mu_i}\mu_i^{y_i}/y_i! = \exp[y_i\theta_i - e^{\theta_i} - \log y_i!]$, and $b'(\theta_i) = b''(\theta_i) = e^{\theta_i} = \mu_i$ (Theorem 4.10).

(b) Since $\log(1-\pi_i) = -\log(1+e^{\theta_i})$,

$$\binom{n_i}{n_iy_i}\pi_i^{n_iy_i}(1-\pi_i)^{n_i-n_iy_i} = \exp\Big[\frac{y_i\theta_i - \log(1+e^{\theta_i})}{1/n_i} + \log\binom{n_i}{n_iy_i}\Big].$$

Then $b'(\theta_i) = e^{\theta_i}/(1+e^{\theta_i}) = \pi_i$ and $b''(\theta_i)a(\phi) = \pi_i(1-\pi_i)/n_i$. ■

For $N(\mu_i, \sigma^2)$: $\theta_i = \mu_i$, $b(\theta) = \theta^2/2$, $a(\phi) = \sigma^2$.

4.4.3 Systematic component and link function

In matrix form $\boldsymbol{\eta} = X\boldsymbol{\beta}$ with $N \times p$ *model matrix* X, and $\eta_i = g(\mu_i)$.

Proposition 4.12 (Canonical link)

The canonical link (the g with $g(\mu_i) = \theta_i$) is $g = (b')^{-1}$. For the Poisson it is the log link; for the binomial it is the logit link.

Proof. By Theorem 4.10, $\mu_i = b'(\theta_i)$ with b' strictly increasing, so $\theta_i = (b')^{-1}(\mu_i)$. Poisson: $b' = \exp$, inverse log. Binomial: $b'(\theta) = e^\theta/(1+e^\theta)$, inverse logit. ■

4.4.4 Likelihood equations for a GLM

The log likelihood

$$L(\boldsymbol{\beta}) = \sum_i \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + \sum_i c(y_i, \phi) \tag{4.12}$$

depends on $\boldsymbol{\beta}$ through $\theta_i(\mu_i)$, $\mu_i = g^{-1}(\eta_i)$, $\eta_i = \sum_j \beta_j x_{ij}$.

Theorem 4.13 (GLM likelihood equations)

For a GLM with exponential dispersion random component,

$$\frac{\partial L_i}{\partial \beta_j} = \frac{(y_i - \mu_i)x_{ij}}{\mathrm{Var}(Y_i)} \frac{\partial \mu_i}{\partial \eta_i}, \tag{4.13}$$

and the likelihood equations are

$$\sum_{i=1}^{N} \frac{(y_i - \mu_i)x_{ij}}{\mathrm{Var}(Y_i)} \frac{\partial \mu_i}{\partial \eta_i} = 0, \qquad j = 1, \ldots, p. \tag{4.14}$$

Proof.

Step 1. By the chain rule, $\partial L_i/\partial\beta_j = (\partial L_i/\partial\theta_i)(\partial\theta_i/\partial\mu_i)(\partial\mu_i/\partial\eta_i)(\partial\eta_i/\partial\beta_j)$.

Step 2. $\partial L_i/\partial\theta_i = (y_i - \mu_i)/a(\phi)$ (proof of Theorem 4.10); $\partial\mu_i/\partial\theta_i = b''(\theta_i) = \mathrm{Var}(Y_i)/a(\phi)$; $\partial\eta_i/\partial\beta_j = x_{ij}$.

Step 3. Multiplying, $a(\phi)$ cancels, giving (4.13); summing over i gives (4.14). ■

The equations depend on the distribution only through μ_i and the *variance function* $v(\mu_i) = \mathrm{Var}(Y_i)$: μ (Poisson), $\mu(1-\mu)$ (Bernoulli), σ^2 (normal).

4.4.5 Likelihood equations for binomial GLMs

Corollary 4.14 (Binomial likelihood equations)

Let $n_iY_i \sim \mathrm{bin}(n_i, \pi_i)$ and $\pi_i = \Phi(\eta_i)$, $\eta_i = \sum_j \beta_j x_{ij}$, for a continuous cdf Φ with density ϕ. The likelihood equations are

$$\sum_i \frac{n_i(y_i - \pi_i)\,x_{ij}}{\pi_i(1-\pi_i)}\,\phi(\eta_i) = 0, \qquad j = 1, \ldots, p. \tag{4.15}$$

For the logit link they reduce to

$$\sum_i n_i(y_i - \pi_i)x_{ij} = 0, \qquad j = 1, \ldots, p. \tag{4.16}$$

Proof. Substitute $\partial\pi_i/\partial\eta_i = \phi(\eta_i)$ and $\mathrm{Var}(Y_i) = \pi_i(1-\pi_i)/n_i$ in (4.14). For the logit, $\partial\eta_i/\partial\pi_i = 1/[\pi_i(1-\pi_i)]$, so $\phi(\eta_i) = \pi_i(1-\pi_i)$ cancels the denominator. ■

4.4.6 Asymptotic covariance matrix of model parameter estimators

Theorem 4.15 (Information matrix of a GLM)

The expected information matrix $\mathcal{I}$ with entries $E[-\partial^2 L(\boldsymbol{\beta})/\partial\beta_h\partial\beta_j]$ is

$$\mathcal{I} = X^\top W X, \qquad W = \operatorname{diag}(w_1,\dots,w_N), \quad w_i = \frac{(\partial\mu_i/\partial\eta_i)^2}{\operatorname{Var}(Y_i)}. \tag{4.17}$$

Proof.

Step 1. By the second Bartlett identity applied to $\boldsymbol{\beta}$, $E(\partial^2 L_i/\partial\beta_h\partial\beta_j) = -E(\partial L_i/\partial\beta_h \cdot \partial L_i/\partial\beta_j)$.

Step 2. By (4.13) and $E(Y_i-\mu_i)^2 = \operatorname{Var}(Y_i)$, the latter equals $x_{ih}x_{ij}(\partial\mu_i/\partial\eta_i)^2/\operatorname{Var}(Y_i) = x_{ih}w_i x_{ij}$.

Step 3. Summing over independent observations gives the (h,j) entry of $X^\top W X$. ■

Hence

$$\widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}}) = \hat{\mathcal{I}}^{-1} = (X^\top \hat{W} X)^{-1}, \tag{4.18}$$

justified by the next theorem.

Theorem 4.16 (Asymptotic normality of the ML estimator)

Let $\boldsymbol{\beta}_0$ be the true value, interior to the parameter space, and write $\boldsymbol{u}(\boldsymbol{\beta}) = \partial L/\partial\boldsymbol{\beta}$, $H(\boldsymbol{\beta}) = \partial^2 L/\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top$, $\mathcal{I}_N(\boldsymbol{\beta}) = X^\top W(\boldsymbol{\beta}) X$. Assume:
(R1) $N^{-1}\mathcal{I}_N(\boldsymbol{\beta}_0) \to \Sigma$, positive definite;
(R2) Lindeberg condition: for each $\boldsymbol{c} \in \mathbb{R}^p$, the terms $\boldsymbol{c}^\top \partial L_i/\partial\boldsymbol{\beta}$ $(i \le N)$, normalized by $(N\boldsymbol{c}^\top\Sigma\boldsymbol{c})^{1/2}$, satisfy Lindeberg's condition;
(R3) $\sup_{\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le\delta} \|N^{-1}H(\boldsymbol{\beta}) + \Sigma\| \xrightarrow{p} 0$ as $N\to\infty$ and then $\delta\to 0$.
Then with probability tending to one the likelihood equations have a root $\hat{\boldsymbol{\beta}}$ with $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$, and

$$\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0}, \Sigma^{-1}), \qquad \text{i.e.} \qquad \hat{\boldsymbol{\beta}} \approx N_p(\boldsymbol{\beta}_0, (X^\top W X)^{-1}).$$

Proof. [Supplement]

Step 1. (Score CLT.) The summands $\partial L_i/\partial\boldsymbol{\beta}$ are independent, mean $\mathbf{0}$, with covariances summing to $\mathcal{I}_N(\boldsymbol{\beta}_0)$. By (R1), (R2), Lindeberg–Feller and Cramér–Wold, $N^{-1/2}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0},\Sigma)$; so $N^{-1}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{p} \mathbf{0}$.

Step 2. (Consistent root.) Let $\lambda > 0$ be the smallest eigenvalue of Σ, and fix small δ so that, by (R3), with probability $\to 1$ all eigenvalues of $-N^{-1}H$ are $\ge \lambda/2$ on $B_\delta = \{\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\| \le \delta\}$. For $\boldsymbol{\beta} \in \partial B_\delta$, Taylor's theorem gives

$$L(\boldsymbol{\beta}) - L(\boldsymbol{\beta}_0) = \boldsymbol{u}(\boldsymbol{\beta}_0)^\top(\boldsymbol{\beta}-\boldsymbol{\beta}_0) + \tfrac{1}{2}(\boldsymbol{\beta}-\boldsymbol{\beta}_0)^\top H(\bar{\boldsymbol{\beta}})(\boldsymbol{\beta}-\boldsymbol{\beta}_0) \le N\delta\Big(\frac{\|\boldsymbol{u}(\boldsymbol{\beta}_0)\|}{N} - \frac{\lambda\delta}{4}\Big),$$

negative with probability $\to 1$ by Step 1. So the maximum of L on the compact B_δ is interior, where $\boldsymbol{u} = \mathbf{0}$; as δ is arbitrary, $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$.

Step 3. (Expansion.) Coordinatewise mean value theorem: $\mathbf{0} = \boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \boldsymbol{u}(\boldsymbol{\beta}_0) + \tilde{H}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)$, rows of $\tilde{H}$ evaluated at intermediate points; by consistency and (R3), $-N^{-1}\tilde{H} \xrightarrow{p} \Sigma$.

Step 4. So $\sqrt{N}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0) = (-N^{-1}\tilde{H})^{-1}N^{-1/2}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0},\Sigma^{-1})$ by Slutsky; and $N^{-1}X^\top\hat{W}X \xrightarrow{p} \Sigma$ by continuity, justifying (4.18). ■

For grouped binomial or Poisson data the same holds with N fixed and $n_i \to \infty$.

4.4.7 Likelihood equations and covariance matrix for Poisson loglinear model

Corollary 4.17 (Poisson loglinear model)

For $\log \boldsymbol{\mu} = X\boldsymbol{\beta}$ with Poisson responses, the likelihood equations are

$$\sum_i (y_i - \mu_i) x_{ij} = 0, \qquad j = 1, \dots, p, \tag{4.19}$$

and $\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\beta}}) = (X^\top \hat{W} X)^{-1}$ with $\hat{W} = \mathrm{diag}(\hat{\mu}_1, \dots, \hat{\mu}_N)$.

Proof. For the log link $\partial \mu_i / \partial \eta_i = \mu_i = \mathrm{Var}(Y_i)$, so (4.14) becomes (4.19) and $w_i = \mu_i^2/\mu_i = \mu_i$. ■

4.5 Inference for generalized linear models

Motivation. Wald, score and LR methods apply to any GLM parameter; LR inference is organized through the deviance, the analogue of the residual sum of squares.

Theorem 4.18 (Wald, likelihood-ratio and score tests)

Under the conditions of Theorem 4.16, for $H_0 : \boldsymbol{\beta} = \boldsymbol{\beta}_0$ each of

$$\begin{aligned} W &= (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)^\top \hat{\mathcal{I}}\, (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) && \text{(Wald)}, \\ \Lambda &= -2[L(\boldsymbol{\beta}_0) - L(\hat{\boldsymbol{\beta}})] && \text{(likelihood ratio)}, \\ S &= \boldsymbol{u}(\boldsymbol{\beta}_0)^\top \mathcal{I}(\boldsymbol{\beta}_0)^{-1} \boldsymbol{u}(\boldsymbol{\beta}_0) && \text{(score)} \end{aligned}$$

converges in distribution to χ^2_p, and the three differ by $o_p(1)$. For nested models $M_0 \subset M_1$, where M_0 imposes q linear restrictions on $\boldsymbol{\beta}$, $\Lambda = -2[L(\hat{\boldsymbol{\beta}}_0) - L(\hat{\boldsymbol{\beta}}_1)] \xrightarrow{d} \chi^2_q$.

Proof. [Supplement] Let $\boldsymbol{Z}_N = N^{-1/2}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} \boldsymbol{Z} \sim N_p(\boldsymbol{0}, \Sigma)$; by Theorem 4.16, $\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = \Sigma^{-1}\boldsymbol{Z}_N + o_p(1)$.

Step 1. (Score.) $S = \boldsymbol{Z}_N^\top (N^{-1}\mathcal{I}_N)^{-1} \boldsymbol{Z}_N \xrightarrow{d} \boldsymbol{Z}^\top \Sigma^{-1} \boldsymbol{Z} \sim \chi^2_p$.

Step 2. (Wald.) $W = [\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)]^\top (N^{-1}\hat{\mathcal{I}}) [\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)] = \boldsymbol{Z}_N^\top \Sigma^{-1} \boldsymbol{Z}_N + o_p(1)$.

Step 3. (LR.) Expanding about $\hat{\boldsymbol{\beta}}$, where $\boldsymbol{u} = \boldsymbol{0}$, $L(\boldsymbol{\beta}_0) = L(\hat{\boldsymbol{\beta}}) + \tfrac{1}{2}(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})^\top H(\bar{\boldsymbol{\beta}})(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})$, so by (R3) $\Lambda = \boldsymbol{Z}_N^\top \Sigma^{-1} \boldsymbol{Z}_N + o_p(1)$.

Step 4. (Nested.) Write M_0 as $\boldsymbol{\beta} = \boldsymbol{\beta}_0 + A\boldsymbol{\gamma}$, A of full rank $p-q$; M_0 has score $A^\top \boldsymbol{u}$ and information $A^\top \mathcal{I} A$. Step 3 for both models gives

$$\Lambda = \boldsymbol{Z}_N^\top [\Sigma^{-1} - A(A^\top \Sigma A)^{-1} A^\top] \boldsymbol{Z}_N + o_p(1).$$

With $\boldsymbol{Z} = \Sigma^{1/2}\boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim N_p(\boldsymbol{0}, I)$, the limit is $\boldsymbol{\varepsilon}^\top (I - P)\boldsymbol{\varepsilon}$ with $P = \Sigma^{1/2} A (A^\top \Sigma A)^{-1} A^\top \Sigma^{1/2}$ a projection of rank $p - q$; so it is χ^2_q. ■

Inverting the tests gives the Wald interval $\hat{\beta}_j \pm z_{\alpha/2} \widehat{\mathrm{se}}(\hat{\beta}_j)$ and the profile-likelihood interval $\{\beta_j : \Lambda(\beta_j) \le \chi^2_1(\alpha)\}$.

4.5.1 Deviance and goodness of fit

Let $\tilde{\theta}_i$ ($\tilde{\mu}_i = y_i$) and $\hat{\theta}_i$ be the saturated and fitted natural parameters. By (4.12) with $a(\phi) = \phi/\omega_i$, $-2[L(\hat{\boldsymbol{\mu}}; \mathbf{y}) - L(\mathbf{y}; \mathbf{y})]$ equals the following divided by ϕ.

Definition (Scaled deviance and deviance)

$$D(\mathbf{y};\hat{\boldsymbol{\mu}}) = 2\sum_i \omega_i\big[y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)\big] \tag{4.20}$$

is the *deviance*, and D/ϕ the *scaled deviance*; they coincide for Poisson and binomial ($\phi = 1$). For the normal linear model $D = \sum_i (y_i - \hat{\mu}_i)^2$.

Theorem 4.19 (Asymptotic distribution of the deviance)

For a binomial or Poisson GLM with a fixed number N of observations whose counts grow ($n_i \to \infty$, or $\mu_i \to \infty$), if the model with p parameters holds then $D(\mathbf{y};\hat{\boldsymbol{\mu}}) \xrightarrow{d} \chi^2_{N-p}$.

Proof. [Supplement] The saturated model is a GLM with N free θ_i, and the working model is the submodel given by $N - p$ linear restrictions $\boldsymbol{\theta} \in \mathrm{col}(X)$ (canonical scale; otherwise reparametrize smoothly). D is the LR statistic for these nested models, so Theorem 4.18 (growing counts as sample size) gives χ^2_{N-p}. ■

4.5.2 Deviance for Poisson models

Proposition 4.20 (Poisson deviance)

For a Poisson GLM,

$$D(\mathbf{y};\hat{\boldsymbol{\mu}}) = 2\sum_i \big[y_i \log(y_i/\hat{\mu}_i) - y_i + \hat{\mu}_i\big]. \tag{4.21}$$

If the model has log link and an intercept, $\sum_i y_i = \sum_i \hat{\mu}_i$ and $D(\mathbf{y};\hat{\boldsymbol{\mu}}) = 2\sum_i y_i \log(y_i/\hat{\mu}_i)$.

Proof. Put $\hat{\theta}_i = \log\hat{\mu}_i$, $b(\hat{\theta}_i) = \hat{\mu}_i$, $\tilde{\theta}_i = \log y_i$, $b(\tilde{\theta}_i) = y_i$, $\omega_i = 1$ in (4.20). With an intercept the first equation of (4.19) is $\sum_i (y_i - \hat{\mu}_i) = 0$, so the linear terms cancel ($0 \log 0 = 0$). ■

For a two-way table under independence this is G^2 of Chapter 3.

4.5.3 Deviance for binomial models: grouped and ungrouped data

Proposition 4.21 (Binomial deviance)

For binomial proportions y_i based on n_i trials,

$$D(\mathbf{y};\hat{\boldsymbol{\mu}}) = 2\sum_i n_i y_i \log\frac{n_i y_i}{n_i \hat{\pi}_i} + 2\sum_i (n_i - n_i y_i)\log\frac{n_i - n_i y_i}{n_i - n_i\hat{\pi}_i}, \tag{4.22}$$

that is, $D = 2\sum \text{observed} \times \log(\text{observed/fitted})$ over the $2N$ cells of successes and failures.

Proof. With $\hat{\theta}_i = \mathrm{logit}\,\hat{\pi}_i$, $b(\hat{\theta}_i) = -\log(1-\hat{\pi}_i)$, $\tilde{\theta}_i = \mathrm{logit}\,y_i$, $b(\tilde{\theta}_i) = -\log(1-y_i)$, $\omega_i = n_i$, (4.20) gives

$$D = 2\sum_i n_i\Big[y_i \log\frac{y_i}{\hat{\pi}_i} + (1-y_i)\log\frac{1-y_i}{1-\hat{\pi}_i}\Big],$$

which is (4.22). ■

Grouped (counts per setting) and ungrouped (0/1) entry give the same ML estimates but different saturated models and deviances; the χ^2 approximation holds only for grouped data with $n_i \to \infty$.

4.5.4 Likelihood-ratio model comparison using the deviance

Proposition 4.22 (Deviance difference)

Let $M_0 \subset M_1$ be nested Poisson or binomial GLMs with fitted values $\hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\mu}}_1$. Then $D(\mathbf{y}; \hat{\boldsymbol{\mu}}_1) \leq D(\mathbf{y}; \hat{\boldsymbol{\mu}}_0)$ and the likelihood-ratio statistic for $H_0 : M_0$ given M_1 is

$$-2\big[L(\hat{\boldsymbol{\mu}}_0; \mathbf{y}) - L(\hat{\boldsymbol{\mu}}_1; \mathbf{y})\big] = D(\mathbf{y}; \hat{\boldsymbol{\mu}}_0) - D(\mathbf{y}; \hat{\boldsymbol{\mu}}_1),$$

asymptotically χ^2 with df equal to the difference in numbers of parameters. With log link and intercept (Poisson) or for binomial data, it equals $2 \sum \text{observed} \times \log(\text{fitted}_1/\text{fitted}_0)$.

Proof. Maximizing over a smaller space gives $L(\hat{\boldsymbol{\mu}}_0) \leq L(\hat{\boldsymbol{\mu}}_1)$; the identity follows by adding and subtracting $L(\mathbf{y}; \mathbf{y})$; the distribution is Theorem 4.18. Subtracting (4.22) (or (4.21) without linear terms) for the two models cancels the log observed terms. ■

The saturated part cancels, so the comparison is valid for ungrouped data too.

4.5.5 Residuals for GLMs

Definition (Deviance and Pearson residuals)

With $D = \sum_i d_i$, $d_i = 2\omega_i[y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)]$, the *deviance residual* is $\sqrt{d_i}\ \operatorname{sgn}(y_i - \hat{\mu}_i)$ and the *Pearson residual* is $e_i = (y_i - \hat{\mu}_i)/[\widehat{\operatorname{Var}}(Y_i)]^{1/2}$.

For independence in a two-way table $\sum e_{ij}^2 = X^2$, $\sum d_{ij} = G^2$. These are less variable than $N(0,1)$; standardizing uses the *hat matrix*

$$\text{HatM} = W^{1/2} X (X^\top W X)^{-1} X^\top W^{1/2}. \tag{4.23}$$

Proposition 4.23 (Leverages and residual covariance)

(a) HatM is symmetric and idempotent with trace p; its diagonal elements (*leverages*) satisfy $0 \leq h_i \leq 1$ and $\sum_i h_i = p$.
(b) To first order, with $V = \operatorname{Cov}(\boldsymbol{Y}) = \operatorname{diag}\{\operatorname{Var}(Y_i)\}$,

$$\operatorname{Cov}(\boldsymbol{Y} - \hat{\boldsymbol{\mu}}) \approx V^{1/2}(I - \text{HatM})V^{1/2}, \qquad \operatorname{Var}(Y_i - \hat{\mu}_i) \approx \operatorname{Var}(Y_i)(1 - h_i).$$

Hence the *standardized Pearson residual* is

$$r_i = \frac{y_i - \hat{\mu}_i}{[\widehat{\operatorname{Var}}(Y_i)(1 - \hat{h}_i)]^{1/2}} = \frac{e_i}{\sqrt{1 - \hat{h}_i}} \qquad \left(\text{Poisson: } \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i(1 - \hat{h}_i)}}\right). \tag{4.24}$$

Proof. [Supplement] (a) With $A = W^{1/2}X$, $\text{HatM} = A(A^\top A)^{-1}A^\top$ is symmetric, $\text{HatM}^2 = \text{HatM}$, $\operatorname{tr}\text{HatM} = \operatorname{tr} I_p = p$, and $h_i = \sum_j h_{ij}^2 \geq h_i^2$ gives $0 \leq h_i \leq 1$.

(b)

Step 1. Let $G = \operatorname{diag}(\partial\eta_i/\partial\mu_i)$; by (4.13), $\boldsymbol{u} = X^\top W G(\boldsymbol{y} - \boldsymbol{\mu})$.

Step 2. By Theorem 4.16, $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \approx (X^\top W X)^{-1} X^\top W G(\boldsymbol{y} - \boldsymbol{\mu})$ and $\hat{\boldsymbol{\mu}} - \boldsymbol{\mu} \approx G^{-1}X(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$, so $\boldsymbol{y} - \hat{\boldsymbol{\mu}} \approx [I - G^{-1}X(X^\top W X)^{-1}X^\top W G](\boldsymbol{y} - \boldsymbol{\mu})$.

Step 3. Since $w_i = 1/(G_{ii}^2 \operatorname{Var}(Y_i))$, $V^{1/2} = G^{-1}W^{-1/2} =: M$, and the bracket is $M(I - \text{HatM})M^{-1}$.

Step 4. With $\operatorname{Cov}(\boldsymbol{y} - \boldsymbol{\mu}) = M^2$ and HatM symmetric idempotent, $\operatorname{Cov}(\boldsymbol{y} - \hat{\boldsymbol{\mu}}) \approx M(I - \text{HatM})M^{-1}M^2M^{-1}(I - \text{HatM})M = M(I - \text{HatM})M$, with ith diagonal entry $\operatorname{Var}(Y_i)(1 - h_i)$.

■

Unlike ordinary regression, leverages depend on the fit through W, so extreme predictor values need not have high leverage.

4.6 Fitting generalized linear models

Motivation. The likelihood equations are nonlinear; Newton–Raphson and Fisher scoring solve them, and Fisher scoring is iteratively reweighted least squares.

4.6.1 Newton–Raphson method

With $\boldsymbol{u}^{(t)}, H^{(t)}$ the score and Hessian at $\boldsymbol{\beta}^{(t)}$, maximizing the quadratic approximation $L(\boldsymbol{\beta}^{(t)}) + \boldsymbol{u}^{(t)\top}(\boldsymbol{\beta}-\boldsymbol{\beta}^{(t)}) + \tfrac{1}{2}(\boldsymbol{\beta}-\boldsymbol{\beta}^{(t)})^\top H^{(t)}(\boldsymbol{\beta}-\boldsymbol{\beta}^{(t)})$ gives the *Newton–Raphson* step (Figure 6)

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - (H^{(t)})^{-1}\boldsymbol{u}^{(t)}. \tag{4.25}$$

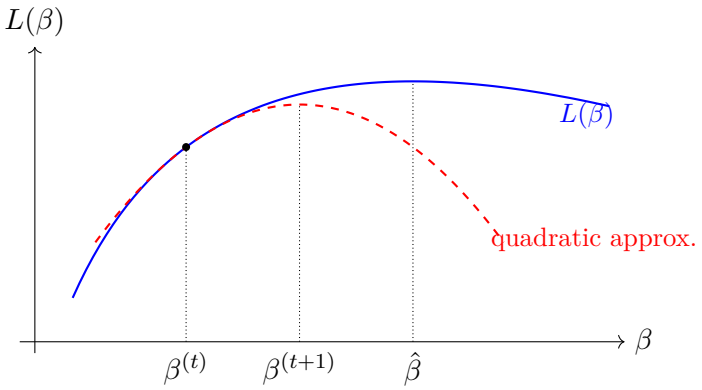


Figure 6: One Newton–Raphson cycle: the next guess maximizes the quadratic approximation at $\beta^{(t)}$.

Example 4.24 (Newton–Raphson for a binomial parameter)

For $y \sim \mathrm{bin}(n,\pi)$, $u = (y-n\pi)/[\pi(1-\pi)]$ and $H = -[y/\pi^2 + (n-y)/(1-\pi)^2]$. The step moves $\pi^{(t)}$ toward y/n; from $\pi^{(0)} = \tfrac{1}{2}$ it reaches y/n at once, otherwise in four or five iterations.

Theorem 4.25 (Quadratic convergence of Newton–Raphson)

Suppose $\boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$, H is Lipschitz on a ball B around $\hat{\boldsymbol{\beta}}$ with constant γ ($\|H(\boldsymbol{a}) - H(\boldsymbol{b})\| \le \gamma\|\boldsymbol{a}-\boldsymbol{b}\|$), and $\|H(\boldsymbol{\beta})^{-1}\| \le M$ on B. If $\boldsymbol{\beta}^{(t)} \in B$, then

$$\|\boldsymbol{\beta}^{(t+1)} - \hat{\boldsymbol{\beta}}\| \le \frac{M\gamma}{2}\,\|\boldsymbol{\beta}^{(t)} - \hat{\boldsymbol{\beta}}\|^2.$$

In particular, if $\boldsymbol{\beta}^{(0)} \in B$ and $\|\boldsymbol{\beta}^{(0)} - \hat{\boldsymbol{\beta}}\| < 2/(M\gamma)$, the iterates converge to $\hat{\boldsymbol{\beta}}$ and the number of correct digits roughly doubles each step.

Proof. [Supplement]

Step 1. With $\boldsymbol{e}_t = \boldsymbol{\beta}^{(t)} - \hat{\boldsymbol{\beta}}$, $\boldsymbol{u}^{(t)} = \boldsymbol{u}^{(t)} - \boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \int_0^1 H(\hat{\boldsymbol{\beta}} + s\boldsymbol{e}_t)\,ds\;\boldsymbol{e}_t$.

Step 2. By (4.25),

$$\boldsymbol{e}_{t+1} = \boldsymbol{e}_t - (H^{(t)})^{-1}\boldsymbol{u}^{(t)} = (H^{(t)})^{-1}\int_0^1 [H(\hat{\boldsymbol{\beta}} + \boldsymbol{e}_t) - H(\hat{\boldsymbol{\beta}} + s\boldsymbol{e}_t)]ds\, \boldsymbol{e}_t.$$

Step 3. The integrand has norm $\leq \gamma(1-s)\|\boldsymbol{e}_t\|$ and $\int_0^1 (1-s)ds = \frac{1}{2}$, so $\|\boldsymbol{e}_{t+1}\| \leq \frac{M\gamma}{2}\|\boldsymbol{e}_t\|^2$.

Step 4. If $\rho = \frac{M\gamma}{2}\|\boldsymbol{e}_0\| < 1$, induction gives $\|\boldsymbol{e}_{t+1}\| \leq \rho\|\boldsymbol{e}_t\|$, the iterates stay in B and $\boldsymbol{e}_t \to \mathbf{0}$, the error being roughly squared each step. ■

4.6.2 Fisher scoring method

Fisher scoring replaces $-H$ (observed information) by the expected information $\mathcal{I}$:

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + (\mathcal{I}^{(t)})^{-1}\boldsymbol{u}^{(t)} \quad \Longleftrightarrow \quad \mathcal{I}^{(t)}\boldsymbol{\beta}^{(t+1)} = \mathcal{I}^{(t)}\boldsymbol{\beta}^{(t)} + \boldsymbol{u}^{(t)}. \tag{4.26}$$

For the binomial parameter $\mathcal{I} = n/[\pi(1-\pi)]$ and one step gives y/n from any start. For a GLM, $\mathcal{I}^{(t)} = X^\top W^{(t)} X$ and $\boldsymbol{u}^{(t)}$ has elements

$$u_j = \sum_{i=1}^{N} \frac{(y_i - \mu_i)x_{ij}}{\mathrm{Var}(Y_i)}\frac{\partial \mu_i}{\partial \eta_i}; \tag{4.27}$$

at convergence $(\mathcal{I}^{(t)})^{-1}$ is the covariance estimate (4.18). For canonical links the two methods coincide (Theorem 4.27); otherwise Fisher scoring has a nonnegative definite $\mathcal{I}$ but need not converge quadratically.

4.6.3 ML as iterative reweighted least squares

For $\boldsymbol{z} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathrm{Cov}(\boldsymbol{\varepsilon}) = V$, the WLS estimator is $(X^\top V^{-1}X)^{-1}X^\top V^{-1}\boldsymbol{z}$.

Theorem 4.26 (Fisher scoring is IRLS)

For a GLM, the Fisher scoring step (4.26) is

$$\boldsymbol{\beta}^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} \boldsymbol{z}^{(t)},$$

the WLS fit of the *working response*

$$z_i^{(t)} = \eta_i^{(t)} + (y_i - \mu_i^{(t)})\frac{\partial \eta_i^{(t)}}{\partial \mu_i^{(t)}} \tag{4.28}$$

on X with weight matrix (inverse covariance) $W^{(t)}$.

Proof.

Step 1. $\mathcal{I}^{(t)}\boldsymbol{\beta}^{(t)} = X^\top W^{(t)} X\boldsymbol{\beta}^{(t)} = X^\top W^{(t)}\boldsymbol{\eta}^{(t)}$.

Step 2. Since $w_i = (\partial\mu_i/\partial\eta_i)^2/\mathrm{Var}(Y_i)$, each term of (4.27) is $x_{ij}w_i(y_i - \mu_i)\,\partial\eta_i/\partial\mu_i$, so $\boldsymbol{u}^{(t)} = X^\top W^{(t)}\boldsymbol{d}^{(t)}$ with $d_i^{(t)} = (y_i - \mu_i^{(t)})\,\partial\eta_i^{(t)}/\partial\mu_i^{(t)}$.

Step 3. The right side of (4.26) is $X^\top W^{(t)}(\boldsymbol{\eta}^{(t)} + \boldsymbol{d}^{(t)}) = X^\top W^{(t)}\boldsymbol{z}^{(t)}$: the WLS normal equations. ■

z_i linearizes the link, $g(y_i) \approx \eta_i + (y_i - \mu_i)\,\partial\eta_i/\partial\mu_i$, and W^{-1} approximates its covariance. Iterating is *iteratively reweighted least squares* (IRLS); a start is $\hat{\boldsymbol{\mu}}^{(0)} = \mathbf{y}$, adjusted where $g(y_i)$ is infinite.

4.6.4 Simplifications for canonical links

Theorem 4.27 (Canonical link properties)

Consider a GLM with the canonical link, $\theta_i = \eta_i = \sum_j \beta_j x_{ij}$, and a common $a(\phi)$ for all i (e.g. Poisson; binomial with all $n_i = 1$). Then:
(a) $\sum_i y_i x_{ij}$, $j = 1, \ldots, p$, are sufficient statistics for $\boldsymbol{\beta}$;
(b) $\partial L_i / \partial \beta_j = (y_i - \mu_i) x_{ij} / a(\phi)$, so the likelihood equations are

$$\sum_i x_{ij} y_i = \sum_i x_{ij} \mu_i, \qquad j = 1, \ldots, p, \qquad \text{i.e.} \quad X^\top \mathbf{y} = X^\top \boldsymbol{\mu}; \tag{4.29}$$

(c) $\partial^2 L / \partial \beta_h \partial \beta_j = -\sum_i x_{ih} x_{ij} b''(\theta_i)/a(\phi)$ does not involve $\mathbf{y}$; hence observed and expected information coincide, $H = -\mathcal{I}$, and Newton–Raphson equals Fisher scoring;
(d) L is concave in $\boldsymbol{\beta}$, strictly if X has full column rank; so a solution of (4.29), if one exists, is the unique ML estimate.
With weights $a(\phi) = \phi/\omega_i$ the same holds with y_i, μ_i replaced by $\omega_i y_i, \omega_i \mu_i$.

Proof.

Step 1. By (4.12) the only term involving both data and parameters is $a(\phi)^{-1} \sum_i y_i \theta_i = a(\phi)^{-1} \sum_j \beta_j (\sum_i y_i x_{ij})$; the factorization theorem gives (a).

Step 2. Here $\partial \mu_i / \partial \eta_i = b''(\theta_i) = \mathrm{Var}(Y_i)/a(\phi)$; inserting into (4.13) gives (b).

Step 3. Since $\partial \mu_i / \partial \beta_h = b''(\theta_i) x_{ih}$, $\partial^2 L_i / \partial \beta_h \partial \beta_j = -x_{ij} x_{ih} b''(\theta_i)/a(\phi)$, free of y_i and equal to its expectation: (c).

Step 4. For $\boldsymbol{c} \in \mathbb{R}^p$, $\boldsymbol{c}^\top H \boldsymbol{c} = -\sum_i b''(\theta_i)(\boldsymbol{x}_i^\top \boldsymbol{c})^2/a(\phi) \le 0$, with equality only if $X\boldsymbol{c} = \mathbf{0}$, impossible for $\boldsymbol{c} \neq \mathbf{0}$ under full rank. A stationary point of a strictly concave function is its unique maximum: (d). ■

Thus (4.29) equates sufficient statistics to their expectations: the normal equations, (4.19), and (4.16) with $n_i = 1$.

Chapter 5. Logistic Regression

Logistic regression is the most important model for categorical responses, used in biomedicine, social science, marketing, credit scoring and genetics. Section 5.1 interprets its parameters, Section 5.2 presents inference, Sections 5.3–5.4 extend to several and to qualitative predictors, and Section 5.5 applies the GLM theory of Chapter 4 to the likelihood equations and their solution.

5.1 Interpreting parameters in logistic regression

Motivation. The logistic model is linear on the logit scale, but people think in probabilities. We need several readings of β: an odds ratio, a local slope $\beta\pi(1-\pi)$ for the probability, and changes in π over the range of x.

For binary Y and predictor X let $\pi(x) = P(Y = 1 \mid X = x)$. The logistic regression model is

$$\pi(x) = \frac{\exp(\alpha+\beta x)}{1+\exp(\alpha+\beta x)}, \qquad \text{equivalently} \qquad \operatorname{logit}\pi(x) = \log\frac{\pi(x)}{1-\pi(x)} = \alpha+\beta x. \tag{5.1}$$

5.1.1 Interpreting β: odds, probabilities, and linear approximations

The sign of β says whether $\pi(x)$ increases or decreases; the curve gets steeper as $|\beta|$ grows and flattens to a constant as $\beta \to 0$, where Y and X are independent. For $\beta > 0$ the curve is a logistic cdf; the symmetry of the logistic density means $\pi(x)$ approaches 1 at the same rate as it approaches 0.

Proposition 5.1 (Odds-ratio interpretation)

Under (5.1) the odds are $\pi(x)/[1-\pi(x)] = e^{\alpha}(e^{\beta})^{x}$; hence for every x,

$$\frac{\pi(x+1)/[1-\pi(x+1)]}{\pi(x)/[1-\pi(x)]} = e^{\beta}:$$

the odds multiply by e^{β} for each unit increase in x, and e^{β} is an odds ratio.

Proof. Exponentiate the logit form: odds $= \exp(\alpha+\beta x)$. The ratio at $x+1$ and x is $\exp(\alpha+\beta x+\beta)/\exp(\alpha+\beta x) = e^{\beta}$. ∎

Proposition 5.2 (Linear approximation (Berkson 1951))

Under (5.1),

$$\frac{d\pi(x)}{dx} = \beta\,\pi(x)[1-\pi(x)].$$

(a) The slope is greatest, $\beta/4$, where $\pi(x) = \frac{1}{2}$, i.e. at $x = -\alpha/\beta$ (the median effective level EL_{50}; in toxicology the LD_{50}). At $\pi(x) = 0.9$ or 0.1 it is 0.09β, and it tends to 0 as $\pi(x) \to 0$ or 1.
(b) $\pi(-\alpha/\beta \pm 1/\beta) = e^{\pm 1}/(1+e^{\pm 1})$, i.e. 0.73 and 0.27: a change of $1/\beta$ in x from the EL_{50} changes π by about 0.23, close to the value $\frac{1}{4}$ suggested by the tangent line.

Proof. [Supplement]

Step 1. With $\eta = \alpha+\beta x$ and $\pi = e^{\eta}/(1+e^{\eta})$,

$$\frac{d\pi}{d\eta} = \frac{e^{\eta}(1+e^{\eta}) - e^{\eta}e^{\eta}}{(1+e^{\eta})^2} = \frac{e^{\eta}}{(1+e^{\eta})^2} = \pi(1-\pi),$$

and $d\eta/dx = \beta$, so $d\pi/dx = \beta\pi(1-\pi)$.

Step 2. The function $\pi(1-\pi)$ on $(0,1)$ is maximized at $\pi = \frac{1}{2}$ with value $\frac{1}{4}$; $\pi(x) = \frac{1}{2}$ iff $\alpha + \beta x = 0$ iff $x = -\alpha/\beta$. Also $0.9 \times 0.1 = 0.09$.

Step 3. At $x = -\alpha/\beta \pm 1/\beta$, $\eta = \pm 1$, so $\pi = e^{\pm 1}/(1+e^{\pm 1}) = 0.731$ or 0.269. ■

So near a point with probability π, a unit change in x changes the probability by roughly $\beta\pi(1-\pi)$ (Figure 7); the approximation is better for smaller changes. Another summary reports $\pi(x)$ at the quartiles of x; the change over the middle half of the data can be compared across predictors with different units. The intercept is rarely of interest, but after centering ($x \mapsto x - \bar{x}$) it is the logit at $\bar{x}$, and centering reduces correlations among estimates in models with quadratic or interaction terms.

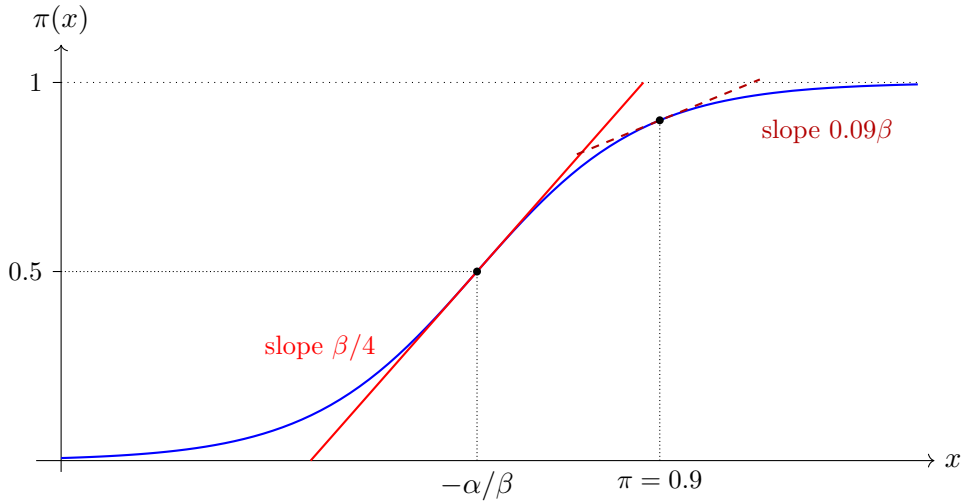


Figure 7: Logistic curve ($\alpha = -3$, $\beta = 1$) with tangent lines: slope $\beta\pi(1-\pi)$ equals $\beta/4$ at $\pi = \frac{1}{2}$ and 0.09β at $\pi = 0.9$.

5.1.2 Looking at the data

Before fitting, check that a logistic curve is plausible. Plotting 0/1 responses against x is uninformative; instead plot sample proportions or sample logits. With y_i successes out of n_i at setting i, the sample logit $\log[y_i/(n_i - y_i)]$ is infinite when $y_i = 0$ or n_i; the adjusted version

$$\log \frac{y_i + \frac{1}{2}}{n_i - y_i + \frac{1}{2}}$$

is the least-biased estimator of this form (bias of order n_i^{-2}, Haldane 1956). The plot should be roughly linear. With continuous x and small n_i, group nearby x values or, better, smooth the data and look for marked departures from an S-shape.

5.1.3 Horseshoe crabs revisited

Example 5.3 (Horseshoe crabs: width and presence of satellites)

For the 173 crabs of Example 4.5, let $Y = 1$ if a female has at least one satellite. The GAM fit and the sample proportions in the eight width categories suggest an increasing S-shaped

trend. The ML fit with $x =$ width (cm) is

$$\hat{\pi}(x) = \frac{\exp(-12.351 + 0.497x)}{1 + \exp(-12.351 + 0.497x)}.$$

Parameter	Estimate	SE	LR 95% limits	Wald χ^2
Intercept	-12.3508	2.6287	$(-17.8097, -7.4573)$	22.07
width	0.4972	0.1017	(0.3084, 0.7090)	23.89

Deviance 194.45 and Pearson $X^2 = 165.14$ on 171 df; log likelihood -97.23.

(a) At the mean width 26.3, $\hat{\pi} = 0.674$; $\hat{\pi} = \frac{1}{2}$ at $x = 12.351/0.497 = 24.8$.
(b) The odds multiply by $e^{0.497} = 1.64$ per cm (a 64% increase).
(c) At $\bar{x}$ the probability rises by about $0.497(0.674)(0.326) = 0.11$ per cm.
(d) At the width quartiles 24.9, 26.1, 27.7, $\hat{\pi} = 0.51,\ 0.65,\ 0.81$: an increase of 0.30 over the middle half.
(e) With weight (kg) instead, $\operatorname{logit}\hat{\pi} = -3.695 + 1.815x$; $\hat{\beta}$'s are not comparable across units, but at the weight quartiles 2.00, 2.35, 2.85 we get $\hat{\pi} = 0.48,\ 0.64,\ 0.81$, an increase of 0.33, similar to width.

5.1.4 Logistic regression with retrospective studies

In case–control studies X is observed for samples of cases ($Y = 1$) and controls ($Y = 0$), so Y is fixed by design. Effects in logistic regression are odds ratios, and odds ratios can be estimated retrospectively.

Theorem 5.4 (Logistic regression under case–control sampling)

Let $Z = 1$ indicate that a subject is sampled, and suppose the sampling probabilities depend only on y: $P(Z = 1 \mid y, x) = \rho_y$ for $y = 0, 1$. If $\operatorname{logit} P(Y = 1 \mid x) = \alpha + \beta x$, then

$$\operatorname{logit} P(Y = 1 \mid Z = 1, x) = \alpha^* + \beta x, \qquad \alpha^* = \alpha + \log(\rho_1/\rho_0).$$

So the slope β is the same as in the prospective model; only the intercept changes.

Proof.

Step 1. By Bayes' theorem,

$$P(Y = 1 \mid Z = 1, x) = \frac{P(Z = 1 \mid Y = 1, x)P(Y = 1 \mid x)}{\sum_{j=0}^{1} P(Z = 1 \mid Y = j, x)P(Y = j \mid x)} = \frac{\rho_1 P(Y = 1 \mid x)}{\rho_0 P(Y = 0 \mid x) + \rho_1 P(Y = 1 \mid x)}.$$

Step 2. Divide numerator and denominator by $P(Y = 0 \mid x)$ and use $P(Y = 1 \mid x)/P(Y = 0 \mid x) = \exp(\alpha + \beta x)$:

$$P(Y = 1 \mid Z = 1, x) = \frac{\rho_1 \exp(\alpha + \beta x)}{\rho_0 + \rho_1 \exp(\alpha + \beta x)} = \frac{\exp[\alpha + \log(\rho_1/\rho_0) + \beta x]}{1 + \exp[\alpha + \log(\rho_1/\rho_0) + \beta x]}.$$

■

If cases are sampled at 10 times the rate of controls, the estimated intercept is $\log 10 = 2.3$ larger than prospectively. For other binary links the effect for Y given X is not recoverable from case–control data; this is a major reason for the popularity of logit models in biomedicine. Matched case–control studies need conditional methods (Section 10.2.5).

Proposition 5.5 (Normal discriminant model implies logistic regression (Cornfield 1962))

Suppose $X \mid Y = i \sim N(\mu_i, \sigma^2)$, $i = 0, 1$, and $P(Y = 1) = \rho$. Then $P(Y = 1 \mid X = x)$ satisfies (5.1) with

$$\beta = \frac{\mu_1 - \mu_0}{\sigma^2}, \qquad \alpha = \log\frac{\rho}{1-\rho} - \frac{\mu_1^2 - \mu_0^2}{2\sigma^2}.$$

If the two variances differ, the logit is quadratic in x.

Proof. [Supplement] Let φ_i be the $N(\mu_i, \sigma_i^2)$ density. By Bayes' theorem,

$$\operatorname{logit} P(Y = 1 \mid x) = \log\frac{\rho\varphi_1(x)}{(1-\rho)\varphi_0(x)} = \log\frac{\rho}{1-\rho} + \log\frac{\sigma_0}{\sigma_1} - \frac{(x-\mu_1)^2}{2\sigma_1^2} + \frac{(x-\mu_0)^2}{2\sigma_0^2}.$$

With $\sigma_0 = \sigma_1 = \sigma$ the x^2 terms cancel and

$$-\frac{(x-\mu_1)^2 - (x-\mu_0)^2}{2\sigma^2} = \frac{(\mu_1 - \mu_0)x}{\sigma^2} - \frac{\mu_1^2 - \mu_0^2}{2\sigma^2},$$

giving the stated α, β. If $\sigma_0 \neq \sigma_1$ the coefficient of x^2 is $\frac{1}{2}(\sigma_0^{-2} - \sigma_1^{-2}) \neq 0$. ∎

So when a population mixes two roughly normal groups with similar spread, the logistic curve fits well; with unequal spreads $\pi(x)$ is nonmonotone (rises then falls, or the reverse).

5.2 Inference for logistic regression

Motivation. ML estimators in logistic regression are asymptotically normal, so the Wald, likelihood-ratio and score triad of Theorem 4.18 applies. The likelihood-ratio versions are generally the most reliable, and goodness of fit needs care when predictors are continuous.

5.2.1 Types of inference

For $\operatorname{logit}\pi(x) = \alpha + \beta x$, the test of independence $H_0 : \beta = 0$ can use

(i) Wald: $z = \hat\beta/\widehat{\mathrm{se}}$, or $z^2 \sim \chi^2_1$;

(ii) likelihood ratio: $-2(L_0 - L_1)$, twice the difference of maximized log likelihoods under H_0 and the full model, χ^2_1;

(iii) score: based on the derivative of the log likelihood at $\beta = 0$, comparing the sufficient statistic for β with its null expectation (Section 5.3.5).

In large samples they agree. The LR test uses the log likelihood at both H_0 and $\hat\beta$ and is preferred; for large $|\beta|$ the Wald test loses power and can behave aberrantly (Hauck and Donner 1977). Inverting a test gives a confidence interval: Wald gives $\hat\beta \pm z_{\alpha/2}\widehat{\mathrm{se}}$; the LR inversion is the profile-likelihood interval.

Proposition 5.6 (Interval for $\pi(x_0)$)

The estimated variance of the fitted logit at x_0 is

$$\widehat{\operatorname{Var}}(\hat\alpha + \hat\beta x_0) = \widehat{\operatorname{Var}}(\hat\alpha) + x_0^2\,\widehat{\operatorname{Var}}(\hat\beta) + 2x_0\,\widehat{\operatorname{Cov}}(\hat\alpha, \hat\beta),$$

and if (l, u) is a $1-\alpha$ interval for $\operatorname{logit}\pi(x_0)$, then $(e^l/(1+e^l),\ e^u/(1+e^u))$ is a $1-\alpha$ interval for $\pi(x_0)$.

Proof. The first formula is the variance of the linear combination $(1, x_0)(\hat{\alpha}, \hat{\beta})^\top$, namely $(1, x_0)\widehat{\mathrm{Cov}}(\hat{\alpha}, \hat{\beta})(1, x_0)^\top$. Since $t \mapsto e^t/(1+e^t)$ is strictly increasing, $l \le \operatorname{logit}\pi(x_0) \le u$ iff $e^l/(1+e^l) \le \pi(x_0) \le e^u/(1+e^u)$, so the coverage is preserved. ■

Equivalently, refit with $x^* = x - x_0$: the intercept and its SE are the fitted logit at x_0 and its SE.

5.2.2 Inference for horseshoe crab data

Example 5.7 (Horseshoe crabs: tests and intervals)

(a) Wald: $z = 0.497/0.102 = 4.9$, $z^2 = 23.9$ ($P < 0.0001$). The maximized log likelihoods are -112.88 under $\beta = 0$ and -97.23 for the model, so the LR statistic is $-2(-112.88 + 97.23) = 31.3$ (df $= 1$), even stronger evidence.
(b) The Wald interval for β is $0.497 \pm 1.96(0.102) = (0.298, 0.697)$; the profile-likelihood interval is (0.308, 0.709). For the odds ratio per cm: $(e^{0.308}, e^{0.709}) = (1.36, 2.03)$: at least a 36% increase and at most a doubling of the odds.
(c) At $x = 26.5$: logit $= -12.351 + 0.497(26.5) = 0.825$, $\hat{\pi} = 0.695$. With $\widehat{\mathrm{Var}}(\hat{\alpha}) = 6.910$, $\widehat{\mathrm{Var}}(\hat{\beta}) = 0.01035$, $\widehat{\mathrm{Cov}} = -0.2668$,

$$\widehat{\mathrm{Var}} = 6.910 + (26.5)^2(0.01035) + 2(26.5)(-0.2668) = 0.038,$$

so the logit interval is $0.825 \pm 1.96\sqrt{0.038} = (0.44, 1.21)$, i.e. (0.61, 0.77) for $\pi(26.5)$.
(d) Only 6 crabs had $x = 26.5$, 4 with satellites: the sample proportion 0.67 has SE $= \sqrt{0.67(0.33)/6} = 0.19$ and score interval (0.30, 0.90), versus model SE $= 0.04$. The model borrows strength from all 173 observations to estimate two parameters; when it approximates the truth decently, its estimates are much more precise (Section 6.4.5).

5.2.3 Checking goodness of fit: ungrouped and grouped data

One way to detect lack of fit compares the model with a more complex one (quadratic term, interaction) by an LR test; if the extra terms do not help, the model gains support. Global tests compare observed counts with fitted values $n_i\hat{\pi}_i$, $n_i(1-\hat{\pi}_i)$ through X^2 or G^2; for a fixed number of predictor settings, as fitted counts grow, they are χ^2 with residual df = (number of settings) − (number of parameters). This requires categorical predictors: as noted in Section 4.5.3, the saturated model differs for grouped and ungrouped data, and the chi-squared limit needs a fixed number of settings.

5.2.4 Goodness of fit of model for horseshoe crabs

Example 5.8 (Horseshoe crabs: goodness of fit)

(a) Adding a quadratic term with centered width gives $\operatorname{logit}\hat{\pi} = 0.618 + 0.533x + 0.040x^2$, with $\widehat{\mathrm{se}} = 0.046$ for the quadratic coefficient; the LR statistic is 0.83 (df $= 1$): no need for it.
(b) Width takes 66 distinct values; treating the data as a 66×2 table violates the chi-squared theory (tiny fitted counts, and more data would add cells). Grouping into the eight width categories of Example 4.5, with fitted values the sums of $\hat{\pi}(x)$ and $1 - \hat{\pi}(x)$ over the crabs in each category:

Width (cm)	Yes	No	Fitted yes	Fitted no
< 23.25	5	9	3.64	10.36
23.25–24.25	4	10	5.31	8.69
24.25–25.25	17	11	13.78	14.22
25.25–26.25	21	18	24.23	14.77
26.25–27.25	15	7	15.94	6.06
27.25–28.25	20	4	19.38	4.62
28.25–29.25	15	3	15.65	2.35
> 29.25	14	0	13.08	0.92

Then $X^2 = 5.3$ and $G^2 = 6.2$ with df $= 8 - 2 = 6$ ($P > 0.4$): no evidence of lack of fit (the theory is still approximate since $\pi(x)$ varies within a category).

5.2.5 Checking goodness of fit with ungrouped data by grouping

With many predictors, grouping on each variable gives a sparse table. Instead, partition the observations by their fitted probabilities.

Definition (Hosmer–Lemeshow statistic)

Order the observations by $\hat{\pi}$ from the ungrouped fit and form g groups of roughly equal size (usually $g = 10$). With y_{ij} the binary outcome and $\hat{\pi}_{ij}$ the fitted probability of observation j in group i ($j = 1, \ldots, n_i$),

$$\mathrm{HL} = \sum_{i=1}^{g} \frac{\left(\sum_j y_{ij} - \sum_j \hat{\pi}_{ij}\right)^2}{\left(\sum_j \hat{\pi}_{ij}\right)\left[1 - \left(\sum_j \hat{\pi}_{ij}\right)/n_i\right]}.$$

Proposition 5.9 (HL as a Pearson statistic)

HL equals the Pearson statistic $\sum(\text{observed} - \text{fitted})^2/\text{fitted}$ for the $g \times 2$ table of observed and fitted counts of successes and failures in the groups.

Proof. [Supplement] In group i let $O_i = \sum_j y_{ij}$, $E_i = \sum_j \hat{\pi}_{ij}$. The failure cell has observed $n_i - O_i$ and fitted $n_i - E_i$, so its deviation is $-(O_i - E_i)$. The two Pearson terms of row i add to

$$(O_i - E_i)^2\Big(\frac{1}{E_i} + \frac{1}{n_i - E_i}\Big) = \frac{(O_i - E_i)^2\, n_i}{E_i(n_i - E_i)} = \frac{(O_i - E_i)^2}{E_i(1 - E_i/n_i)}.$$

Summing over i gives HL. ■

Reference distribution and limitations

HL is not asymptotically chi-squared in the usual sense, since observations within a group do not share a success probability; Hosmer and Lemeshow found by simulation that when the number of distinct covariate patterns equals n, the null distribution is approximately χ^2_{g-2}. Ties in $\hat{\pi}$ make the grouping somewhat arbitrary, so software values differ slightly. For the crab width model with $g = 10$, HL $= 3.5$ on 8 df: a decent fit. Like other global statistics, HL has low power against specific alternatives and does not say what is wrong; comparing with a more complex model is more informative, and diagnostics (Section 6.2) locate problems.

5.3 Logit models with categorical predictors

Motivation. Qualitative predictors enter through dummy variables. Different codings change the individual parameters but not the fit or the odds ratios; for ordered factors, scores lead to the linear logit model and the Cochran–Armitage trend test.

5.3.1 ANOVA-type representation of factors

For a single factor X with I categories, let $y_i \sim \text{bin}(n_i, \pi_i)$ in row i of an $I \times 2$ table. The model

$$\log \frac{\pi_i}{1-\pi_i} = \alpha + \beta_i, \qquad i = 1, \dots, I, \tag{5.2}$$

resembles one-way ANOVA and has one redundant parameter, so a constraint is imposed, e.g. $\beta_I = 0$.

Proposition 5.10 (Coding invariance and saturation)

(a) Model (5.2) is saturated: for any $\{\pi_i\} \subset (0,1)$ there are parameters satisfying it, and its fitted values are the sample proportions.
(b) Under any constraint, $\alpha + \beta_i = \operatorname{logit} \pi_i$ and $\beta_a - \beta_b$ is the log odds ratio for rows a, b; these and the fit are the same for every coding. With $\beta_I = 0$, $\alpha = \operatorname{logit} \pi_I$ and β_i is the log odds ratio of row i versus row I.
(c) No effect, $\beta_1 = \cdots = \beta_I$, is statistical independence of X and Y.

Proof.

(a) Take $\alpha = \operatorname{logit} \pi_I$ and $\beta_i = \operatorname{logit} \pi_i - \operatorname{logit} \pi_I$. The model has I free parameters for I binomials, so it is saturated and $\hat{\pi}_i = y_i/n_i$.

(b) If $(\alpha, \boldsymbol{\beta})$ satisfies the model, so does $(\alpha - c, \beta_i + c)$ for any c, and every constraint selects one such c; $\alpha + \beta_i$ and $\beta_a - \beta_b$ do not depend on c. In particular $\beta_a - \beta_b = \operatorname{logit} \pi_a - \operatorname{logit} \pi_b$, a log odds ratio.

(c) Equal β_i means equal π_i across rows. ∎

5.3.2 Dummy variables in logit models

Equivalently, with dummy variables $x_i = 1$ for row i and 0 otherwise ($i = 1, \dots, I-1$),

$$\operatorname{logit} \pi = \alpha + \beta_1 x_1 + \cdots + \beta_{I-1} x_{I-1};$$

no dummy for category I corresponds to $\beta_I = 0$ (some software instead sets $\beta_1 = 0$). *Effect coding* imposes $\sum_i \beta_i = 0$; for $I = 2$ it uses $x = 1$ in category 1 and $x = -1$ in category 2, and then the log odds ratio is $\beta_1 - \beta_2 = 2\beta_1$, whereas with a 0/1 dummy it is β_1. A single $\hat{\beta}_i$ is meaningful only in comparison with another category.

5.3.3 Alcohol and infant malformation example revisited

Example 5.11 (Maternal alcohol consumption and malformation)

Alcohol (drinks/day)	Present	Absent	Logit	Observed prop.	Fitted (linear logit)
0	48	17066	−5.87	0.0028	0.0026
< 1	38	14464	−5.94	0.0026	0.0030
1–2	5	788	−5.06	0.0063	0.0041
3–5	1	126	−4.84	0.0079	0.0091
≥ 6	1	37	−3.61	0.0263	0.0231

For (5.2) with malformation as response, $\hat{\alpha} + \hat{\beta}_i$ are the sample logits, e.g. $\log(48/17066) = -5.87$. With $\beta_5 = 0$: $\hat{\alpha} = -3.61$, $\hat{\beta}_1 = -2.26$; with $\beta_1 = 0$: $\hat{\alpha} = -5.87$. Apart from a slight reversal in the first two rows, the logits increase with alcohol. Independence has $\hat{\alpha} = \log(93/32481) = -5.86$; testing it (df $= 4$) gives $X^2 = 12.1$ ($P = 0.02$) but $G^2 = 6.2$ ($P = 0.19$): mixed signals, since the counts are very unbalanced (exact conditional P-values 0.03 and 0.13). Both ignore the ordering of alcohol, and one case more or less in the last rows changes those proportions substantially.

5.3.4 Linear logit model for $I \times 2$ tables

For ordered categories with scores $x_1, \ldots, x_I$ the *linear logit model*

$$\operatorname{logit} \pi_i = \alpha + \beta x_i \tag{5.3}$$

lies between independence ($\beta = 0$) and the saturated model (5.2).

Example 5.12 (Alcohol: linear logit fit)

With scores (0, 0.5, 1.5, 4.0, 7.0) (the last somewhat arbitrary): $\hat{\alpha} = -5.9605$ (0.1154) and $\hat{\beta} = 0.3166$ (0.1254), profile-likelihood interval (0.0187, 0.5236). The estimated odds of malformation multiply by $e^{0.317} = 1.37$ per drink per day. Fit: $G^2 = 1.95$, $X^2 = 2.05$, df $= 3$; log likelihood -635.60. The fitted proportions are in the table of Example 5.11.

5.3.5 Cochran–Armitage trend test

For an $I \times 2$ table with ordered rows and independent $y_i \sim \text{bin}(n_i, \pi_i)$, Cochran (1954) and Armitage (1955) partitioned the Pearson statistic using the linear probability model $\pi_i = \alpha + \beta x_i$ fitted by (weighted) least squares. Let $n = \sum n_i$, $p_i = y_i/n_i$, $p = \sum y_i/n$ and $\bar{x} = \sum n_i x_i/n$. The prediction equation is $\hat{\pi}_i = p + b(x_i - \bar{x})$ with

$$b = \frac{\sum_i n_i (p_i - p)(x_i - \bar{x})}{\sum_i n_i (x_i - \bar{x})^2}.$$

Theorem 5.13 (Partition of the Pearson statistic)

Let $X^2(I) = \sum_i n_i (p_i - p)^2 / [p(1-p)]$ be the Pearson statistic for independence in the $I \times 2$ table. Then

$$X^2(I) = z^2 + X^2(L),$$

where

$$X^2(L) = \frac{\sum_i n_i(p_i - \hat{\pi}_i)^2}{p(1-p)}, \qquad z^2 = \frac{b^2 \sum_i n_i(x_i - \bar{x})^2}{p(1-p)} = \frac{\left[\sum_i (x_i - \bar{x}) y_i\right]^2}{p(1-p) \sum_i n_i (x_i - \bar{x})^2}.$$

Under independence z^2 is asymptotically χ^2_1 (the *Cochran–Armitage trend test* of $\beta = 0$); when the linear probability model holds, $X^2(L)$ is asymptotically χ^2_{I-2} (a test of linearity).

Proof. [Supplement]

Step 1. (The Pearson form.) The cells of row i have observed y_i, $n_i - y_i$ and fitted $n_i p$, $n_i(1-p)$, with deviations $\pm(y_i - n_i p)$. As in Proposition 5.9 the two terms add to $(y_i - n_i p)^2/[n_i p(1-p)] = n_i(p_i - p)^2/[p(1-p)]$.

Step 2. (Decomposition.) Write $p_i - p = (p_i - \hat{\pi}_i) + (\hat{\pi}_i - p)$. Then

$$\sum n_i (p_i - p)^2 = \sum n_i (p_i - \hat{\pi}_i)^2 + \sum n_i (\hat{\pi}_i - p)^2 + 2 \sum n_i (p_i - \hat{\pi}_i)(\hat{\pi}_i - p).$$

Step 3. (Cross term vanishes.) Since $\hat{\pi}_i - p = b(x_i - \bar{x})$,

$$\sum n_i (p_i - \hat{\pi}_i)(x_i - \bar{x}) = \sum n_i (p_i - p)(x_i - \bar{x}) - b \sum n_i (x_i - \bar{x})^2 = 0$$

by the definition of b.

Step 4. (Middle term.) $\sum n_i(\hat{\pi}_i - p)^2 = b^2 \sum n_i (x_i - \bar{x})^2$. Since $\sum n_i (x_i - \bar{x}) = 0$, $\sum n_i (p_i - p)(x_i - \bar{x}) = \sum (y_i - n_i p)(x_i - \bar{x}) = \sum (x_i - \bar{x}) y_i$, which gives the second form of z^2. Dividing Step 2 by $p(1-p)$ yields the partition.

Step 5. (Null distribution of z^2.) Under independence $E \sum (x_i - \bar{x}) Y_i = \pi \sum n_i (x_i - \bar{x}) = 0$ and $\mathrm{Var} \sum (x_i - \bar{x}) Y_i = \pi(1-\pi) \sum n_i (x_i - \bar{x})^2$. By the CLT and $p \xrightarrow{p} \pi$ (Slutsky), $z \xrightarrow{d} N(0,1)$. The limit of $X^2(L)$ follows from the chi-squared theory for the fit of a 2-parameter model to I binomials. ■

Proposition 5.14 (Cochran–Armitage = score test)

The score statistic for $H_0 : \beta = 0$ in the linear logit model (5.3) equals z^2 of Theorem 5.13.

Proof. [Supplement]

Step 1. Under H_0 the MLE is $\tilde{\alpha} = \mathrm{logit}\, p$, so $\tilde{\pi}_i = p$. By (4.16) in count form the score components at $(\tilde{\alpha}, 0)$ are

$$u_\alpha = \sum_i (y_i - n_i p) = 0, \qquad u_\beta = \sum_i x_i (y_i - n_i p) = \sum_i (x_i - \bar{x}) y_i.$$

Step 2. The information at $(\tilde{\alpha}, 0)$ is $X^\top \mathrm{diag}\{n_i p(1-p)\} X$:

$$\mathcal{I} = p(1-p) \begin{pmatrix} n & \sum n_i x_i \\ \sum n_i x_i & \sum n_i x_i^2 \end{pmatrix},$$

and by the formula for the inverse of a 2×2 matrix,

$$[\mathcal{I}^{-1}]_{\beta\beta} = \frac{n}{p(1-p)\left[n \sum n_i x_i^2 - (\sum n_i x_i)^2\right]} = \frac{1}{p(1-p) \sum n_i (x_i - \bar{x})^2}.$$

Step 3. The score statistic $\boldsymbol{u}^\top \mathcal{I}^{-1} \boldsymbol{u}$ with $u_\alpha = 0$ is $u_\beta^2 [\mathcal{I}^{-1}]_{\beta\beta} = z^2$. ■

The statistic also equals $M^2 = (n-1) r^2$ of Section 3.4 for $J = 2$, with $n - 1$ replaced by n; for $I = 2$, $X^2(L) = 0$ and $z^2 = X^2(I)$.

Example 5.15 (Alcohol: trend test)

With the scores above, $z^2 = 6.6$ ($P = 0.010$), and $X^2(I) = 12.1 = 6.6 + 5.5$, where $X^2(L) = 5.5$ (df $= 3$) shows only slight departure from linearity. The correlation is only $r = 0.014$ ($n = 32{,}574$), but r is a poor descriptive measure for such unbalanced tables. The Wald statistic is $(0.317/0.125)^2 = 6.4$ ($P = 0.012$) and the LR statistic 4.25 ($P = 0.039$); given the highly unbalanced counts the LR approach is safest, and the profile interval (0.02, 0.52) is preferable to the Wald interval (0.07, 0.56). Exact methods (Section 6.7.4) are relevant even though n is huge.

5.4 Multiple logistic regression

Motivation. With several predictors each coefficient is a conditional effect, controlling for the others. Comparing nested models by deviance differences is the main tool for deciding which predictors and interactions are needed.

For predictors $\mathbf{x} = (x_1, \ldots, x_p)$,

$$\operatorname{logit} \pi(\mathbf{x}) = \alpha + \beta_1 x_1 + \cdots + \beta_p x_p, \qquad \pi(\mathbf{x}) = \frac{\exp(\alpha + \beta_1 x_1 + \cdots + \beta_p x_p)}{1 + \exp(\alpha + \beta_1 x_1 + \cdots + \beta_p x_p)}. \tag{5.4}$$

e^{β_j} is the multiplicative effect on the odds of a unit increase in x_j, at fixed levels of the other predictors.

5.4.1 Logit models for multiway contingency tables

With binary X and Z coded by dummies ($x_1 = z_1 = 1$, $x_2 = z_2 = 0$) and independent binomials at each (i, k), consider

$$\operatorname{logit} P(Y = 1) = \alpha + \beta_1 x_i + \beta_2 z_k. \tag{5.5}$$

Proposition 5.16 (Main-effects model and homogeneous association)

In (5.5), at each level z_k of Z,

$$\operatorname{logit} P(Y = 1 \mid x = 1, z_k) - \operatorname{logit} P(Y = 1 \mid x = 0, z_k) = \beta_1,$$

so e^{β_1} is the conditional XY odds ratio, the same at each level of Z (homogeneous association). $\beta_1 = 0$ is conditional independence of X and Y given Z.

Proof. $(\alpha + \beta_1 + \beta_2 z_k) - (\alpha + \beta_2 z_k) = \beta_1$, free of z_k; a difference of logits is a log odds ratio. If $\beta_1 = 0$ the conditional odds ratios all equal 1. ■

No interaction thus means additivity on the logit scale; on another scale (probit, identity) interaction may appear or disappear, and in some applications another scale is natural. An ANOVA-type form handles factors with any numbers of categories:

$$\operatorname{logit} P(Y = 1) = \alpha + \beta_i^X + \beta_k^Z, \tag{5.6}$$

with one redundant parameter per factor (e.g. $\beta_I^X = \beta_K^Z = 0$; superscripts are labels). Conditional independence of X and Y is $\beta_1^X = \cdots = \beta_I^X$. For two categories each, (5.6) with $\beta_2^X = \beta_2^Z = 0$ is (5.5) with $\beta_1^X = \beta_1$, $\beta_1^Z = \beta_2$.

5.4.2 AIDS and AZT example

Example 5.17 (AIDS symptoms, AZT use and race)

338 veterans with faltering immune systems were randomized to immediate AZT or to AZT only after severe T-cell weakness; the response is development of AIDS symptoms over 3 years.

		Symptoms	
Race	AZT use	Yes	No
White	Yes	14	93
	No	32	81
Black	Yes	11	52
	No	12	43

In (5.5) let $x = 1$ for immediate AZT and $z = 1$ for whites. Fit:

Parameter	Estimate	SE	Wald χ^2
Intercept	-1.0736	0.2629	16.67
AZT	-0.7195	0.2790	6.65
race	0.0555	0.2886	0.04

(a) The estimated conditional odds ratio for AZT is $e^{-0.7195} = 0.487$: for each race the odds of symptoms are about halved by immediate AZT. Wald interval $\exp[-0.720 \pm 1.96(0.279)] = (0.28, 0.84)$; profile interval (0.279, 0.835). Race: 1.057, interval (0.60, 1.86).
(b) $H_0 : \beta_1 = 0$ (conditional independence of AZT and symptoms given race): LR statistic 6.9 (df $= 1$, $P = 0.01$); Wald $(-0.720/0.279)^2 = 6.65$.
(c) Codings: for (last $= 0$, first $= 0$, sum $= 0$) the intercept is $(-1.074, -1.738, -1.406)$, AZT-yes $(-0.720, 0, -0.360)$, white $(0.055, 0, 0.028)$. In each, intercept + AZT-yes + white $= -1.738$, so $\hat{\pi} = e^{-1.738}/(1+e^{-1.738}) = 0.15$ for whites with immediate AZT, and $\hat{\beta}_1^X - \hat{\beta}_2^X = -0.720$ always.
(d) Fitted probabilities (95% limits): white/AZT 0.150 (0.099, 0.220); white/no 0.265 (0.197, 0.348); black/AZT 0.143 (0.087, 0.225); black/no 0.255 (0.170, 0.364).

5.4.3 Goodness of fit as a likelihood-ratio test

Write $G^2(M_0 \mid M_1) = -2(L_0 - L_1)$ for the LR statistic testing M_0 within M_1. The goodness-of-fit statistic $G^2(M)$ is the case $M_0 = M$, $M_1 =$ saturated, with df = (number of binomials) − (number of parameters). By Proposition 4.22, with L_S the saturated log likelihood,

$$G^2(M_0 \mid M_1) = -2(L_0 - L_S) - [-2(L_1 - L_S)] = G^2(M_0) - G^2(M_1).$$

Example 5.18 (AIDS: goodness of fit)

Whites with immediate AZT have $\hat{\pi} = 0.150$, so the fitted counts are $107(0.150) = 16.0$ and $107(0.850) = 91.0$; similarly for all 8 cells. Then $G^2 = 1.38$, $X^2 = 1.39$, with df $= 4 - 3 = 1$ ($P > 0.2$). Since the model has no interaction, this is also a test of homogeneous odds ratios; adding the interaction gives the saturated model. For H_0: no race effect, $G^2(M_0) - G^2(M_1) = 1.42 - 1.38 = 0.04$.

The comparison statistic is often nearly chi-squared even when the separate $G^2(M_i)$ are not (continuous predictors, small fitted values), provided the models differ by few parameters.

5.4.4 Horseshoe crab example revisited

Example 5.19 (Horseshoe crabs: width and color)

Color has four observed categories (medium light, medium, medium dark, dark), a surrogate for age. With dummies c_1, c_2, c_3 for the first three (dark is the reference),

$$\operatorname{logit}\pi = \alpha + \beta_1 c_1 + \beta_2 c_2 + \beta_3 c_3 + \beta_4 x. \tag{5.7}$$

Parameter	Estimate	SE	LR 95% limits	χ^2
intercept	−12.7151	2.7618	(−18.4564, −7.5788)	21.20
c_1	1.3299	0.8525	(−0.2738, 3.1354)	2.43
c_2	1.4023	0.5484	(0.3527, 2.5260)	6.54
c_3	1.1061	0.5921	(−0.0279, 2.3138)	3.49
width	0.4680	0.1055	(0.2713, 0.6870)	19.66

Deviance 187.46, Pearson 168.66 on 168 df; log likelihood −93.73.

For dark crabs $\operatorname{logit}\hat\pi = -12.715 + 0.468x$; for medium-light, $-11.385 + 0.468x$. At $x = 26.3$, $\hat\pi = 0.399$ (dark) and 0.715 (medium light). Without interaction the curves for the colors are horizontal shifts of each other and never cross; a 1-cm increase multiplies the odds by $e^{0.468} = 1.60$ for every color. Medium-light versus dark: $e^{1.330} = 3.8$ at any width (at 26.3: $2.51/0.66 = 3.8$).

5.4.5 Model comparison

Example 5.20 (Horseshoe crabs: testing color and interaction)

$H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ (no color effect given width): $-2(L_0 - L_1) = 7.0$, df = 3, $P = 0.07$: slight evidence of a color effect. Adding the three color×width products (equivalent to separate logistic fits per color, so color comparisons depend on width) gives LR = 4.4, df = 3, $P = 0.22$: weak evidence of interaction.

5.4.6 Quantitative treatment of ordinal predictor

Example 5.21 (Horseshoe crabs: color scores)

With scores $c = 1, 2, 3, 4$,

$$\operatorname{logit}\pi = \alpha + \beta_1 c + \beta_2 x, \qquad \hat\beta_1 = -0.509\ (0.224), \quad \hat\beta_2 = 0.458\ (0.104).$$

Each step darker multiplies the odds by $e^{-0.509} = 0.60$ at fixed width. Against (5.7), LR = 1.7 (df = 2, $P = 0.44$): the color parameters are consistent with a linear trend in the scores. The qualitative estimates (1.33, 1.40, 1.11, 0) suggest the binary scoring dark (0) versus other (1); against (5.7) it gives LR = 0.5 (df = 2), and the fit is

$$\operatorname{logit}\hat\pi = -12.980 + 1.300c + 0.478x \qquad (\widehat{\text{se}}:\ 0.526,\ 0.104). \tag{5.8}$$

At a given width, non-dark crabs have $e^{1.300} = 3.7$ times the odds of dark crabs. All three models say dark crabs are least likely to have satellites; deciding the best scoring needs more data. Treating an ordinal predictor quantitatively, when it fits, is simpler and gives more powerful one-df tests (Section 6.4).

5.4.7 Standardized and probability-based interpretations

To compare quantitative predictors with different units, one can fit with standardized predictors $(x_j - \bar{x}_j)/s_{x_j}$, or equivalently multiply $\hat{\beta}_j$ by s_{x_j}: the effect of a one-standard-deviation change. For a probability scale, Proposition 5.2 extends: at a setting with $\hat{\pi}$, a unit increase in x_j (others fixed) changes the probability by about $\hat{\beta}_j \hat{\pi}(1-\hat{\pi})$. More precisely, set the other predictors at their means and report $\hat{\pi}$ at the quartiles of x_j; for a dummy predictor report $\hat{\pi}$ at its two values.

Example 5.22 (Horseshoe crabs: effects on the probability scale)

For (5.8) at $\hat{\pi} = 0.5$, a 1-cm width increase raises π by about $0.478(0.5)(0.5) = 0.12$, considerable since 1 cm is under half a standard deviation. With $\bar{c} = 0.873$, $\hat{\pi} = 0.51$ at the lower width quartile 24.9 and 0.80 at the upper quartile 27.7. At $x = 26.3$, $\hat{\pi} = 0.40$ for dark and 0.71 for other colors.

Variable	Estimate	SE	Comparison	Change in probability
No interaction model				
Intercept	−12.980	2.727		
Color (0 dark, 1 other)	1.300	0.526	$(1, 0)$ at $\bar{x}$	$0.31 = 0.71 - 0.40$
Width x (cm)	0.478	0.104	(UQ, LQ) at $\bar{c}$	$0.29 = 0.80 - 0.51$
Interaction model				
Intercept	−5.854	6.694		
Color	−6.958	7.318		
Width x	0.200	0.262	(UQ, LQ) at $c = 0$	$0.13 = 0.43 - 0.30$
Width × color	0.322	0.286	(UQ, LQ) at $c = 1$	$0.29 = 0.84 - 0.55$

The width effect appears larger for lighter crabs, but the interaction is not significant.

5.5 Fitting logistic regression models

Motivation. Logistic regression is a canonical-link binomial GLM, so the theory of Sections 4.4–4.6 specializes neatly: the likelihood is concave, Newton–Raphson coincides with Fisher scoring and IRLS, and finite estimates exist exactly when the two outcome groups overlap.

Let $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})$ be setting i $(i = 1, \ldots, N)$, with $x_{i1} = 1$ for the intercept, and let y_i be the number of successes in n_i trials (with continuous predictors typically $N = n$ and $n_i = 1$). The model is

$$\pi(\mathbf{x}_i) = \frac{\exp\left(\sum_{j=1}^{p} \beta_j x_{ij}\right)}{1 + \exp\left(\sum_{j=1}^{p} \beta_j x_{ij}\right)}. \tag{5.9}$$

5.5.1 Likelihood equations

Theorem 5.23 (Logistic log likelihood and likelihood equations)

For independent $y_i \sim \text{bin}(n_i, \pi(\mathbf{x}_i))$ under (5.9),

$$L(\boldsymbol{\beta}) = \sum_j \Big(\sum_i y_i x_{ij}\Big)\beta_j - \sum_i n_i \log\Big[1 + \exp\Big(\sum_j \beta_j x_{ij}\Big)\Big], \tag{5.10}$$

so $\{\sum_i y_i x_{ij}\}$ are sufficient, and the likelihood equations are

$$\sum_i y_i x_{ij} - \sum_i n_i \hat{\pi}_i x_{ij} = 0, \quad j = 1, \dots, p, \qquad \text{i.e.} \qquad X^\top \mathbf{y} = X^\top \hat{\boldsymbol{\mu}}, \tag{5.11}$$

where $\hat{\mu}_i = n_i \hat{\pi}_i$.

Proof.

Step 1. The joint mass function is proportional to

$$\prod_i \pi_i^{y_i} (1-\pi_i)^{n_i - y_i} = \Big\{ \prod_i (1-\pi_i)^{n_i} \Big\} \exp\Big[\sum_i y_i \log \frac{\pi_i}{1-\pi_i} \Big].$$

Step 2. With $\operatorname{logit} \pi_i = \sum_j \beta_j x_{ij}$, the exponent is $\sum_i y_i \sum_j \beta_j x_{ij} = \sum_j \beta_j \sum_i y_i x_{ij}$, and $1 - \pi_i = [1 + \exp(\sum_j \beta_j x_{ij})]^{-1}$. Taking logs gives (5.10); the factorization theorem gives sufficiency.

Step 3. Differentiating,

$$\frac{\partial L}{\partial \beta_j} = \sum_i y_i x_{ij} - \sum_i n_i x_{ij} \frac{\exp(\sum_k \beta_k x_{ik})}{1 + \exp(\sum_k \beta_k x_{ik})} = \sum_i (y_i - n_i \pi_i) x_{ij}.$$

■

These are the binomial GLM equations (4.16) written with counts, an instance of Theorem 4.27: for canonical links the likelihood equations equate sufficient statistics to their estimated expected values (they are the normal equations of ordinary regression for the identity link). They are nonlinear and need iteration.

5.5.2 Asymptotic covariance matrix of parameter estimators

Theorem 5.24 (Information matrix for logistic regression)

$$-\frac{\partial^2 L}{\partial \beta_a \partial \beta_b} = \sum_i x_{ia} x_{ib}\, n_i \pi_i (1 - \pi_i),$$

which does not involve $\mathbf{y}$; so observed and expected information coincide, and

$$\widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}}) = \{ X^\top \operatorname{diag}[n_i \hat{\pi}_i (1 - \hat{\pi}_i)] X \}^{-1}. \tag{5.12}$$

Proof. From Step 3 above, $\partial L / \partial \beta_a = \sum_i (y_i - n_i \pi_i) x_{ia}$, and by Proposition 5.2, Step 1, $\partial \pi_i / \partial \beta_b = \pi_i (1 - \pi_i) x_{ib}$. Hence $\partial^2 L / \partial \beta_a \partial \beta_b = -\sum_i n_i \pi_i (1 - \pi_i) x_{ia} x_{ib}$. This is free of y_i, so it equals its expectation, and Theorem 4.15 gives (5.12): the GLM form $(X^\top \hat{W} X)^{-1}$ with $w_i = n_i \pi_i (1 - \pi_i)$. ■

The square roots of the diagonal of (5.12) are the standard errors of $\hat{\boldsymbol{\beta}}$.

5.5.3 Distribution of probability estimators

For a setting $\mathbf{x}$, the fitted logit $\mathbf{x}^\top \hat{\boldsymbol{\beta}}$ has estimated variance $\mathbf{x}^\top \widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}}) \mathbf{x}$, so

$$\mathbf{x}^\top \hat{\boldsymbol{\beta}} \pm z_{\alpha/2} \sqrt{\mathbf{x}^\top \widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}}) \mathbf{x}}$$

is a large-sample interval for $\operatorname{logit} \pi(\mathbf{x})$; applying $t \mapsto e^t/(1+e^t)$ to the endpoints gives one for $\pi(\mathbf{x})$ (Proposition 5.6).

5.5.4 Newton–Raphson method applied to logistic regression

Proposition 5.25 (Newton–Raphson / IRLS for logistic regression)

With $\pi_i^{(t)} = \exp(\mathbf{x}_i^\top \boldsymbol{\beta}^{(t)})/[1+\exp(\mathbf{x}_i^\top \boldsymbol{\beta}^{(t)})]$ and $\mu_i^{(t)} = n_i \pi_i^{(t)}$, the Newton–Raphson step is

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \{X^\top \mathrm{diag}[n_i \pi_i^{(t)}(1-\pi_i^{(t)})] X\}^{-1} X^\top (\mathbf{y} - \boldsymbol{\mu}^{(t)}). \tag{5.13}$$

Equivalently $\boldsymbol{\beta}^{(t+1)} = (X^\top V_t^{-1} X)^{-1} X^\top V_t^{-1} \boldsymbol{z}^{(t)}$, with $V_t = \mathrm{diag}\{1/[n_i \pi_i^{(t)}(1-\pi_i^{(t)})]\}$ and

$$z_i^{(t)} = \log \frac{\pi_i^{(t)}}{1-\pi_i^{(t)}} + \frac{y_i - n_i \pi_i^{(t)}}{n_i \pi_i^{(t)}(1-\pi_i^{(t)})}. \tag{5.14}$$

At convergence, $-H^{-1}$ is the estimated covariance (5.12).

Proof. By Theorems 5.23 and 5.24, $\boldsymbol{u}^{(t)} = X^\top(\mathbf{y} - \boldsymbol{\mu}^{(t)})$ and $H^{(t)} = -X^\top \mathrm{diag}[n_i \pi_i^{(t)}(1-\pi_i^{(t)})] X$; substitute in (4.25). Since $H = -\mathcal{I}$ (canonical link), this is also Fisher scoring, hence IRLS by Theorem 4.26. For the logit, $\partial \eta_i / \partial \mu_i = 1/[n_i \pi_i (1-\pi_i)]$ when $\mu_i = n_i \pi_i$, so the working response (4.28) is (5.14), and $W = V_t^{-1}$. ■

The elements of V_t are the estimated asymptotic variances of sample logits (Section 3.1.6), and $\boldsymbol{z}^{(t)}$ is the linearized logit of the sample data. Starting from $\boldsymbol{\beta}^{(0)}$ (e.g. $\mathbf{0}$), the iterates converge to $\hat{\boldsymbol{\beta}}$ (Walker and Duncan 1967).

5.5.5 Convergence and existence of finite estimates

Theorem 5.26 (Concavity and uniqueness)

If X has full column rank, $L(\boldsymbol{\beta})$ in (5.10) is strictly concave; hence the ML estimate, when it exists, is unique, and any root of (5.11) is it.

Proof. By Theorem 5.24, $\boldsymbol{c}^\top H \boldsymbol{c} = -\sum_i n_i \pi_i (1-\pi_i)(\mathbf{x}_i^\top \boldsymbol{c})^2$ with every $n_i \pi_i (1-\pi_i) > 0$. It is < 0 unless $X\boldsymbol{c} = \mathbf{0}$, i.e. $\boldsymbol{c} = \mathbf{0}$. A strictly concave function has at most one maximizer, and a stationary point of a concave function is a global maximum. ■

Theorem 5.27 (Existence of the ML estimate: overlap versus separation)

Write the data as n individual binary observations $(\mathbf{x}_i, y_i)$ and let $s_i = 2y_i - 1 \in \{-1, 1\}$. Assume X has full column rank. Call the data *separated* if some $\boldsymbol{d} \neq \mathbf{0}$ satisfies $s_i \mathbf{x}_i^\top \boldsymbol{d} \geq 0$ for all i, with strict inequality for at least one i (a hyperplane with all $y = 1$ on one side and all $y = 0$ on the other, ties on it allowed). Otherwise the data *overlap*.
(a) If the data are separated, no ML estimate exists: L increases indefinitely along $\boldsymbol{d}$ and some $|\hat{\beta}_j|$ is "infinite".
(b) If the data overlap, a finite ML estimate exists, and it is unique.

Proof. [Supplement]

Step 1. (A convenient form.) For one binary observation, $y\eta - \log(1+e^\eta) = -\log(1+e^{-s\eta})$ (check $y = 1$ and $y = 0$). Hence

$$L(\boldsymbol{\beta}) = -\sum_i \log[1 + \exp(-s_i \mathbf{x}_i^\top \boldsymbol{\beta})] < 0.$$

Each term is a strictly increasing function of $s_i \mathbf{x}_i^\top \boldsymbol{\beta}$.

Step 2. (a) Suppose $\boldsymbol{d}$ separates and $\hat{\boldsymbol{\beta}}$ were a maximizer. For $t > 0$, $s_i \mathbf{x}_i^\top (\hat{\boldsymbol{\beta}} + t\boldsymbol{d}) \geq s_i \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}$ for all i, with strict inequality for some i. By Step 1, $L(\hat{\boldsymbol{\beta}} + t\boldsymbol{d}) > L(\hat{\boldsymbol{\beta}})$, a contradiction. So the supremum

is not attained; along $\boldsymbol{\beta} + t\boldsymbol{d}$ the likelihood keeps increasing as $t \to \infty$ (under complete separation, $s_i \mathbf{x}_i^\top \boldsymbol{d} > 0$ for all i, it tends to the perfect-fit value $L = 0$).

Step 3. (b) Overlap and full rank imply: for every unit vector $\boldsymbol{d}$, $m(\boldsymbol{d}) = \min_i s_i \mathbf{x}_i^\top \boldsymbol{d} < 0$. Indeed, otherwise $s_i \mathbf{x}_i^\top \boldsymbol{d} \geq 0$ for all i; not separated forces all equalities, so $X\boldsymbol{d} = \mathbf{0}$, contradicting full rank.

Step 4. m is continuous on the compact unit sphere, so $\max_{\|\boldsymbol{d}\|=1} m(\boldsymbol{d}) = -c < 0$. For $\boldsymbol{\beta} = r\boldsymbol{d}$ with $r > 0$, $\|\boldsymbol{d}\| = 1$, choose i with $s_i \mathbf{x}_i^\top \boldsymbol{d} \leq -c$; then by Step 1

$$L(r\boldsymbol{d}) \leq -\log[1 + \exp(rc)] \leq -rc.$$

Step 5. Hence $\{\boldsymbol{\beta} : L(\boldsymbol{\beta}) \geq L(\mathbf{0})\} \subset \{\|\boldsymbol{\beta}\| \leq -L(\mathbf{0})/c\}$ is bounded; it is closed by continuity, hence compact and nonempty. The continuous L attains its maximum on it, which is a global maximum. Uniqueness is Theorem 5.26. ■

Similar results hold for the probit and some other links (Silvapulle 1981). In practice software rarely detects $\hat{\beta} = \pm\infty$: after a few cycles the log likelihood looks flat, convergence criteria are met, and enormous standard errors are reported.

Example 5.28 (Perfect discrimination)

Let $y = 0$ at $x = 10, 20, 30, 40$ and $y = 1$ at $x = 60, 70, 80, 90$. The ideal fit has $\hat{\pi} = 0$ for $x \leq 40$ and $\hat{\pi} = 1$ for $x \geq 60$. Taking $\alpha = -50\beta$ (so $\hat{\pi}(50) = 0.5$) and letting $\beta \to \infty$ gives ever larger likelihood approaching a perfect fit (Figure 8). PROC GENMOD reports $\operatorname{logit} \hat{\pi} = -192.2 + 3.8x$ with standard errors 8.0×10^8 and 1.5×10^7, numerical artifacts of $\hat{\beta} = \infty$.

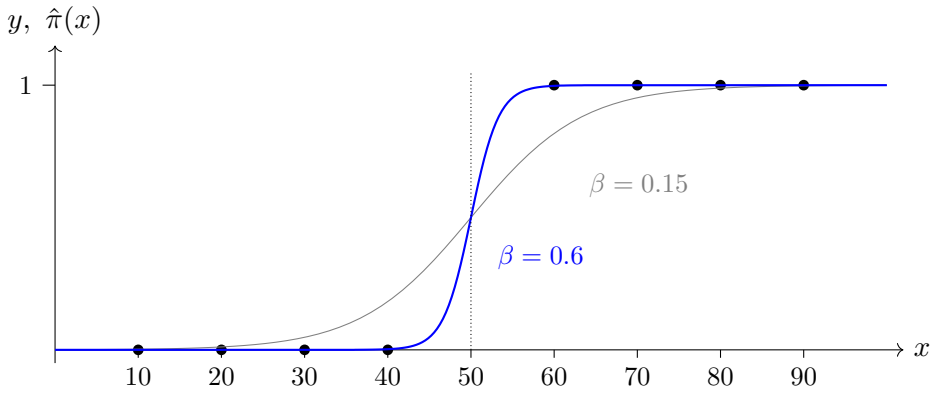


Figure 8: Complete separation: with $\alpha = -50\beta$ the fitted curve sharpens toward a step at $x = 50$ as $\beta \to \infty$, and the likelihood increases without attaining a maximum.

Chapter 6. Building and Applying Logistic Regression Models

6.1 Strategies in model selection

Motivation. With many candidate logistic models we need one complex enough to fit but simple enough to interpret, chosen by tests, AIC and subject-matter reasoning.

Confirmatory studies compare a few models fixed in advance; exploratory searches only generate hypotheses. Start by looking at each predictor's marginal effect. Rough guidelines: at least 10 outcomes of each type per predictor; and beware multicollinearity, which can make each of several correlated predictors look unimportant given the others.

6.1.1 Horseshoe crab example revisited

Example 6.1 (Horseshoe crabs: all main effects)

For 173 female crabs, $y = 1$ if she has a satellite; predictors color C (4 categories), spine S (3), weight and width W. The main-effects model (dummies c_1, c_2, c_3, s_1, s_2) has LR statistic 40.6 against the null ($df = 7$, $P < 0.0001$), yet no single effect is clearly significant:

Parameter	Estimate	SE	Parameter	Estimate	SE
Intercept	−9.273	3.838	color 2	1.506	0.567
weight	0.826	0.704	color 3	1.120	0.593
width	0.263	0.195	spine 1	−0.400	0.503
color 1	1.609	0.936	spine 2	−0.496	0.629

This signals multicollinearity (corr(weight, width) = 0.887), so we keep W and the factors C, S. Models are named by highest-order terms ($C+S+W$, $C+S*W$, ...), always including the corresponding main effects.

6.1.2 Stepwise procedures

Forward selection adds the term with the smallest P-value at each stage (*stepwise* also re-tests earlier terms); *backward elimination* starts complex and removes the term with the largest P-value until every deletion significantly worsens the fit. Add or drop a factor as a whole, never single dummies. Backward is usually preferred, since forward may stop early through one low-power test. Selected effects tend to be overestimated, and a variable central to the study may be kept even if not significant.

6.1.3 Backward elimination for the horseshoe crab example

With a continuous predictor G^2 is not approximately χ^2, but deviance differences between nested models are LR statistics with approximate χ^2 null distributions.

Example 6.2 (Horseshoe crabs: backward elimination)

Selected rows of the book's Table 6.2:

	Model	G^2	df	AIC	Comparison	Difference
1	$C{*}S{*}W$	170.44	152	212.4	–	–
2	$C{*}S + C{*}W + S{*}W$	173.68	155	209.7	(2)–(1)	3.2 ($df = 3$)
3c	$C{*}S + C{*}W$	173.69	157	205.7	(3c)–(2)	0.0 ($df = 2$)
4b	$W + C{*}S$	177.61	160	203.6	(4b)–(3c)	3.9 ($df = 3$)
5	$C + S + W$	186.61	166	200.6	(5)–(4b)	9.0 ($df = 6$)
6c	$C + W$	187.46	168	197.5	(6c)–(5)	0.8 ($df = 2$)
7a	C	212.06	169	220.1	(7a)–(6c)	24.5 ($df = 1$)
7b	W	194.45	171	198.5	(7b)–(6c)	7.0 ($df = 3$)
8	$C = \text{dark} + W$	187.96	170	194.0	(8)–(6c)	0.5 ($df = 2$)
9	None	225.76	172	227.8	(9)–(8)	37.8 ($df = 2$)

All interactions drop out, then S; removing C from $C + W$ costs 7.0 ($df = 3$, $P = 0.07$) and removing W costs 24.5. A single dark/not-dark dummy loses almost nothing: final model width + dark.

6.1.4 AIC, model selection, and the "correct" model

No model is exactly true; a parsimonious model may estimate quantities of interest better than a complex one. AIC judges a model by the expected closeness of its fit to the truth.

Definition (AIC)

For a model with p parameters and maximized log likelihood L_M, $\text{AIC} = -2(L_M - p) = -2L_M + 2p$; smaller is better.

Proposition 6.3 (AIC and the adjusted deviance)

Let models for a categorical response be fitted to the same data, with N observations in grouped form (so the saturated model has N parameters and the residual degrees of freedom are $df = N - p$). Then

$$\text{AIC} = [G^2 - 2\,df] + \text{const},$$

where the constant $-2L_S + 2N$ is the same for all models. Hence ordering models by AIC is the same as ordering them by $G^2 - 2\,df$.

Proof. $G^2 = -2(L_M - L_S)$ gives $-2L_M = G^2 - 2L_S$; with $p = N - df$, $\text{AIC} = [G^2 - 2\,df] + (2N - 2L_S)$, and the last term depends only on the data. ∎

Theorem 6.4 (Akaike's bias correction)

Let $Y_1, \ldots, Y_n$ be i.i.d. with density $f(\cdot;\theta_0)$ belonging to a regular parametric model with p-dimensional parameter, $\hat\theta$ the MLE and $\ell(\theta) = \sum_i \log f(Y_i;\theta)$. Let $Y_1^*, \ldots, Y_n^*$ be an independent copy of the data and $\ell^*(\theta)$ its log likelihood. Then, to first order,

$$E[\ell(\hat\theta)] - E[\ell^*(\hat\theta)] \approx p.$$

Thus $-2\ell(\hat\theta) + 2p = \text{AIC}$ is an approximately unbiased estimate of $-2E[\ell^*(\hat\theta)]$, the expected out-of-sample deviance (equivalently, of $2n$ times the Kullback–Leibler discrepancy of $f(\cdot;\hat\theta)$ from the truth, up to a constant).

Proof. [Supplement] Let I be the per-observation Fisher information, so $\sqrt{n}(\hat\theta-\theta_0)\xrightarrow{d}N(0,I^{-1})$.

Step 1. Expanding about $\hat\theta$ ($\ell'(\hat\theta)=0$), $\ell(\theta_0)\approx\ell(\hat\theta)-\frac{n}{2}(\hat\theta-\theta_0)^\top I(\hat\theta-\theta_0)$; since $n(\hat\theta-\theta_0)^\top I(\hat\theta-\theta_0)\xrightarrow{d}\chi^2_p$, $E[\ell(\hat\theta)]\approx E[\ell(\theta_0)]+p/2$.

Step 2. Expanding ℓ^* about θ_0, and using that $\ell^{*\prime}(\theta_0)$ has mean zero and is independent of $\hat\theta$,

$$E[\ell^*(\hat\theta)]\approx E[\ell^*(\theta_0)]-\tfrac{n}{2}E[(\hat\theta-\theta_0)^\top I(\hat\theta-\theta_0)]=E[\ell^*(\theta_0)]-\tfrac{p}{2}.$$

Step 3. $\ell(\theta_0)$ and $\ell^*(\theta_0)$ are equal in distribution, so subtracting gives $E[\ell(\hat\theta)]-E[\ell^*(\hat\theta)]\approx p$, i.e. $E[-2\ell(\hat\theta)+2p]\approx-2E[\ell^*(\hat\theta)]$. Finally $-2E[\ell^*(\theta)]/n=2\,\mathrm{KL}(f_{\theta_0}\|f_\theta)-2E\log f(Y;\theta_0)$, whose last term is model-free. ■

Example 6.5 (Horseshoe crabs: AIC)

AIC is smallest for $C+W$ (197.5) among C,S,W models, and smaller still for the dark-indicator model (194.0), though that model was suggested by the $C+W$ fit.

6.1.5 Using causal hypotheses to guide model building

A time ordering suggests a sequence of logit models, each response explained by earlier variables.

Example 6.6 (Divorce and pre/extramarital sex)

G = gender, P = premarital sex, E = extramarital sex, M = marital status:

	Women				Men			
PMS	Yes		No		Yes		No	
EMS	Yes	No	Yes	No	Yes	No	Yes	No
Divorced	17	54	36	214	28	60	17	68
Still married	4	25	4	322	11	42	4	130

For $G\to P\to E\to M$: response P on none, $G^2=75.3$ ($df=1$; G–P odds ratio 0.27); E on none / P / $G+P$, $G^2=48.9/2.9/0.0$ ($df=3/2/1$), weak evidence for a direct G effect; M on $E+P$ / $E{*}P$ / $E{*}P+G$, $G^2=18.2/5.2/0.7$ ($df=5/4/3$), so an $E\times P$ interaction is needed. The causal reasoning led close to, but not exactly to, a well-fitting model.

6.1.6 New model-building strategies for data mining

In huge data sets almost every effect is significant, so tests are of little use and predictive-power summaries (Section 6.2) matter more.

6.2 Logistic regression diagnostics

Motivation. A global fit statistic can hide where a model fails; residuals and influence measures examine the fit observation by observation.

6.2.1 Pearson, deviance, and standardized residuals

Let $y_i\sim\mathrm{bin}(n_i,\pi_i)$ at setting $i=1,\dots,N$, with fitted $\hat\pi_i$ and leverage $\hat h_i$, the ith diagonal element of $\hat W^{1/2}X(X^\top\hat WX)^{-1}X^\top\hat W^{1/2}$, $\hat W=\mathrm{diag}(n_i\hat\pi_i(1-\hat\pi_i))$.

Definition (Residuals for binomial GLMs)

(a) Pearson residual:

$$e_i = \frac{y_i - n_i\hat{\pi}_i}{\sqrt{n_i\hat{\pi}_i(1-\hat{\pi}_i)}}. \tag{6.1}$$

(b) Standardized (adjusted) Pearson residual: $r_i = e_i/\sqrt{1-\hat{h}_i}$.
(c) Deviance residual: $\sqrt{d_i}\ \mathrm{sgn}(y_i - n_i\hat{\pi}_i)$, with $d_i = 2[y_i \log \frac{y_i}{n_i\hat{\pi}_i} + (n_i - y_i)\log \frac{n_i - y_i}{n_i - n_i\hat{\pi}_i}]$.

Thus $X^2 = \sum_i e_i^2$ and $G^2 = \sum_i d_i$.

Theorem 6.7 (Approximate distribution of residuals)

Suppose the logistic model holds with p parameters and the n_i are large. Then, approximately,
(a) $\mathrm{Var}(y_i - n_i\hat{\pi}_i) \approx n_i\pi_i(1-\pi_i)(1-h_i)$, so $\mathrm{Var}(e_i) \approx 1 - h_i < 1$ while r_i is approximately $N(0,1)$;
(b) $E(\sum_i e_i^2) \approx N - p = df$, so $E(\sum_i e_i^2)/N \approx df/N < 1$.

Proof. [Supplement] Write $\mu = (n_i\pi_i)$, $W = \mathrm{diag}(n_i\pi_i(1-\pi_i)) = \mathrm{Cov}(y)$, $A = X^\top W X$.

Step 1. The logit likelihood equations are $X^\top(y - \hat{\mu}) = 0$, and $\partial\mu/\partial\beta^\top = WX$, so $\hat{\mu} - \mu \approx WX(\hat{\beta} - \beta)$ and $0 \approx X^\top(y-\mu) - A(\hat{\beta} - \beta)$, i.e. $\hat{\beta} - \beta \approx A^{-1}X^\top(y - \mu)$.

Step 2. Hence $y - \hat{\mu} \approx (I - WXA^{-1}X^\top)(y - \mu)$ and

$$\mathrm{Cov}(y - \hat{\mu}) \approx (I - WXA^{-1}X^\top)\, W\, (I - XA^{-1}X^\top W) = W - WXA^{-1}X^\top W.$$

Step 3. So $\mathrm{Cov}(W^{-1/2}(y - \hat{\mu})) \approx I - H$, $H = W^{1/2}XA^{-1}X^\top W^{1/2}$ with diagonal (h_i). The components of $W^{-1/2}(y - \hat{\mu})$ are the e_i (with π_i for $\hat{\pi}_i$, a negligible change), proving (a); normality follows from that of y, and $r_i = e_i/\sqrt{1-h_i}$ has unit variance.

Step 4. H is a projection of rank p, so $E\sum e_i^2 \approx \mathrm{tr}(I - H) = N - p$. ■

With $n_i = 1$ (continuous predictors) residuals take two values and are uninformative; group observations where possible.

6.2.2 Heart disease example

Example 6.8 (Blood pressure and heart disease)

Framingham men, eight blood-pressure categories (scores $111.5, \ldots, 191.5$), response: heart disease within six years.

Blood pressure	Sample size	Heart disease	Fitted		Std. residual	
			Indep.	Linear	Indep.	Linear
< 117	156	3	10.8	5.2	−2.62	−1.11
117–126	252	17	17.4	10.6	−0.12	2.37
127–136	284	12	19.7	15.1	−2.02	−0.95
137–146	271	16	18.8	18.1	−0.74	−0.57
147–156	139	12	9.6	11.6	0.84	0.13
157–166	85	8	5.9	8.9	0.93	−0.33
167–186	99	16	6.9	14.2	3.76	0.65
> 186	43	8	3.0	8.4	3.07	−0.18

Independence fits poorly ($G^2 = 30.0$, $df = 7$) with residuals increasing in x; the linear logit model removes the trend ($G^2 = 5.9$, $df = 6$), leaving only $r_2 = 2.37$, unsurprising among

eight.

6.2.3 Graduate admissions example

Example 6.9 (Admissions at the University of Florida)

For 23 departments, the model with no gender effect, $\operatorname{logit}\pi_{ik} = \alpha + \beta_k^D$, fits rather poorly ($G^2 = 44.7$, $df = 23$). The standardized residuals for (female, admitted) are large in astronomy (2.87; 6 of 6 females vs. 3 of 11 males admitted), geography (2.17) and psychology (-2.27); without these three the model fits well ($G^2 = 24.4$, $df = 20$). Adding a gender effect barely helps ($G^2 = 42.4$, $df = 22$), since the deviations go in different directions. Its conditional odds ratio 1.19 (favouring females) versus the marginal 0.94 is Simpson's paradox.

Proposition 6.10 (Residuals under conditional independence)

In a $2 \times 2 \times K$ table with $y_{ik} \sim \text{bin}(n_{ik}, \pi_{ik})$, the model $\operatorname{logit}\pi_{ik} = \alpha + \beta_k$ has $\hat\pi_{1k} = \hat\pi_{2k} = y_{+k}/n_{+k}$, and in each stratum the raw residuals satisfy $y_{1k} - n_{1k}\hat\pi_{1k} = -(y_{2k} - n_{2k}\hat\pi_{2k})$. Consequently the two standardized residuals in stratum k are equal in magnitude and opposite in sign, whereas the two ordinary Pearson residuals need not be.

Proof.

Step 1. The likelihood equation for β_k is $\sum_i y_{ik} = \sum_i n_{ik}\hat\pi_{ik}$, and $\hat\pi_{1k} = \hat\pi_{2k}$, so $\hat\pi_{ik} = y_{+k}/n_{+k}$.

Step 2. Then $y_{1k} - n_{1k}\hat\pi_{1k} = (n_{2k}y_{1k} - n_{1k}y_{2k})/n_{+k} = -(y_{2k} - n_{2k}\hat\pi_{2k})$.

Step 3. Negatives of each other have the same standard error, so the standardized residuals are $\pm$ the same number; the Pearson residuals divide instead by $\sqrt{n_{ik}\hat\pi_{ik}(1-\hat\pi_{ik})}$, which differs for $i = 1, 2$ when $n_{1k} \neq n_{2k}$. ■

6.2.4 Influence diagnostics for logistic regression

Deleting an observation is equivalent to adding a parameter that fits it perfectly. Leverage depends on the fit through $\hat W$, so an extreme x_i with $\hat\pi_i$ near 0 or 1 can have small leverage. Influence measures: *Dfbeta* (change in $\hat\beta_j$ on deleting i, over its SE); the *confidence interval displacement* c_i; and the drops ΔX_i^2, ΔG_i^2 in the fit statistics.

Theorem 6.11 (One-step influence approximations)

Let $A = X^\top \hat W X$, $w_i = n_i\hat\pi_i(1-\hat\pi_i)$, $h_i = w_i x_i^\top A^{-1} x_i$, and let $\hat\beta_{(i)}$ be the one-step Newton approximation (started from $\hat\beta$) to the estimate without observation i. Then

$$\hat\beta - \hat\beta_{(i)} = \frac{A^{-1}x_i\,(y_i - n_i\hat\pi_i)}{1-h_i}, \qquad c_i = (\hat\beta - \hat\beta_{(i)})^\top A(\hat\beta - \hat\beta_{(i)}) = \frac{e_i^2 h_i}{(1-h_i)^2},$$

and the score statistic for the extra parameter that fits observation i perfectly equals $r_i^2 = e_i^2/(1-h_i)$, the usual approximation to ΔX_i^2.

Proof. [Supplement]

Step 1. The full score vanishes at $\hat\beta$, so the reduced-data score is $U_{(i)}(\hat\beta) = -x_i(y_i - n_i\hat\pi_i)$, with information $A - w_i x_i x_i^\top$; one Newton step gives $\hat\beta_{(i)} = \hat\beta + (A - w_i x_i x_i^\top)^{-1} U_{(i)}(\hat\beta)$.

Step 2. By Sherman–Morrison,

$$(A - w_i x_i x_i^\top)^{-1} x_i = A^{-1}x_i + \frac{w_i A^{-1}x_i x_i^\top A^{-1}x_i}{1 - w_i x_i^\top A^{-1} x_i} = \frac{A^{-1}x_i}{1-h_i},$$

giving $\hat\beta - \hat\beta_{(i)}$.

Step 3. Then $c_i = (y_i - n_i\hat\pi_i)^2 x_i^\top A^{-1}x_i/(1-h_i)^2 = (y_i - n_i\hat\pi_i)^2 h_i/\{w_i(1-h_i)^2\} = e_i^2 h_i/(1-h_i)^2$.

Step 4. Add γ with covariate $\mathbf{e}_i$. At $(\hat\beta, 0)$, $U_\gamma = y_i - n_i\hat\pi_i$, $I_{\gamma\gamma} = w_i$, $I_{\gamma\beta} = w_i x_i^\top$, $I_{\beta\beta} = A$; the efficient information is $w_i - w_i^2 x_i^\top A^{-1}x_i = w_i(1-h_i)$, so the score statistic is $(y_i - n_i\hat\pi_i)^2/\{w_i(1-h_i)\} = r_i^2$. Fitting γ equals deleting i, so this approximates the drop in X^2 (and G^2). ■

Example 6.12 (Heart disease: influence)

For the linear logit model of Example 6.8, category 2 has by far the largest Dfbeta (-1.14), c (2.26), ΔX^2 (5.64) and ΔG^2 (5.04); all others are below 1.4.

6.2.5 Summarizing predictive power: R and R-squared measures

No R^2 analog is universally useful. Proposals:

(a) the correlation $r(y, \hat\pi)$ (crabs: 0.402 for W, 0.452 for $C+W$, 0.526 for $C*S*W$, the last inflated by its many parameters);

(b) $1 - \sum(y_i - \hat\pi_i)^2 / \sum(y_i - \bar y)^2$;

(c) likelihood-based measures.

Proposition 6.13 (Likelihood R^2 measures)

Let $L_0 \le L_M \le L_S$ be the maximized log likelihoods of the null (intercept-only), the given and the saturated models. Then $(L_M - L_0)/(L_S - L_0) \in [0,1]$, and:
(a) for n ungrouped Bernoulli observations, $L_S = 0$ and the measure reduces to McFadden's $D = (L_0 - L_M)/L_0$, where $L_0 = n[\bar y \log \bar y + (1-\bar y)\log(1-\bar y)]$;
(b) for N grouped binomial counts it equals $D^* = \{G^2(0) - G^2(M)\}/G^2(0)$.

Proof. The models are nested, so $L_0 \le L_M \le L_S \le 0$, giving the range.

(a) The saturated fit $\hat\pi_i = y_i \in \{0,1\}$ makes every term $y_i \log\hat\pi_i + (1-y_i)\log(1-\hat\pi_i)$ zero, so $L_S = 0$; the null fit $\hat\pi_i = \bar y$ gives L_0.

(b) $G^2(M) = -2(L_M - L_S)$ and $G^2(0) = -2(L_0 - L_S)$; take the ratio. ■

D^* is misleading: it equals 1 for any perfectly fitting model and tends to 1 as $n \to \infty$ whenever the model holds, confounding fit with predictive power; use D on ungrouped data.

6.2.6 Summarizing predictive power: classification tables and ROC curves

A *classification table* cross-classifies y with $\hat y = \mathbb{1}\{\hat\pi_i > \pi_0\}$, summarized by sensitivity and specificity; it discards information and depends on the arbitrary π_0.

Definition (ROC curve and concordance index)

The *ROC curve* plots sensitivity against 1−specificity as π_0 ranges over $[0,1]$. With n_1, n_0 the numbers of 1's and 0's, the *concordance index*

$$c = \frac{1}{n_1 n_0} \sum_{i:\, y_i=1} \sum_{j:\, y_j=0} \Big[\mathbb{1}\{\hat\pi_i > \hat\pi_j\} + \tfrac{1}{2}\mathbb{1}\{\hat\pi_i = \hat\pi_j\} \Big]$$

is the proportion of (case, non-case) pairs in which the case has the larger fitted probability.

Proposition 6.14 (Area under the ROC curve)

The area under the empirical ROC curve (with linear interpolation across ties) equals the concordance index c.

Proof. [Supplement]

Step 1. Lower π_0 from 1 to 0. Passing a distinct fitted value v moves the curve from (a,b) by $(m_0(v)/n_0, m_1(v)/n_1)$, where $m_y(v)$ counts observations with response y and $\hat\pi = v$, and $b = \#\{i : y_i = 1, \hat\pi_i > v\}/n_1$.

Step 2. The trapezoid added has area

$$\frac{m_0(v)}{n_0}\Big(b + \frac{m_1(v)}{2n_1}\Big) = \frac{1}{n_0 n_1} \sum_{j:\, y_j=0,\ \hat\pi_j = v} \Big[\#\{i : y_i = 1, \hat\pi_i > v\} + \tfrac{1}{2}\#\{i : y_i = 1, \hat\pi_i = v\} \Big].$$

Step 3. Summing over v runs over every non-case j once, giving c. ∎

$c = 0.5$ is random guessing. Crabs: $c = 0.639$ (color), 0.742 (width), 0.771 (width and color).

6.3 Inference about conditional associations in $2 \times 2 \times K$ tables

Motivation. Comparing two treatments on a binary outcome across strata is a core application; logit models, the CMH test and the Mantel–Haenszel estimator are closely linked tools.

Example 6.15 (Eight-center clinical trial)

Drug vs. control cream for curing an infection (μ_{11k}, $\mathrm{Var}(n_{11k})$: null hypergeometric moments of drug successes):

	Drug		Control				
Center	Succ.	Fail.	Succ.	Fail.	Odds ratio	μ_{11k}	$\mathrm{Var}(n_{11k})$
1	11	25	10	27	1.19	10.36	3.79
2	16	4	22	10	1.82	14.62	2.47
3	14	5	7	12	4.80	10.50	2.41
4	2	14	1	16	2.29	1.45	0.70
5	6	11	0	12	∞	3.52	1.20
6	1	10	0	10	∞	0.52	0.25
7	1	4	1	8	2.0	0.71	0.42
8	4	2	6	1	0.33	4.62	0.62

6.3.1 Using logit models to test conditional independence

For binary Y, binary X ($x_1 = 1$, $x_2 = 0$) and control Z, let $\pi_{ik} = P(Y = 1 \mid X = i, Z = k)$ and

$$\operatorname{logit} \pi_{ik} = \alpha + \beta x_i + \beta_k^Z, \qquad i = 1, 2,\ k = 1, \ldots, K, \tag{6.2}$$

with common conditional odds ratio e^β. Conditional independence $H_0 : \beta = 0$ is tested by Wald or by LR against

$$\operatorname{logit} \pi_{ik} = \alpha + \beta_k^Z, \tag{6.3}$$

both $df = 1$. The goodness-of-fit test of (6.3) ($df = K$) is less powerful when the association is similar across strata.

6.3.2 Cochran–Mantel–Haenszel test of conditional independence

Given both margins of each partial table, n_{11k} determines the table.

Proposition 6.16 (Null hypergeometric moments)

Under conditional independence, given the margins of partial table k, n_{11k} is hypergeometric with

$$\mu_{11k} = E(n_{11k}) = \frac{n_{1+k} n_{+1k}}{n_{++k}}, \qquad \operatorname{Var}(n_{11k}) = \frac{n_{1+k} n_{2+k} n_{+1k} n_{+2k}}{n_{++k}^2 (n_{++k} - 1)}.$$

Proof. [Supplement] Drop k, let $n = n_{++}$, $q = n_{+1}/n$. Under H_0 the n_{+1} successes are a random subset of the n subjects. With I_s the success indicator of row-1 subject s, $n_{11} = \sum_s I_s$, $P(I_s = 1) = q$ and $P(I_s = I_t = 1) = n_{+1}(n_{+1} - 1)/\{n(n-1)\}$ for $s \neq t$.

Step 1. $E(n_{11}) = n_{1+} q = n_{1+} n_{+1}/n$.

Step 2. $\operatorname{Var}(I_s) = q(1-q)$ and $\operatorname{Cov}(I_s, I_t) = -q(1-q)/(n-1)$, so

$$\operatorname{Var}(n_{11}) = n_{1+} q(1-q)\Big[1 - \frac{n_{1+} - 1}{n - 1}\Big] = n_{1+} q(1-q) \frac{n - n_{1+}}{n - 1} = \frac{n_{1+} n_{2+} n_{+1} n_{+2}}{n^2 (n-1)}.$$

■

Definition (CMH statistic)

$$\mathrm{CMH} = \frac{\big[\sum_k (n_{11k} - \mu_{11k})\big]^2}{\sum_k \operatorname{Var}(n_{11k})}, \tag{6.4}$$

approximately χ^2_1 under H_0. Cochran's version uses the variance $n_{1+k} n_{2+k} n_{+1k} n_{+2k}/n_{++k}^3$.

Deviations accumulate when the association has the same direction in every stratum. The hypergeometric formulation also covers retrospective studies and randomized trials.

Theorem 6.17 (CMH as a score test for the logit model)

For model (6.2) with independent binomial rows:
(a) the (unconditional) score statistic for $H_0 : \beta = 0$ is Cochran's statistic;
(b) the score statistic for β based on the conditional likelihood given all row and column margins is exactly CMH. In particular CMH and Cochran differ only by the factors $(n_{++k} - 1)/n_{++k}$ in the variance and are asymptotically equivalent to the Wald and LR tests of $\beta = 0$.

Proof. [Supplement] Absorb α: $\operatorname{logit} \pi_{ik} = \beta x_i + \gamma_k$.

Step 1. (Score.) From $L = \sum_{ik}[y_{ik}(\beta x_i + \gamma_k) - n_{ik}\log(1+e^{\beta x_i+\gamma_k})]$, $\partial L/\partial\beta = \sum_k (y_{1k} - n_{1k}\pi_{1k})$ and $\partial L/\partial\gamma_k = \sum_i (y_{ik} - n_{ik}\pi_{ik})$. Under H_0, $\hat\pi_{1k} = \hat\pi_{2k} = n_{+1k}/n_{++k}$ ($y_{ik} = n_{i1k}$), so $U = \sum_k (n_{11k} - \mu_{11k})$.

Step 2. (Information.) With $v_k = \pi_k(1-\pi_k)$: $I_{\beta\beta} = \sum_k n_{1+k}v_k$, $I_{\beta\gamma_k} = n_{1+k}v_k$, $I_{\gamma_k\gamma_k} = n_{++k}v_k$, $I_{\gamma_k\gamma_l} = 0$ ($k \neq l$). The efficient information is

$$I_{\beta\beta} - \sum_k \frac{I^2_{\beta\gamma_k}}{I_{\gamma_k\gamma_k}} = \sum_k v_k \frac{n_{1+k}n_{2+k}}{n_{++k}};$$

with $\hat v_k = n_{+1k}n_{+2k}/n^2_{++k}$, U^2 over this is Cochran's statistic: (a).

Step 3. (Conditional score.) Given all margins,

$$P(n_{11k} = t_k,\ k = 1,\dots,K) = \prod_k \frac{c_k(t_k)e^{\beta t_k}}{\sum_u c_k(u)e^{\beta u}},$$

a one-parameter exponential family with statistic $T = \sum_k n_{11k}$, score $T - E_\beta(T)$ and information $\mathrm{Var}_\beta(T)$. At $\beta = 0$ these are $\sum_k (n_{11k} - \mu_{11k})$ and $\sum_k \mathrm{Var}(n_{11k})$ (Proposition 6.16), so the statistic is CMH: (b). The last claim is standard score/Wald/LR asymptotics. ■

6.3.3 Multicenter clinical trial example

Example 6.18 (Clinical trial: tests)

All centers but the last show a positive association: CMH = 6.38 ($P = 0.012$); model (6.2) gives $\hat\beta = 0.777$ (SE 0.307), Wald 6.42, LR 6.67.

6.3.4 Estimation of common odds ratio

Under (6.2) the ML estimate of $\theta = e^\beta$ is $e^{\hat\beta}$; non-model alternatives are Woolf's estimator and the Mantel–Haenszel estimator.

Definition (Mantel–Haenszel estimator)

$$\hat\theta_{MH} = \frac{\sum_k n_{11k}n_{22k}/n_{++k}}{\sum_k n_{12k}n_{21k}/n_{++k}} = \frac{\sum_k R_k}{\sum_k S_k} = \frac{R}{S}. \tag{6.5}$$

Theorem 6.19 (Unbiased estimating equation for $\hat\theta_{MH}$)

If the rows of each partial table are independent binomials and the odds ratio equals θ in every stratum, then $E(R - \theta S) = 0$. Consequently $\hat\theta_{MH}$ is consistent both as $n \to \infty$ with K fixed and as $K \to \infty$ with bounded strata sizes.

Proof. [Supplement]

Step 1. With independent $n_{11k} \sim \mathrm{bin}(n_{1+k}, \pi_{1k})$, $n_{21k} \sim \mathrm{bin}(n_{2+k}, \pi_{2k})$: $E(n_{11k}n_{22k}) = n_{1+k}n_{2+k}\pi_{1k}(1-\pi_{2k})$ and $E(n_{12k}n_{21k}) = n_{1+k}n_{2+k}(1-\pi_{1k})\pi_{2k}$.

Step 2. Since $\theta = \pi_{1k}(1-\pi_{2k})/\{(1-\pi_{1k})\pi_{2k}\}$, $E(R_k - \theta S_k) = 0$; sum over k.

Step 3. Under either scheme, with $m = n$ or $m = K$, R/m and S/m are averages of independent terms with bounded variances, so $(R - \theta S)/m \xrightarrow{p} 0$ while S/m stays bounded away from 0 (nondegenerate strata). Hence $\hat\theta_{MH} - \theta = (R - \theta S)/S \xrightarrow{p} 0$. ■

Theorem 6.20 (Unconditional ML breaks down for matched pairs)

In model (6.2) with one matched pair per stratum, let b (c) be the number of pairs in which only the first (only the second) member has $y = 1$. Then the unconditional ML estimator is $\hat{\beta} = 2\log(b/c)$, whereas $\hat{\theta}_{MH} = b/c$. As $K \to \infty$, $\log(b/c) \xrightarrow{p} \beta$, so $\hat{\beta} \xrightarrow{p} 2\beta$.

Proof. [Supplement]

Step 1. A concordant pair's likelihood is maximized at $\gamma_k = \pm\infty$ with value 1: no information on β.

Step 2. With $u = e^{\gamma_k}$, $\psi = e^{\beta}$, the two discordant types have likelihoods $\frac{u\psi}{(1+u\psi)(1+u)}$ and $\frac{u}{(1+u\psi)(1+u)}$, both maximized at $u = \psi^{-1/2}$ (log-derivative $1/u - \psi/(1+u\psi) - 1/(1+u) = 0 \iff u^2\psi = 1$), with values $\psi/(1+\psi^{1/2})^2$ and $1/(1+\psi^{1/2})^2$.

Step 3. The profile log likelihood $b\beta - 2(b+c)\log(1+e^{\beta/2})$ has derivative $b-(b+c)e^{\beta/2}/(1+e^{\beta/2})$, zero at $e^{\beta/2} = b/c$: $\hat{\beta} = 2\log(b/c)$.

Step 4. $R_k = \frac{1}{2}$ in a b-type pair, $S_k = \frac{1}{2}$ in a c-type pair, both 0 otherwise, so $\hat{\theta}_{MH} = b/c$.

Step 5. Given discordance, a pair is b-type with probability $\pi_1(1-\pi_2)/\{\pi_1(1-\pi_2)+(1-\pi_1)\pi_2\} = \psi/(1+\psi)$, so $b/c \xrightarrow{p} \psi$. ■

The Robins–Breslow–Greenland variance estimate, valid under both asymptotic schemes, is

$$\hat{\sigma}^2(\log\hat{\theta}_{MH}) = \frac{\sum_k (n_{11k}+n_{22k})R_k/n_{++k}}{2R^2} + \frac{\sum_k (n_{12k}+n_{21k})S_k/n_{++k}}{2S^2} + \frac{\sum_k \left[(n_{11k}+n_{22k})S_k + (n_{12k}+n_{21k})R_k\right]/n_{++k}}{2RS}.$$

Example 6.21 (Clinical trial: common odds ratio)

For Example 6.15, $\hat{\theta}_{MH} = 2.13$, $\log\hat{\theta}_{MH} = 0.758$ ($\hat{\sigma} = 0.303$), 95% CI $(1.18, 3.87)$; the Wald and LR intervals for e^{β} are $(1.19, 3.97)$ and $(1.20, 4.02)$.

Neither $\hat{\theta}_{MH}$ nor CMH requires exactly equal odds ratios; both work well when the effects mostly share a direction.

6.3.5 Testing homogeneity of odds ratios

Homogeneity $\theta_{XY(1)} = \cdots = \theta_{XY(K)}$ is model (6.2), tested by its G^2 or X^2 with $df = K - 1$.

Proposition 6.22 (Fitted counts under a common odds ratio)

Fix the margins of stratum k and a value $\theta > 0$. There is a unique m_k with $\max(0, n_{1+k} + n_{+1k} - n_{++k}) < m_k < \min(n_{1+k}, n_{+1k})$ such that the table with $(1,1)$ entry m_k and the given margins has odds ratio θ:

$$\frac{m_k\,(n_{2+k} - n_{+1k} + m_k)}{(n_{1+k} - m_k)(n_{+1k} - m_k)} = \theta.$$

Proof. [Supplement] The other cells $n_{1+k} - m$, $n_{+1k} - m$, $n_{2+k} - n_{+1k} + m$ are all positive exactly on the stated open interval. There the left side is continuous, strictly increasing in m, tends to 0 at the lower end and to $+\infty$ at the upper end; by the intermediate value theorem there is exactly one root (of a quadratic in m). ■

Definition (Breslow–Day statistic)

With m_k from Proposition 6.22 at $\theta = \hat{\theta}_{MH}$ and $v_k \approx [1/m_k + 1/(n_{1+k} - m_k) + 1/(n_{+1k} - m_k) + 1/(n_{2+k} - n_{+1k} + m_k)]^{-1}$, BD $= \sum_k (n_{11k} - m_k)^2/v_k$ is approximately χ^2_{K-1} under homogeneity.

Example 6.23 (Clinical trial: homogeneity)

$G^2 = 9.7$, $X^2 = 8.0$ ($df = 7$): no evidence against a common odds ratio.

6.3.6 Summarizing heterogeneity in odds ratios

Exactly equal odds ratios are unrealistic; a *random-effects* logit model treats the stratum log odds ratios as a sample from $N(\mu, \sigma^2)$ (Chapter 12).

Chapter 7. Logit Models for Multinomial Responses

7.1 Nominal responses: baseline-category logit models

Motivation. Many responses have more than two unordered categories (food type, brand, party). Pairing each category with a baseline gives $J-1$ simultaneous logit equations that together describe all pairwise odds, extending binary logistic regression.

Let Y have J categories and $\pi_j(\mathbf{x}) = P(Y = j \mid \mathbf{x})$, $\sum_j \pi_j(\mathbf{x}) = 1$. At a fixed setting $\mathbf{x}$, the counts in the J categories are multinomial with probabilities $\{\pi_1(\mathbf{x}), \ldots, \pi_J(\mathbf{x})\}$. Of the $\binom{J}{2}$ pairwise logits, any suitable $J-1$ determine the rest.

7.1.1 Baseline-category logits

Definition (Baseline-category logit model)

With category J as baseline,

$$\log \frac{\pi_j(\mathbf{x})}{\pi_J(\mathbf{x})} = \alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x}, \qquad j = 1, \ldots, J-1. \tag{7.1}$$

The effects $\boldsymbol{\beta}_j$ depend on which category is paired with the baseline.

Proposition 7.1 (All pairwise logits)

Under (7.1), for any categories a, b,

$$\log \frac{\pi_a(\mathbf{x})}{\pi_b(\mathbf{x})} = (\alpha_a - \alpha_b) + (\boldsymbol{\beta}_a - \boldsymbol{\beta}_b)^\top \mathbf{x},$$

with the convention $\alpha_J = 0$, $\boldsymbol{\beta}_J = \mathbf{0}$. Hence the choice of baseline is arbitrary: changing it reparametrizes but does not change the model or its fit.

Proof. $\log(\pi_a/\pi_b) = \log(\pi_a/\pi_J) - \log(\pi_b/\pi_J)$; substitute (7.1) (the convention makes the formula valid when a or b equals J). The set of probability vectors described is the same for every baseline, since each parametrization is a linear reparametrization of the others. ■

With categorical predictors and nonsparse data, X^2 and G^2 check fit; with continuous predictors or sparse data they remain valid for comparing nested models differing by few terms.

7.1.2 Alligator food choice example

Example 7.2 (Alligator food choice)

For 219 alligators from four Florida lakes, the response is the primary food type by volume (fish, invertebrate, reptile, bird, other); predictors are lake L, gender G and size S (≤ 2.3 m vs. > 2.3 m). The data are sparse (219 observations in 80 cells), so G^2 is more reliable for comparing models than for testing fit. Comparisons give $G^2[(G) \mid (\cdot)] = 2.1$ and $G^2[(L+S) \mid (G+L+S)] = 2.2$, each on $df = 4$, so we collapse over gender. For the collapsed table, $(L+S)$ has $G^2 = 17.1$, $X^2 = 15.0$ ($df = 12$), whereas (L) has 38.2 and (S) 66.2; both L and S matter. Observed counts (fitted values for $L+S$ in parentheses):

Lake	Size	Fish	Invert.	Reptile	Bird	Other
Hancock	≤ 2.3	23 (20.9)	4 (3.6)	2 (1.9)	2 (2.7)	8 (9.9)
	> 2.3	7 (9.1)	0 (0.4)	1 (1.1)	3 (2.3)	5 (3.1)
Oklawaha	≤ 2.3	5 (5.2)	11 (12.0)	1 (1.5)	0 (0.2)	3 (1.1)
	> 2.3	13 (12.8)	8 (7.0)	6 (5.5)	1 (0.8)	0 (1.9)
Trafford	≤ 2.3	5 (4.4)	11 (12.4)	2 (2.1)	1 (0.9)	5 (4.2)
	> 2.3	8 (8.6)	7 (5.6)	6 (5.9)	3 (3.1)	5 (5.8)
George	≤ 2.3	16 (18.5)	19 (16.9)	1 (0.5)	2 (1.2)	3 (3.8)
	> 2.3	17 (14.5)	1 (3.1)	0 (0.5)	1 (1.8)	3 (2.2)

Only two of the 40 standardized residuals exceed 2 in absolute value. With fish as baseline, dummies $s = 1$ for small size and z_H, z_O, z_T for Hancock, Oklawaha, Trafford (George is the reference), the ML estimates (SE) are:

Logit	Intercept	Size ≤ 2.3	Hancock	Oklawaha	Trafford
$\log(\pi_I/\pi_F)$	-1.55	1.46 (0.40)	-1.66 (0.61)	0.94 (0.47)	1.12 (0.49)
$\log(\pi_R/\pi_F)$	-3.31	-0.35 (0.58)	1.24 (1.19)	2.46 (1.12)	2.94 (1.12)
$\log(\pi_B/\pi_F)$	-2.09	-0.63 (0.64)	0.70 (0.78)	-0.65 (1.20)	1.09 (0.84)
$\log(\pi_O/\pi_F)$	-1.90	0.33 (0.45)	0.83 (0.56)	0.01 (0.78)	1.52 (0.62)

Given lake, small alligators have estimated odds of invertebrates rather than fish $e^{1.46} = 4.3$ times those of large alligators (Wald 95% CI $e^{1.46 \pm 1.96(0.396)} = (2.0, 9.3)$). Those odds are relatively higher in Trafford and Oklawaha and lower in Hancock than in George. Other pairs follow by subtraction (Proposition 7.1), e.g.

$$\begin{aligned}\log(\hat{\pi}_I/\hat{\pi}_O) &= (-1.55 + 1.46s - 1.66z_H + 0.94z_O + 1.12z_T) \\ &\quad - (-1.90 + 0.33s + 0.83z_H + 0.01z_O + 1.52z_T) \\ &= 0.35 + 1.13s - 2.48z_H + 0.93z_O - 0.39z_T.\end{aligned}$$

7.1.3 Estimating response probabilities

Proposition 7.3 (Response probabilities)

Model (7.1) is equivalent to

$$\pi_j(\mathbf{x}) = \frac{\exp(\alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x})}{1 + \sum_{h=1}^{J-1} \exp(\alpha_h + \boldsymbol{\beta}_h^\top \mathbf{x})}, \qquad j = 1, \ldots, J, \tag{7.2}$$

with $\alpha_J = 0$, $\boldsymbol{\beta}_J = \mathbf{0}$. For $J = 2$ this is binary logistic regression.

Proof. From (7.1), $\pi_j = \pi_J \exp(\eta_j)$ with $\eta_j = \alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x}$, valid also for $j = J$ with $\eta_J = 0$. Summing over j, $1 = \pi_J \sum_{h=1}^{J} e^{\eta_h} = \pi_J(1 + \sum_{h<J} e^{\eta_h})$. Solve for π_J and substitute. Conversely (7.2) gives $\pi_j/\pi_J = e^{\eta_j}$. ■

The parameters of the baseline category are set to zero for identifiability: adding a common vector to all $(\alpha_j, \boldsymbol{\beta}_j)$, $j = 1, \ldots, J$, leaves (7.2) unchanged.

Example 7.4 (Alligators: fitted probabilities)

For a large alligator in Lake Hancock ($s = 0$, $z_H = 1$),

$$\hat{\pi}_I = \frac{e^{-1.55-1.66}}{1 + e^{-1.55-1.66} + e^{-3.31+1.24} + e^{-2.09+0.70} + e^{-1.90+0.83}} = 0.023,$$

and similarly $\hat{\pi}_R = 0.072$, $\hat{\pi}_B = 0.141$, $\hat{\pi}_O = 0.194$, $\hat{\pi}_F = 0.570$. With a quantitative predictor (actual length), plotting $\hat{\pi}_j(x)$ against x is informative; with $J > 2$ a single $\hat{\pi}_j(x)$ need not be monotone in x.

7.2 Ordinal responses: cumulative logit models

Motivation. When the categories are ordered, a model that uses the ordering is more parsimonious and has more power (Section 6.4). Cumulative logits with a common effect across cutpoints give the most popular such model, motivated by an underlying continuous latent response.

7.2.1 Cumulative logits

Definition (Cumulative logits)

With $P(Y \le j \mid \mathbf{x}) = \pi_1(\mathbf{x}) + \cdots + \pi_j(\mathbf{x})$,

$$\operatorname{logit} P(Y \le j \mid \mathbf{x}) = \log \frac{P(Y \le j \mid \mathbf{x})}{1 - P(Y \le j \mid \mathbf{x})} = \log \frac{\pi_1(\mathbf{x}) + \cdots + \pi_j(\mathbf{x})}{\pi_{j+1}(\mathbf{x}) + \cdots + \pi_J(\mathbf{x})}, \quad j = 1, \ldots, J-1.$$

Each cumulative logit uses all J categories; a model for one of them alone is a binary logit model for the collapsing $\{1, \ldots, j\}$ vs. $\{j+1, \ldots, J\}$.

7.2.2 Proportional odds model

Definition (Proportional odds model)

$$\operatorname{logit} P(Y \le j \mid \mathbf{x}) = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}, \qquad j = 1, \ldots, J-1, \tag{7.3}$$

with intercepts $\alpha_1 < \alpha_2 < \cdots < \alpha_{J-1}$ (since $P(Y \le j \mid \mathbf{x})$ increases in j) and a single effect vector $\boldsymbol{\beta}$ common to all cutpoints.

Proposition 7.5 (Parallel curves and cumulative odds ratios)

Under (7.3):
(a) for a single predictor and $\beta \neq 0$, the curve of $P(Y \le k \mid x)$ is the curve of $P(Y \le j \mid x)$ shifted horizontally: $P(Y \le k \mid X = x) = P(Y \le j \mid X = x + (\alpha_k - \alpha_j)/\beta)$;
(b) for any $\mathbf{x}_1, \mathbf{x}_2$ and every j, the cumulative odds ratio is

$$\frac{P(Y \le j \mid \mathbf{x}_1)/P(Y > j \mid \mathbf{x}_1)}{P(Y \le j \mid \mathbf{x}_2)/P(Y > j \mid \mathbf{x}_2)} = \exp[\boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)],$$

the same for all j and proportional (on the log scale) to the distance between $\mathbf{x}_1$ and $\mathbf{x}_2$;
(c) the conditional distributions are stochastically ordered: for each pair $\mathbf{x}_1, \mathbf{x}_2$, either $P(Y \le j \mid \mathbf{x}_1) \le P(Y \le j \mid \mathbf{x}_2)$ for all j, or $\ge$ for all j.

Proof. Let $F(z) = e^z/(1+e^z)$, so $P(Y \le j \mid \mathbf{x}) = F(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x})$.

(a) $F(\alpha_j + \beta[x + (\alpha_k - \alpha_j)/\beta]) = F(\alpha_k + \beta x)$.

(b) The log of the cumulative odds ratio is the difference of logits, $(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_1) - (\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_2) = \boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)$.

(c) F is increasing, so $P(Y \le j \mid \mathbf{x}_1) \le P(Y \le j \mid \mathbf{x}_2)$ iff $\boldsymbol{\beta}^\top \mathbf{x}_1 \le \boldsymbol{\beta}^\top \mathbf{x}_2$, a condition not involving j. ■

With one predictor, e^β is the cumulative odds ratio for $x_1 - x_2 = 1$, the same for every collapsing of Y into $(\le j, > j)$: hence McCullagh's name *proportional odds*. The response curves for $j = 1, 2, 3$ ($J = 4$) have exactly the same shape (Figure 9).

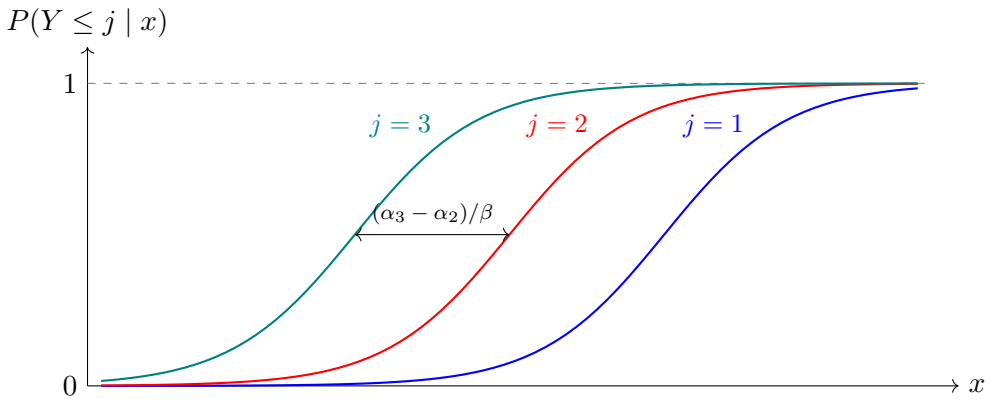


Figure 9: Proportional odds model with $J = 4$ and $\beta > 0$: the three cumulative probability curves are logistic curves with a common slope, horizontally displaced from one another.

Because the $J - 1$ curves are constrained to share a shape, the fit differs from separate binary fits. With y_{ij} indicators of the response of subject i, the likelihood is

$$\prod_{i=1}^{n} \prod_{j=1}^{J} \left[\frac{\exp(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_i)}{1 + \exp(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_i)} - \frac{\exp(\alpha_{j-1} + \boldsymbol{\beta}^\top \mathbf{x}_i)}{1 + \exp(\alpha_{j-1} + \boldsymbol{\beta}^\top \mathbf{x}_i)} \right]^{y_{ij}} \tag{7.4}$$

with $\alpha_0 = -\infty$, $\alpha_J = +\infty$; it is maximized by Fisher scoring.

7.2.3 Mental impairment example

Example 7.6 (Mental impairment)

A random sample of 40 adults in Alachua County, Florida. Y = mental impairment (1 well, 2 mild symptom formation, 3 moderate symptom formation, 4 impaired); x_1 = life events index (number and severity of important events in the past three years); x_2 = SES (1 high, 0 low). Listed as (x_2, x_1) within each response category:

Well	(1,1) (1,9) (1,4) (1,3) (0,2) (1,0) (0,1) (1,3) (1,3) (1,7) (0,1) (0,2)
Mild	(1,5) (0,6) (1,3) (0,1) (1,8) (1,2) (0,5) (1,5) (1,9) (0,3) (1,3) (1,1)
Moderate	(0,0) (1,4) (0,3) (0,9) (1,6) (0,4) (0,3)
Impaired	(1,8) (1,2) (1,7) (0,5) (0,4) (0,4) (1,8) (0,8) (0,9)

The fit of logit $P(Y \le j \mid \mathbf{x}) = \alpha_j + \beta_1 x_1 + \beta_2 x_2$:

Parameter	Estimate	SE	LR 95% CI
α_1	-0.282	0.642	$(-1.56,\ 0.98)$
α_2	1.213	0.661	$(-0.05,\ 2.57)$
α_3	2.209	0.721	$(0.86,\ 3.71)$
life (β_1)	-0.319	0.121	$(-0.57,\ -0.09)$
SES (β_2)	1.111	0.611	$(-0.06,\ 2.35)$

At the mean life events $\bar{x}_1 = 4.275$ and low SES,

$$\hat{P}(Y=1) = \frac{\exp[-0.282 - 0.319(4.275)]}{1+\exp[-0.282 - 0.319(4.275)]} = 0.16.$$

The cumulative probability from the "well" end decreases with life events and is higher at high SES: given life events, the odds of impairment below any fixed level at high SES are $e^{1.111} = 3.0$ times those at low SES. In probability terms, at $\bar{x}_1$, $\hat{P}(Y=1)$ is 0.37 (high SES) vs. 0.16 (low SES); between the life-events quartiles 2.0 and 6.5 it falls from 0.55 to 0.22 (high SES) and from 0.28 to 0.09 (low SES). The score test of proportional odds (separate β's for each logit, 4 extra parameters) gives 2.33, $df = 4$, $P = 0.68$.

7.2.4 More complex models

Interactions, factors and so on enter as in binary logistic regression; only the intercept is replaced by $J-1$ cutpoint intercepts. For the mental impairment data, adding the interaction gives

$$\text{logit}\,\hat{P}(Y \le j \mid \mathbf{x}) = \hat{\alpha}_j - 0.420x_1 + 0.371x_2 + 0.181x_1x_2,$$

interaction SE 0.238: the life-events effect is -0.420 at low SES and -0.239 at high SES, not significantly different.

When proportional odds fails

Replacing $\boldsymbol{\beta}$ by $\boldsymbol{\beta}_j$ gives nonparallel curves, which necessarily cross for some $\mathbf{x}$, violating the ordering of cumulative probabilities. A small P-value in the proportional odds test may reflect a large n more than important differences among the $\hat{\boldsymbol{\beta}}_j$, and the simpler model may have smaller MSE even if biased. If the fit is poor in practical terms, options include:
(1) a link with a nonsymmetric response curve (complementary log-log);
(2) extra terms such as interactions;
(3) dispersion parameters;
(4) separate effects for some predictors only (partial proportional odds);
(5) a baseline-category logit model, using the ordering informally in interpretation.

7.3 Ordinal responses: cumulative link models

Motivation. The latent-variable argument works for any error distribution, so the logit can be replaced by other inverse cdfs. Probit and complementary log-log links give cumulative models with their own interpretations, and a scale component handles dispersion differences.

Definition (Cumulative link model)

For a continuous cdf G,

$$G^{-1}[P(Y \le j \mid \mathbf{x})] = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}, \qquad j = 1, \ldots, J-1. \tag{7.5}$$

Model (7.5) arises from crude measurement of a latent Y^* following a location family $G(y^* - \boldsymbol{\beta}^\top \mathbf{x})$; the α_j are cutpoints on a standardized latent scale (using $-\boldsymbol{\beta}$ instead of $+\boldsymbol{\beta}$ only flips signs).

7.3.1 Types of cumulative links

(a) *Cumulative probit* ($G = \Phi$): appropriate when Y^* is normal. In $\Phi^{-1}[P(Y \le j)] = \alpha_j - \boldsymbol{\beta}^\top \mathbf{x}$, a unit increase in x_k raises $E(Y^*)$ by β_k standard deviations of ε. Logit and probit fits are similar; logit parameters are easier to interpret.

(b) *Complementary log-log* ($G(z) = 1 - \exp(-e^z)$, from an extreme-value latent distribution): $\log\{-\log[1 - P(Y \le j \mid \mathbf{x})]\} = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}$. It is the grouped-time version of the proportional hazards model and is sometimes called a *proportional hazards* model. $P(Y \le j)$ approaches 1 faster than it approaches 0. The log-log link suits the case where cloglog holds for the reversed categories (reversing the order of the categories turns G into $1 - G(-z)$).

Proposition 7.7 (Power interpretation of the cumulative cloglog model)

Under the cumulative complementary log-log model, for all j,

$$P(Y > j \mid \mathbf{x}_1) = P(Y > j \mid \mathbf{x}_2)^{\exp[\boldsymbol{\beta}^\top (\mathbf{x}_1 - \mathbf{x}_2)]}.$$

Proof. $P(Y > j \mid \mathbf{x}) = 1 - G(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}) = \exp\left[-e^{\alpha_j} e^{\boldsymbol{\beta}^\top \mathbf{x}}\right]$. Hence

$$\log P(Y > j \mid \mathbf{x}_1) = -e^{\alpha_j} e^{\boldsymbol{\beta}^\top \mathbf{x}_2} e^{\boldsymbol{\beta}^\top (\mathbf{x}_1 - \mathbf{x}_2)} = e^{\boldsymbol{\beta}^\top (\mathbf{x}_1 - \mathbf{x}_2)} \log P(Y > j \mid \mathbf{x}_2);$$

exponentiate. ■

7.3.2 Estimation for cumulative link models

Cumulative link models are multivariate GLMs. McCullagh's Fisher scoring algorithm uses the likelihood written as in (7.4) with G in place of the logistic cdf; for large n the maximum is unique, and the log likelihood is concave for the logit, probit and complementary log-log links, so iterations converge rapidly.

7.3.3 Life table example

Example 7.8 (U.S. life lengths, 1981)

Percent distribution of life length by gender and race (fitted cloglog values in parentheses):

	Males		Females	
Life length	White	Black	White	Black
0–20	2.4 (2.4)	3.6 (4.4)	1.6 (1.2)	2.7 (2.3)
20–40	3.4 (3.5)	7.5 (6.4)	1.4 (1.9)	2.9 (3.4)
40–50	3.8 (4.4)	8.3 (7.7)	2.2 (2.4)	4.4 (4.3)
50–65	17.5 (16.7)	25.0 (26.1)	9.9 (9.6)	16.3 (16.3)
Over 65	72.9 (73.0)	55.6 (55.4)	84.9 (84.9)	73.7 (73.7)

The life-length cdf rises slowly at young ages and sharply at old ages, suggesting the cloglog link (also implied by an exponentially increasing hazard, the Gompertz extreme-value law). For G (1 female) and R (1 black),

$$\log\{-\log[1 - P(Y \le j \mid G = g, R = r)]\} = \alpha_j + \beta_1 g + \beta_2 r, \qquad \hat{\beta}_1 = -0.658,\ \hat{\beta}_2 = 0.626.$$

(Fit statistics are irrelevant: these are population values.) By Proposition 7.7, $P(Y > j \mid G = 0, r) = P(Y > j \mid G = 1, r)^{\exp(0.658)}$: given race, men's survival proportion beyond any age is women's raised to the power 1.93; given gender, blacks' is whites' to the power $e^{0.626} = 1.87$. If white women survive beyond some age with probability π, white men and black women do so with probability about π^2, and black men with about π^4.

Chapter 8. Loglinear Models for Contingency Tables

Loglinear models are Poisson GLMs with log link for the cell counts of a contingency table. They treat all cross-classified variables symmetrically, as joint responses, and their purpose is to describe association and interaction patterns. With one response it is simpler to use logit models; Section 8.5 shows that logit models with categorical predictors are special loglinear models. Throughout, $\{n_{ij}\}$ or $\{n_{ijk}\}$ are observed counts, $\{\mu\}$ expected counts and $\{\pi\}$ cell probabilities; a "+" subscript denotes summation over that index.

8.1 Loglinear models for two-way tables

Motivation. For an $I \times J$ table the familiar independence structure $\pi_{ij} = \pi_{i+}\pi_{+j}$ is multiplicative, so on the log scale it is additive, like a two-way ANOVA without interaction. Adding the "interaction" terms gives a model that can describe any association.

Let $\{\mu_{ij}\}$ be expected counts: either $\mu_{ij} = n\pi_{ij}$ under multinomial sampling, or $\mu_{ij} = E(Y_{ij})$ for $N = IJ$ independent Poisson counts. Formulas in terms of μ_{ij} cover both cases. Denote the row variable by X and the column variable by Y.

8.1.1 Independence model

Definition (Loglinear model of independence)

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y, \tag{8.1}$$

with identifiability constraints such as $\lambda_I^X = \lambda_J^Y = 0$ (baseline coding) or $\sum_i \lambda_i^X = \sum_j \lambda_j^Y = 0$.

Theorem 8.1 (Independence as a loglinear model)

For positive $\{\mu_{ij}\}$ with $\mu_{ij} = n\pi_{ij}$, the variables X and Y are independent if and only if (8.1) holds for some parameters. The ML fitted values are $\hat{\mu}_{ij} = n_{i+}n_{+j}/n$.

Proof.

Step 1. If $\pi_{ij} = \pi_{i+}\pi_{+j}$, then $\log \mu_{ij} = \log n + \log \pi_{i+} + \log \pi_{+j}$. Put $\lambda_i^X = \log \pi_{i+} - \log \pi_{I+}$, $\lambda_j^Y = \log \pi_{+j} - \log \pi_{+J}$ and $\lambda = \log n + \log \pi_{I+} + \log \pi_{+J}$; this is (8.1) with baseline constraints.

Step 2. Conversely, if $\mu_{ij} = a_i b_j$ with $a_i = e^{\lambda + \lambda_i^X}$, $b_j = e^{\lambda_j^Y}$, then $\mu_{i+} = a_i b_+$, $\mu_{+j} = a_+ b_j$, $\mu_{++} = a_+ b_+$, so

$$\pi_{ij} = \frac{a_i b_j}{a_+ b_+} = \frac{a_i b_+}{a_+ b_+} \cdot \frac{a_+ b_j}{a_+ b_+} = \pi_{i+}\pi_{+j}.$$

Step 3. The fitted values $n_{i+}n_{+j}/n$ satisfy the model and reproduce the row and column totals; by Birch's result (Theorem 8.19) they are the ML estimates. ∎

Hence the Pearson X^2 and G^2 tests of independence are goodness-of-fit tests of the loglinear model (8.1), with $\text{df} = (I-1)(J-1)$.

8.1.2 Interpretation of parameters

A loglinear GLM takes the N cell counts, not the n subjects, as its observations, and does not distinguish response from explanatory variables. For interpretation it nevertheless helps to form a logit.

Proposition 8.2 (Independence means a constant logit)

Let $J = 2$. Under (8.1),

$$\operatorname{logit} P(Y = 1 \mid X = i) = \lambda_1^Y - \lambda_2^Y \qquad \text{for every } i,$$

so the odds of column 1 equal $\exp(\lambda_1^Y - \lambda_2^Y)$ in every row.

Proof. $P(Y = 1 \mid X = i)/P(Y = 2 \mid X = i) = \mu_{i1}/\mu_{i2}$, hence the logit equals

$$\log \mu_{i1} - \log \mu_{i2} = (\lambda + \lambda_i^X + \lambda_1^Y) - (\lambda + \lambda_i^X + \lambda_2^Y) = \lambda_1^Y - \lambda_2^Y,$$

which does not depend on i. ■

In general, differences of main-effect parameters for one variable are log odds of one category versus another on that variable.

8.1.3 Saturated model

Definition (Saturated two-way model)

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY}. \tag{8.2}$$

The association terms λ_{ij}^{XY} (coefficients of products of dummy variables) measure departures from independence; with baseline constraints $\lambda_{Ij}^{XY} = \lambda_{iJ}^{XY} = 0$.

Theorem 8.3 (Saturated model and odds ratios)

(a) Model (8.2) has IJ nonredundant parameters and describes every positive table $\{\mu_{ij}\}$ exactly.
(b) Every odds ratio is a contrast of the association parameters; for rows a, c and columns b, d,

$$\log \frac{\mu_{ab}\mu_{cd}}{\mu_{ad}\mu_{cb}} = \lambda_{ab}^{XY} + \lambda_{cd}^{XY} - \lambda_{ad}^{XY} - \lambda_{cb}^{XY}.$$

In particular, for a 2×2 table $\log \theta = \lambda_{11}^{XY} + \lambda_{22}^{XY} - \lambda_{12}^{XY} - \lambda_{21}^{XY}$.
(c) Independence holds iff all $\lambda_{ij}^{XY} = 0$; the test has df $= (I - 1)(J - 1)$.

Proof.

Step 1. Counting under baseline constraints: $1 + (I - 1) + (J - 1) + (I - 1)(J - 1) = IJ$.

Step 2. Given positive $\{\mu_{ij}\}$, set $h_{ij} = \log \mu_{ij}$ and define

$$\lambda = h_{IJ}, \quad \lambda_i^X = h_{iJ} - h_{IJ}, \quad \lambda_j^Y = h_{Ij} - h_{IJ}, \quad \lambda_{ij}^{XY} = h_{ij} - h_{iJ} - h_{Ij} + h_{IJ}.$$

These satisfy the baseline constraints and $\lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY} = h_{ij}$, so every positive table is represented (uniquely, since the map is invertible).

Step 3. For (b), substitute (8.2): in $h_{ab} + h_{cd} - h_{ad} - h_{cb}$ the terms λ, $\lambda_a^X, \lambda_c^X, \lambda_b^Y, \lambda_d^Y$ each appear once with $+$ and once with $-$, leaving the stated contrast.

Step 4. For (c), by step 2 all $\lambda_{ij}^{XY} = 0$ iff $h_{ij} - h_{iJ} - h_{Ij} + h_{IJ} = 0$ for all i, j, i.e. all odds ratios with the last row and column are 1, which is equivalent to independence. The $(I - 1)(J - 1)$ free association parameters give the df. ■

Hierarchical models

A model is *hierarchical* if whenever it contains a term it also contains all lower-order terms built from the same variables (e.g. λ_{ij}^{XY} requires λ_i^X, λ_j^Y). Nonhierarchical models such as $\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_{ij}^{XY}$ have fits and interpretations that depend on the coding (with baseline coding it forces $\log \mu_{Ij} = \lambda$ in every column) and are rarely sensible. In a model with λ^{XY}, interpretation should focus on the highest-order terms, not on main effects.

8.1.4 Alternative parameter constraints

Instead of $\lambda_{Ij}^{XY} = \lambda_{iJ}^{XY} = 0$ one may use $\sum_i \lambda_{ij}^{XY} = \sum_j \lambda_{ij}^{XY} = 0$. Only contrasts such as the one in Theorem 8.3(b) are invariant. For a 2×2 table with $\log \theta = 2.0$: baseline coding gives $\lambda_{11}^{XY} = 2.0$ and the others 0; sum-to-zero coding gives $\lambda_{11}^{XY} = \lambda_{22}^{XY} = 0.5$, $\lambda_{12}^{XY} = \lambda_{21}^{XY} = -0.5$. Both give $\log \theta = 2.0$. Baseline coding has the advantage that some parameters in a set may have infinite estimates while others stay finite.

8.1.5 Multinomial models for cell probabilities

Proposition 8.4 (Poisson to multinomial)

If Y_{ij} are independent Poisson with $\log \mu_{ij}$ given by (8.2), then conditional on $\sum Y_{ij} = n$ the counts are multinomial$(n, \{\pi_{ij}\})$ with

$$\pi_{ij} = \frac{\exp(\lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY})}{\sum_a \sum_b \exp(\lambda + \lambda_a^X + \lambda_b^Y + \lambda_{ab}^{XY})}, \tag{8.3}$$

in which λ cancels.

Proof. Let $\mu_{++} = \sum \mu_{ij}$; the total is Poisson(μ_{++}). For counts $\{n_{ij}\}$ summing to n,

$$P(\{Y_{ij} = n_{ij}\} \mid Y_{++} = n) = \frac{\prod_{ij} e^{-\mu_{ij}} \mu_{ij}^{n_{ij}} / n_{ij}!}{e^{-\mu_{++}} \mu_{++}^n / n!} = \frac{n!}{\prod n_{ij}!} \prod_{ij} \Big(\frac{\mu_{ij}}{\mu_{++}} \Big)^{n_{ij}},$$

since $\prod e^{-\mu_{ij}} = e^{-\mu_{++}}$ and $\sum n_{ij} = n$. This is multinomial with $\pi_{ij} = \mu_{ij}/\mu_{++}$, which is (8.3); the common factor e^{λ} cancels. ∎

The intercept λ only describes the total sample size, which is random under Poisson sampling but fixed under multinomial sampling.

8.2 Loglinear models for independence and interaction in three-way tables

Motivation. Three-way tables admit several distinct kinds of independence (mutual, joint, conditional) and of association (homogeneous or not). Each one corresponds to a loglinear model in which certain interaction terms are absent, so the model symbol tells you the independence structure.

The following lemma turns every independence statement into a statement about which terms appear.

Lemma 8.5 (Generating classes and factorization)

Let a hierarchical loglinear model have generating class $\{C_1, \dots, C_m\}$ (the highest-order terms, i.e. the model symbol; e.g. (XZ, YZ) has $C_1 = \{X, Z\}$, $C_2 = \{Y, Z\}$). Then a positive table $\{\mu\}$ satisfies the model if and only if

$$\mu(\mathbf{x}) = \prod_{r=1}^{m} \psi_r(\mathbf{x}_{C_r})$$

for some positive functions ψ_r that depend on the cell $\mathbf{x}$ only through the coordinates in C_r.

Proof. [Supplement]

Step 1. Every term of the model involves a subset of some C_r, hence is a function of $\mathbf{x}_{C_r}$. Grouping terms by a generator containing them, $\log \mu = \sum_r g_r(\mathbf{x}_{C_r})$; exponentiate to get the product form.

Step 2. Conversely, by Theorem 8.3 (step 2, applied to the variables in C_r, and its obvious extension to more than two variables) any function $g(\mathbf{x}_C)$ decomposes as a constant plus main effects plus interactions of every subset of C. Since the model is hierarchical, all these subsets are terms of the model. Summing the decompositions of $\log \psi_r$ gives $\log \mu$ in the model's form. ■

8.2.1 Types of independence

Consider an $I \times J \times K$ table of three responses X, Y, Z with probabilities $\{\pi_{ijk}\}$ (or Poisson means $\{\mu_{ijk}\}$).

Definition (Three kinds of independence)

(i) *Mutual independence*: $\pi_{ijk} = \pi_{i++}\pi_{+j+}\pi_{++k}$ for all i, j, k.
(ii) *Joint independence* of Y from (X, Z): $\pi_{ijk} = \pi_{i+k}\pi_{+j+}$, i.e. two-way independence between Y and the IK combinations of (X, Z).
(iii) *Conditional independence* of X and Y given Z: $\pi_{ij|k} = \pi_{i+|k}\pi_{+j|k}$ for all k, where $\pi_{ij|k} = P(X = i, Y = j \mid Z = k)$; equivalently

$$\pi_{ijk} = \pi_{i+k}\pi_{+jk}/\pi_{++k} \qquad \text{for all } i, j, k. \tag{8.4}$$

Theorem 8.6 (Independence structures as loglinear models)

For positive tables:
(a) mutual independence $\iff$ (X, Y, Z): $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z$;
(b) Y jointly independent of (X, Z) $\iff$ (XZ, Y): $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ik}^{XZ}$;
(c) $X \perp Y \mid Z$ $\iff$ (XZ, YZ): $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ}$.
Moreover (a) $\Rightarrow$ (b) $\Rightarrow$ (c).

Proof. By Lemma 8.5 it suffices to match each independence statement with a product form. Write $\mu = n\pi$.

Step 1 (c). If (8.4) holds, $\mu_{ijk} = \psi_1(i,k)\psi_2(j,k)$ with $\psi_1 = n\pi_{i+k}$ and $\psi_2 = \pi_{+jk}/\pi_{++k}$. Conversely, if $\mu_{ijk} = A_{ik}B_{jk}$, then $\mu_{i+k} = A_{ik}B_{+k}$, $\mu_{+jk} = A_{+k}B_{jk}$ and $\mu_{++k} = A_{+k}B_{+k}$, whence

$$\frac{\mu_{i+k}\mu_{+jk}}{\mu_{++k}} = \frac{A_{ik}B_{+k}\, A_{+k}B_{jk}}{A_{+k}B_{+k}} = A_{ik}B_{jk} = \mu_{ijk},$$

which is (8.4) after dividing by n.

Step 2 (b). If $\pi_{ijk} = \pi_{i+k}\pi_{+j+}$ we have a product $\psi_1(i,k)\psi_2(j)$. Conversely if $\mu_{ijk} = A_{ik}B_j$ then $\mu_{i+k} = A_{ik}B_+$, $\mu_{+j+} = A_{++}B_j$, $\mu_{+++} = A_{++}B_+$, and $\mu_{i+k}\mu_{+j+}/\mu_{+++} = A_{ik}B_j$, i.e. $\pi_{ijk} = \pi_{i+k}\pi_{+j+}$.

Step 3 (a). Same computation with $\mu_{ijk} = A_iB_jC_k$: $\mu_{i++}\mu_{+j+}\mu_{++k}/\mu_{+++}^2 = A_iB_jC_k$.

Step 4. The implications follow because the model in (a) is a special case of (b) (set $\lambda^{XZ} = 0$), which is a special case of (c) (set $\lambda^{YZ} = 0$). ■

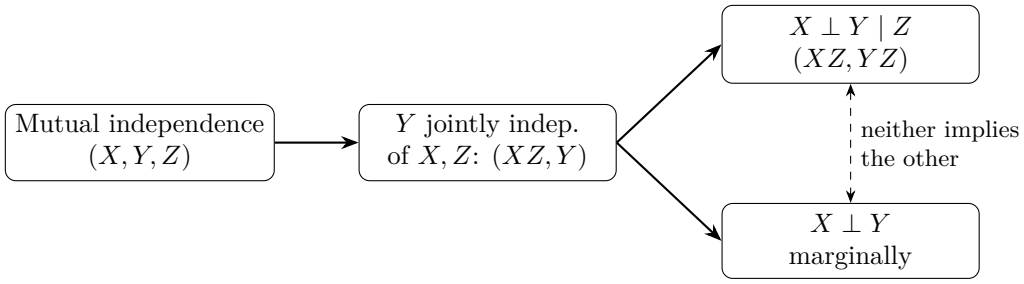


Figure 10: Relationships among types of XY independence (book Figure 8.1).

Conditional independence does not imply marginal independence, and conversely (Simpson's paradox). Both hold when one of the stronger structures applies (Figure 10).

8.2.2 Homogeneous association and three-factor interaction

Definition (Homogeneous association and the saturated model)

The model of *homogeneous association* (no three-factor interaction) is

$$\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ij}^{XY} + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ}, \tag{8.5}$$

symbol (XY, XZ, YZ); equivalently $\pi_{ijk} = \psi_{ij}\phi_{jk}\omega_{ik}$. The general (saturated) model adds λ_{ijk}^{XYZ}, symbol (XYZ).

The saturated model has

$$1+(I-1)+(J-1)+(K-1)+(I-1)(J-1)+(I-1)(K-1)+(J-1)(K-1)+(I-1)(J-1)(K-1) = IJK$$

parameters, one per cell (expand $IJK = [(I-1)+1][(J-1)+1][(K-1)+1]$). Unlike the models of Theorem 8.6, (8.5) has no closed-form expression of ψ, ϕ, ω in terms of margins.

Model symbol	Association terms	Interpretation
(X, Y, Z)	none	mutual independence
(XY, Z)	λ^{XY}	Z jointly independent of X, Y
(XY, YZ)	$\lambda^{XY}, \lambda^{YZ}$	$X \perp Z \mid Y$
(XY, XZ, YZ)	all two-factor	homogeneous association
(XYZ)	all, incl. λ^{XYZ}	saturated

8.2.3 Interpreting model parameters

At level k of Z the XY conditional association is described by the $(I-1)(J-1)$ local odds ratios

$$\theta_{ij(k)} = \frac{\mu_{ijk}\,\mu_{i+1,j+1,k}}{\mu_{i,j+1,k}\,\mu_{i+1,j,k}}, \tag{8.6}$$

and similarly $\theta_{i(j)k}$ for XZ and $\theta_{(i)jk}$ for YZ. Conditional independence of X, Y given Z means all $\theta_{ij(k)} = 1$.

Theorem 8.7 (Homogeneous association)

(a) Under (XY, XZ, YZ),

$$\log \theta_{ij(k)} = \lambda_{ij}^{XY} + \lambda_{i+1,j+1}^{XY} - \lambda_{i,j+1}^{XY} - \lambda_{i+1,j}^{XY} \quad \text{for all } k, \tag{8.7}$$

and similarly for XZ and YZ odds ratios.
(b) Conversely, a positive table satisfies (XY, XZ, YZ) if and only if $\theta_{ij(1)} = \cdots = \theta_{ij(K)}$ for all i, j. Equivalently, iff the XZ odds ratios are the same at every j, or iff the YZ odds ratios are the same at every i.

Proof. [Supplement]

Step 1. For (a), substitute (8.5) into $\log \theta_{ij(k)}$. Each of $\lambda, \lambda^X, \lambda^Y, \lambda_k^Z, \lambda_{ik}^{XZ}, \lambda_{i+1,k}^{XZ}, \lambda_{jk}^{YZ}, \lambda_{j+1,k}^{YZ}$ appears once with each sign and cancels; only the XY contrast remains, and it does not involve k.

Step 2. For (b), suppose $\theta_{ij(k)} = \theta_{ij}$ does not depend on k. Let $h_{ijk} = \log \mu_{ijk}$. Telescoping the local log odds ratios over the rectangle $[1, i] \times [1, j]$,

$$h_{ijk} - h_{1jk} - h_{i1k} + h_{11k} = \sum_{a<i} \sum_{b<j} \log \theta_{ab} =: g(i, j),$$

which is free of k. Therefore

$$h_{ijk} = g(i, j) + [h_{1jk} - h_{11k}] + h_{i1k},$$

a sum of a function of (i, j), a function of (j, k) and a function of (i, k). By Lemma 8.5 the table satisfies (XY, XZ, YZ).

Step 3. The symmetric statements follow by the same argument with the roles of the variables permuted; since each is equivalent to (XY, XZ, YZ), they are equivalent to one another. ■

The three-factor term describes how an odds ratio changes across the third variable. For a $2 \times 2 \times 2$ table, direct substitution of the saturated model gives

$$\log \frac{\theta_{11(1)}}{\theta_{11(2)}} = (\lambda_{111}^{XYZ} + \lambda_{221}^{XYZ} - \lambda_{121}^{XYZ} - \lambda_{211}^{XYZ}) - (\lambda_{112}^{XYZ} + \lambda_{222}^{XYZ} - \lambda_{122}^{XYZ} - \lambda_{212}^{XYZ}),$$

which equals λ_{111}^{XYZ} when second-category parameters are set to 0. When X, Y are binary and second-level parameters are zero, $\log \theta_{11(k)} = \lambda_{11}^{XY}$.

8.2.4 Alcohol, cigarette, and marijuana use example

Example 8.8 (Alcohol, cigarette, marijuana use)

A 1992 survey of 2276 high-school seniors near Dayton, Ohio, recorded whether they had ever used alcohol (A), cigarettes (C) and marijuana (M).

		Observed: M		Fit of (AC, AM, CM)	
A	C	yes	no	yes	no
yes	yes	911	538	910.4	538.6
yes	no	44	456	44.6	455.4
no	yes	3	43	3.6	42.4
no	no	2	279	1.4	279.6

Estimated odds ratios under several models:

	Conditional			Marginal		
Model	AC	AM	CM	AC	AM	CM
(A, C, M)	1.0	1.0	1.0	1.0	1.0	1.0
(AC, M)	17.7	1.0	1.0	17.7	1.0	1.0
(AM, CM)	1.0	61.9	25.1	2.7	61.9	25.1
(AC, AM, CM)	7.8	19.8	17.3	17.7	61.9	25.1
(ACM), M=yes	13.8	24.3	17.5	17.7	61.9	25.1
(ACM), M=no	7.7	13.5	9.7			

A term absent from a model gives a conditional odds ratio 1.0; e.g. under (AM, CM),

$$\frac{909.24 \times 0.24}{45.76 \times 4.76} = \frac{438.84 \times 179.84}{555.16 \times 142.16} = 1.0,$$

yet the fitted marginal AC odds ratio is 2.7: conditional independence does not imply marginal independence. Under (AC, AM, CM) the AC conditional odds ratio is 7.8 at both levels of M, $= \exp(\hat{\lambda}_{11}^{AC} + \hat{\lambda}_{22}^{AC} - \hat{\lambda}_{12}^{AC} - \hat{\lambda}_{21}^{AC})$. Estimated odds ratios depend heavily on the model, so model selection matters.

8.3 Inference for loglinear models

Motivation. Once a loglinear model is fitted, we need to know whether it fits, which association terms are needed, and how large the associations are. The tools are the usual deviance tests and Wald/likelihood intervals for GLMs.

8.3.1 Chi-squared goodness-of-fit tests

$X^2 = \sum(n - \hat{\mu})^2/\hat{\mu}$ and $G^2 = 2\sum n \log(n/\hat{\mu})$ test model fit, with df = number of cells − number of nonredundant parameters. Model selection can also use AIC $= -2$(maximized log likelihood − number of parameters), equivalently minimizing $G^2 - 2\,\text{df}$.

Example 8.9 (Goodness of fit for the student survey)

Model	G^2	X^2	df	P-value
(A, C, M)	1286.0	1411.4	4	< 0.001
(A, CM)	534.2	505.6	3	< 0.001
(C, AM)	939.6	824.2	3	< 0.001
(M, AC)	843.8	704.9	3	< 0.001
(AC, AM)	497.4	443.8	2	< 0.001
(AC, CM)	92.0	80.8	2	< 0.001
(AM, CM)	187.8	177.6	2	< 0.001
(AC, AM, CM)	0.4	0.4	1	0.54
(ACM)	0.0	0.0	0	–

Only the homogeneous association model fits; it also minimizes AIC.

8.3.2 Inference about conditional associations

Tests about a term compare nested models. The likelihood-ratio statistic $-2(L_0 - L_1)$ equals $G^2(M_0) - G^2(M_1)$, the difference of deviances. For XY conditional independence within (XY, XZ, YZ), $H_0 : \lambda_{ij}^{XY} = 0$ for all $(I-1)(J-1)$ parameters, and the statistic is

$$G^2[(XZ, YZ) \mid (XY, XZ, YZ)] = G^2(XZ, YZ) - G^2(XY, XZ, YZ), \qquad \text{df} = (I-1)(J-1).$$

This is a model-based alternative to the generalized CMH test.

Example 8.10 (Conditional associations in the student survey)

For AC conditional independence, $G^2[(AM, CM) \mid (AC, AM, CM)] = 187.8 - 0.4 = 187.4$, df $= 1$, $P < 0.001$. The other two pairwise terms are likewise needed (LR statistics 91.6 for AM, 497.0 for CM). With baseline coding (last category $= 0$), (AC, AM, CM) gives

$$\hat{\lambda}_{11}^{AC} = 2.054 \text{ (SE} = 0.174), \quad \hat{\lambda}_{11}^{AM} = 2.986 \text{ (0.465)}, \quad \hat{\lambda}_{11}^{CM} = 2.848 \text{ (0.164)}.$$

For binary variables these are conditional log odds ratios. The 95% Wald interval for the AC conditional odds ratio is $\exp(2.054 \pm 1.96 \times 0.174) = (5.5, 11.0)$; for AM it is (8.0, 49.2) and for CM (12.5, 23.8). All associations are strong, and the marginal ones are stronger still: controlling for one response moderates the association between the other two.

With large samples, significance alone is not informative; confidence intervals show whether associations are practically important.

8.4 Loglinear models for higher dimensions

Motivation. The three-way ideas extend directly to four or more variables. The difficulties are the large number of possible terms, which makes model selection harder, and the large number of cells, which can make tables sparse.

8.4.1 Four-way contingency tables

For variables W, X, Y, Z, the model with all two-factor terms,

$$\log \mu_{hijk} = \lambda + \lambda_h^W + \lambda_i^X + \lambda_j^Y + \lambda_k^Z$$

$$+\lambda_{hi}^{WX}+\lambda_{hj}^{WY}+\lambda_{hk}^{WZ}+\lambda_{ij}^{XY}+\lambda_{ik}^{XZ}+\lambda_{jk}^{YZ},$$

symbol (WX, WY, WZ, XY, XZ, YZ), has each pair conditionally dependent with the same odds ratios at every combination of levels of the other two variables. Removing a two-factor term means conditional independence of that pair given the other two variables. Under (WXY, WZ, XZ, YZ), the WX, WY, XY associations may vary across levels of the remaining variable of the triple, while every association involving Z is homogeneous. The saturated model adds all three-factor terms and λ^{WXYZ}.

Proposition 8.11 (Odds ratios in models without a variable in higher-order terms)

If a variable V appears in no term of order three or more, then for every other variable U the conditional UV odds ratios are the same at every combination of levels of the remaining variables, and they equal the exponentiated contrast of λ^{UV}.

Proof. [Supplement] Fix levels of the remaining variables and form a UV odds ratio $\mu_{ab\cdot}\mu_{cd\cdot}/(\mu_{ad\cdot}\mu_{cb\cdot})$ (U levels a, c; V levels b, d). In the log, any term not involving V appears with U at level a once with each sign (in μ_{ab} and μ_{ad}), and likewise at level c, so it cancels. Any term not involving U cancels the same way. The only terms involving both U and V are, by hypothesis, the two-factor term λ^{UV}, which does not depend on the other variables. Hence the log odds ratio is $\lambda_{ab}^{UV}+\lambda_{cd}^{UV}-\lambda_{ad}^{UV}-\lambda_{cb}^{UV}$ at every level combination. ■

8.4.2 Automobile accident example

Example 8.12 (Automobile accidents and seat belts)

In 1991, 68,694 passengers in auto accidents in Maine were classified by gender (G), location (L: urban, rural), seat-belt use (S) and injury (I).

Gender	Location	Seat belt	Injury no	Injury yes
Female	Urban	No	7287	996
		Yes	11587	759
	Rural	No	3246	973
		Yes	6134	757
Male	Urban	No	10381	812
		Yes	10969	380
	Rural	No	6123	1084
		Yes	6693	513

Goodness of fit:

Model	G^2	df	P
(G, I, L, S)	2792.8	11	< 0.0001
(GI, GL, GS, IL, IS, LS)	23.4	5	< 0.001
(GIL, GIS, GLS, ILS)	1.3	1	0.25
(GIL, GS, IS, LS)	18.6	4	0.001
(GIS, GL, IL, LS)	22.8	4	< 0.001
(GLS, GI, IL, IS)	7.5	4	0.11
(ILS, GI, GL, GS)	20.6	4	< 0.001

Estimated conditional odds ratios:

Odds ratio	(GI, GL, GS, IL, IS, LS)	(GLS, GI, IL, IS)
GI	0.58	0.58
IL	2.13	2.13
IS	0.44	0.44
GL (S=no / yes)	1.23 / 1.23	1.33 / 1.17
GS (L=urban / rural)	0.63 / 0.63	0.66 / 0.58
LS (G=female / male)	1.09 / 1.09	1.17 / 1.03

Under the pairwise model the IS conditional log odds ratio is -0.814 (SE 0.028), so the 95% interval for the odds ratio is $\exp(-0.814 \pm 1.96 \times 0.028) = (0.42, 0.47)$: wearing a seat belt more than halves the odds of injury at each gender–location combination. Injury is more likely in rural accidents and for females; males have 0.63 times the odds of females of wearing seat belts. In (GLS, GI, IL, IS) the variable I is in no three-factor term, so (Proposition 8.11) its odds ratios with G, L, S are homogeneous; the GL, GS, LS odds ratios vary with the third variable, e.g. $0.66 = (7273.2 \times 10{,}959.2)/(11{,}632.6 \times 10{,}358.9)$ for urban accidents.

8.4.3 Large samples and statistical versus practical significance

(GLS, GI, IL, IS) fits significantly better than the pairwise model: $23.4 - 7.5 = 15.9$, df $= 1$, $P = 0.0001$. Yet the fitted GL, GS, LS odds ratios are similar at both levels of the third variable, so the three-factor interaction is weak; the significance reflects the huge n. With large samples, focus on estimation; the simpler pairwise model is adequate for most purposes.

8.4.4 Dissimilarity index

Definition (Dissimilarity index)

For cell counts $n_i = np_i$ and fitted values $\hat{\mu}_i = n\hat{\pi}_i$,

$$\hat{\Delta} = \frac{\sum_i |n_i - \hat{\mu}_i|}{2n} = \frac{1}{2}\sum_i |p_i - \hat{\pi}_i|.$$

It estimates the population index $\Delta = \frac{1}{2}\sum |\pi_i - \pi_i^{\text{model}}|$.

Proposition 8.13 (Meaning of $\hat{\Delta}$)

If $\sum \hat{\mu}_i = n$ (true for any loglinear model with an intercept), then $0 \le \hat{\Delta} \le 1$, and $n\hat{\Delta}$ is the minimum number of observations that must be moved to other cells to make the observed table equal to the fitted one.

Proof. [Supplement]

Step 1. $\sum |p_i - \hat{\pi}_i| \le \sum p_i + \sum \hat{\pi}_i = 2$, so $\hat{\Delta} \le 1$; clearly $\hat{\Delta} \ge 0$.

Step 2. Let $d_i = n_i - \hat{\mu}_i$. Since $\sum d_i = 0$, $\sum d_i^+ = \sum d_i^- = \frac{1}{2}\sum |d_i| = n\hat{\Delta}$.

Step 3. Moving one observation changes $\sum d_i^+$ by at most 1, so at least $\sum d_i^+$ moves are necessary. Moving d_i^+ observations out of each cell with surplus into cells with deficits (total deficit $\sum d_i^-$, the same amount) is sufficient (allowing fractional "observations" since $\hat{\mu}$ need not be integer). Hence the minimum is $n\hat{\Delta}$. ■

Values $\hat{\Delta} < 0.02$ or 0.03 indicate that the data follow the model closely in practical terms even if

the model is rejected; $\hat{\Delta}$ tends to overestimate Δ when Δ is near 0, especially for small n. For the accident data, $\hat{\Delta} = 0.008$ for the pairwise model and 0.003 for (GLS, GI, IL, IS): moving less than 1% of the data yields a perfect fit.

8.5 Loglinear–logit model connection

Motivation. Loglinear models treat variables symmetrically, logit models single out a response. Forming a logit from a loglinear model helps interpret it, and every logit model with categorical predictors is equivalent to a loglinear model that contains the full interaction among the predictors.

8.5.1 Using logit models to interpret loglinear models

Proposition 8.14 (Logit form of (XY, XZ, YZ))

If Y is binary and (XY, XZ, YZ) holds, then

$$\text{logit}\, P(Y = 1 \mid X = i, Z = k) = \alpha + \beta_i^X + \beta_k^Z, \tag{8.8}$$

with $\alpha = \lambda_1^Y - \lambda_2^Y$, $\beta_i^X = \lambda_{i1}^{XY} - \lambda_{i2}^{XY}$, $\beta_k^Z = \lambda_{1k}^{YZ} - \lambda_{2k}^{YZ}$.

Proof. The logit is $\log(\mu_{i1k}/\mu_{i2k})$. Subtracting (8.5) at $j = 2$ from (8.5) at $j = 1$, the terms $\lambda, \lambda_i^X, \lambda_k^Z, \lambda_{ik}^{XZ}$ cancel, leaving

$$(\lambda_1^Y - \lambda_2^Y) + (\lambda_{i1}^{XY} - \lambda_{i2}^{XY}) + (\lambda_{1k}^{YZ} - \lambda_{2k}^{YZ}).$$

■

The λ^{XZ} terms, which describe association among explanatory variables, cancel: the logit model $(X + Z)$ says nothing about it.

8.5.2 Auto accident example revisited

Example 8.15 (Injury as a response)

For (GLS, GI, IL, IS), forming the logit on injury cancels every term without the index i of I, including λ^{GLS}, giving the logit model $(G + L + S)$:

$$\text{logit}\, P(I = 1 \mid G = g, L = l, S = s) = \alpha + \beta_g^G + \beta_l^L + \beta_s^S, \qquad \beta_s^S = \lambda_{1s}^{IS} - \lambda_{2s}^{IS}.$$

Fitted values, G^2, residual df and standardized residuals are identical for the two models. The seat-belt log odds ratio is $\hat{\beta}_1^S - \hat{\beta}_2^S = -0.817$ in the logit model and $\hat{\lambda}_{11}^{IS} + \hat{\lambda}_{22}^{IS} - \hat{\lambda}_{12}^{IS} - \hat{\lambda}_{21}^{IS} = -0.817$ in the loglinear model. The loglinear model treats the 16 counts as Poisson; the logit model treats the GLS margin as fixed and the 8 counts $\{n_{g1ls}\}$ as binomial.

8.5.3 Correspondence between loglinear and logit models

Forming the logit on Y in (XY, YZ) also gives (8.8), but (XY, YZ) is *not* equivalent to the logit model: it additionally forces $X \perp Z \mid Y$. The equivalent loglinear model must contain the most general interaction among the explanatory variables.

Theorem 8.16 (Loglinear–logit equivalence)

Let Y be the response and $\mathbf{x}$ index the combinations of the explanatory variables. Consider a loglinear model containing the full interaction term among the explanatory variables, $\log \mu_{\mathbf{x}j} = \gamma_{\mathbf{x}} + \eta_{\mathbf{x}j}(\boldsymbol{\beta})$, where $\gamma_{\mathbf{x}}$ is unrestricted and $\eta_{\mathbf{x}j}$ collects the terms involving Y (with $\eta_{\mathbf{x}J} = 0$). Then the ML fitted values under Poisson sampling coincide with those of the baseline-category logit model $\log(\pi_{j|\mathbf{x}}/\pi_{J|\mathbf{x}}) = \eta_{\mathbf{x}j}(\boldsymbol{\beta})$ fitted with the margins $n_{\mathbf{x}+}$ fixed, and the ML estimates of $\boldsymbol{\beta}$ agree.

Proof. [Supplement]

Step 1. Reparametrize: $\mu_{\mathbf{x}+} = e^{\gamma_{\mathbf{x}}} \sum_j e^{\eta_{\mathbf{x}j}}$ and $\pi_{j|\mathbf{x}} = e^{\eta_{\mathbf{x}j}} / \sum_h e^{\eta_{\mathbf{x}h}}$, so $\mu_{\mathbf{x}j} = \mu_{\mathbf{x}+}\pi_{j|\mathbf{x}}$. Since $\gamma_{\mathbf{x}}$ is free, $(\{\mu_{\mathbf{x}+}\}, \boldsymbol{\beta})$ ranges over $(0,\infty)^{\#\mathbf{x}} \times \{\text{all } \boldsymbol{\beta}\}$, a product set (variation independence).

Step 2. The Poisson log likelihood splits:

$$\sum_{\mathbf{x},j}(n_{\mathbf{x}j}\log\mu_{\mathbf{x}j} - \mu_{\mathbf{x}j}) = \sum_{\mathbf{x}}(n_{\mathbf{x}+}\log\mu_{\mathbf{x}+} - \mu_{\mathbf{x}+}) + \sum_{\mathbf{x},j} n_{\mathbf{x}j}\log\pi_{j|\mathbf{x}}(\boldsymbol{\beta}).$$

Step 3. The two parts depend on separate parameters, so they are maximized separately: $\hat{\mu}_{\mathbf{x}+} = n_{\mathbf{x}+}$, and $\hat{\boldsymbol{\beta}}$ maximizes the second part, which is exactly the product-multinomial (binomial if $J = 2$) log likelihood of the logit model. Hence $\hat{\mu}_{\mathbf{x}j} = n_{\mathbf{x}+}\hat{\pi}_{j|\mathbf{x}}$ in both models. ■

Loglinear symbol	Logit model	Logit symbol
(Y, XZ)	α	$(-)$
(XY, XZ)	$\alpha + \beta_i^X$	(X)
(YZ, XZ)	$\alpha + \beta_k^Z$	(Z)
(XY, YZ, XZ)	$\alpha + \beta_i^X + \beta_k^Z$	$(X + Z)$
(XYZ)	$\alpha + \beta_i^X + \beta_k^Z + \beta_{ik}^{XZ}$	$(X*Z)$

With several response variables the loglinear approach is more general; with one response, logit models are more natural.

8.5.4 Generalized loglinear model

Write the counts as a vector $\mathbf{n} = (n_1, \ldots, n_N)'$ with means $\boldsymbol{\mu}$. An ordinary loglinear model is

$$\log \boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}. \tag{8.9}$$

For instance, independence in a 2×2 table with $\lambda_2^X = \lambda_2^Y = 0$ is

$$\begin{pmatrix} \log\mu_{11} \\ \log\mu_{12} \\ \log\mu_{21} \\ \log\mu_{22} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} \lambda \\ \lambda_1^X \\ \lambda_1^Y \end{pmatrix}.$$

The *generalized loglinear model* is $\mathbf{C}\log(\mathbf{A}\boldsymbol{\mu}) = \mathbf{X}\boldsymbol{\beta}$. With $\mathbf{A} = \mathbf{I}_4$, $\mathbf{C} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$, $\mathbf{X} = (1,1)'$ and $\boldsymbol{\beta} = \alpha$, it states that the logits in both rows equal α, i.e. independence again. The class includes logit models and, later, models for marginal distributions.

8.6 Loglinear model fitting: likelihood equations

Motivation. Why are fitted values of loglinear models obtained by matching marginal tables? Because the Poisson model is an exponential family whose sufficient statistics are those margins. This section derives the sufficient statistics, likelihood equations, uniqueness, closed-form fits, degrees of freedom and the asymptotic covariance matrix.

Derivations use Poisson sampling, which needs no constraint on parameters; Theorem 8.21 transfers them to multinomial sampling.

8.6.1 Minimal sufficient statistics

The kernel of the Poisson log likelihood for a three-way table is

$$L(\boldsymbol{\mu}) = \sum_{i,j,k} n_{ijk} \log \mu_{ijk} - \sum_{i,j,k} \mu_{ijk}. \tag{8.10}$$

Substituting the saturated model,

$$\begin{aligned} L = n\lambda &+ \sum_i n_{i++}\lambda_i^X + \sum_j n_{+j+}\lambda_j^Y + \sum_k n_{++k}\lambda_k^Z + \sum_{i,j} n_{ij+}\lambda_{ij}^{XY} + \sum_{i,k} n_{i+k}\lambda_{ik}^{XZ} \\ &+ \sum_{j,k} n_{+jk}\lambda_{jk}^{YZ} + \sum_{i,j,k} n_{ijk}\lambda_{ijk}^{XYZ} - \sum_{i,j,k} \exp(\lambda + \lambda_i^X + \cdots + \lambda_{ijk}^{XYZ}). \end{aligned} \tag{8.11}$$

Theorem 8.17 (Sufficient statistics are marginal tables)

For a hierarchical loglinear model, the minimal sufficient statistics are the marginal tables corresponding to the terms in the model symbol: e.g. $\{n_{i++}\}, \{n_{+j+}\}, \{n_{++k}\}$ for (X, Y, Z); $\{n_{ij+}\}, \{n_{++k}\}$ for (XY, Z); $\{n_{ij+}\}, \{n_{+jk}\}$ for (XY, YZ); $\{n_{ij+}\}, \{n_{i+k}\}, \{n_{+jk}\}$ for (XY, XZ, YZ).

Proof.

Step 1. Setting the absent parameters to zero in (8.11), the joint distribution is an exponential family whose canonical statistics are the coefficients of the remaining parameters, i.e. the marginal tables of all terms present. By the factorization theorem these are sufficient; minimality holds because, after removing redundant parameters, the family is full rank.

Step 2. A lower-order margin is a sum of a higher-order one (e.g. $n_{i++} = \sum_j n_{ij+}$), so only the margins of the highest-order terms (the generators) are needed. ■

For the saturated model the full table is sufficient: no data reduction.

8.6.2 Likelihood equations for loglinear models

In the matrix form (8.9), $\log \mu_i = \sum_j x_{ij}\beta_j$ and

$$L(\boldsymbol{\beta}) = \sum_i n_i \Big(\sum_j x_{ij}\beta_j\Big) - \sum_i \exp\Big(\sum_j x_{ij}\beta_j\Big). \tag{8.12}$$

Theorem 8.18 (Likelihood equations match margins)

(a) The likelihood equations are

$$\mathbf{X}'\mathbf{n} = \mathbf{X}'\hat{\boldsymbol{\mu}}\,: \tag{8.13}$$

sufficient statistics equal their fitted (expected) values.
(b) For a hierarchical model, (8.13) is equivalent to equality of the observed and fitted marginal tables in the model symbol. For (XZ, YZ):

$$\hat{\mu}_{i+k} = n_{i+k} \text{ for all } i,k, \qquad \hat{\mu}_{+jk} = n_{+jk} \text{ for all } j,k. \tag{8.14}$$

Proof.

Step 1. Since $\partial \exp(\sum_j x_{ij}\beta_j)/\partial\beta_j = x_{ij}\mu_i$,

$$\frac{\partial L}{\partial \beta_j} = \sum_i n_i x_{ij} - \sum_i \mu_i x_{ij}, \qquad j = 1, \ldots, p.$$

Setting these to 0 gives (8.13).

Step 2. (8.13) says $\mathbf{v}'\mathbf{n} = \mathbf{v}'\hat{\boldsymbol{\mu}}$ for every $\mathbf{v}$ in the column space of $\mathbf{X}$. By Lemma 8.5, this column space is the set of functions $\sum_r g_r(\mathbf{x}_{C_r})$, which is spanned by the indicator vectors $\mathbf{v} = \mathbb{1}\{\mathbf{x}_{C_r} = \mathbf{c}\}$ of the cells of each generator's margin. For such $\mathbf{v}$, $\mathbf{v}'\mathbf{n}$ is the marginal count at $\mathbf{c}$. Hence (8.13) is equivalent to equality of all generator margins. For (XZ, YZ) this is (8.14). ■

Thus fitted values are smoothed versions of the data that agree with them on certain margins but whose associations follow the model.

8.6.3 Birch's results for loglinear models

Theorem 8.19 (Birch)

There is at most one $\hat{\boldsymbol{\mu}}$ that both satisfies the loglinear model and matches the data in the minimal sufficient statistics, and if it exists it is the ML estimate. Conversely, any ML estimate has these two properties.

Proof. [Supplement]

Step 1. Write $\boldsymbol{\eta} = \log \boldsymbol{\mu} \in \mathcal{V} = \text{col}(\mathbf{X})$. As a function of $\boldsymbol{\eta}$, $L = \mathbf{n}'\boldsymbol{\eta} - \sum_i e^{\eta_i}$ is strictly concave on $\mathbb{R}^N$: its Hessian is $-\text{diag}(e^{\eta_i})$, negative definite. Its restriction to the subspace $\mathcal{V}$ is still strictly concave.

Step 2. A point $\boldsymbol{\eta} \in \mathcal{V}$ is stationary for $L|_{\mathcal{V}}$ iff the gradient $\mathbf{n} - \boldsymbol{\mu}$ is orthogonal to $\mathcal{V}$, i.e. $\mathbf{X}'(\mathbf{n} - \boldsymbol{\mu}) = \mathbf{0}$, which is margin matching by Theorem 8.18.

Step 3. For a differentiable concave function, a stationary point is a global maximum; for a strictly concave one, there is at most one maximizer. Hence a $\boldsymbol{\mu}$ in the model that matches the margins is the unique MLE, and conversely the MLE (an interior maximizer) is stationary, so it matches the margins. ■

For example, for independence in a two-way table, $\hat{\mu}_{ij} = n_{i+}n_{+j}/n$ satisfies the model and has $\hat{\mu}_{i+} = n_{i+}$, $\hat{\mu}_{+j} = n_{+j}$; by Birch it is the MLE.

8.6.4 Direct versus iterative calculation of fitted values

Theorem 8.20 (Closed-form (direct) estimates)

The ML fitted values for three-way models other than (XY, XZ, YZ) are:

Model	Probabilistic form	Fitted value $\hat{\mu}_{ijk}$
(X, Y, Z)	$\pi_{i++}\pi_{+j+}\pi_{++k}$	$n_{i++}n_{+j+}n_{++k}/n^2$
(XY, Z)	$\pi_{ij+}\pi_{++k}$	$n_{ij+}n_{++k}/n$
(XY, XZ)	$\pi_{ij+}\pi_{i+k}/\pi_{i++}$	$n_{ij+}n_{i+k}/n_{i++}$
(XY, XZ, YZ)	$\psi_{ij}\phi_{jk}\omega_{ik}$	iterative (IPF)
(XYZ)	none	n_{ijk}

(other models by symmetry, e.g. (XZ, YZ): $\hat{\mu}_{ijk} = n_{i+k}n_{+jk}/n_{++k}$).

Proof. Take (XZ, YZ); the others are identical in structure.

Step 1. By Theorem 8.6, under the model $\mu_{ijk} = \mu_{i+k}\mu_{+jk}/\mu_{++k}$.

Step 2. Define $\hat{\mu}_{ijk} = n_{i+k}n_{+jk}/n_{++k}$. It has the product form $A_{ik}B_{jk}$, so it satisfies the model.

Step 3. Summing over j: $\hat{\mu}_{i+k} = n_{i+k}n_{++k}/n_{++k} = n_{i+k}$; summing over i: $\hat{\mu}_{+jk} = n_{+jk}$. Thus it matches the sufficient margins (8.14).

Step 4. By Birch's theorem it is the MLE. For (X, Y, Z), (XY, Z), (XY, XZ) the displayed fits likewise have the factorized form and reproduce the sufficient margins (e.g. for (XY, XZ): $\sum_k n_{ij+}n_{i+k}/n_{i++} = n_{ij+}$ and $\sum_j n_{ij+}n_{i+k}/n_{i++} = n_{i+k}$). For (XYZ), $\hat{\mu} = \mathbf{n}$ trivially. ■

Models with closed-form estimates are called *direct* (or *decomposable*: fitted values are products and ratios of fitted margins). Unsaturated models containing all two-factor terms, such as (XY, XZ, YZ), have no direct estimates. Software uses iterative methods for all models anyway.

8.6.5 Connection between multinomial and Poisson loglinear models

Theorem 8.21 (Poisson and multinomial give the same estimates)

Write the Poisson model as $\log \mu_i = \alpha + \mathbf{x}_i'\boldsymbol{\beta}$ (α the intercept). Then the Poisson log likelihood decomposes as

$$L(\alpha, \boldsymbol{\beta}) = \Big[\sum_i n_i \log \pi_i(\boldsymbol{\beta})\Big] + [n \log \tau - \tau], \qquad \pi_i(\boldsymbol{\beta}) = \frac{e^{\mathbf{x}_i'\boldsymbol{\beta}}}{\sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}}}, \tag{8.15}$$

where $\tau = \sum_i \mu_i$. The first bracket is the multinomial log likelihood given n, the second the Poisson log likelihood of n. Consequently $\hat{\boldsymbol{\beta}}$ and its asymptotic covariance are the same under Poisson and multinomial sampling.

Proof.

Step 1. $L = \sum n_i \log \mu_i - \sum \mu_i = \sum n_i(\alpha + \mathbf{x}_i'\boldsymbol{\beta}) - \tau = n\alpha + \sum n_i \mathbf{x}_i'\boldsymbol{\beta} - \tau$.

Step 2. Since $\tau = e^{\alpha} \sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}}$, $\alpha = \log \tau - \log \sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}}$. Substituting,

$$L = \sum_i n_i \mathbf{x}_i'\boldsymbol{\beta} - n \log \sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}} + n \log \tau - \tau = \sum_i n_i \log \pi_i(\boldsymbol{\beta}) + (n \log \tau - \tau).$$

Step 3. The map $(\alpha, \boldsymbol{\beta}) \mapsto (\tau, \boldsymbol{\beta})$ is one-to-one onto $(0, \infty) \times \mathbb{R}^p$, and $\boldsymbol{\beta}$ enters only the first bracket. Hence $\hat{\boldsymbol{\beta}}$ maximizes the multinomial log likelihood, $\hat{\tau} = n$, and the information matrix is block diagonal in $(\tau, \boldsymbol{\beta})$, so the $\boldsymbol{\beta}$ block of the inverse information is the same for both sampling schemes. ■

The same argument with several independent multinomial samples works as long as the model contains the term for the margin fixed by the design (e.g. λ^{XZ} when $\{n_{i+k}\}$ are fixed), which

forces $\hat{\mu}_{i+k} = n_{i+k}$; this is Theorem 8.16. For a single multinomial sample, $\widehat{\text{cov}}(\hat{\boldsymbol{\beta}}) = \{\mathbf{X}'[\text{diag}(\hat{\boldsymbol{\mu}}) - \hat{\boldsymbol{\mu}}\hat{\boldsymbol{\mu}}'/n]\mathbf{X}\}^{-1}$, where $\mathbf{X}$ omits the intercept column; it equals the Poisson covariance $\{\mathbf{X}'\,\text{diag}(\hat{\boldsymbol{\mu}})\mathbf{X}\}^{-1}$ with the row and column for α deleted.

Chapter 9. Building and Extending Loglinear–Logit Models

This chapter turns the loglinear models of Chapter 8 into a working toolkit: association graphs that display conditional independence and tell us when a table may be collapsed (Section 9.1), model selection and comparison (Section 9.2), residual diagnostics (Section 9.3), models that use the ordering of categories (Section 9.4), and Poisson regression for rates (Section 9.5).

9.1 Association graphs and collapsibility

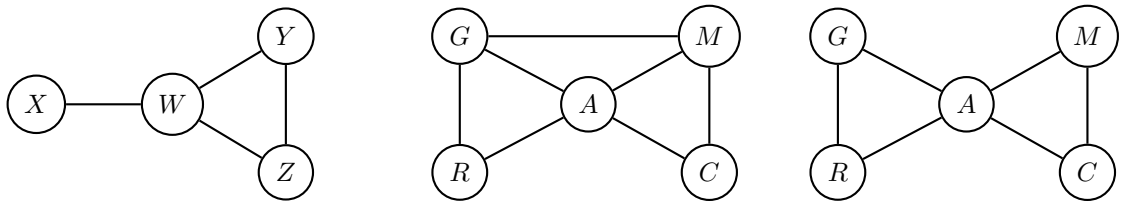
Motivation. With four or five variables it is hard to read conditional independence off a model formula. Drawing the model as a graph turns these questions into questions about paths, and it also tells us when a table can be collapsed over some variables without changing the associations of interest.

9.1.1 Association graphs

Definition (Association graph, path, separation)

The *association graph* of a loglinear model has a vertex for each variable and an edge between two variables if and only if the model contains a term involving both, i.e. if they are conditionally associated. A *path* is a sequence of edges leading from one variable to another. A set S of variables *separates* X and Y if every path between X and Y passes through a vertex of S.

Models with the same pairwise associations have the same graph; e.g. (WX, WY, WZ, YZ) and (WX, WYZ) share Figure 11.



(WX, WY, WZ, YZ) model 6: $(AC, AM, CM, AG, AR, GM, GR)$ model 7: (AC, AM, CM, AG, AR, GR)

Figure 11: Association graphs. Left: book Figure 9.1; W separates X from $\{Y, Z\}$. Middle and right: two models for the five-way student survey (Example 9.5); in model 7, A separates $\{G, R\}$ from $\{C, M\}$.

Theorem 9.1 (Separation implies conditional independence)

In a hierarchical loglinear model, if a set S of variables separates X and Y in the association graph, then X and Y are conditionally independent given S (after marginalizing over all other variables). More generally, if S separates sets A_0 and B_0, then $A_0 \perp B_0 \mid S$.

Proof. [Supplement]

Step 1. Let V be the set of all variables, A the set of vertices reachable from X by paths avoiding S, and $B = V \setminus (A \cup S)$. Since S separates X and Y, $Y \in B$.

Step 2. Every term T of the model lies in $A \cup S$ or in $B \cup S$. Indeed, if T contained $a \in A$ and $b \in B$, then the hierarchical model would contain the two-factor term $\{a, b\} \subseteq T$, so a–b would be an edge and b would be reachable from X avoiding S, i.e. $b \in A$, a contradiction.

Step 3. Grouping terms accordingly, $\mu(\mathbf{x}) = f(\mathbf{x}_A, \mathbf{x}_S)\, g(\mathbf{x}_B, \mathbf{x}_S)$ (assign terms inside S to f). Hence, for fixed $\mathbf{x}_S$,

$$P(\mathbf{x}_A, \mathbf{x}_B \mid \mathbf{x}_S) = \frac{f(\mathbf{x}_A, \mathbf{x}_S)\, g(\mathbf{x}_B, \mathbf{x}_S)}{\sum_{\mathbf{a}} f(\mathbf{a}, \mathbf{x}_S) \sum_{\mathbf{b}} g(\mathbf{b}, \mathbf{x}_S)},$$

which factors into a function of $\mathbf{x}_A$ times a function of $\mathbf{x}_B$: $\mathbf{x}_A \perp \mathbf{x}_B \mid \mathbf{x}_S$.

Step 4. Summing over the coordinates of A other than X and the coordinates of B other than Y preserves the product form, so $X \perp Y \mid S$. The statement for sets $A_0 \subseteq A$, $B_0 \subseteq B$ is the same computation. ■

In Figure 11 (left), W separates X and Y, so $X \perp Y \mid W$ (and also given $\{W, Z\}$); likewise $X \perp Z \mid W$.

9.1.2 Collapsibility in three-way contingency tables

Theorem 9.2 (Collapsibility conditions)

In a three-way table, the XY marginal odds ratios equal the XY conditional odds ratios (at every level of Z) if Z and X are conditionally independent given Y, or Z and Y are conditionally independent given X; that is, under models (XY, YZ) or (XY, XZ) (or simpler models).

Proof. [Supplement]

Step 1. Under (XY, XZ), by Lemma 8.5, $\mu_{ijk} = A_{ij} C_{ik}$. The conditional odds ratio at level k is

$$\frac{A_{ij} C_{ik}\, A_{i'j'} C_{i'k}}{A_{ij'} C_{ik}\, A_{i'j} C_{i'k}} = \frac{A_{ij} A_{i'j'}}{A_{ij'} A_{i'j}},$$

free of k. The marginal table is $\mu_{ij+} = A_{ij} C_{i+}$, and the row factors $C_{i+}, C_{i'+}$ cancel in the odds ratio, giving the same value.

Step 2. Under (XY, YZ), $\mu_{ijk} = A_{ij} B_{jk}$ and $\mu_{ij+} = A_{ij} B_{j+}$; now column factors cancel, with the same conclusion. ■

Example 9.3 (Collapsibility in the student survey)

In (AM, CM) (graph A–M–C), C is conditionally independent of A, so the AM association is collapsible over C: fitted conditional and marginal AM odds ratios both equal 61.9 (Example 8.8); similarly CM (25.1). The AC association is not collapsible, because M is linked to both A and C: conditional AC odds ratio 1.0, marginal 2.7. In (AC, AM, CM) no collapsibility condition holds, and every pair has quite different conditional and marginal associations.

9.1.3 Collapsibility and logit models

In a multicenter clinical trial with binary treatment x, binary response Y and centers Z, the model $\operatorname{logit} P(Y = 1 \mid X = i, Z = k) = \alpha + \beta x_i + \beta_k^Z$ corresponds to (XY, XZ, YZ), so the common conditional odds ratio e^{β} typically differs from the odds ratio of the marginal 2×2 table. The simpler model without center effects, $\operatorname{logit} = \alpha + \beta x_i$, states $Y \perp Z \mid X$; it is equivalent to (XY, XZ), which is collapsible over Z. So when center effects are negligible the treatment effect is approximately the marginal odds ratio.

9.1.4 Collapsibility and association graphs for multiway tables

Theorem 9.4 (Parametric collapsibility, Bishop et al.)

Partition the variables into disjoint sets A, B, C such that B separates A and C in the association graph. After collapsing the table over the variables in C, the parameters relating variables within A, and those relating variables in A to variables in B, are unchanged.

Proof. [Supplement]

Step 1. No term contains variables from both A and C (that would create an A–C edge, a path avoiding B). Put all terms meeting A into f and all others into g; then each term of f lies in $A \cup B$ and each term of g lies in $B \cup C$:

$$\mu(\mathbf{a}, \mathbf{b}, \mathbf{c}) = f(\mathbf{a}, \mathbf{b})\, g(\mathbf{b}, \mathbf{c}), \qquad \log f = \sum_{T \cap A \neq \emptyset} \lambda^T.$$

Step 2. Collapsing over C: $\mu(\mathbf{a}, \mathbf{b}) = f(\mathbf{a}, \mathbf{b})\, G(\mathbf{b})$ with $G(\mathbf{b}) = \sum_{\mathbf{c}} g(\mathbf{b}, \mathbf{c})$. Thus

$$\log \mu(\mathbf{a}, \mathbf{b}) = \sum_{T \cap A \neq \emptyset} \lambda^T + \log G(\mathbf{b}).$$

Step 3. $\log G(\mathbf{b})$ decomposes into terms involving only variables of B. Since the decomposition of $\log \mu(\mathbf{a}, \mathbf{b})$ into main-effect and interaction terms is unique under fixed (e.g. baseline) constraints, every term involving a variable of A in the collapsed table equals the corresponding λ^T of the full model. ■

For (WX, WY, WZ, YZ): with $A = \{X\}$, $B = \{W\}$, $C = \{Y, Z\}$, the WX association is unchanged after collapsing over Y, Z; with $A = \{Y, Z\}$, $B = \{W\}$, $C = \{X\}$, the associations among W, Y, Z are unchanged after collapsing over X. A variable independent of all others can be collapsed over without affecting other terms. When B has several variables the parameters are unchanged but their ML estimates may differ slightly; estimates also agree ("commutativity of fitting and collapsing") if the model contains the highest-order term relating the variables of B to each other.

9.2 Model selection and comparison

Motivation. Even with five variables there are thousands of hierarchical loglinear models. We need principles (which terms must be included) and tools (likelihood-ratio comparisons of nested models) to find a model that fits well but is simple enough to interpret.

9.2.1 Considerations in model selection

A model should be complex enough to fit well yet simple to interpret, smoothing rather than overfitting. Guidelines:

(i) Distinguish responses from explanatory variables and concentrate on terms linking responses and linking explanatory variables to responses.

(ii) Include the most general interaction among the explanatory variables, and any margin fixed by the sampling design; then the likelihood equations reproduce these totals. For Table 8.8 with responses I, S and predictors G, L, the model should contain GL (so the fit has 20,629 urban women as observed), i.e. be at least as complex as (GL, S, I).

(iii) With a single categorical response, use the equivalent logit model.

(iv) For exploration, fit the model with all single-factor terms, then all two-factor terms, then all three-factor terms, and so on, to locate the needed complexity; automatic searches (backward elimination) should be used with care.

9.2.2 Model building for the Dayton student survey

Example 9.5 (Five-way student survey)

The survey of Example 8.8 also recorded gender (G) and race (R):

		Race = white				Race = other			
		Female		Male		Female		Male	
A	C	M=yes	no	yes	no	yes	no	yes	no
yes	yes	405	268	453	228	23	23	30	19
yes	no	13	218	28	201	2	19	1	18
no	yes	1	17	1	17	0	1	1	8
no	no	1	117	1	133	0	12	0	17

Treating A, C, M as responses and G, R as explanatory, every model contains GR.

Model	G^2	df
1. Mutual independence $+ GR$	1325.1	25
2. Homogeneous association (all two-factor terms)	15.3	16
3. All three-factor terms	5.3	6
4g. (2) $- CR$	15.8	17
5. $(AC, AM, CM, AG, AR, GM, GR, MR)$	16.7	18
6. $(AC, AM, CM, AG, AR, GM, GR)$	19.9	19
7. (AC, AM, CM, AG, AR, GR)	28.8	20

Model 3 improves on model 2 only by $15.3 - 5.3 = 10.0$ on 10 df, so three-factor terms are dropped. Backward elimination from model 2 (removing each of the 9 candidate pairs, models 4a–4i, e.g. removing AC gives 201.2 and removing CM gives 513.5) first removes CR (increase 0.5, df 1), then CG (+0.9), then MR (+3.2); removing AG next would add 5.3 ($P = 0.02$), so we stop at model 6. (These P-values are not literal, since the data chose the tests; many counts are small, but G^2 differences remain useful.)

In model 6 (Figure 11, middle) every path from C to $\{G, R\}$ goes through $\{A, M\}$: given alcohol and marijuana use, cigarette use is independent of gender and race. By Theorem 9.4, the AC and CM conditional associations equal those of (AC, AM, CM) fitted to the ACM table collapsed over G, R. Model 7 drops GM; then A separates $\{G, R\}$ from $\{C, M\}$, so all associations among A, C, M equal those of (AC, AM, CM) in the collapsed table. Model 7 fits reasonably ($G^2 = 28.8$, df $= 20$, $\hat{\Delta} = 0.036$), so one may collapse over gender and race to study A, C, M; the full model additionally estimates the effects of G and R on the responses.

9.2.3 Loglinear model comparison statistics

Let M_0 be a special case of M_1, with fitted values $\hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\mu}}_1$.

Theorem 9.6 (Partitioning G^2)

$$G^2(M_0) = G^2(M_1) + G^2(M_0 \mid M_1), \tag{9.1}$$

where the likelihood-ratio statistic for M_0 against M_1 has the form

$$G^2(M_0 \mid M_1) = G^2(M_0) - G^2(M_1) = 2\sum_i \hat{\mu}_{1i} \log \frac{\hat{\mu}_{1i}}{\hat{\mu}_{0i}}, \tag{9.2}$$

which is the G^2 statistic for M_0 with $\hat{\boldsymbol{\mu}}_1$ playing the role of the data.

Proof.

Step 1. $G^2(M) = 2\sum n_i \log(n_i/\hat{\mu}_i)$, so

$$G^2(M_0) - G^2(M_1) = 2\sum_i n_i(\log \hat{\mu}_{1i} - \log \hat{\mu}_{0i}).$$

Step 2. Write $\log \hat{\boldsymbol{\mu}}_1 = \mathbf{X}_1\hat{\boldsymbol{\beta}}_1$ and, since $M_0 \subseteq M_1$, $\log \hat{\boldsymbol{\mu}}_0 = \mathbf{X}_0\hat{\boldsymbol{\beta}}_0 = \mathbf{X}_1\hat{\boldsymbol{\beta}}_0^*$, where $\hat{\boldsymbol{\beta}}_0^*$ is $\hat{\boldsymbol{\beta}}_0$ with zeros appended for the extra parameters. Then

$$2\mathbf{n}'(\log \hat{\boldsymbol{\mu}}_1 - \log \hat{\boldsymbol{\mu}}_0) = 2\,\mathbf{n}'\mathbf{X}_1(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_0^*) = 2\,\hat{\boldsymbol{\mu}}_1'\mathbf{X}_1(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_0^*),$$

using the likelihood equations $\mathbf{X}_1'\mathbf{n} = \mathbf{X}_1'\hat{\boldsymbol{\mu}}_1$ for M_1 (Theorem 8.18). This is $2\sum \hat{\mu}_{1i}\log(\hat{\mu}_{1i}/\hat{\mu}_{0i})$.

Step 3. $G^2(M_0 \mid M_1)$ has the G^2 form. (Note also $\sum \hat{\mu}_{0i} = \sum \hat{\mu}_{1i} = n$ when both models contain an intercept.) Equation (9.1) is the definition rearranged. ■

$G^2(M_0 \mid M_1)$ depends on the data only through $\hat{\boldsymbol{\mu}}_1$, i.e. through the sufficient statistics of M_1. The Pearson difference $X^2(M_0) - X^2(M_1)$ need not be nonnegative; the natural Pearson analogue is

$$X^2(M_0 \mid M_1) = \sum_i \frac{(\hat{\mu}_{1i} - \hat{\mu}_{0i})^2}{\hat{\mu}_{0i}}. \tag{9.3}$$

When M_0 holds, both are asymptotically χ^2 with $\mathrm{df} = \mathrm{df}(M_0) - \mathrm{df}(M_1)$, and they behave similarly even in fairly sparse tables (Haberman). When M_1 holds but M_0 does not, $G^2(M_1)$ remains χ^2 while the others grow without bound.

9.2.4 Partitioning chi-squared with model comparisons

Applying (9.1) repeatedly to a nested sequence $M_1 \supset M_2 \supset \cdots$ partitions G^2 into components that are asymptotically independent. Example: for a $2 \times J$ table let M_j state that columns $1, \ldots, j$ form an independence table. Then

$$G^2(M_J) = G^2(M_2) + G^2(M_3 \mid M_2) + \cdots + G^2(M_J \mid M_{J-1}),$$

and substituting the fitted values shows that $G^2(M_j \mid M_{j-1})$ is the G^2 for independence in the 2×2 table whose first column combines columns $1, \ldots, j-1$ and whose second column is column j. This justifies the partitioning of Section 3.3.3.

With s planned tests for nested models, using level $1-(1-\alpha)^{1/s}$ for each keeps the overall asymptotic type I error at most α (the tests being asymptotically independent). For $s = 3$ and $\alpha = 0.05$, use $1 - 0.95^{1/3} = 0.017$ per test.

9.2.5 Identical marginal and conditional tests of independence

When both models are direct, they contain the independence linkages that guarantee collapsibility, and the conditional test reduces to a marginal one (Sundberg, 1975): if two direct models differ

only by one pairwise term, $G^2(M_0 \mid M_1)$ equals G^2 for independence in the marginal table of that pair.

Proposition 9.7 (A conditional test that equals a marginal test)

$G^2[(X,Y,Z) \mid (XY,Z)]$ equals $G^2(X,Y)$ for testing independence in the marginal XY table.

Proof. By Theorem 8.20, $\hat{\mu}_{1,ijk} = n_{ij+}n_{++k}/n$ and $\hat{\mu}_{0,ijk} = n_{i++}n_{+j+}n_{++k}/n^2$, so $\hat{\mu}_1/\hat{\mu}_0 = n\,n_{ij+}/(n_{i++}n_{+j+})$ does not depend on k. By (9.2),

$$G^2 = 2\sum_{i,j,k} \frac{n_{ij+}n_{++k}}{n} \log \frac{n\,n_{ij+}}{n_{i++}n_{+j+}} = 2\sum_{i,j} n_{ij+} \log \frac{n_{ij+}}{n_{i++}n_{+j+}/n},$$

after summing n_{++k}/n over k. This is G^2 for independence in the $\{n_{ij+}\}$ table. ■

This is expected: under (XY,Z) the XY association is collapsible over Z.

9.3 Diagnostics for checking models

Motivation. G^2 tells us whether a model fits; residuals tell us where it fails, cell by cell.

9.3.1 Residuals for loglinear models

Definition (Pearson and standardized residuals)

For cell i the *Pearson residual* is

$$e_i = \frac{n_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}, \qquad \sum_i e_i^2 = X^2. \tag{9.4}$$

The *standardized Pearson residual* is $r_i = e_i/\sqrt{1-\hat{h}_i}$, where $\hat{h}_i$ is the leverage (diagonal of the estimated hat matrix).

Pearson residuals have asymptotic variances below 1 (on average residual df/number of cells); r_i is asymptotically $N(0,1)$ and is preferable. Deviance residuals are an alternative.

9.3.2 Student survey example revisited

For the five-way survey with all two-factor terms, the only large standardized residual is 3.2 (fitted 3.1 for a count of 8); for model 6 it is 3.3 (fitted 2.9) in the same cell: nonwhite males who smoked cigarettes but used neither alcohol nor marijuana. Given the many cells and large n, neither model looks problematic.

9.3.3 Correspondence between loglinear and logit residuals

Proposition 9.8 (Logit versus loglinear Pearson residuals)

For equivalent logit and loglinear models with binary response, the logit-model Pearson residual for a success count equals the loglinear Pearson residual for that cell divided by $\sqrt{1-\hat{\pi}}$; hence it is larger in absolute value.

Proof. Let the cell have count y out of N trials in its binomial row. The fitted values agree: $\hat{\mu} = N\hat{\pi}$ (Theorem 8.16). The logit residual is $(y - N\hat{\pi})/\sqrt{N\hat{\pi}(1-\hat{\pi})}$ and the loglinear residual is $(y-\hat{\mu})/\sqrt{\hat{\mu}} = (y - N\hat{\pi})/\sqrt{N\hat{\pi}}$. Their ratio is $1/\sqrt{1-\hat{\pi}} > 1$. ■

After standardization by their own estimated standard errors, the two kinds of residual are identical, another reason to prefer standardized residuals.

9.4 Modeling ordinal associations

Motivation. Nominal loglinear models ignore category order: permuting rows gives the same fit. For two ordinal variables the saturated model is useless and independence often fails mostly in the corner cells; a single "trend" parameter captures this.

Example 9.9 (Premarital sex and teenage birth control)

1991 General Social Survey. Rows: premarital sex is always wrong (1), almost always wrong (2), wrong only sometimes (3), not wrong at all (4). Columns: teenage birth control should be available, strongly disagree (1) to strongly agree (4). Entries: count, (independence fit), [standardized residual], {L×L fit}.

	SD	D	A	SA
1	81 (42.4) [7.6] {80.9}	68 (51.2) [3.1] {67.6}	60 (86.4) [−4.1] {69.4}	38 (67.0) [−4.8] {29.1}
2	24 (16.0) [2.3] {20.8}	26 (19.3) [1.8] {23.1}	29 (32.5) [−0.8] {31.5}	14 (25.2) [−2.8] {17.6}
3	18 (30.1) [−2.7] {24.4}	41 (36.3) [1.0] {36.1}	74 (61.2) [2.2] {65.7}	42 (47.4) [−1.0] {48.8}
4	36 (70.6) [−6.1] {33.0}	57 (85.2) [−4.6] {65.1}	161 (143.8) [2.4] {157.4}	157 (111.4) [6.8] {155.5}

Independence fits poorly, $G^2(I) = 127.6$, df $= 9$; the corner residuals show a positive trend.

9.4.1 Linear-by-linear association in two-way tables

Definition (Linear-by-linear association (L×L) model)

With ordered row scores $u_1 \le \cdots \le u_I$ and column scores $v_1 \le \cdots \le v_J$,

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \beta u_i v_j, \tag{9.5}$$

the special case $\lambda_{ij}^{XY} = \beta u_i v_j$ of the saturated model: one association parameter instead of $(I-1)(J-1)$. Independence is $\beta = 0$. With $u_i = i$, $v_j = j$ it is the *uniform association* model.

Theorem 9.10 (Odds ratios under L×L)

Under (9.5), for rows $a < c$ and columns $b < d$,

$$\log \frac{\mu_{ab}\mu_{cd}}{\mu_{ad}\mu_{cb}} = \beta(u_c - u_a)(v_d - v_b). \tag{9.6}$$

With unit-spaced scores all local odds ratios equal e^{β}. If $\beta > 0$, then Y tends to increase with X and μ_{ij} exceeds its independence value where X, Y are both high or both low.

Proof.

Step 1. By Theorem 8.3(b) the log odds ratio equals $\lambda_{ab}^{XY} + \lambda_{cd}^{XY} - \lambda_{ad}^{XY} - \lambda_{cb}^{XY}$ with $\lambda_{ij}^{XY} = \beta u_i v_j$:

$$\beta(u_a v_b + u_c v_d - u_a v_d - u_c v_b) = \beta(u_c - u_a)(v_d - v_b).$$

Step 2. For adjacent rows and columns with unit spacing, $(u_{i+1} - u_i)(v_{j+1} - v_j) = 1$.

Step 3. For $\beta > 0$ all odds ratios with $a < c$, $b < d$ exceed 1, which is positive (likelihood-ratio type) dependence. The deviation $\beta u_i v_j$ from independence is largest in absolute value at the corners. ■

The deviation $\beta u_i v_j$ is linear in the column scores within a row (slope βu_i) and linear in the row scores within a column, hence the name. Scores are often midpoints of an underlying continuous scale, or standardized (mean 0, variance 1 under the margins); then β is a log odds ratio for unit standard-deviation changes, and for an underlying bivariate normal with correlation ρ, $\beta \approx \rho/(1-\rho^2)$ ($\approx \rho$ for weak association).

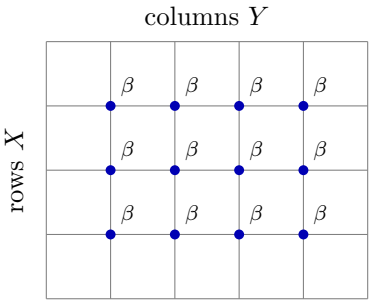


Figure 12: Uniform association: every 2×2 table of adjacent rows and columns has log odds ratio β (book Figure 9.2).

9.4.2 Corresponding logit model for adjacent responses

Proposition 9.11 (Adjacent-categories logit form)

Let $\pi_{j|i} = P(Y = j \mid X = i)$. Under (9.5),

$$\log \frac{\pi_{j+1|i}}{\pi_{j|i}} = (\lambda_{j+1}^{Y} - \lambda_{j}^{Y}) + \beta(v_{j+1} - v_j)u_i,$$

and for unit-spaced $\{v_j\}$ this is $\alpha_j + \beta u_i$: the same linear effect of X on all $J-1$ adjacent-categories logits.

Proof. $\pi_{j+1|i}/\pi_{j|i} = \mu_{i,j+1}/\mu_{ij}$; subtracting (9.5) at j from (9.5) at $j+1$, λ and λ_i^X cancel, leaving the stated expression. ■

9.4.3 Likelihood equations and model fitting

Theorem 9.12 (Likelihood equations for L×L)

The ML fit of (9.5) satisfies

$$\hat{\mu}_{i+} = n_{i+}, \qquad \hat{\mu}_{+j} = n_{+j}, \qquad \sum_{i,j} u_i v_j \hat{\mu}_{ij} = \sum_{i,j} u_i v_j n_{ij}.$$

Hence the fitted and observed distributions have the same marginal means and variances of the scores and the same correlation between them. The residual df are $IJ - I - J$.

Proof.

Step 1. The Poisson log likelihood is

$$L = n\lambda + \sum_i n_{i+}\lambda_i^X + \sum_j n_{+j}\lambda_j^Y + \beta \sum_{i,j} u_i v_j n_{ij} - \sum_{i,j} \exp(\lambda + \lambda_i^X + \lambda_j^Y + \beta u_i v_j).$$

Differentiating with respect to λ_i^X, λ_j^Y and β and equating to zero gives the three sets of equations (the sufficient statistics are the margins and $\sum u_i v_j n_{ij}$).

Step 2. Dividing by n, the fitted proportions $\hat{\pi}_{ij}$ and the sample proportions p_{ij} have equal margins, hence equal means and variances of u and of v, and equal $\sum u_i v_j \hat{\pi}_{ij} = \sum u_i v_j p_{ij}$, hence equal covariance and correlation.

Step 3. The model has $1 + (I-1) + (J-1) + 1$ parameters, so df $= IJ - (I + J)$, unsaturated except for 2×2 tables. ■

The fitted counts therefore reproduce the trend of the data. Fitting uses Newton–Raphson.

9.4.4 Sex opinions example

Example 9.13 (L×L fit to Example 9.9)

With scores $1, 2, 3, 4$ for both variables, $G^2(\text{L}\times\text{L}) = 11.5$, df $= 8$, a dramatic improvement over 127.6 on 9 df, especially in the corners. $\hat{\beta} = 0.286$ (SE 0.028): local odds ratio $e^{0.286} = 1.33$, 95% CI $\exp(0.286 \pm 1.96 \times 0.028) = (1.26, 1.41)$. Nonlocal odds ratios are larger: for the four corner cells, by (9.6),

$$\exp[0.286(4-1)(4-1)] = 13.1 = \frac{80.9 \times 155.5}{29.1 \times 33.0}.$$

Rescaled equally spaced scores (e.g. rows $2, 4, 6, 8$) give the same fit with $\hat{\beta}$ rescaled (0.143). Scores $1, 2, 4, 5$ (treating categories 2 and 3 as farther apart) give $G^2 = 8.8$, $\hat{\beta} = 0.146$. Standardized scores give $\hat{\beta} = 0.374$, and solving $\beta = \rho/(1-\rho^2)$ gives $\hat{\rho} = 0.333$ for an underlying bivariate normal.

9.4.5 Directed ordinal test of independence

Within L×L, independence is $H_0 : \beta = 0$, tested by

$$G^2(I \mid \text{L} \times \text{L}) = G^2(I) - G^2(\text{L} \times \text{L}), \qquad \text{df} = 1.$$

For Example 9.9: $127.6 - 11.5 = 116.1$; the Wald statistic $(\hat{\beta}/\text{SE})^2 = 102.5$; the score statistic is the correlation statistic $M^2 = (n-1)r^2 = 112.6$ of Section 3.4.1.

Why the ordinal test is more powerful

When L×L holds, $G^2(I \mid \mathrm{L}\times\mathrm{L})$ and $G^2(I)$ have (asymptotically) the same noncentrality, but the former has df $= 1$ instead of $(I-1)(J-1)$. For a fixed noncentrality, the power of a χ^2 test decreases as df increase, so the directed test is more powerful, and the advantage grows with I and J.

9.5 Poisson regression for rates

Motivation. When events are counted over different amounts of exposure (time, space, population), the rate, not the raw count, is the natural response. A loglinear model with an offset models the rate while keeping the Poisson count as the observation.

9.5.1 Analyzing rates using loglinear models with offsets

Proposition 9.14 (Rate model with offset)

For counts n_i with exposure (index) t_i, the model

$$\log(\mu_i/t_i) = \alpha + \beta x_i \tag{9.7}$$

is the Poisson loglinear model $\log \mu_i = \log t_i + \alpha + \beta x_i$, in which $\log t_i$ is an offset (a predictor whose coefficient is fixed at 1). Equivalently $\mu_i = t_i \exp(\alpha + \beta x_i)$: the mean is proportional to the exposure, and e^β is the multiplicative effect of a unit increase in x on the rate.

Proof. Add $\log t_i$ to both sides of (9.7) and exponentiate. The ratio of rates at $x+1$ and x is $e^{\alpha+\beta(x+1)}/e^{\alpha+\beta x} = e^\beta$. ■

The identity link $\mu_i/t_i = \alpha + \beta x_i$, i.e. $\mu_i = \alpha t_i + \beta x_i t_i$, needs no offset: it is a Poisson GLM with identity link, predictors t_i and $x_i t_i$ and no intercept. Effects are additive rather than multiplicative; fitting may fail if negative fitted values arise.

9.5.2 Modeling death rates for heart valve operations

Example 9.15 (Heart valve replacement)

109 patients, classified by valve type and age; time at risk is the total follow-up (months) of all patients in the cell, those who died and those censored.

Age		Aortic	Mitral
< 55	Deaths	4	1
	Time at risk	1259	2082
	Death rate	0.0032	0.0005
55+	Deaths	7	9
	Time at risk	1417	1647
	Death rate	0.0049	0.0055

A binary model for the $2 \times 2 \times 2$ table (died yes/no) would be wrong, since patients had different exposures (3 to 97 months). With dummies a (age 55+) and v (mitral),

$$\log(\mu_{ij}/t_{ij}) = \alpha + \beta_1 a_i + \beta_2 v_j. \tag{9.8}$$

Estimates: $\hat{\beta}_1 = 1.221$ (SE 0.514), $\hat{\beta}_2 = -0.330$ (SE 0.438). Given valve type, the older group's death rate is $e^{1.221} = 3.4$ times the younger group's; 95% Wald CI $\exp(1.221 \pm 1.96 \times 0.514) = (1.2, 9.3)$, likelihood-ratio CI $(1.3, 10.4)$. Fit: $G^2 = 3.2$, $X^2 = 3.1$, df $= 1$; the model without valve type has $G^2 = 3.8$ (df 2); models without age fit poorly. Fitted death rates: < 55: 0.0018 (aortic), 0.0013 (mitral); 55+: 0.0062, 0.0044. The identity-link model fits with $G^2 = 1.1$ (df 1); its $\hat{\beta}_1 = 0.0040$ (SE 0.0014) is an additive difference in death rates. Only 21 of 109 patients died, so estimates are imprecise, but every patient contributes through the times at risk.

Chapter 10. Models for Matched Pairs

Matched pairs (the same subject measured twice, a case matched with a control, two raters judging one slide) give dependent responses whose cross-classification is a square $I \times I$ table. We study marginal models, subject-specific (conditional) models, symmetry-type loglinear models and agreement.

10.1 Comparing dependent proportions

Motivation. Two proportions measured on the same subjects are correlated, so the independent-sample variance is wrong; the correct test of equal margins uses only the discordant pairs.

For n pairs let π_{ab} be the probability of outcome a for observation 1 and b for observation 2, n_{ab} the count, $p_{ab} = n_{ab}/n$, and $\{n_{ab}\} \sim \text{multinomial}(n; \{\pi_{ab}\})$.

Definition (Marginal homogeneity)

The square table satisfies **marginal homogeneity** if $\pi_{a+} = \pi_{+a}$ for all a.

For 2×2 tables $\pi_{1+} - \pi_{+1} = \pi_{12} - \pi_{21}$, so marginal homogeneity means $\pi_{12} = \pi_{21}$.

10.1.1 Inference for dependent proportions

Let $\delta = \pi_{+1} - \pi_{1+}$ and $d = p_{+1} - p_{1+} = p_{21} - p_{12}$.

Theorem 10.1 (Variance of a difference of dependent proportions)

Under multinomial sampling,

$$\begin{aligned} n\,\mathrm{Var}(d) &= \pi_{1+}(1-\pi_{1+}) + \pi_{+1}(1-\pi_{+1}) - 2(\pi_{11}\pi_{22} - \pi_{12}\pi_{21}) \\ &= (\pi_{12}+\pi_{21}) - (\pi_{12}-\pi_{21})^2. \end{aligned}$$

Hence $d \pm z_{\alpha/2}\,\hat\sigma(d)$ with $\hat\sigma^2(d) = \big[(p_{12}+p_{21}) - (p_{12}-p_{21})^2\big]/n$ is a large-sample confidence interval for δ.

Proof.

Step 1. Multinomial moments: $n\,\mathrm{Var}(p_{ab}) = \pi_{ab}(1-\pi_{ab})$, $n\,\mathrm{Cov}(p_{ab}, p_{cd}) = -\pi_{ab}\pi_{cd}$ for distinct cells. Each margin is binomial, and

$$n\,\mathrm{Cov}(p_{11}+p_{21},\, p_{11}+p_{12}) = \pi_{11}(1-\pi_{11}-\pi_{12}-\pi_{21}) - \pi_{12}\pi_{21} = \pi_{11}\pi_{22} - \pi_{12}\pi_{21},$$

so $\mathrm{Var}(p_{+1}) + \mathrm{Var}(p_{1+}) - 2\,\mathrm{Cov}(p_{+1}, p_{1+})$ gives the first line.

Step 2. Directly, $n\,\mathrm{Var}(p_{21}-p_{12}) = \pi_{21}(1-\pi_{21}) + \pi_{12}(1-\pi_{12}) + 2\pi_{12}\pi_{21}$, which is the second line.

Step 3. By the multivariate CLT and consistency of $\hat\sigma(d)$, Slutsky's theorem gives the interval. ■

Under $H_0 : \delta = 0$ the variance becomes $(\pi_{12}+\pi_{21})/n$, estimated by $\hat\sigma_0^2(d) = (n_{12}+n_{21})/n^2$.

Theorem 10.2 (McNemar's test is the score test)

The score statistic for $H_0 : \pi_{12} = \pi_{21}$ in the multinomial model is z_0^2, where

$$z_0 = \frac{d}{\hat{\sigma}_0(d)} = \frac{n_{21} - n_{12}}{(n_{21} + n_{12})^{1/2}}; \tag{10.1}$$

z_0^2 is asymptotically χ_1^2 under H_0 (**McNemar's test**).

Proof.

Step 1. Reparametrize by $(\pi_{11}, \pi_{22}, \sigma, q)$ with $\sigma = \pi_{12} + \pi_{21}$, $q = \pi_{21}/\sigma$, subject only to $\pi_{11} + \pi_{22} + \sigma = 1$; H_0 is $q = \frac{1}{2}$. With $n^* = n_{12} + n_{21}$,

$$L = n_{11} \log \pi_{11} + n_{22} \log \pi_{22} + n^* \log \sigma + n_{21} \log q + n_{12} \log(1 - q),$$

and q separates from the constrained parameters, so the information is block diagonal and the score test uses only the q-block.

Step 2. $U(q) = n_{21}/q - n_{12}/(1-q)$ and $\mathcal{I}(q) = n\sigma/[q(1-q)]$. At $q = \frac{1}{2}$, $\hat{\sigma} = n^*/n$: $U = 2(n_{21} - n_{12})$, $\mathcal{I} = 4n^*$, and $U^2/\mathcal{I} = (n_{21} - n_{12})^2/n^* = z_0^2$. Also $d/\hat{\sigma}_0(d) = [(n_{21} - n_{12})/n]/[(n^*)^{1/2}/n]$ gives (10.1); the χ_1^2 limit is standard score asymptotics. ■

The diagonal counts do not affect the test, though all pairs enter d and its standard error.

10.1.2 Prime Minister approval rating example

Example 10.3 (Prime Minister approval)

1600 British voters rated the Prime Minister twice, six months apart: $(n_{11}, n_{12}, n_{21}, n_{22}) = (794, 150, 86, 570)$ (1 = approve). Then $p_{1+} = 0.59$, $p_{+1} = 0.55$, $\hat{\sigma}(d) = 0.0095$, and a 95% interval for $\pi_{+1} - \pi_{1+}$ is $(-0.06, -0.02)$. McNemar's $z_0 = (86 - 150)/236^{1/2} = -4.17$: approval dropped.

10.1.3 Increased precision with dependent samples

Independent samples have $n \operatorname{Var}(\hat{\pi}_1 - \hat{\pi}_2) = \pi_1(1 - \pi_1) + \pi_2(1 - \pi_2)$, the first form of Theorem 10.1 without the covariance term. Matched data usually have positive dependence ($\pi_{11}\pi_{22} > \pi_{12}\pi_{21}$), which reduces $\operatorname{Var}(d)$: in Example 10.3 independent samples would give SE 0.0175 instead of 0.0095.

10.1.4 Small-sample test comparing matched proportions

Proposition 10.4 (Exact conditional test)

Conditional on $n^* = n_{12} + n_{21}$, $n_{21} \sim \text{bin}(n^*, \pi_{21}/(\pi_{12} + \pi_{21}))$. Under $H_0 : \pi_{12} = \pi_{21}$ this is $\text{bin}(n^*, \frac{1}{2})$, and the normal approximation to it gives exactly McNemar's z_0.

Proof.

Step 1. A multinomial conditioned on the total of a subset of cells is multinomial on that subset with renormalized probabilities; for the two cells $(1, 2), (2, 1)$ this gives the binomial, with parameter $\frac{1}{2}$ under H_0.

Step 2. With mean $n^*/2$ and variance $n^*/4$, $z = (n_{21} - \frac{1}{2}n^*)/(\frac{1}{4}n^*)^{1/2} = (n_{21} - n_{12})/(n^*)^{1/2} = z_0$. ■

For Example 10.3, $P(\text{bin}(236, \frac{1}{2}) \geq 150) = 0.00002$ (one-sided).

10.1.5 Connection between McNemar and Cochran–Mantel–Haenszel tests

Display each pair as its own 2×2 stratum (rows: occasion; columns: outcome; row totals $(1, 1)$). This $2 \times 2 \times n$ **subject-specific table** contrasts with the **population-averaged table** of pairs. Equal success probability at both occasions for every subject means conditional independence given subject, which implies marginal homogeneity.

Theorem 10.5 (McNemar as a CMH statistic)

The Cochran–Mantel–Haenszel statistic for conditional independence in the $2 \times 2 \times n$ subject-specific table equals $(n_{21} - n_{12})^2/(n_{21} + n_{12})$.

Proof.

Step 1. Let $c_k \in \{0, 1, 2\}$ be the number of successes in stratum k. The hypergeometric moments are $E(n_{11k}) = c_k/2$ and $\text{Var}(n_{11k}) = c_k(2 - c_k)/4$.

Step 2. Concordant strata ($c_k = 0, 2$) have zero variance and $n_{11k} = E(n_{11k})$, so they contribute nothing.

Step 3. Discordant strata have variance $\frac{1}{4}$ and $n_{11k} - \frac{1}{2} = +\frac{1}{2}$ (n_{12} strata) or $-\frac{1}{2}$ (n_{21} strata), so

$$\text{CMH} = \frac{\left[\frac{1}{2}(n_{12} - n_{21})\right]^2}{\frac{1}{4}(n_{12} + n_{21})} = \frac{(n_{21} - n_{12})^2}{n_{21} + n_{12}}.$$

■

With more categories or occasions, generalized CMH tests with one stratum per subject test marginal homogeneity (Section 11.1.4).

10.2 Conditional logistic regression for binary matched pairs

Motivation. A logit model with one parameter per subject, with those parameters conditioned away, yields the estimator n_{21}/n_{12} of a subject-specific odds ratio.

10.2.1 Marginal versus conditional models for matched pairs

Let (Y_1, Y_2) be a subject's pair (1 = success) and $x_1 = 0$, $x_2 = 1$.

Definition (Marginal and conditional models)

A **marginal** (population-averaged) model describes the margins, e.g. $\text{logit}\, P(Y_t = 1) = \alpha + \beta x_t$. A **conditional** (subject-specific) model has link $P(Y_{it} = 1) = \alpha_i + \beta x_t$, $i = 1, \ldots, n$.

In the marginal logit model $\hat{\beta} = \log[p_{+1}p_{2+}/(p_{+2}p_{1+})]$; in the conditional model β is the association in the subject-stratified table.

Proposition 10.6 (Subject-specific versus population-averaged effects)

Let α_i be drawn from a distribution F in the conditional model.
(a) With the identity link, $P(Y_2 = 1) - P(Y_1 = 1) = \beta$; the subject-specific and population-

averaged effects coincide.
(b) With the logit link, the population-averaged probabilities are $P(Y_t = 1) = E_F[e^{\alpha+\beta x_t}/(1+e^{\alpha+\beta x_t})]$, which in general is not of the form $e^{a+bx_t}/(1+e^{a+bx_t})$ with $b = \beta$.

Proof.

(a) Average $(\alpha_i + \beta) - \alpha_i = \beta$ over F.

(b) The first statement is total probability. Take $\alpha = \pm c$ with probability $\frac{1}{2}$ each and $\beta > 0$. Then $E[\operatorname{expit}(\alpha)] = \frac{1}{2}$, so $b = \operatorname{logit}\left(\frac{1}{2}[\operatorname{expit}(c+\beta) + \operatorname{expit}(\beta - c)]\right) \to \operatorname{logit}(\frac{1}{2}) = 0$ as $c \to \infty$, while β is fixed; so $b \neq \beta$ in general. ■

10.2.2 A logit model with subject-specific probabilities

The Cox–Rasch model is

$$\operatorname{logit} P(Y_{it} = 1) = \alpha_i + \beta x_t, \qquad x_1 = 0,\ x_2 = 1, \tag{10.2}$$

so each subject's odds of success at occasion 2 are e^{β} times those at occasion 1, with responses independent given the parameters.

Proposition 10.7 (Heterogeneity induces positive association)

Suppose $\alpha_1, \ldots, \alpha_n$ are i.i.d. from F and, given α_i, Y_{i1}, Y_{i2} are independent with the probabilities of (10.2). Then $\operatorname{Cov}(Y_{i1}, Y_{i2}) \geq 0$ for every β, with equality if and only if F is degenerate.

Proof. [Supplement]

Step 1. With $f(a) = \operatorname{expit}(a)$, $g(a) = \operatorname{expit}(a+\beta)$, conditional independence gives $\operatorname{Cov}(Y_{i1}, Y_{i2}) = \operatorname{Cov}(f(\alpha), g(\alpha))$, $\alpha \sim F$.

Step 2. For an independent copy α', f, g strictly increasing give $(f(\alpha) - f(\alpha'))(g(\alpha) - g(\alpha')) \geq 0$, whose expectation is $2\operatorname{Cov}(f(\alpha), g(\alpha))$.

Step 3. Equality forces $\alpha = \alpha'$ a.s., possible for independent copies only if F is degenerate. ■

So the model accounts for within-pair dependence, but its many α_i ruin ordinary ML (Proposition 10.9); the remedy is *conditional ML*.

10.2.3 Conditional ML inference for binary matched pairs

Theorem 10.8 (Conditional ML estimator of the subject-specific odds ratio)

In model (10.2) let $S_i = y_{i1} + y_{i2}$.
(a) S_i is sufficient for α_i. Given $S_i = 0$ or $S_i = 2$ the pair is determined; given $S_i = 1$,

$$P(Y_{i1} = 0, Y_{i2} = 1 \mid S_i = 1) = \frac{e^{\beta}}{1+e^{\beta}}, \qquad P(Y_{i1} = 1, Y_{i2} = 0 \mid S_i = 1) = \frac{1}{1+e^{\beta}}.$$

(b) The conditional likelihood is $\ell_c(\beta) = (e^{\beta})^{n_{21}}/(1+e^{\beta})^{n^*}$ with $n^* = n_{12} + n_{21}$, i.e. $n_{21} \mid n^* \sim \operatorname{bin}(n^*, e^{\beta}/(1+e^{\beta}))$.
(c) The conditional ML estimator and its standard error are

$$\hat{\beta} = \log \frac{n_{21}}{n_{12}}, \qquad \mathrm{SE} = \left(\frac{1}{n_{21}} + \frac{1}{n_{12}}\right)^{1/2}.$$

Proof.

Step 1. The joint mass function is proportional to $\exp\left(\sum_i \alpha_i S_i + \beta \sum_i y_{i2}\right)$, an exponential family with S_i sufficient for α_i.

Step 2. Given $S_i \in \{0, 2\}$ the pair is fixed. Given $S_i = 1$, with $c_i = (1 + e^{\alpha_i})(1 + e^{\alpha_i+\beta})$, $P(0,1) = e^{\alpha_i+\beta}/c_i$ and $P(1,0) = e^{\alpha_i}/c_i$, so $P(0,1 \mid S_i = 1) = e^{\beta}/(1 + e^{\beta})$.

Step 3. Multiplying over the n^* discordant pairs gives $\ell_c(\beta) = e^{\beta n_{21}}/(1 + e^{\beta})^{n^*}$, a binomial likelihood.

Step 4. $\log \ell_c$ is strictly concave with derivative $n_{21} - n^* e^{\beta}/(1 + e^{\beta})$, zero at $e^{\hat{\beta}} = n_{21}/n_{12}$.

Step 5. The information $n^* e^{\beta}/(1 + e^{\beta})^2$ equals $n_{21}n_{12}/n^*$ at $\hat{\beta}$; its inverse is $1/n_{21} + 1/n_{12}$. ■

The conditional test of $\beta = 0$ is the binomial test of Proposition 10.4.

Proposition 10.9 (Inconsistency of unconditional ML)

Treating $\alpha_1, \ldots, \alpha_n$ as fixed parameters, the unconditional ML estimator of β in (10.2) is $2\log(n_{21}/n_{12})$, which converges to 2β.

Proof. [Supplement]

Step 1. A concordant pair is fitted perfectly by $\alpha_i \to \pm\infty$ (likelihood 1 for any β) and drops out.

Step 2. A $(1,0)$ pair has likelihood $p_1(1 - p_2)$, $p_1 = \text{expit}(\alpha_i)$, $p_2 = \text{expit}(\alpha_i + \beta)$; the α_i-score $(1 - p_1) - p_2$ vanishes at $\alpha_i = -\beta/2$, giving $\text{expit}(-\beta/2)^2$. A $(0,1)$ pair profiles to $\text{expit}(\beta/2)^2$.

Step 3. The profile log-likelihood $2n_{21}\log\text{expit}(\beta/2) + 2n_{12}\log\text{expit}(-\beta/2)$ is that of Theorem 10.8 in $\beta/2$, so $\hat{\beta}/2 = \log(n_{21}/n_{12})$, which tends to β by the law of large numbers. ■

10.2.4 Random effects in binary matched-pairs model

Treating $\alpha_i \sim N(\mu, \sigma^2)$ and integrating out gives a three-parameter GLMM (Chapter 12), which also yields $\hat{\beta} = \log(n_{21}/n_{12})$ when the sample log odds ratio is nonnegative.

Proposition 10.10 (Mantel–Haenszel estimator)

Applied to the n subject-specific 2×2 tables, the Mantel–Haenszel estimator of the common odds ratio comparing observation 2 with observation 1 equals n_{21}/n_{12}.

Proof. [Supplement] With rows (observation 2, observation 1) and columns (success, failure), $n_{11k} = y_{k2}$, $n_{12k} = 1 - y_{k2}$, $n_{21k} = y_{k1}$, $n_{22k} = 1 - y_{k1}$, $n_{++k} = 2$, so

$$\hat{\theta}_{MH} = \frac{\sum_k n_{11k}n_{22k}/n_{++k}}{\sum_k n_{12k}n_{21k}/n_{++k}} = \frac{\sum_k y_{k2}(1 - y_{k1})}{\sum_k (1 - y_{k2})y_{k1}} = \frac{n_{21}}{n_{12}}.$$

■

	$Y_2 = 1$	$Y_2 = 0$
$Y_1 = 1$	n_{11}	n_{12}
$Y_1 = 0$	n_{21}	n_{22}

grey: $S_i \in \{0, 2\}$, no information on β

orange: $S_i = 1$, $n_{21} \mid n^* \sim \text{bin}\big(n^*, e^{\beta}/(1 + e^{\beta})\big)$

Figure 10.1. Only the discordant cells carry conditional information about β.

10.2.5 Logistic regression for matched case–control studies

In a matched case–control study each case ($Y = 1$) is matched with a control ($Y = 0$) and both are measured on X. With

$$\operatorname{logit} P(Y_{it} = 1) = \alpha_i + \beta x_{it}, \tag{10.3}$$

e^{β} is an XY odds ratio and hence estimable from retrospective data.

Example 10.11 (Diabetes and myocardial infarction)

144 Navajo MI cases were matched with 144 controls on age and gender. In 37 pairs only the case had diabetes, in 16 only the control, in 9 both and in 82 neither; so $e^{\hat\beta} = 37/16 = 2.3$, $\hat\beta = 0.84$.

10.2.6 Conditional ML for matched pairs with multiple predictors

With p predictors the model is

$$\operatorname{logit} P(Y_{it} = 1) = \alpha_i + \beta_1 x_{1it} + \cdots + \beta_p x_{pit} = \alpha_i + x_{it}'\boldsymbol{\beta}. \tag{10.4}$$

Theorem 10.12 (Conditional likelihood as no-intercept logistic regression)

In model (10.4), for pairs with $S_i = 1$,

$$P(Y_{i1} = 0, Y_{i2} = 1 \mid S_i = 1) = \frac{e^{x_{i2}'\boldsymbol{\beta}}}{e^{x_{i1}'\boldsymbol{\beta}} + e^{x_{i2}'\boldsymbol{\beta}}} = \frac{e^{x_i^{*\prime}\boldsymbol{\beta}}}{1 + e^{x_i^{*\prime}\boldsymbol{\beta}}}, \qquad x_i^* = x_{i2} - x_{i1}.$$

Hence:
(a) the conditional ML estimate of $\boldsymbol{\beta}$ is obtained by fitting an ordinary logistic regression, with no intercept, to the discordant pairs only, with response $y_i^* = 1$ for $(0,1)$ pairs, $y_i^* = 0$ for $(1,0)$ pairs and predictor x_i^*;
(b) a predictor with $x_{ji1} = x_{ji2}$ for all i (a between-pair effect) cancels from the conditional likelihood, so its coefficient cannot be estimated by conditional ML.

Proof.

Step 1. As in Theorem 10.8, the factors e^{α_i} and $c_i = (1 + e^{\alpha_i + x_{i1}'\boldsymbol{\beta}})(1 + e^{\alpha_i + x_{i2}'\boldsymbol{\beta}})$ cancel, leaving $e^{x_{i2}'\boldsymbol{\beta}}/(e^{x_{i1}'\boldsymbol{\beta}} + e^{x_{i2}'\boldsymbol{\beta}})$; divide by $e^{x_{i1}'\boldsymbol{\beta}}$.

Step 2. The conditional likelihood $\prod_{S_i=1} \operatorname{expit}(x_i^{*\prime}\boldsymbol{\beta})^{y_i^*}\left[1 - \operatorname{expit}(x_i^{*\prime}\boldsymbol{\beta})\right]^{1-y_i^*}$ is the no-intercept logistic likelihood, proving (a).

Step 3. If $x_{ji1} = x_{ji2}$ for all i, every x_i^* has jth component 0 and β_j disappears, proving (b). ■

In Example 10.11 (control $t = 1$, case $t = 2$) $y_i^* = 1$ and $x_i^* = -1$ (16 pairs) or $+1$ (37 pairs), again giving $\hat\beta = \log(37/16)$.

10.2.7 Marginal models and conditional models: extensions

Both kinds of model extend to multinomial responses and to matched sets or clusters. By Theorem 10.12(b), conditional ML estimates only within-cluster effects; random-effects models (Chapter 12) have no such restriction.

10.3 Marginal models for square contingency tables

Motivation. For $I > 2$ categories the two margins are compared by models acting directly on them: cumulative logits for ordinal and baseline-category logits for nominal responses.

For a square table with $\pi_{ab} = P(Y_1 = a, Y_2 = b)$, marginal homogeneity is $P(Y_1 = a) = P(Y_2 = a)$ for all a.

10.3.1 Marginal models for ordinal classifications

The cumulative logit model

$$\operatorname{logit} P(Y_t \le j) = \alpha_j + \beta x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.5}$$

($x_1 = 0$, $x_2 = 1$) has proportional odds; $\beta > 0$ means Y_1 tends to be higher, and marginal homogeneity is $\beta = 0$. ML fitting maximizes the multinomial likelihood in $\{\pi_{ab}\}$ subject to constraints on the margins; the fit has df $= 2(I-1) - I = I - 2$.

10.3.2 Premarital and extramarital sex example

Example 10.13 (Premarital and extramarital sex)

In the 1989 GSS, 475 subjects rated premarital and extramarital sex on 1 = always wrong, ..., 4 = not wrong at all. The cumulative marginal proportions are $(0.307, 0.389, 0.611)$ (premarital) and $(0.815, 0.918, 0.987)$ (extramarital). Model (10.5) gives $\hat{\beta} = 2.51$ (SE 0.13): far more permissive views on premarital sex. Marginal homogeneity has $G^2 = 348.1$ (df = 3), model (10.5) $G^2 = 35.1$ (df = 2); see Section 10.4.7 for better models.

10.3.3 Marginal models for nominal classifications

The baseline-category model

$$\log \frac{P(Y_t = j)}{P(Y_t = I)} = \alpha_j + \beta_j x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.6}$$

is saturated for the margins; marginal homogeneity is $\beta_1 = \cdots = \beta_{I-1} = 0$, tested by G^2 or X^2 with df $= I - 1$. A non-iterative alternative uses $d_a = p_{+a} - p_{a+}$ and $\mathbf{d} = (d_1, \ldots, d_{I-1})'$.

Proposition 10.14 (Bhapkar and Stuart statistics)

(a) The covariance matrix $\mathbf{V}$ of $\sqrt{n}\,\mathbf{d}$ has elements

$$v_{ab} = -(\pi_{ab} + \pi_{ba}) - (\pi_{+a} - \pi_{a+})(\pi_{+b} - \pi_{b+}), \qquad a \ne b,$$
$$v_{aa} = \pi_{+a} + \pi_{a+} - 2\pi_{aa} - (\pi_{+a} - \pi_{a+})^2.$$

(b) Under marginal homogeneity the Wald (Bhapkar) statistic $W = n\mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d}$ is asymptotically χ^2_{I-1}. Stuart's statistic $W_0 = n\mathbf{d}'\hat{\mathbf{V}}_0^{-1}\mathbf{d}$ uses the null version $\hat{v}_{ab0} = -(p_{ab} + p_{ba})$, $\hat{v}_{aa0} = p_{+a} + p_{a+} - 2p_{aa}$, and

$$W = \frac{W_0}{1 - W_0/n}.$$

(c) For $I = 2$, W_0 is McNemar's statistic z_0^2.

Proof. [Supplement]

Step 1. $\mathbf{d}$ is the mean of i.i.d. $\mathbf{X}_k$ with $X_{ka} = \mathbb{1}\{Y_{k2} = a\} - \mathbb{1}\{Y_{k1} = a\}$. For $a \neq b$, $E(X_a X_b) = -(\pi_{ab} + \pi_{ba})$, and $E(X_a^2) = \pi_{+a} + \pi_{a+} - 2\pi_{aa}$; subtracting $E(X_a)E(X_b)$ gives (a).

Step 2. By the CLT and consistency of $\hat{\mathbf{V}}$, under H_0 ($E\mathbf{d} = \mathbf{0}$) $W \xrightarrow{d} \chi^2_{I-1}$.

Step 3. $\hat{\mathbf{V}} = \hat{\mathbf{V}}_0 - \mathbf{d}\mathbf{d}'$. By Sherman–Morrison with $q = \mathbf{d}'\hat{\mathbf{V}}_0^{-1}\mathbf{d} = W_0/n$,

$$\mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d} = q + \frac{q^2}{1-q} = \frac{q}{1-q},$$

so $W = W_0/(1 - W_0/n)$.

Step 4. For $I = 2$, $d_1 = p_{21} - p_{12}$ and $\hat{v}_{110} = p_{12} + p_{21}$, so $W_0 = (n_{21} - n_{12})^2/(n_{12} + n_{21})$. ■

For ordered categories a single-df ordinal test is often more powerful.

10.3.4 Migration example

Example 10.15 (Migration 1980–1985)

Region of residence (NE, MW, S, W) in 1980 and 1985 for 55,981 U.S. residents; about 95% lie on the diagonal, and, e.g., 124 moved NE→W but 63 W→NE. Marginal homogeneity has $G^2 = 240.8$ (df = 3), Bhapkar's $W = 236.5$. The marginal proportions $(0.218, 0.260, 0.330, 0.191)$ in 1980 and $(0.213, 0.253, 0.339, 0.194)$ in 1985 differ by -0.005, -0.007, 0.009, 0.003 (each ± 0.001 by Theorem 10.1): significant only because n is huge.

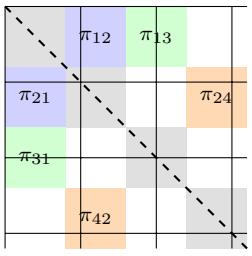
10.4 Symmetry, quasi-symmetry, and quasi-independence

Motivation. Loglinear models for the joint table: quasi-symmetry fits well, has a subject-specific meaning and yields a likelihood-ratio test of marginal homogeneity.

Definition (Symmetry)

An $I \times I$ distribution satisfies **symmetry** if $\pi_{ab} = \pi_{ba}$ for all $a \neq b$.

Symmetry implies marginal homogeneity ($\pi_{a+} = \sum_b \pi_{ba} = \pi_{+a}$); for $I > 2$ the converse fails.



grey: diagonal, fitted exactly by S, QS, QI

mirror cells $(a,b), (b,a)$: equal under S; under QS, $\pi_{ab}/\pi_{ba} = e^{\tau_b - \tau_a}$

Figure 10.2. Mirror pairs of a square table ($I = 4$).

10.4.1 Symmetry as logit and loglinear models

For $\pi_{ab} > 0$, symmetry is $\log(\pi_{ab}/\pi_{ba}) = 0$ $(a < b)$, or, with $\mu_{ab} = n\pi_{ab}$,

$$\log \mu_{ab} = \lambda + \lambda_a + \lambda_b + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}. \tag{10.7}$$

Theorem 10.16 (Fitting and testing symmetry)

For Poisson or multinomial counts:
(a) the ML fit of symmetry is $\hat{\mu}_{ab} = (n_{ab} + n_{ba})/2$ (so $\hat{\mu}_{aa} = n_{aa}$);
(b) the residual df is $I(I-1)/2$;
(c) the Pearson statistic is Bowker's statistic

$$X^2 = \sum_{a<b} \frac{(n_{ab} - n_{ba})^2}{n_{ab} + n_{ba}},$$

which for $I = 2$ is McNemar's z_0^2;
(d) the standardized Pearson residuals are $r_{ab} = (n_{ab} - n_{ba})/(n_{ab} + n_{ba})^{1/2} = -r_{ba}$, and $\sum_{a<b} r_{ab}^2 = X^2$.

Proof.

Step 1. (a) Writing the model as $\log \mu_{ab} = \lambda_{ab}$ with $\lambda_{ab} = \lambda_{ba}$, the Poisson log-likelihood splits into $(n_{ab} + n_{ba}) \log \mu - 2\mu$ for each pair $a < b$ and $n_{aa} \log \mu_{aa} - \mu_{aa}$ for each diagonal cell; maximize each. The multinomial fit is the same.

Step 2. (b) df $= I^2 - I - \binom{I}{2} = I(I-1)/2$.

Step 3. (c) With $m = (n_{ab}+n_{ba})/2$, both cells of a pair have $(n-m)^2 = (n_{ab}-n_{ba})^2/4$, contributing $2(n_{ab} - n_{ba})^2/(4m)$; diagonal cells contribute 0.

Step 4. (d) Each pair is fitted by one parameter with design $(1, 1)'$ and equal weights, so the leverage is $h = \frac{1}{2}$ and $r_{ab} = (n_{ab} - m)/\sqrt{m/2} = (n_{ab} - n_{ba})/(n_{ab} + n_{ba})^{1/2}$; squaring and summing gives (c). ■

10.4.2 Quasi-symmetry

Definition (Quasi-symmetry)

The **quasi-symmetry** (QS) model lets the main effects in (10.7) differ:

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}. \tag{10.8}$$

Symmetry is $\lambda_a^X = \lambda_a^Y$; independence is $\lambda_{ab} = 0$.

Since $\lambda_a^X + \lambda_b^Y = (\lambda_a^X + \lambda_b^X) + (\lambda_b^Y - \lambda_b^X)$, QS can be written

$$\log \mu_{ab} = \lambda'_{ab} + \tau_b, \qquad \lambda'_{ab} = \lambda'_{ba}, \quad \tau_b = \lambda_b^Y - \lambda_b^X, \quad \tau_I = 0. \tag{10.9}$$

Proposition 10.17 (Likelihood equations for quasi-symmetry)

The ML fit of QS satisfies

$$\hat{\mu}_{a+} = n_{a+}, \qquad \hat{\mu}_{+b} = n_{+b}, \qquad \hat{\mu}_{ab} + \hat{\mu}_{ba} = n_{ab} + n_{ba} \quad (a \le b);$$

one of the first two sets is implied by the other two. In particular $\hat{\mu}_{aa} = n_{aa}$, and the residual df is $(I-1)(I-2)/2$.

Proof.

Step 1. Poisson likelihood equations equate the sufficient statistics (row totals, column totals, symmetric sums $n_{ab} + n_{ba}$, $a \le b$) to their expectations; $a = b$ gives $\hat{\mu}_{aa} = n_{aa}$.

Step 2. Given the third set and the row equations, $\hat{\mu}_{+a} = \sum_b (n_{ab} + n_{ba} - \hat{\mu}_{ab}) = n_{a+} + n_{+a} - \hat{\mu}_{a+} = n_{+a}$.

Step 3. Form (10.9) has $I(I+1)/2 + (I-1)$ parameters, so df $= (I-1)(I-2)/2$. ■

QS has the multiplicative form

$$\pi_{ab} = \alpha_a \beta_b \gamma_{ab}, \qquad \gamma_{ab} = \gamma_{ba}, \tag{10.10}$$

with symmetry the case $\alpha_a = \beta_a$.

Theorem 10.18 (Odds-ratio characterization of quasi-symmetry)

Let all $\pi_{ab} > 0$.
(a) QS holds if and only if

$$\frac{\pi_{ab}\pi_{II}}{\pi_{aI}\pi_{Ib}} = \frac{\pi_{ba}\pi_{II}}{\pi_{bI}\pi_{Ia}} \qquad \text{for all } a, b. \tag{10.11}$$

(b) Under QS the local odds ratios satisfy $\theta_{ab} = \theta_{ba}$, where $\theta_{ab} = \pi_{ab}\pi_{a+1,b+1}/(\pi_{a,b+1}\pi_{a+1,b})$.

Proof. [Supplement]

Step 1. ($\Rightarrow$) Under (10.10) the α's and β's cancel, leaving $\gamma_{ab}\gamma_{II}/(\gamma_{aI}\gamma_{Ib})$ and $\gamma_{ba}\gamma_{II}/(\gamma_{bI}\gamma_{Ia})$, equal by symmetry of γ.

Step 2. ($\Leftarrow$) Put $\alpha_a = \pi_{aI}$, $\beta_b = \pi_{Ib}/\pi_{II}$, $\gamma_{ab} = \pi_{ab}\pi_{II}/(\pi_{aI}\pi_{Ib})$; then $\pi_{ab} = \alpha_a\beta_b\gamma_{ab}$ and (10.11) says $\gamma_{ab} = \gamma_{ba}$.

Step 3. (b) Similarly $\theta_{ab} = \gamma_{ab}\gamma_{a+1,b+1}/(\gamma_{a,b+1}\gamma_{a+1,b})$, which is unchanged when a, b are swapped since γ is symmetric. ■

The meaning of QS comes from the subject-specific version of (10.6):

$$\log \frac{P(Y_{it} = j)}{P(Y_{it} = I)} = \alpha_{ij} + \beta_j x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.12}$$

with $x_1 = 0$, $x_2 = 1$, $\alpha_{iI} = \beta_I = 0$, and responses independent given the subject.

Theorem 10.19 (Subject-specific logit models imply quasi-symmetry)

If (10.12) holds and subjects are sampled from any distribution F of $(\alpha_{i1}, \ldots, \alpha_{i,I-1})$, then the population-averaged table satisfies QS with $\tau_j = \lambda_j^Y - \lambda_j^X = \beta_j$ (constraint $\tau_I = 0$).

Proof. [Supplement]

Step 1. With $D_1(\boldsymbol{\alpha}) = \sum_k e^{\alpha_k}$, $D_2(\boldsymbol{\alpha}) = \sum_k e^{\alpha_k + \beta_k}$, conditional independence gives $P(Y_1 = a, Y_2 = b \mid \boldsymbol{\alpha}) = e^{\alpha_a} e^{\alpha_b + \beta_b}/[D_1(\boldsymbol{\alpha}) D_2(\boldsymbol{\alpha})]$.

Step 2. Averaging over F, $\pi_{ab} = e^{\beta_b}\gamma_{ab}$ with $\gamma_{ab} = E_F[e^{\alpha_a + \alpha_b}/(D_1 D_2)] = \gamma_{ba}$. This is (10.10) with $\alpha_a \equiv 1$, i.e. (10.9) with $\tau_b = \beta_b$. ■

Moreover (Conaway 1989) the conditional ML estimates of $\{\beta_j\}$ equal $\hat{\lambda}_j^Y - \hat{\lambda}_j^X$ from the QS fit.

10.4.3 Quasi-independence

Definition (Quasi-independence)

A square table satisfies **quasi-independence** (QI) if the classifications are independent given that the row and column outcomes differ:

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \delta_a \, I(a = b). \tag{10.13}$$

The likelihood equations $\hat{\mu}_{a+} = n_{a+}$, $\hat{\mu}_{+a} = n_{+a}$, $\hat{\mu}_{aa} = n_{aa}$ fit the diagonal exactly; off-diagonal rectangle odds ratios equal 1, and $\text{df} = (I-1)^2 - I$ $(I \geq 3)$.

Proposition 10.20 (Quasi-independence within quasi-symmetry)

QI is the special case of QS in which the γ_{ab}, $a \neq b$, in (10.10) are all equal. For $I = 3$ the two models are equivalent.

Proof. [Supplement]

Step 1. QI is (10.10) with $\gamma_{ab} = 1$ off the diagonal and $\gamma_{aa} = e^{\delta_a}$; conversely a constant off-diagonal γ is absorbed into α, and γ_{aa} into δ_a.

Step 2. For $I = 3$, the positive numbers $c_1 = (\gamma_{12}\gamma_{13}/\gamma_{23})^{1/2}$, $c_2 = (\gamma_{12}\gamma_{23}/\gamma_{13})^{1/2}$, $c_3 = (\gamma_{13}\gamma_{23}/\gamma_{12})^{1/2}$ satisfy $c_a c_b = \gamma_{ab}$ $(a \neq b)$. Then $\pi_{ab} = (\alpha_a c_a)(\beta_b c_b)$ off the diagonal, which is QI; with Step 1 the models coincide. ■

10.4.4 Migration revisited

Example 10.21 (Migration: loglinear models)

For Example 10.15, independence ($G^2 = 125{,}923$) and symmetry ($G^2 = 243.6$, df $= 6$) fail; QI is better ($G^2 = 69.5$, df $= 5$) but misfits NE→S and W→S; QS fits well ($G^2 = 3.0$, df $= 3$). Its effects are $\hat{\tau} = (-0.672, -0.623, 0.122)$ (W baseline): a person's odds of living in the South rather than the West were $e^{0.122} = 1.13$ times higher in 1985 than in 1980.

10.4.5 Marginal homogeneity and quasi-symmetry

Theorem 10.22 (Caussinus)

For a table with all $\pi_{ab} > 0$,

$$\text{quasi-symmetry} + \text{marginal homogeneity} \iff \text{symmetry}.$$

Proof.

Step 1. ($\Leftarrow$) Symmetry is QS with $\alpha_a = \beta_a$ and implies marginal homogeneity.

Step 2. ($\Rightarrow$) Write $\pi_{ab} = \kappa_a \gamma^*_{ab}$ with $\kappa_a = \alpha_a/\beta_a$, $\gamma^*_{ab} = \beta_a\beta_b\gamma_{ab} = \gamma^*_{ba} > 0$. Marginal homogeneity $\kappa_j \sum_b \gamma^*_{jb} = \sum_a \kappa_a \gamma^*_{aj}$ reads

$$\kappa_j = \sum_a \kappa_a \, w_{aj}, \qquad w_{aj} = \frac{\gamma^*_{aj}}{\sum_b \gamma^*_{bj}} > 0, \quad \sum_a w_{aj} = 1.$$

Step 3. For j with $\kappa_j = \min_a \kappa_a$, a positive-weight average equal to the minimum forces all $\kappa_a = \kappa$. Then $\pi_{ab} = \kappa\gamma^*_{ab} = \pi_{ba}$. ■

Corollary 10.23 (Conditional test of marginal homogeneity)

Given that QS holds, marginal homogeneity is equivalent to symmetry, i.e. to $\lambda_a^X = \lambda_a^Y$ (all $\tau_a = 0$). It is tested by

$$G^2(S \mid QS) = G^2(S) - G^2(QS), \qquad \text{df} = \tfrac{I(I-1)}{2} - \tfrac{(I-1)(I-2)}{2} = I - 1.$$

Proof. The equivalence is Theorem 10.22. In (10.9) symmetry is the nested submodel $\tau_1 = \cdots = \tau_{I-1} = 0$, so the deviance difference is the LR statistic, with df from Theorem 10.16(b) and Proposition 10.17. ■

For the migration data $G^2(S \mid QS) = 240.6$ (df $= 3$), close to $G^2 = 240.8$ and $W = 236.5$ of Example 10.15.

10.4.6 Ordinal quasi-symmetry model

With scores $u_1 \le \cdots \le u_I$, the **ordinal quasi-symmetry** model is

$$\log \mu_{ab} = \lambda + \lambda_a + \lambda_b + \beta u_b + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}, \tag{10.14}$$

QS with $\tau_b = \beta u_b$; symmetry is $\beta = 0$. Equivalently

$$\log \frac{\pi_{ab}}{\pi_{ba}} = \beta(u_b - u_a), \qquad a < b, \tag{10.15}$$

a no-intercept logit model for $n_{ab} \mid (n_{ab} + n_{ba})$ with predictor $u_b - u_a$.

Proposition 10.24 (Ordinal quasi-symmetry and marginal homogeneity)

Let $u_1 < \cdots < u_I$ and all $\pi_{ab} > 0$.
(a) The likelihood equations are $\sum_a u_a \hat\mu_{a+} = \sum_a u_a n_{a+}$, $\sum_b u_b \hat\mu_{+b} = \sum_b u_b n_{+b}$ and $\hat\mu_{ab} + \hat\mu_{ba} = n_{ab} + n_{ba}$ $(a \le b)$. The fitted margins need not equal the observed ones, but the fitted row and column means equal the observed means.
(b) Under (10.14), the column mean exceeds the row mean if $\beta > 0$ and is smaller if $\beta < 0$; hence ordinal QS + marginal homogeneity = symmetry, and $H_0 : \beta = 0$ tests marginal homogeneity (df $= 1$).

Proof. [Supplement]

Step 1. (a) The sufficient statistics of $\log \mu_{ab} = \lambda'_{ab} + \beta u_b$ are the symmetric sums and $\sum_b u_b n_{+b}$, giving the second and third sets. Summing $(u_a + u_b)$ times the third set over $a \le b$ gives $\sum_{a,b}(u_a + u_b)\hat\mu_{ab} = \sum_{a,b}(u_a + u_b)n_{ab}$; subtracting the second set gives the first.

Step 2. (b) By (10.15),

$$\sum_{a,b}(u_b - u_a)\pi_{ab} = \sum_{a<b}(u_b - u_a)\pi_{ba}\left(e^{\beta(u_b - u_a)} - 1\right),$$

every term having the sign of β. Equal means force $\beta = 0$, i.e. symmetry; the converse is immediate. ■

10.4.7 Premarital and extramarital sex revisited

Example 10.25 (Premarital and extramarital sex: symmetry models)

For Example 10.13, symmetry fails ($G^2 = 402.2$, df $= 6$); QS ($G^2 = 1.4$, df $= 3$) and ordinal QS with scores $1, \ldots, 4$ ($G^2 = 2.1$, df $= 5$, $\hat{\beta} = -2.86$) fit well. By (10.15), a premarital response x categories more permissive than the extramarital one is $e^{2.86x}$ times as likely as the reverse.

10.4.8 Other ordinal models for square tables

The **conditional symmetry** model (McCullagh 1978)

$$\log(\pi_{ab}/\pi_{ba}) = \tau \qquad \text{for all } a < b \tag{10.16}$$

describes $\pi_{ab} > \pi_{ba}$ for all $a < b$ (or the reverse); symmetry is $\tau = 0$. The **quasi-uniform association** model

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \beta u_a u_b + \delta_a I(a = b) \tag{10.17}$$

adds linear-by-linear association off the diagonal to QI; it is a special case of QS. For Example 10.13, $\hat{\tau} = -4.130$ (SE 0.451) and $\hat{\beta} = 0.632$ (SE 0.106).

10.5 Measuring agreement between observers

Motivation. Agreement between two raters is a different property from association and needs diagonal-oriented models and indices such as kappa.

Example 10.26 (Pathologists' diagnoses)

Two pathologists A and B classified 118 slides into four ordered categories (1 = negative, ..., 4 = invasive carcinoma). The diagonal counts are $(22, 7, 36, 10)$; the most common disagreements are B rating 3 when A rated 2 (14) or 4 (17). Independence fits poorly ($G^2 = 118.0$, df $= 9$), with large positive diagonal residuals.

10.5.1 Agreement: departures from independence

With π_{ab} the probability that A chooses a and B b, $\sum_a \pi_{aa}$ is the agreement probability. Strong agreement requires strong association but not conversely (A might always rate one category higher). Independence is the baseline of chance agreement.

10.5.2 Using quasi-independence to analyze agreement

For two subjects rated a and b by one observer and b, a or a, b by the other, the odds of agreement are

$$\tau_{ab} = \frac{\pi_{aa}\pi_{bb}}{\pi_{ab}\pi_{ba}} = \frac{\mu_{aa}\mu_{bb}}{\mu_{ab}\mu_{ba}}. \tag{10.18}$$

Proposition 10.27 (Agreement odds under QI and QS)

Under QI, $\tau_{ab} = \exp(\delta_a + \delta_b)$. Under QS, $\tau_{ab} = \exp(\lambda_{aa} + \lambda_{bb} - \lambda_{ab} - \lambda_{ba})$.

Proof. [Supplement] In $\log \mu_{aa} + \log \mu_{bb} - \log \mu_{ab} - \log \mu_{ba}$ the main effects $\lambda_a^X + \lambda_a^Y + \lambda_b^X + \lambda_b^Y$ cancel against those of the off-diagonal cells, leaving $\delta_a + \delta_b$ under (10.13) and $\lambda_{aa} + \lambda_{bb} - \lambda_{ab} - \lambda_{ba}$ under (10.8). ■

For Example 10.26, QI has $G^2 = 13.2$ (df $= 5$) and $\hat{\tau}_{23} = e^{2.5} = 12.3$.

10.5.3 Quasi-symmetry and agreement modeling

QS also permits association given disagreement and fits Example 10.26 well ($G^2 = 1.0$, df $= 2$, $\hat{\tau}_{23} = 10.7$). Pairs with $n_{ab} + n_{ba} = 0$ are fitted by 0 and should be removed for correct df. Symmetry fits poorly ($G^2 = 39.2$, df $= 5$), and $G^2(S \mid QS) = 38.2$ (df $= 3$) shows strong marginal heterogeneity.

10.5.4 Kappa measure of agreement

Definition (Cohen's kappa)

With $P_o = \sum_a \pi_{aa}$ and $P_e = \sum_a \pi_{a+}\pi_{+a}$ (agreement expected under independence),

$$\kappa = \frac{\sum_a \pi_{aa} - \sum_a \pi_{a+}\pi_{+a}}{1 - \sum_a \pi_{a+}\pi_{+a}} = \frac{P_o - P_e}{1 - P_e}.$$

$\kappa = 0$ at chance agreement, $\kappa = 1$ at perfect agreement.

Theorem 10.28 (Asymptotic variance of sample kappa)

Under multinomial sampling, $\sqrt{n}(\hat{\kappa} - \kappa) \xrightarrow{d} N(0, \sigma_\kappa^2)$, where

$$\sigma_\kappa^2 = \frac{P_o(1-P_o)}{(1-P_e)^2} + \frac{2(1-P_o)\left[2P_oP_e - \sum_a \pi_{aa}(\pi_{a+} + \pi_{+a})\right]}{(1-P_e)^3} + \frac{(1-P_o)^2\left[\sum_a \sum_b \pi_{ab}(\pi_{b+} + \pi_{+a})^2 - 4P_e^2\right]}{(1-P_e)^4}.$$

The estimated variance of $\hat{\kappa}$ (Fleiss, Cohen and Everitt 1969) substitutes sample proportions and divides by n.

Proof. [Supplement]

Step 1. By the delta method for the multinomial, $\sigma^2 = \sum_{a,b} \pi_{ab} g_{ab}^2 - \left(\sum_{a,b} \pi_{ab} g_{ab}\right)^2$ with $g_{ab} = \partial\kappa/\partial\pi_{ab}$.

Step 2. With $\kappa = 1 - (1-P_o)/(1-P_e)$, $\partial P_o/\partial\pi_{ab} = I(a=b)$ and $\partial P_e/\partial\pi_{ab} = \pi_{+a} + \pi_{b+}$,

$$g_{ab} = \frac{I(a=b)(1-P_e) - (1-P_o)(\pi_{+a} + \pi_{b+})}{(1-P_e)^2}.$$

Step 3. Since $\sum_{a,b} \pi_{ab}(\pi_{+a} + \pi_{b+}) = 2P_e$, $\sum \pi_{ab} g_{ab} = [P_o(1-P_e) - 2P_e(1-P_o)]/(1-P_e)^2$. With $s = \sum_a \pi_{aa}(\pi_{a+} + \pi_{+a})$ and $t = \sum_{a,b} \pi_{ab}(\pi_{+a} + \pi_{b+})^2$,

$$(1-P_e)^4 \sum_{a,b} \pi_{ab} g_{ab}^2 = P_o(1-P_e)^2 - 2(1-P_e)(1-P_o)s + (1-P_o)^2 t.$$

Step 4. Subtracting the square of the first moment,

$$(1-P_e)^4 \sigma_\kappa^2 = P_o(1-P_o)(1-P_e)^2 + 2(1-P_e)(1-P_o)(2P_oP_e - s) + (1-P_o)^2(t - 4P_e^2),$$

which is the stated formula. ■

For Example 10.26, $P_o = 0.636$, $P_e = 0.281$, $\hat{\kappa} = 0.493$ (SE 0.057).

10.5.5 Weighted kappa: quantifying disagreement

Definition (Weighted kappa)

With weights $0 \le w_{ab} \le 1$, $w_{aa} = 1$, $w_{ab} = w_{ba}$,

$$\kappa_w = \frac{\sum_a \sum_b w_{ab}\pi_{ab} - \sum_a \sum_b w_{ab}\pi_{a+}\pi_{+b}}{1 - \sum_a \sum_b w_{ab}\pi_{a+}\pi_{+b}}.$$

Common choices are $w_{ab} = 1 - |a-b|/(I-1)$ and (Fleiss–Cohen) $w_{ab} = 1 - (a-b)^2/(I-1)^2$.

Both kappas depend strongly on the margins, so models of the agreement structure are more informative than a single index.

10.5.6 Extensions to multiple observers

With R observers one can model the $\binom{R}{2}$ pairwise marginal tables simultaneously, use generalized kappas, or use latent-class models (Chapter 13).

Chapter 11. Analyzing Repeated Categorical Response Data

Repeated categorical responses arise over time (longitudinal studies), under several conditions (crossover trials) or within *clusters* (teeth, litters, families); within-cluster observations are dependent. This chapter extends Chapter 10 to T responses with covariates: comparing margins (Section 11.1), ML marginal models (Section 11.2), *generalized estimating equations* (Sections 11.3–11.4) and transitional models (Section 11.5).

11.1 Comparing marginal distributions: multiple responses

Motivation. The T marginal distributions usually matter more than the joint dependence, so we extend the marginal analyses of Chapter 10 to $T > 2$.

11.1.1 Binary marginal models and marginal homogeneity

For binary $(Y_1, \ldots, Y_T)$ the marginal logit model

$$\operatorname{logit} P(Y_t = 1) = \alpha + \beta_t, \qquad t = 1, \ldots, T, \tag{11.1}$$

($\beta_T = 0$) extends Section 10.2.1. Let $\pi_{\mathbf{i}} = P(Y_1 = i_1, \ldots, Y_T = i_T)$ be the cell probabilities of the 2^T table. Model (11.1) is saturated, and **marginal homogeneity** is $\beta_1 = \cdots = \beta_T$; fitting it by ML is awkward because the likelihood involves the 2^T joint probabilities. With $p_{\mathbf{i}} = n_{\mathbf{i}}/n$ and $\hat{\pi}^{MH}$ the constrained ML fit,

$$-2\big[L(\hat{\pi}^{MH}) - L(\mathbf{p})\big] = 2\sum_{\mathbf{i}} n_{\mathbf{i}} \log\left(p_{\mathbf{i}}/\hat{\pi}^{MH}_{\mathbf{i}}\right) \tag{11.2}$$

is asymptotically χ^2_{T-1} under H_0.

11.1.2 Crossover drug comparison example

Example 11.1 (Crossover study of three drugs)

46 subjects used drugs A, B, C, each response favorable or unfavorable. The proportions favorable are $(0.61, 0.61, 0.35)$, and (11.2) equals 5.95 (df $= 2$, $P = 0.05$). A Bonferroni interval ($z = 2.39$) for A vs C is $0.261 \pm 2.39(0.108) = (0.002, 0.520)$ (Theorem 10.1), the same for B vs C.

11.1.3 Modeling margins of a multicategory response

For I categories, the saturated model $\log[P(Y_t = j)/P(Y_t = I)] = \beta_{tj}$ gives a marginal homogeneity test ($\beta_{1j} = \cdots = \beta_{Tj}$) of the form (11.2) with df $= (T-1)(I-1)$. For ordinal responses,

$$\operatorname{logit} P(Y_t \le j) = \alpha_j + \beta_t, \qquad t = 1, \ldots, T, \quad j = 1, \ldots, I-1, \tag{11.3}$$

($\beta_T = 0$) gives a test of $\beta_1 = \cdots = \beta_T$ with df $= T - 1$.

11.1.4 Wald and generalized CMH score tests of marginal homogeneity

Let $p_j(t)$ be the sample proportion in category j at occasion t, $d_j(t) = p_j(t) - \sum_s p_j(s)/T$, and $\mathbf{d}$ the vector of $d_j(t)$, $t \le T-1$, $j \le I-1$. Bhapkar's statistic

$$W = n\,\mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d} \tag{11.4}$$

is asymptotically $\chi^2_{(I-1)(T-1)}$, generalizing Proposition 10.14.

Generalized CMH statistics treat each subject as a $T \times I$ stratum. Conditional independence given subject (exchangeable responses) implies marginal homogeneity; for $I = T = 2$ the test is McNemar's (Theorem 10.5), and for $I = 2$, $T > 2$ it is Cochran's Q.

Theorem 11.2 (Cochran's Q as a generalized CMH statistic)

For binary responses, let R_k be the number of successes of subject k, C_t the number of successes at occasion t and $N = \sum_t C_t = \sum_k R_k$. The generalized CMH statistic (general association) for the $T \times 2 \times n$ subject-stratified table is

$$Q = \frac{T(T-1)\sum_{t=1}^{T}(C_t - N/T)^2}{TN - \sum_k R_k^2},$$

which is asymptotically χ^2_{T-1} under marginal homogeneity. For $T = 2$, $Q = (n_{21}-n_{12})^2/(n_{21}+n_{12})$.

Proof. [Supplement]

Step 1. Under conditional independence with fixed stratum margins, subject k's R_k successes are allocated to the T occasions completely at random, independently over subjects; the CMH statistic is the quadratic form in $\mathbf{C} = (C_1, \ldots, C_T)'$ about its null mean with a generalized inverse of its null covariance.

Step 2. With $c_k = (R_k/T)(1 - R_k/T)$, sampling without replacement gives $E(Y_{kt}) = R_k/T$, $\mathrm{Var}(Y_{kt}) = c_k$, $\mathrm{Cov}(Y_{kt}, Y_{ks}) = -c_k/(T-1)$, i.e. $\mathrm{Cov}(\mathbf{Y}_k) = \frac{T}{T-1}c_k\mathbf{P}$ with $\mathbf{P} = \mathbf{I} - \frac{1}{T}\mathbf{J}$.

Step 3. Summing over subjects, $E(\mathbf{C}) = (N/T)\mathbf{1}$ and $\mathrm{Cov}(\mathbf{C}) = \kappa\mathbf{P}$ with

$$\kappa = \frac{T}{T-1}\sum_k c_k = \frac{TN - \sum_k R_k^2}{T(T-1)}.$$

Step 4. $\mathbf{P}$ is the projection onto $\{\mathbf{1}'\mathbf{v} = 0\}$, its own generalized inverse, and contains $\mathbf{C} - E\mathbf{C}$ in its range, so the quadratic form is $\kappa^{-1}\sum_t (C_t - N/T)^2 = Q$. Since $\mathrm{rank}\,\mathbf{P} = T-1$, the CLT gives the χ^2_{T-1} limit.

Step 5. For $T = 2$, concordant subjects contribute 0 to $2N - \sum_k R_k^2$, which equals $n_{12} + n_{21}$, and $\sum_t (C_t - N/2)^2 = (n_{12} - n_{21})^2/2$; hence $Q = (n_{12} - n_{21})^2/(n_{12} + n_{21})$. ∎

11.2 Marginal modeling: maximum likelihood approach

Motivation. Marginal logit models describe how the margins depend on covariates; ML maximizes the full multinomial likelihood subject to constraints on the margins.

11.2.1 Longitudinal mental depression example

Example 11.3 (Mental depression)

Subjects with mild or severe depression ($s = 0, 1$) were randomized to a standard or new drug ($d = 0, 1$) and classified normal or abnormal at weeks 1, 2, 4 ($t = 0, 1, 2$, $\log_2$ week); $n = 340$. The proportion normal rises over time, faster for the new drug (from 0.18 to 0.83 for severe/new versus 0.21 to 0.46 for severe/standard). Main effects only gives $G^2 = 34.6$ (df = 8); adding drug × time,

$$\text{logit}\, P(Y_t = 1) = \alpha + \beta_1 s + \beta_2 d + \beta_3 t + \beta_4 dt,$$

gives $G^2 = 4.2$ (df = 7), with $\hat{\beta}_3 = 0.48$ (SE 0.12), $\hat{\beta}_4 = 1.01$ (SE 0.18), $\hat{\beta}_1 = -1.29$ (SE 0.14) and $\hat{\beta}_2 = -0.06$ (SE 0.22).

11.2.2 Modeling a repeated multinomial response

For I categories, model the $I - 1$ marginal logits (baseline-category or cumulative) at each t as $\text{logit}_j(t) = \alpha_j + \boldsymbol{\beta}_j' \mathbf{x}_t$, with $\boldsymbol{\beta}_j = \boldsymbol{\beta}$ under proportional odds; check whether effects of $\mathbf{x}$ change over t.

11.2.3 Insomnia example

Example 11.4 (Insomnia clinical trial)

Patients reported time to fall asleep (4 ordered categories) before ($t = 0$) and after ($t = 1$) two weeks of active treatment ($x = 1$) or placebo ($x = 0$). The proportional odds model

$$\text{logit}\, P(Y_t \le j) = \alpha_j + \beta_1 t + \beta_2 x + \beta_3 tx \tag{11.5}$$

has $G^2 = 8.0$ (df = 6), $\hat{\beta}_1 = 1.074$ (SE 0.162), $\hat{\beta}_2 = 0.046$ (SE 0.236), $\hat{\beta}_3 = 0.662$ (SE 0.244): the active-vs-placebo cumulative odds ratio is 1.04 initially and $e^{0.708} = 2.03$ at follow-up.

11.2.4 Comparisons that control for initial response

If the initial margins differ between groups, their marginal changes may differ even with identical conditional follow-up distributions; then compare follow-up responses given the initial response y_1,

$$\text{logit}\, P(Y_2 \le j) = \alpha_j + \beta_1 x + \beta_2 y_1, \tag{11.6}$$

an ordinal analogue of ANCOVA and a simple transitional model (Section 11.5).

11.3 Marginal modeling: generalized estimating equations (GEE) approach

Motivation. ML marginal fitting is impractical for large T or many covariates; GEE needs only the marginal mean, a variance function and a working correlation, and still estimates $\boldsymbol{\beta}$ consistently.

11.3.1 Generalized estimating equation methodology: basic ideas

For $(Y_1, \ldots, Y_T)$ (T may vary by subject), the quasi-likelihood idea is applied to each margin: a link for $\mu_t = E(Y_t)$, a variance function $v(\mu_t)$, and a *working correlation* for $\{Y_t\}$. The estimates solve the **generalized estimating equations** (Liang and Zeger 1986).

Definition (Working correlation structures)

For cluster i with T_i observations, a working correlation matrix $\mathbf{R}(\boldsymbol{\alpha})$ specifies $\mathrm{Corr}(Y_{it}, Y_{is})$:
(i) *independence*: $\mathbf{R} = \mathbf{I}$;
(ii) *exchangeable*: $\mathrm{Corr}(Y_{it}, Y_{is}) = \alpha$ for all $t \neq s$;
(iii) *autoregressive* AR(1): $\mathrm{Corr}(Y_{it}, Y_{is}) = \alpha^{|t-s|}$;
(iv) *unstructured*: a separate α_{ts} for each pair.

The key properties, proved in Section 11.4:

(a) If the mean model is correct, GEE estimators are consistent *even if the working correlation is wrong.*

(b) Naive (model-based) standard errors are then wrong; GEE uses the empirical dependence to compute *robust* (sandwich) standard errors.

Even the independence structure gives good estimates with corrected standard errors; a good working correlation improves efficiency. Compare several structures for sensitivity; exchangeable is a sensible default, unstructured costs efficiency for large T.

Limitations of GEE

There is no likelihood, hence no likelihood-ratio tests; inference uses Wald statistics with the sandwich covariance. With few clusters sandwich SEs tend to be too small and variable, and quasi-score tests (Boos 1992; Rotnitzky and Jewell 1990) may be more trustworthy.

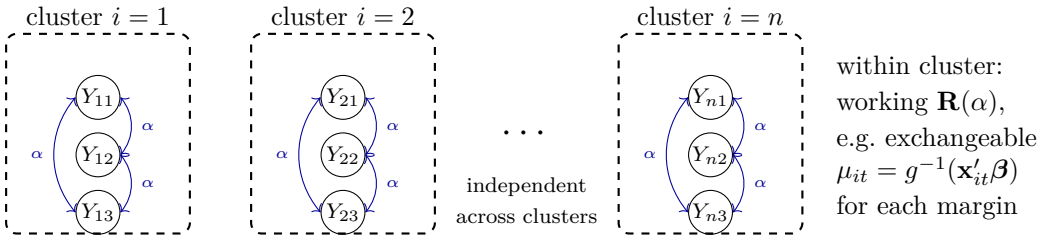


Figure 11.1. n independent clusters with dependent observations; GEE asymptotics are in n.

11.3.2 Longitudinal mental depression example

Example 11.5 (Depression: GEE fit)

For Example 11.3, GEE is close to ML for every working correlation: exchangeable gives $\hat{\beta}_3 = 0.48$ (SE 0.12), $\hat{\beta}_4 = 1.02$ (SE 0.19). Under independence the estimates (intercept -0.028, severe -1.314, new drug -0.060, time 0.482, drug $\times$ time 1.017) are those of ordinary logistic regression on 3×340 observations; the naive SEs $(0.164, 0.146, 0.222, 0.115, 0.189)$ and robust SEs $(0.174, 0.146, 0.229, 0.120, 0.188)$ nearly agree because $\hat{\alpha}_{\text{exch}} = -0.003$.

11.3.3 GEE approach for multinomial responses: insomnia example

For multinomial responses (Lipsitz, Kim and Zhao 1994) one chooses a working correlation for each pair of categories across occasions and uses the $(I-1)\times(I-1)$ multinomial covariance at each occasion.

Example 11.6 (Insomnia: GEE fit)

For (11.5) with independence working correlation, $\hat\beta_1 = 1.038$ (SE 0.168), $\hat\beta_2 = 0.034$ (SE 0.238), $\hat\beta_3 = 0.708$ (SE 0.244), similar to ML.

11.4 Quasi-likelihood and its GEE multivariate extension: details

Motivation. Consistency under a wrong correlation and the sandwich both follow from one fact about mean-zero estimating functions (Lemma 11.8).

For subject i let y_i have mean μ_i, variance function $v(\mu_i)$, and $\eta_i = g(\mu_i) = \mathbf{x}_i'\boldsymbol\beta$. The **quasi-likelihood** (QL) estimates solve

$$\mathbf{u}(\boldsymbol\beta) = \sum_{i=1}^{n} \Big(\frac{\partial \mu_i}{\partial \boldsymbol\beta}\Big) \frac{y_i - \mu_i}{v(\mu_i)} = \mathbf{0}, \qquad \mu_i = g^{-1}(\mathbf{x}_i'\boldsymbol\beta). \tag{11.7}$$

Proposition 11.7 (Motivations for the quasi-score equations)

(a) If y_i has a natural exponential family distribution with variance function v, (11.7) are the GLM likelihood equations.
(b) With $v(\mu_i)$ replaced by known constants v_i, (11.7) are the normal equations for minimizing $\sum_i (y_i-\mu_i)^2/v_i$.
(c) Since $\partial\mu_i/\partial\beta_j = (\partial\mu_i/\partial\eta_i)x_{ij}$, (11.7) depend only on the link and the variance function.

Proof. [Supplement] (a) For $f(y;\theta) = \exp\{y\theta - b(\theta) + c(y)\}$, $\mu = b'(\theta)$ and $\partial\mu/\partial\theta = b''(\theta) = v(\mu)$, so

$$\frac{\partial [y_i\theta_i - b(\theta_i)]}{\partial \beta_j} = (y_i - \mu_i)\frac{\partial\theta_i}{\partial\mu_i}\frac{\partial\mu_i}{\partial\beta_j} = \frac{y_i-\mu_i}{v(\mu_i)}\frac{\partial\mu_i}{\partial\beta_j}.$$

(b) $\partial/\partial\beta_j$ of $\sum_i (y_i-\mu_i)^2/v_i$ is $-2\sum_i (y_i-\mu_i)v_i^{-1}\partial\mu_i/\partial\beta_j$.

(c) Chain rule. ∎

11.4.1 Properties of quasi-likelihood estimators

Definition (Unbiased estimating function)

A function $\mathbf{h}(\mathbf{y};\boldsymbol\beta)$ is an **unbiased estimating function** if $E_{\boldsymbol\beta}[\mathbf{h}(\mathbf{Y};\boldsymbol\beta)] = \mathbf{0}$ for all $\boldsymbol\beta$; the equations $\mathbf{h}(\mathbf{y};\boldsymbol\beta)=\mathbf{0}$ that define $\hat{\boldsymbol\beta}$ are **estimating equations**.

The quasi-score is unbiased whenever $E(y_i) = \mu_i(\boldsymbol\beta)$, whatever the true variance.

Lemma 11.8 (Asymptotics of estimating equations)

Let clusters $i = 1,\dots,n$ be independent, $\boldsymbol\beta \in \Theta \subseteq \mathbb{R}^p$ open, with true value $\boldsymbol\beta_0$, and let $\boldsymbol\gamma$ be a nuisance parameter estimated by $\hat{\boldsymbol\gamma} \xrightarrow{p} \boldsymbol\gamma^*$. Let $\mathbf{U}_n(\boldsymbol\beta;\boldsymbol\gamma) = \sum_{i=1}^n \boldsymbol\psi_i(\boldsymbol\beta;\boldsymbol\gamma)$ and $\mathbf{H}_i =$

$-\partial\boldsymbol{\psi}_i/\partial\boldsymbol{\beta}'$. Assume:
(E1) $E\,\boldsymbol{\psi}_i(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)=\mathbf{0}$ for all i;
(E2) $n^{-1}\sum_i \mathrm{Cov}\left(\boldsymbol{\psi}_i(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)\right)\to\mathbf{B}$, positive definite, and $\sup_i E\|\boldsymbol{\psi}_i(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)\|^{2+\delta}<\infty$ for some $\delta>0$;
(E3) $\boldsymbol{\psi}_i$ is continuously differentiable in $\boldsymbol{\beta}$, and on neighbourhoods $\mathcal{N}$ of $\boldsymbol{\beta}_0$ and $\mathcal{M}$ of $\boldsymbol{\gamma}^*$, $\sup_{\mathcal{N}\times\mathcal{M}}\|n^{-1}\sum_i \mathbf{H}_i(\boldsymbol{\beta};\boldsymbol{\gamma})-\mathbf{A}(\boldsymbol{\beta},\boldsymbol{\gamma})\|\xrightarrow{p}0$ for a continuous $\mathbf{A}(\cdot,\cdot)$ with $\mathbf{A}=\mathbf{A}(\boldsymbol{\beta}_0,\boldsymbol{\gamma}^*)$ non-singular;
(E4) $n^{-1/2}[\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})-\mathbf{U}_n(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)]\xrightarrow{p}\mathbf{0}$.
Then:
(i) for every sufficiently small $r>0$, with probability tending to one, $\mathbf{U}_n(\boldsymbol{\beta};\hat{\boldsymbol{\gamma}})=\mathbf{0}$ has exactly one root $\hat{\boldsymbol{\beta}}$ in the ball $\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le r$, and $\hat{\boldsymbol{\beta}}\xrightarrow{p}\boldsymbol{\beta}_0$;
(ii) $\sqrt{n}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0)\xrightarrow{d}N(\mathbf{0},\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1\prime})$;
(iii) if moreover $n^{-1}\sum_i \boldsymbol{\psi}_i\boldsymbol{\psi}_i'$ converges in probability, uniformly on $\mathcal{N}\times\mathcal{M}$, to a continuous limit equal to $\mathbf{B}$ at $(\boldsymbol{\beta}_0,\boldsymbol{\gamma}^*)$, then with $\hat{\mathbf{A}}=n^{-1}\sum_i \mathbf{H}_i(\hat{\boldsymbol{\beta}};\hat{\boldsymbol{\gamma}})$ and $\hat{\mathbf{B}}=n^{-1}\sum_i \boldsymbol{\psi}_i\boldsymbol{\psi}_i'(\hat{\boldsymbol{\beta}};\hat{\boldsymbol{\gamma}})$, the sandwich $\hat{\mathbf{A}}^{-1}\hat{\mathbf{B}}\hat{\mathbf{A}}^{-1\prime}\xrightarrow{p}\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1\prime}$.

Proof. [Supplement]

Step 1. Set-up. Let $c=\|\mathbf{A}^{-1}\|$. Choose $r>0$ and a neighbourhood $\mathcal{M}'\subseteq\mathcal{M}$ of $\boldsymbol{\gamma}^*$ with $\bar{B}_r=\{\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le r\}\subseteq\mathcal{N}$ and $\|\mathbf{A}(\boldsymbol{\beta},\boldsymbol{\gamma})-\mathbf{A}\|\le 1/(4c)$ on $\bar{B}_r\times\mathcal{M}'$. Let ε_n be the supremum in (E3) and

$$E_n=\Big\{\hat{\boldsymbol{\gamma}}\in\mathcal{M}',\ \ \varepsilon_n\le\tfrac{1}{4c},\ \ \|n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})\|\le\tfrac{r}{2c}\Big\}.$$

By (E1)–(E2) and Chebyshev, $n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)\xrightarrow{p}\mathbf{0}$, hence by (E4) also with $\hat{\boldsymbol{\gamma}}$; so $P(E_n)\to1$.

Step 2. Linearization. For $\boldsymbol{\beta}\in\bar{B}_r$ (convex),

$$n^{-1}\mathbf{U}_n(\boldsymbol{\beta};\hat{\boldsymbol{\gamma}})=n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})-\mathbf{A}_n(\boldsymbol{\beta})(\boldsymbol{\beta}-\boldsymbol{\beta}_0),\qquad \mathbf{A}_n(\boldsymbol{\beta})=\int_0^1 n^{-1}\sum_i \mathbf{H}_i(\boldsymbol{\beta}_0+s(\boldsymbol{\beta}-\boldsymbol{\beta}_0);\hat{\boldsymbol{\gamma}})\,ds,$$

and on E_n, $\|\mathbf{A}_n(\boldsymbol{\beta})-\mathbf{A}\|\le\varepsilon_n+\frac{1}{4c}\le\frac{1}{2c}$ on $\bar{B}_r$.

Step 3. Existence. The fixed points of $F(\boldsymbol{\beta})=\boldsymbol{\beta}+\mathbf{A}^{-1}n^{-1}\mathbf{U}_n(\boldsymbol{\beta};\hat{\boldsymbol{\gamma}})$ are roots, and by Step 2

$$F(\boldsymbol{\beta})-\boldsymbol{\beta}_0=\mathbf{A}^{-1}n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})+\mathbf{A}^{-1}(\mathbf{A}-\mathbf{A}_n(\boldsymbol{\beta}))(\boldsymbol{\beta}-\boldsymbol{\beta}_0),$$

so on E_n, $\|F(\boldsymbol{\beta})-\boldsymbol{\beta}_0\|\le c\cdot\frac{r}{2c}+c\cdot\frac{1}{2c}\cdot r=r$. By Brouwer's theorem F has a fixed point $\hat{\boldsymbol{\beta}}\in\bar{B}_r$.

Step 4. Uniqueness. Two roots $\boldsymbol{\beta}_1,\boldsymbol{\beta}_2\in\bar{B}_r$ give $\mathbf{0}=\tilde{\mathbf{A}}(\boldsymbol{\beta}_1-\boldsymbol{\beta}_2)$ with $\|\tilde{\mathbf{A}}-\mathbf{A}\|\le\frac{1}{2c}$; since $\|\mathbf{A}^{-1}(\tilde{\mathbf{A}}-\mathbf{A})\|\le\frac12$, $\tilde{\mathbf{A}}$ is invertible (Neumann series) and $\boldsymbol{\beta}_1=\boldsymbol{\beta}_2$.

Step 5. Consistency. The argument holds for every $r'\le r$, and by uniqueness the roots coincide, so $P(\|\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0\|\le r')\to1$ for all $r'>0$. This proves (i).

Step 6. Asymptotic normality. On E_n, $\sqrt{n}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0)=\mathbf{A}_n(\hat{\boldsymbol{\beta}})^{-1}\,n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})$. By (E3), continuity and consistency, $\mathbf{A}_n(\hat{\boldsymbol{\beta}})\xrightarrow{p}\mathbf{A}$. By (E4), $n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})=n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)+o_p(1)$, a normalized sum of independent mean-zero vectors; for fixed $\boldsymbol{\lambda}\neq\mathbf{0}$ Lyapunov's condition

$$\frac{\sum_i E|\boldsymbol{\lambda}'\boldsymbol{\psi}_i|^{2+\delta}}{\left(\sum_i \mathrm{Var}(\boldsymbol{\lambda}'\boldsymbol{\psi}_i)\right)^{1+\delta/2}}=\frac{O(n)}{\left(n\,\boldsymbol{\lambda}'\mathbf{B}\boldsymbol{\lambda}\,(1+o(1))\right)^{1+\delta/2}}\to0$$

holds, so by Cramér–Wold $n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)\xrightarrow{d}N(\mathbf{0},\mathbf{B})$, and Slutsky gives (ii).

Step 7. Sandwich. $\hat{\mathbf{A}}\xrightarrow{p}\mathbf{A}$ as in Step 6 and $\hat{\mathbf{B}}\xrightarrow{p}\mathbf{B}$ by the extra assumption; continuity of inversion and multiplication proves (iii). ∎

Theorem 11.9 (Consistency and asymptotic normality of QL estimators)

Suppose the mean model $g(\mu_i) = \mathbf{x}_i'\boldsymbol{\beta}$ is correct at $\boldsymbol{\beta}_0$ (the variance function v may be wrong) and that conditions (E2), (E3) and the condition in (iii) of Lemma 11.8 hold for $\boldsymbol{\psi}_i = (\partial\mu_i/\partial\boldsymbol{\beta})(y_i - \mu_i)/v(\mu_i)$ (they hold, e.g., for bounded covariates, smooth g and v, $v > 0$, and bounded fourth moments of y_i). Then the QL estimator is consistent and asymptotically normal, with

$$\operatorname{Cov}(\hat{\boldsymbol{\beta}}) \approx \mathbf{V}\Big[\sum_i \frac{\partial\mu_i}{\partial\boldsymbol{\beta}} \frac{\operatorname{Var}(Y_i)}{v(\mu_i)^2} \frac{\partial\mu_i}{\partial\boldsymbol{\beta}'}\Big]\mathbf{V}, \qquad \mathbf{V} = \Big[\sum_i \frac{\partial\mu_i}{\partial\boldsymbol{\beta}} \frac{1}{v(\mu_i)} \frac{\partial\mu_i}{\partial\boldsymbol{\beta}'}\Big]^{-1}. \tag{11.8}$$

If $\operatorname{Var}(Y_i) = v(\mu_i)$, this reduces to $\operatorname{Cov}(\hat{\boldsymbol{\beta}}) \approx \mathbf{V}$, the GLM formula.

Proof.

Step 1. (E1) holds since $E(y_i) = \mu_i(\boldsymbol{\beta}_0)$ and the coefficient of $y_i - \mu_i$ is nonrandom; (E4) is vacuous.

Step 2. $\mathbf{H}_i = v(\mu_i)^{-1}(\partial\mu_i/\partial\boldsymbol{\beta})(\partial\mu_i/\partial\boldsymbol{\beta}') - (\partial[v(\mu_i)^{-1}\partial\mu_i/\partial\boldsymbol{\beta}]/\partial\boldsymbol{\beta}')(y_i - \mu_i)$; the second term has mean zero at $\boldsymbol{\beta}_0$, giving $\mathbf{A} = \lim n^{-1}\sum_i v(\mu_i)^{-1}(\partial\mu_i/\partial\boldsymbol{\beta})(\partial\mu_i/\partial\boldsymbol{\beta}')$, and $\operatorname{Cov}(\boldsymbol{\psi}_i) = v(\mu_i)^{-2}\operatorname{Var}(Y_i)(\partial\mu_i/\partial\boldsymbol{\beta})(\partial\mu_i/\partial\boldsymbol{\beta}')$ gives $\mathbf{B}$.

Step 3. Lemma 11.8(ii) gives $\operatorname{Cov}(\hat{\boldsymbol{\beta}}) \approx n^{-1}\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$, which is (11.8); if $\operatorname{Var}(Y_i) = v(\mu_i)$ then $\mathbf{B} = \mathbf{A}$ and it collapses to $\mathbf{V}$. ■

11.4.2 GEE methodology: technical details

For cluster i let $\mathbf{y}_i = (y_{i1}, \dots, y_{iT_i})'$, $\boldsymbol{\mu}_i = E(\mathbf{Y}_i)$, $\mathbf{X}_i$ the $T_i \times p$ matrix with rows $\mathbf{x}_{it}'$, and $\eta_{it} = g(\mu_{it}) = \mathbf{x}_{it}'\boldsymbol{\beta}$. Each margin has

$$f(y_{it}; \theta_{it}, \phi) = \exp\{[y_{it}\theta_{it} - b(\theta_{it})]/\phi + c(y_{it}, \phi)\},$$

so $\mu_{it} = b'(\theta_{it})$ and $\operatorname{Var}(Y_{it}) = \phi\, b''(\theta_{it})$.

Definition (Generalized estimating equations)

Let $\mathbf{B}_i = \operatorname{diag}(b''(\theta_{i1}), \dots, b''(\theta_{iT_i}))$ and let $\mathbf{R}(\boldsymbol{\alpha})$ be a working correlation matrix. The **working covariance** is

$$\mathbf{V}_i = \phi\, \mathbf{B}_i^{1/2}\mathbf{R}(\boldsymbol{\alpha})\mathbf{B}_i^{1/2} \tag{11.9}$$

(= $\operatorname{Cov}(\mathbf{Y}_i)$ if $\mathbf{R}$ is the true correlation). Let $\boldsymbol{\Delta}_i = \operatorname{diag}(\partial\theta_{it}/\partial\eta_{it})$ (the identity for the canonical link) and $\mathbf{D}_i = \partial\boldsymbol{\mu}_i/\partial\boldsymbol{\beta}' = \mathbf{B}_i\boldsymbol{\Delta}_i\mathbf{X}_i$ ($T_i \times p$). The GEE estimator $\hat{\boldsymbol{\beta}}$ solves

$$\mathbf{u}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \mathbf{D}_i'\mathbf{V}_i^{-1}(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}. \tag{11.10}$$

$\mathbf{D}_i = \mathbf{B}_i\boldsymbol{\Delta}_i\mathbf{X}_i$ is the chain rule, and the scalar ϕ cancels from (11.10).

Proposition 11.10 (Independence working correlation gives the GLM fit)

With $\mathbf{R} = \mathbf{I}$, (11.10) becomes $\sum_i \mathbf{X}_i'\boldsymbol{\Delta}_i(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ (for the canonical link, $\sum_i \mathbf{X}_i'(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$): the ordinary GLM likelihood equations treating all y_{it} as independent.

Proof. With $\mathbf{V}_i = \phi\mathbf{B}_i$, $\mathbf{D}_i'\mathbf{V}_i^{-1} = \mathbf{X}_i'\boldsymbol{\Delta}_i\mathbf{B}_i(\phi\mathbf{B}_i)^{-1} = \phi^{-1}\mathbf{X}_i'\boldsymbol{\Delta}_i$; drop $1/\phi$ and compare with Proposition 11.7(a) summed over all (i, t). ■

Definition (Moment estimators of the working parameters)

With Pearson residuals $e_{it} = (y_{it} - \hat\mu_{it})/b''(\hat\theta_{it})^{1/2}$, $N = \sum_i T_i$ (Liang and Zeger 1986):

$$\hat\phi = \frac{1}{N-p}\sum_i\sum_t e_{it}^2, \qquad \hat\alpha_{\text{exch}} = \frac{\sum_i \sum_{t<s} e_{it}e_{is}}{\hat\phi\big[\sum_i T_i(T_i-1)/2 - p\big]},$$

$$\hat\alpha_{\text{AR1}} = \frac{\sum_i \sum_{t\le T_i-1} e_{it}e_{i,t+1}}{\hat\phi\big[\sum_i (T_i-1) - p\big]}, \qquad \hat R_{ts} = \frac{1}{n\hat\phi}\sum_i e_{it}e_{is} \ \ \text{(unstructured)}.$$

The algorithm alternates a modified Fisher scoring step for $\boldsymbol\beta$ given $(\boldsymbol\alpha, \phi)$,

$$\boldsymbol\beta^{(k+1)} = \boldsymbol\beta^{(k)} + \Big[\sum_i \mathbf D_i'\mathbf V_i^{-1}\mathbf D_i\Big]^{-1}\sum_i \mathbf D_i'\mathbf V_i^{-1}(\mathbf y_i - \boldsymbol\mu_i),$$

with moment updates of $(\boldsymbol\alpha, \phi)$ from the current residuals.

Theorem 11.11 (Liang–Zeger: consistency and asymptotic normality of GEE)

Assume:
(G1) the marginal mean model is correct: $E(\mathbf Y_i) = \boldsymbol\mu_i(\boldsymbol\beta_0)$ for all i (the working correlation may be wrong);
(G2) $\sqrt n(\hat{\boldsymbol\alpha} - \boldsymbol\alpha^*) = O_p(1)$ for some $\boldsymbol\alpha^*$ (not necessarily a "true" value);
(G3) conditions (E2), (E3) and the condition in (iii) of Lemma 11.8 hold for $\boldsymbol\psi_i(\boldsymbol\beta;\boldsymbol\alpha) = \mathbf D_i'\mathbf V_i^{-1}(\boldsymbol\alpha)(\mathbf y_i - \boldsymbol\mu_i)$, and $n^{-1}\sum_i \partial\boldsymbol\psi_i(\boldsymbol\beta_0;\boldsymbol\alpha)/\partial\boldsymbol\alpha' \xrightarrow{p} \mathbf 0$ uniformly for $\boldsymbol\alpha$ near $\boldsymbol\alpha^*$.
Then, as the number of clusters $n \to \infty$, $\hat{\boldsymbol\beta} \xrightarrow{p} \boldsymbol\beta_0$ and $\sqrt n(\hat{\boldsymbol\beta} - \boldsymbol\beta_0) \xrightarrow{d} N(\mathbf 0, \mathbf V_G)$, where $\mathbf V_G = \lim_n \mathbf V_{G,n}$,

$$\mathbf V_{G,n} = n\Big[\sum_i \mathbf D_i'\mathbf V_i^{-1}\mathbf D_i\Big]^{-1}\Big[\sum_i \mathbf D_i'\mathbf V_i^{-1}\operatorname{Cov}(\mathbf Y_i)\mathbf V_i^{-1}\mathbf D_i\Big]\Big[\sum_i \mathbf D_i'\mathbf V_i^{-1}\mathbf D_i\Big]^{-1},$$

evaluated at $(\boldsymbol\beta_0, \boldsymbol\alpha^*)$. The sandwich estimator $\hat{\mathbf V}_{G,n}/n$, which substitutes $\hat{\boldsymbol\beta}$, $\hat{\boldsymbol\alpha}$, $\hat\phi$ and replaces $\operatorname{Cov}(\mathbf Y_i)$ by $(\mathbf y_i - \hat{\boldsymbol\mu}_i)(\mathbf y_i - \hat{\boldsymbol\mu}_i)'$, is consistent for $\operatorname{Cov}(\hat{\boldsymbol\beta})$.

Proof. [Supplement] We verify the hypotheses of Lemma 11.8 with nuisance $\boldsymbol\gamma = \boldsymbol\alpha$ (ϕ cancels from (11.10)).

Step 1. (E1). At $\boldsymbol\beta_0$, $\mathbf D_i$ and $\mathbf V_i(\boldsymbol\alpha^*)$ are nonrandom and $E(\mathbf y_i - \boldsymbol\mu_i) = \mathbf 0$ by (G1), so $E\boldsymbol\psi_i(\boldsymbol\beta_0;\boldsymbol\alpha^*) = \mathbf 0$ *for any* $\boldsymbol\alpha^*$: a wrong working correlation does not destroy consistency.

Step 2. $\mathbf A$ and $\mathbf B$. Differentiating,

$$\mathbf H_i = \mathbf D_i'\mathbf V_i^{-1}\mathbf D_i - \sum_j \Big[\frac{\partial(\mathbf D_i'\mathbf V_i^{-1})}{\partial\beta_j}(\mathbf y_i - \boldsymbol\mu_i)\Big]\mathbf e_j',$$

whose second term has mean zero at $\boldsymbol\beta_0$, so $\mathbf A = \lim n^{-1}\sum_i \mathbf D_i'\mathbf V_i^{-1}\mathbf D_i$; and $\operatorname{Cov}(\boldsymbol\psi_i) = \mathbf D_i'\mathbf V_i^{-1}\operatorname{Cov}(\mathbf Y_i)\mathbf V_i^{-1}\mathbf D_i$ averages to $\mathbf B$. (E2), (E3) hold by (G3).

Step 3. (E4). By the mean value theorem, for some $\bar{\boldsymbol\alpha}$ between $\hat{\boldsymbol\alpha}$ and $\boldsymbol\alpha^*$ (componentwise),

$$n^{-1/2}\big[\mathbf U_n(\boldsymbol\beta_0;\hat{\boldsymbol\alpha}) - \mathbf U_n(\boldsymbol\beta_0;\boldsymbol\alpha^*)\big] = \Big[n^{-1}\sum_i \frac{\partial\boldsymbol\psi_i(\boldsymbol\beta_0;\bar{\boldsymbol\alpha})}{\partial\boldsymbol\alpha'}\Big]\sqrt n(\hat{\boldsymbol\alpha} - \boldsymbol\alpha^*).$$

Each $\partial\boldsymbol\psi_i/\partial\alpha_k = \mathbf D_i'(\partial\mathbf V_i^{-1}/\partial\alpha_k)(\mathbf y_i - \boldsymbol\mu_i)$ has mean zero at $\boldsymbol\beta_0$; by (G3) the bracket is $o_p(1)$ and by (G2) the last factor is $O_p(1)$, so the product is $o_p(1)$: estimating $\boldsymbol\alpha$ has no first-order effect.

Step 4. Lemma 11.8(i)–(ii) gives consistency and $\sqrt n(\hat{\boldsymbol\beta} - \boldsymbol\beta_0) \xrightarrow{d} N(\mathbf 0, \mathbf A^{-1}\mathbf B\mathbf A^{-1})$ with $\mathbf A^{-1}\mathbf B\mathbf A^{-1} = \lim \mathbf V_{G,n}$. Lemma 11.8(iii) gives consistency of the sandwich with $\hat{\mathbf B} = n^{-1}\sum_i \mathbf D_i'\mathbf V_i^{-1}(\mathbf y_i - \hat{\boldsymbol\mu}_i)(\mathbf y_i -$

$\hat{\boldsymbol{\mu}}_i)'\mathbf{V}_i^{-1}\mathbf{D}_i$ at $(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\alpha}})$; replacing $\hat{\mathbf{A}}$ by $n^{-1}\sum \mathbf{D}_i'\mathbf{V}_i^{-1}\mathbf{D}_i$ at the estimates changes it by $o_p(1)$ (Step 2). ■

Corollary 11.12 (Model-based covariance)

If the working covariance is correct, $\mathrm{Cov}(\mathbf{Y}_i) = \mathbf{V}_i$, then $\mathbf{V}_{G,n}/n = \left(\sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}\mathbf{D}_i\right)^{-1}$.

Proof. Substituting $\mathrm{Cov}(\mathbf{Y}_i) = \mathbf{V}_i$, the middle factor equals $\sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}\mathbf{D}_i$ and cancels one outer inverse. ■

Theorem 11.13 (Optimality of the true covariance as weight)

Consider all estimating equations $\sum_i \mathbf{D}_i'\mathbf{W}_i(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ with nonrandom positive definite weights $\mathbf{W}_i$, and let $\boldsymbol{\Sigma}_i = \mathrm{Cov}(\mathbf{Y}_i)$. Their asymptotic covariance satisfies, in the Loewner order,

$$\mathbf{G}^{-1}\mathbf{M}\mathbf{G}^{-1} \;\geq\; \mathbf{F}^{-1}, \qquad \mathbf{G} = \sum_i \mathbf{D}_i'\mathbf{W}_i\mathbf{D}_i, \quad \mathbf{M} = \sum_i \mathbf{D}_i'\mathbf{W}_i\boldsymbol{\Sigma}_i\mathbf{W}_i\mathbf{D}_i, \quad \mathbf{F} = \sum_i \mathbf{D}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{D}_i,$$

with equality when $\mathbf{W}_i = \boldsymbol{\Sigma}_i^{-1}$. So a correct working covariance gives the most efficient GEE estimator.

Proof. [Supplement]

Step 1. With $\mathbf{K}_i = \begin{pmatrix}\mathbf{W}_i\mathbf{D}_i & \boldsymbol{\Sigma}_i^{-1}\mathbf{D}_i\end{pmatrix}$, the matrix $\mathbf{K}_i'\boldsymbol{\Sigma}_i\mathbf{K}_i = \begin{pmatrix}\mathbf{D}_i'\mathbf{W}_i\boldsymbol{\Sigma}_i\mathbf{W}_i\mathbf{D}_i & \mathbf{D}_i'\mathbf{W}_i\mathbf{D}_i \\ \mathbf{D}_i'\mathbf{W}_i\mathbf{D}_i & \mathbf{D}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{D}_i\end{pmatrix}$ is positive semidefinite; summing, $\left(\begin{smallmatrix}\mathbf{M} & \mathbf{G} \\ \mathbf{G} & \mathbf{F}\end{smallmatrix}\right) \geq 0$.

Step 2. Since $\mathbf{F} > 0$, the Schur complement $\mathbf{M} - \mathbf{G}\mathbf{F}^{-1}\mathbf{G} \geq 0$.

Step 3. Congruence by the symmetric $\mathbf{G}^{-1}$ gives $\mathbf{G}^{-1}\mathbf{M}\mathbf{G}^{-1} \geq \mathbf{F}^{-1}$; if $\mathbf{W}_i = \boldsymbol{\Sigma}_i^{-1}$ then $\mathbf{G} = \mathbf{M} = \mathbf{F}$. ■

Correlation is an awkward association measure for binary data

The Fréchet bounds $\max(0, p_t + p_s - 1) \leq P(Y_t = Y_s = 1) \leq \min(p_t, p_s)$ restrict $\mathrm{Corr}(Y_t, Y_s)$ to a range depending on the means (e.g. at most $1/3$ when $p_t = 0.1$, $p_s = 0.5$). Odds ratios are variation-independent of the means; *alternating logistic regressions* (Carey, Zeger and Diggle 1993) alternate a GEE step for $\boldsymbol{\beta}$ with a logistic step for the log odds ratios.

11.5 Markov chains: transitional modeling

Motivation. Modelling each response given earlier ones factors the joint distribution into transitions, fitted by ordinary multinomial or logistic models.

11.5.1 Transitional models

For a discrete-time, discrete-state process $\{Y_t\}$, **transitional models** describe Y_t given $y_0, \ldots, y_{t-1}$ (and covariates), via

$$f(y_0, \ldots, y_T) = f(y_0)f(y_1 \mid y_0)f(y_2 \mid y_0, y_1)\cdots f(y_T \mid y_0, \ldots, y_{T-1}).$$

11.5.2 First-order Markov chains

Definition (Markov chain)

$\{Y_t\}$ is a (first-order) **Markov chain** if, for all t, Y_{t+1} given $(Y_0, \dots, Y_t)$ has the same distribution as Y_{t+1} given Y_t alone. It is a **kth-order** chain if Y_{t+1} given $(Y_0, \dots, Y_t)$ depends only on $(Y_{t-k+1}, \dots, Y_t)$. The **transition probabilities** are $\pi_{j|i}(t) = P(Y_t = j \mid Y_{t-1} = i)$, with $\sum_j \pi_{j|i}(t) = 1$; they form the one-step $I \times I$ transition matrix.

Theorem 11.14 (Markov factorization and its loglinear form)

(a) $\{Y_0, \dots, Y_T\}$ is a Markov chain if and only if

$$f(y_0, \dots, y_T) = f(y_0) f(y_1 \mid y_0) f(y_2 \mid y_1) \cdots f(y_T \mid y_{T-1}). \tag{11.11}$$

(b) A positive joint distribution is that of a Markov chain if and only if it satisfies the loglinear model $(Y_0Y_1, Y_1Y_2, \dots, Y_{T-1}Y_T)$, i.e. $\log \pi(y_0, \dots, y_T) = \sum_{t=1}^{T} h_t(y_{t-1}, y_t)$ for some functions h_t.

Proof. [Supplement]

Step 1. (a) The Markov property turns the general factorization into (11.11). Conversely, summing (11.11) over $y_{t+1}, \dots, y_T$ gives $f(y_0, \dots, y_t) = f(y_0) \prod_{s \le t} f(y_s \mid y_{s-1})$, and the ratio of these for $t+1$ and t is $f(y_{t+1} \mid y_t)$.

Step 2. (b, $\Rightarrow$) Take logs of (11.11), absorbing $\log f(y_0)$ into h_1.

Step 3. (b, $\Leftarrow$) If $\pi(\mathbf{y}) = \prod_t g_t(y_{t-1}, y_t)$, let $G_t(y_t) = \sum_{y_{t+1}, \dots, y_T} \prod_{s > t} g_s(y_{s-1}, y_s)$ ($G_T \equiv 1$). Then

$$P(y_{t+1} \mid y_0, \dots, y_t) = \frac{g_{t+1}(y_t, y_{t+1}) G_{t+1}(y_{t+1})}{G_t(y_t)},$$

since the factors $\prod_{s \le t} g_s$ cancel; this depends only on (y_t, y_{t+1}). ∎

Adding three-factor terms $(Y_{t-2}Y_{t-1}Y_t)$ gives a second-order chain.

Theorem 11.15 (ML estimation of transition probabilities)

Let $n_{ij}(t)$ be the number of transitions from state i at time $t-1$ to state j at time t, $n_{i+}(t) = \sum_j n_{ij}(t)$, and $\{n_{i0}\}$ the initial counts, multinomial with probabilities $\{\pi_{i0}\}$. For n independent subjects following a Markov chain, the likelihood is proportional to

$$\Big[\prod_{i=1}^{I} \pi_{i0}^{n_{i0}}\Big] \prod_{t=1}^{T} \prod_{i=1}^{I} \prod_{j=1}^{I} \pi_{j|i}(t)^{n_{ij}(t)}, \tag{11.12}$$

and the ML estimates are $\hat{\pi}_{j|i}(t) = n_{ij}(t)/n_{i+}(t)$ (Anderson and Goodman 1957). For a time-homogeneous chain, $\pi_{j|i}(t) \equiv \pi_{j|i}$, the ML estimate is $\hat{\pi}_{j|i} = \sum_t n_{ij}(t) / \sum_t n_{i+}(t)$.

Proof.

Step 1. Each subject contributes $\pi_{y_0 0} \prod_t \pi_{y_t|y_{t-1}}(t)$ by (11.11); grouping factors gives (11.12).

Step 2. [Supplement] The likelihood factors into separate multinomial kernels $\prod_j \pi_{j|i}(t)^{n_{ij}(t)}$ with unrelated parameters, each maximized separately.

Step 3. For a kernel with counts n_j, $m = \sum_j n_j > 0$, Jensen's inequality gives

$$\sum_j n_j \log \pi_j - \sum_j n_j \log \frac{n_j}{m} = m \sum_j \frac{n_j}{m} \log \frac{\pi_j}{n_j/m} \le m \log \sum_j \pi_j = 0,$$

with equality iff $\pi_j = n_j/m$.

Step 4. Under time homogeneity the kernels for fixed i merge, with counts $\sum_t n_{ij}(t)$. ■

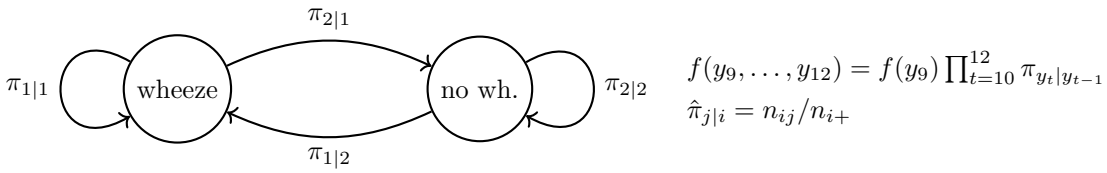


Figure 11.2. A two-state first-order Markov chain.

11.5.3 Respiratory illness example

Example 11.16 (Wheeze at ages 9–12)

Children in a Harvard air-pollution study were examined annually at ages 9–12 for wheeze (2^4 table, $n = 1019$). The first-order chain ($G^2 = 122.9$, df $= 8$) and second-order chain ($G^2 = 23.9$, df $= 4$) fit poorly, perhaps due to subject heterogeneity. All six pairwise associations fit well ($G^2 = 1.5$, df $= 5$); a common log odds ratio for pairs one year apart (1.75) and another for pairs further apart (1.04) also fits ($G^2 = 2.3$, df $= 9$).

11.5.4 Transitional models with explanatory variables

With covariates, $f(y_1,\ldots,y_T;\mathbf{x}) = f(y_1;\mathbf{x})\prod_{t\geq 2} f(y_t \mid y_1,\ldots,y_{t-1};\mathbf{x})$, and for binary y the **regressive logistic model** (Bonney 1987) is

$$f(y_t \mid y_1,\ldots,y_{t-1};\mathbf{x}_t) = \frac{\exp\left[y_t(\alpha + \beta_1 y_1 + \cdots + \beta_{t-1}y_{t-1} + \boldsymbol{\beta}'\mathbf{x}_t)\right]}{1+\exp(\alpha + \beta_1 y_1 + \cdots + \beta_{t-1}y_{t-1} + \boldsymbol{\beta}'\mathbf{x}_t)}, \qquad y_t = 0, 1.$$

First-order Markov structure drops $y_1,\ldots,y_{t-2}$. The meaning of $\boldsymbol{\beta}$ depends on which past responses are conditioned on, unlike marginal effects.

Proposition 11.17 (Fitting transitional models by ordinary logistic regression)

Conditional on the first response(s), the likelihood of a regressive logistic model is the product, over subjects and transitions, of Bernoulli likelihoods with the stated logits. Hence the conditional ML estimates are obtained by fitting an ordinary logistic regression that treats each transition as a separate independent observation, with previous responses as predictors.

Proof. [Supplement] Subject i contributes $f(y_{i1};\mathbf{x})\prod_{t\geq 2} f(y_{it} \mid y_{i1},\ldots,y_{i,t-1};\mathbf{x}_{it})$. Conditioning on the first factor, each remaining factor is $\text{expit}(\eta_{it})^{y_{it}}[1-\text{expit}(\eta_{it})]^{1-y_{it}}$ with η_{it} linear in the parameters and known regressors (past responses, $\mathbf{x}_{it}$). Independence across subjects makes the product over all (i,t) a logistic-regression likelihood with one observation per transition. ■

11.5.5 Child's respiratory illness and maternal smoking

Example 11.18 (Respiratory illness and maternal smoking)

Children were assessed annually at ages 7–10 for respiratory illness; $s = 1$ if the mother smoked. Using three transitions per child (12 binomials over (s, t, y_{t-1})), the ML fit is

$$\text{logit}\,\hat{P}(Y_t = 1) = -0.293 + 0.296s - 0.243t + 2.211y_{t-1}, \qquad t = 8, 9, 10,$$

with SEs $(0.846, 0.156, 0.095, 0.158)$ and $G^2 = 3.1$ (df $= 8$). The previous response has a strong effect; the smoking effect is marginal (LR 3.55, $P = 0.06$).